1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

K60 MLB

REV	ECN	DESCRIPTION OF REVISION	CK APPD	DATE
				2011-02-08

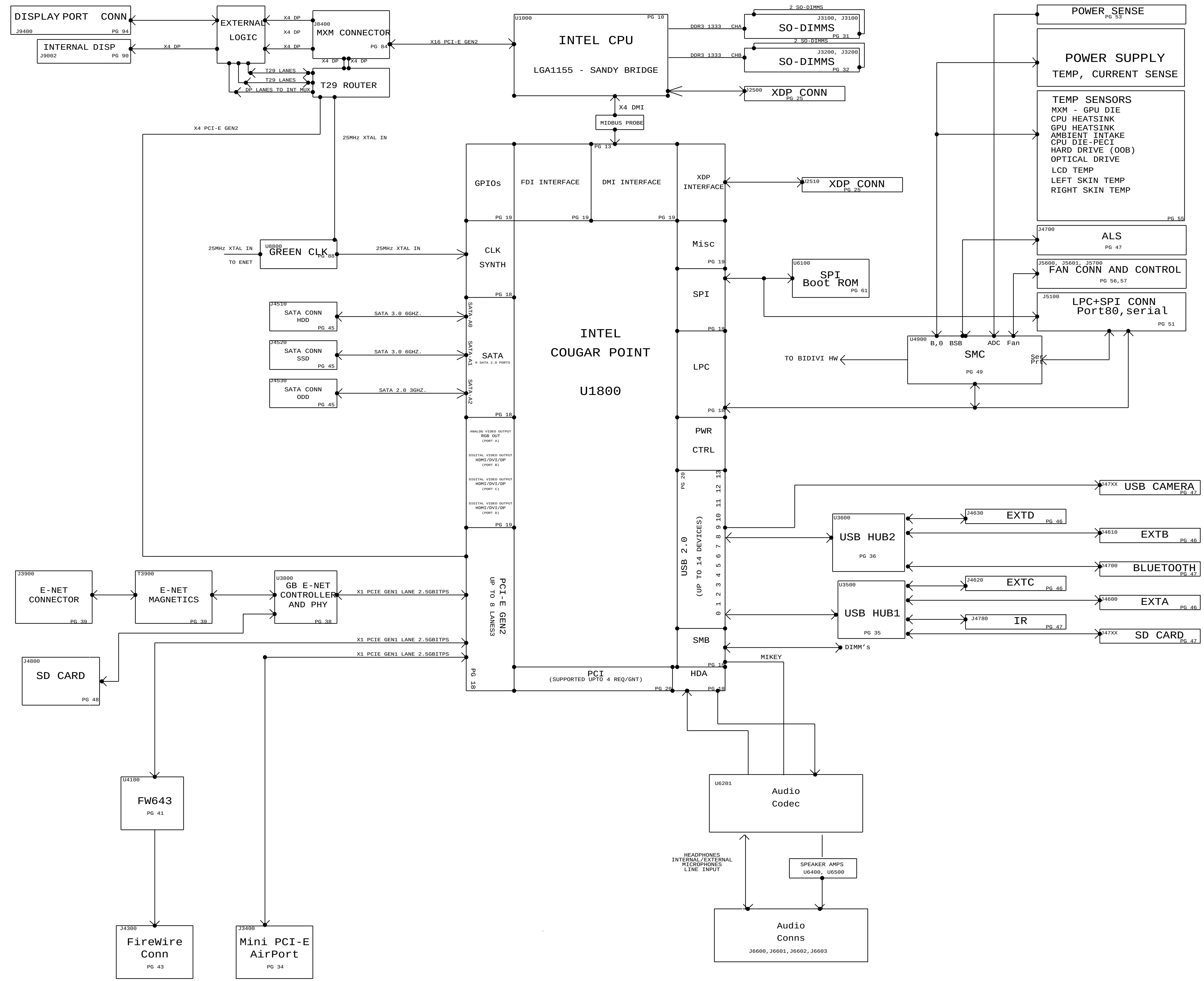
LAST_MODIFIED=Tue Feb 8 14:39:56 2011

DRAWING
TITLE=K22
ABBREV=DRAWING
LAST_MODIFIED=Tue Feb 8 14:39:56 2011

DRAWING TITLE: SCH,K60,MLB

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D
REVISION: 11.1.0
BRANCH:



SYNC_MASTER=K60_SIJI SYNC_DATE=01/06/2011

PAGE TITLE
System Block Diagram

Apple Inc.

DRAWING NUMBER	SIZE
051-8115	D

REVISION 11.1.0

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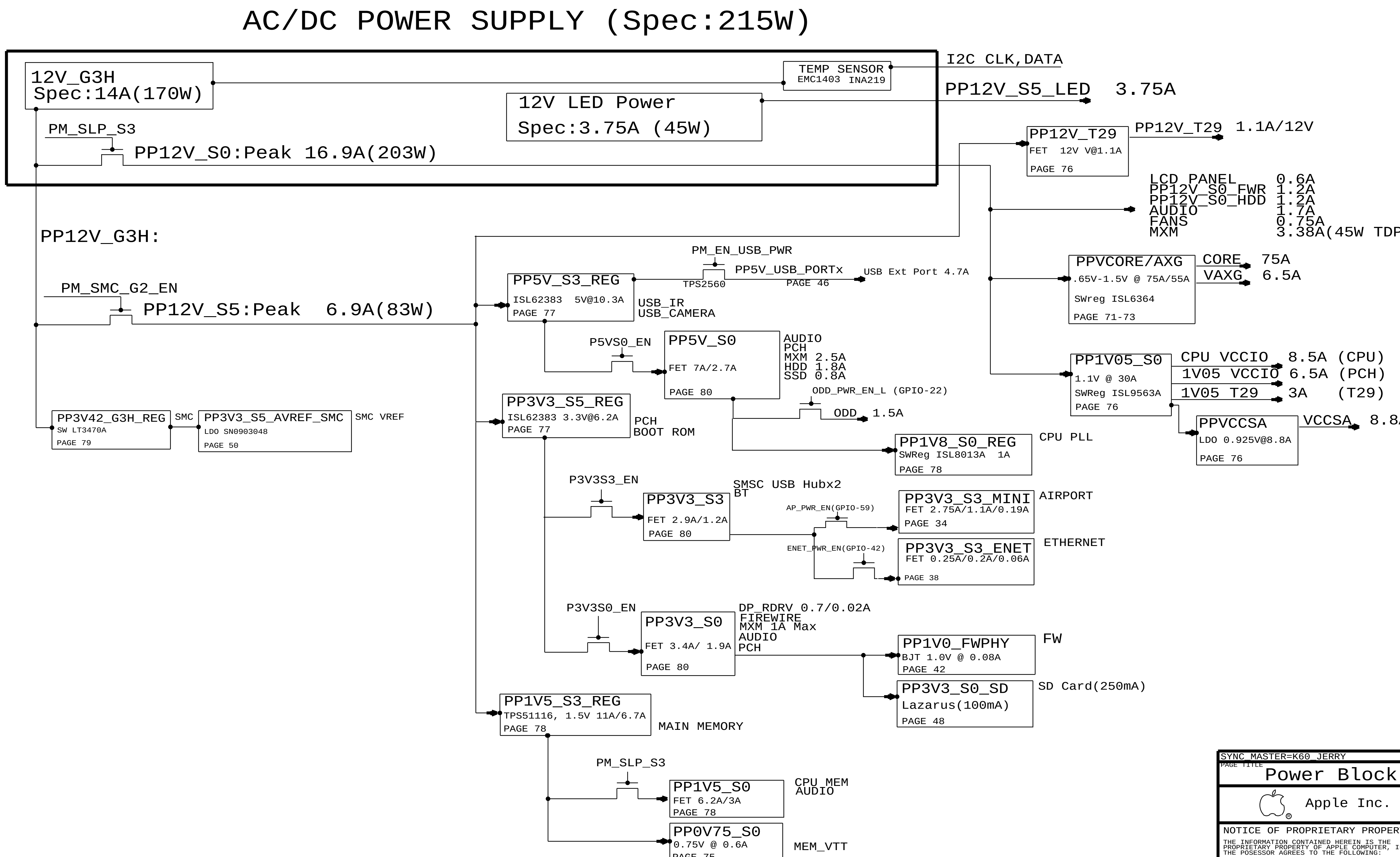

SYNC_MASTER=K60_JERRY SYNC_DATE=01/06/2011

PAGE TITLE

Power Block Diagram

Apple Inc.

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-0801	PCBA,MLB,DEV,K60	DEVELOPMENT,DEV_GROUP
639-1767	PCBA,MLB,K60,2.5G,4C,PRQ,P2_ODD	K60,2P5GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,YES_DBG
639-1820	PCBA,MLB,K60,2.7G,4C,PRQ,P2_ODD	K60,2P7GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,YES_DBG
639-1821	PCBA,MLB,K60,2.8G,4C,PRQ,P2_ODD	K60,2P8GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,YES_DBG
639-2160	PCBA,MLB,K60,2.5G,4C,PRQ,P2_ODD,NO_DBG	K60,2P5GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,NO_DBG
639-2159	PCBA,MLB,K60,2.7G,4C,PRQ,P2_ODD,NO_DBG	K60,2P7GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,NO_DBG
639-2161	PCBA,MLB,K60,2.8G,4C,PRQ,P2_ODD,NO_DBG	K60,2P8GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P2,NO_DBG
639-2118	PCBA,MLB,K60,2.5G,4C,PRQ,P1_ODD	K60,2P5GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,YES_DBG
639-2122	PCBA,MLB,K60,2.7G,4C,PRQ,P1_ODD	K60,2P7GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,YES_DBG
639-2119	PCBA,MLB,K60,2.8G,4C,PRQ,P1_ODD	K60,2P8GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,YES_DBG
639-2132	PCBA,MLB,K60,2.5G,4C,PRQ,P1_ODD,NO_DBG	K60,2P5GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,NO_DBG
639-2133	PCBA,MLB,K60,2.7G,4C,PRQ,P1_ODD,NO_DBG	K60,2P7GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,NO_DBG
639-2134	PCBA,MLB,K60,2.8G,4C,PRQ,P1_ODD,NO_DBG	K60,2P8GHZ_SNB_CPU_PRQ,BASIC1,BASIC2,CPUVCORE-3PH,ODD_SATA:P1,NO_DBG

BOM GROUPS

BOM GROUP	BOM OPTIONS
BASIC1	COMMON,ALTERNATE,MXM,FCIM,CPU_1V5_SENSE,CPU_VCCSA_SENSE,1V05_PCH_SENSE,HUB_USX2061,PRODUCTION,VAXG,SSD
BASIC2	AP,BT,IR,T29
DEV_GROUP	VREFMRGN_A,VREFMRGN_B,DIMM_1V5_SENSE
YES_DBG	XDP,XDP_CONN,XDP_CPU_BPM,MOJOMUX:YES,LPCPLUS:YES
NO_DBG	MOJOMUX:NO,LPCPLUS:NO

CPU SOCKET & ILM SUB-BOMS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0071	1	SOCKET,LGA1155,CPU-LF	U1000	CRITICAL	TYCO_SOCKET
604-1474	1	ASSY,PURCHASED,ILM,TYCO	ILM	CRITICAL	TYCO_SOCKET
511S0073	1	SOCKET,LGA1155,CPU-LF	U1000	CRITICAL	MOLEX_SOCKET
604-1161	1	ASSY,PURCHASED,ILM,MOLEX	ILM	CRITICAL	MOLEX_SOCKET

BOM NUMBER	BOM NAME	BOM OPTIONS
085-2452	SUB ASSY,CPU SOCKET,K60,TYCO	TYCO_SOCKET
085-2453	SUB ASSY,CPU SOCKET,K60,MOLEX	MOLEX_SOCKET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
085-2452	1	TYCO CPU SOCKET AND ILM	SKT_ILM	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
085-2453	085-2452		SKT_ILM	MOLEX ALTERNATE

BOARD STACK-UP

TOP	SIGNAL
2	GROUND
3	SIGNAL
4	POWER
5	POWER
6	SIGNAL
7	GROUND
BOTTOM	SIGNAL

COMMON

	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
	337S4088	1	IC,COUGAR POINT,SLJ4F,BD82Z68,PRQ,B3	U1800	CRITICAL	
	353S3055	1	IC,PI3VEDP212,X2 DP MUX,QFN	U9390	CRITICAL	
	338S0753	1	IC,FW643,1394B_PCIE,PHY/LINK	U4100	CRITICAL	
	825-7122	1	MLB LABEL,48.0X4.8	X14	CRITICAL	
	343S0534	1	IC,BCM57765,ENET&SD,8X8	U3900	CRITICAL	
RAW: 335S0807	341T0184	1	FLASH,EFI BOOTROM,K60/K62	U6100	CRITICAL	
RAW: 335S0539	341T0328	1	SFLASH ENET 2MBIT,CIV	U3990	CRITICAL	
	338S0945	1	T29 ROUTER, IC ASSP	U9700	CRITICAL	T29
RAW: 335S0550	341T0257	1	IC,T29,SERIAL EEPROM	U9790	CRITICAL	T29
RAW: 337S3997	341T0326	1	IC,MCU,32B,LPC1112A,16KB/2KB,HVQFN25	U9330	CRITICAL	T29
RAW: 335S0709	341T0330	1	IC,MXM SYS ROM,24C02	U8570	CRITICAL	
RAW: 338S0878	341T0185	1	IC,SMC,K60	U4900	CRITICAL	K60

CPUS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4043	1	SNB,SR00S,PRQ,D2,2.5,65W,4+1,6M,LGA	CPU	CRITICAL	2P5GHZ_SNB_CPU_PRQ
337S4062	1	SNB,SR009,PRQ,D2,2.7,65W,4+1,6M,LGA	CPU	CRITICAL	2P7GHZ_SNB_CPU_PRQ
337S4061	1	SNB,SR00E,PRQ,D2,2.8,65W,4+1,8M,LGA	CPU	CRITICAL	2P8GHZ_SNB_CPU_PRQ

K60 PARTS

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-8115	1	SCH,MLB,K60	SCH1		K60
820-2641	1	PCBF,MLB,K60	MLB1		K60

K60 ALTERNATE PARTS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
128S0298	128S0293			330UF
371S0679	371S0652			PIN DIODE
377S0107	377S0066			USB DIODE
376S0972	376S0612			ROHM TRA-BJT

SYNC_MASTER=K60_AARON SYNC_DATE=N/A

PAGE TITLE: BOM Configuration

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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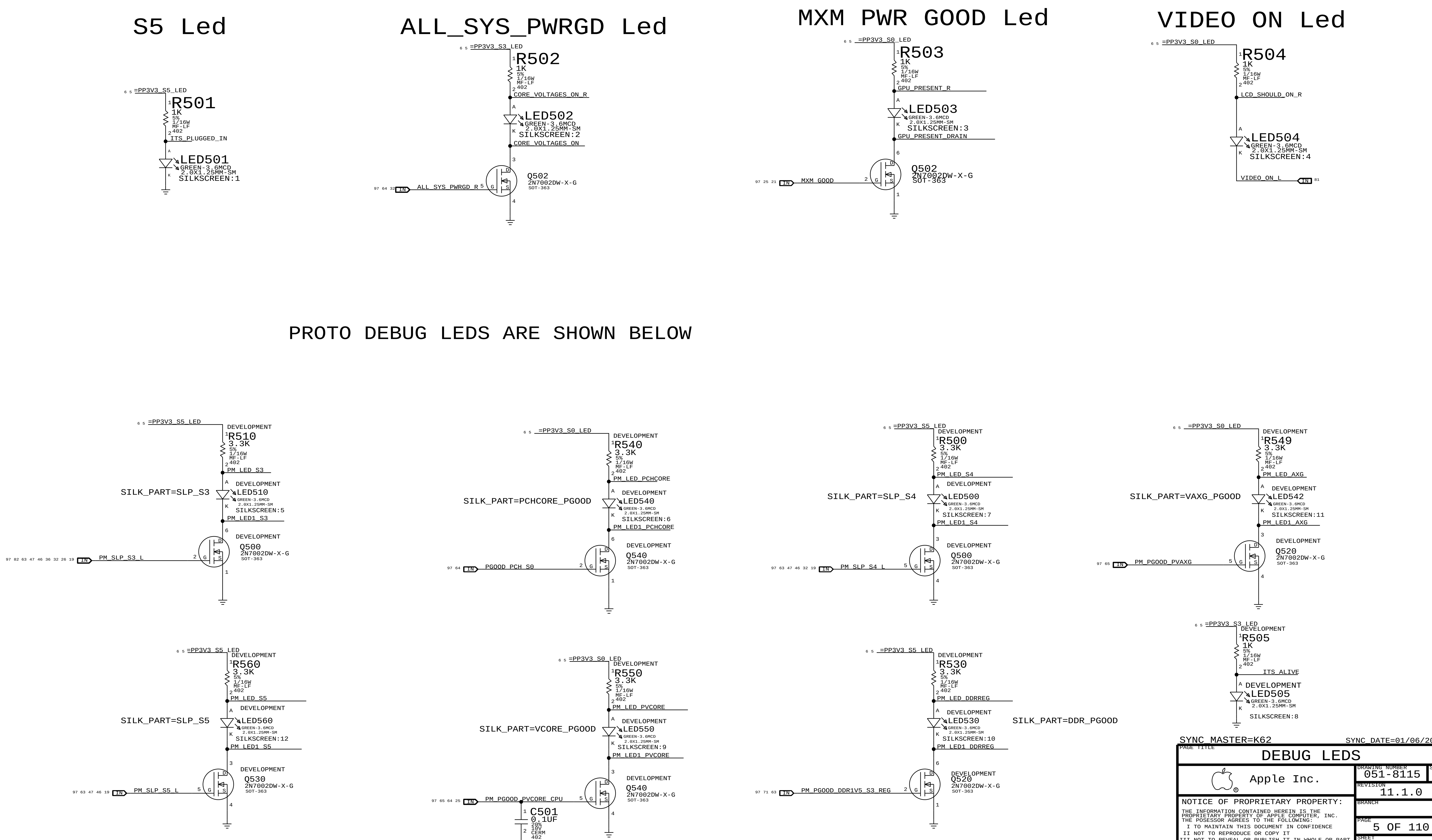

SYNC_MASTER=K62
SYNC_DATE=01/06/2011

PAGE TITLE
DEBUG LEDS

Apple Inc.

DRAWING NUMBER 051-8115
SIZE D
REVISION 11.1.0
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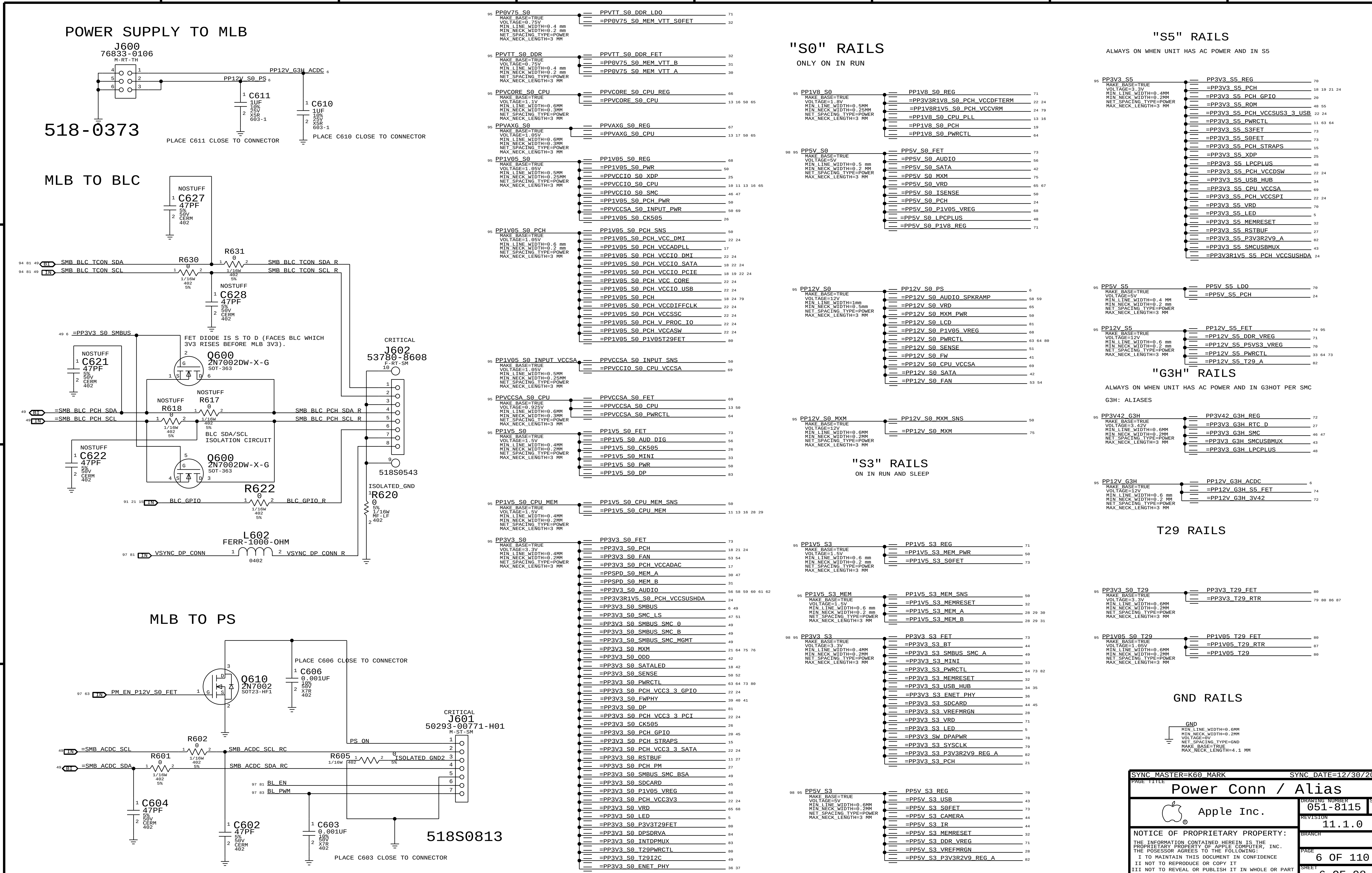

SYNC_MASTER=K60_MARK SYNC_DATE=12/30/2010

PAGE TITLE
Power Conn / Alias

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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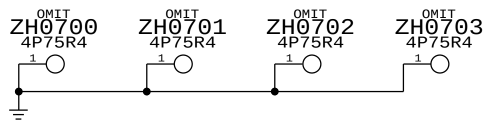
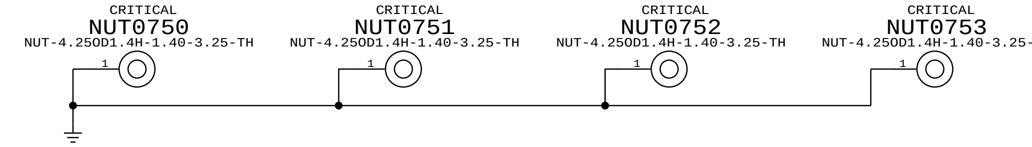
CPU Heatsink

4mm Plated Holes (998-0850)



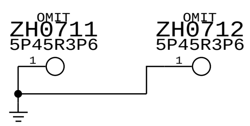
DIMM CONNECTOR NUTS

Nuts (805-9582)



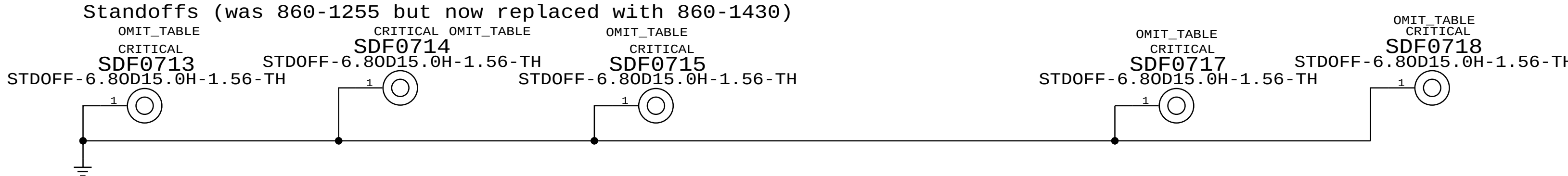
PCH HEATSINK

MOUNTING HOLES (998-0873, 998-0976)



Rear Cover

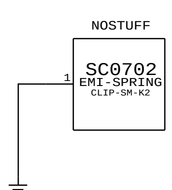
Standoffs (was 860-1255 but now replaced with 860-1430)



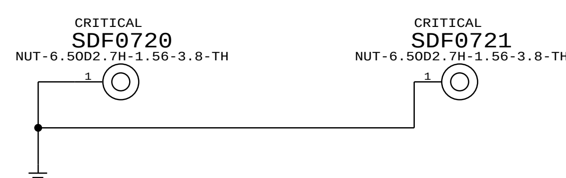
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
860-1430	5	STANDOFF,MLB,K60/K62	SDF0713,SDF0714,SDF0715,SDF0717,SDF0718	

For EMC

EMC Spring (870-1577); Near DIMMs



MXM STANDOFFS (835-0272)



SYNC_MASTER=K74_MASTER SYNC_DATE=N/A

PAGE TITLE

Holes

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

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8 7 6 5 4 3 2 1

UNUSED CPU SIGNALS

- TP_CPU_RSVD<16..1> = NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CPU_RSVD<46..19> = NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCI ALIASES

- TP_PCI_AD<31..0> = NC_PCI_AD<31..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCI_C_BE_L<3..0> = NC_PCI_C_BE_L<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCI_PAR = NC_PCI_PAR MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCI_RESET_L = NC_PCI_RESET_L MAKE_BASE=TRUE NO_TEST=TRUE
- TP_LPC_DREQ0_L = NC_LPC_DREQ0_L MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MEM ALIASES

- TP_MEM_A_DQ_CB<7..0> = NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_MEM_A_DQS_N<8> = NC_MEM_A_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_MEM_A_DQS_P<8> = NC_MEM_A_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_MEM_B_DQ_CB<7..0> = NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_MEM_B_DQS_N<8> = NC_MEM_B_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_MEM_B_DQS_P<8> = NC_MEM_B_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED MISC ALIASES

- TP_HDA_SDIN1 = NC_HDA_SDIN1 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_HDA_SDIN2 = NC_HDA_SDIN2 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_HDA_SDIN3 = NC_HDA_SDIN3 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_PWM0 = NC_PCH_PWM0 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_PWM1 = NC_PCH_PWM1 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_PWM2 = NC_PCH_PWM2 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_PWM3 = NC_PCH_PWM3 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_SST = NC_PCH_SST MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_CL_CLK1 = NC_PCH_CL_CLK1 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_CL_DATA1 = NC_PCH_CL_DATA1 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_CL_RST1 = NC_PCH_CL_RST1 MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED PCIE ALIASES

- TP_PCIE_CLK100M_PE5P = NC_PCIE_CLK100M_PE5P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE5N = NC_PCIE_CLK100M_PE5N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE6P = NC_PCIE_CLK100M_PE6P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE6N = NC_PCIE_CLK100M_PE6N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE7P = NC_PCIE_CLK100M_PE7P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE7N = NC_PCIE_CLK100M_PE7N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PE_RX_N<3..0> = NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PE_RX_P<3..0> = NC_PE_RXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PE_TX_N<3..0> = NC_PE_TXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PE_TX_P<3..0> = NC_PE_TXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_D2R_PERN4 = NC_PCIE_D2R_PERN4 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_D2R_PERP4 = NC_PCIE_D2R_PERP4 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_R2D_PETN4 = NC_PCIE_R2D_PETN4 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_R2D_PETP4 = NC_PCIE_R2D_PETP4 MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE4P = NC_PCIE_CLK100M_PE4P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCIE_CLK100M_PE4N = NC_PCIE_CLK100M_PE4N MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED USB ALIASES

- TP_USB_1N = NC_USB_1N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_1P = NC_USB_1P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_2N = NC_USB_2N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_2P = NC_USB_2P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_3N = NC_USB_3N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_3P = NC_USB_3P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_4N = NC_USB_4N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_4P = NC_USB_4P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_5N = NC_USB_5N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_5P = NC_USB_5P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_6N = NC_USB_6N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_6P = NC_USB_6P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_7N = NC_USB_7N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_7P = NC_USB_7P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_10N = NC_USB_10N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_10P = NC_USB_10P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_11N = NC_USB_11N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_11P = NC_USB_11P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_12N = NC_USB_12N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_12P = NC_USB_12P MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_13N = NC_USB_13N MAKE_BASE=TRUE NO_TEST=TRUE
- TP_USB_13P = NC_USB_13P MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED DISPLAY ALIASES

- TP_CRT_IG_DDC_CLK = NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_DDC_DATA = NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_RED = NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_GREEN = NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_BLUE = NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_HSYNC = NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CRT_IG_VSYNC = NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_MLN<3..0> = NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_MLP<3..0> = NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_AUX_N = NC_DP_IG_B_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_AUX_P = NC_DP_IG_B_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_HPD = NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_DDC_CLK = NC_DP_IG_B_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_B_DDC_DATA = NC_DP_IG_B_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_MLN<3..0> = NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_MLP<3..0> = NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_AUX_N = NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_AUX_P = NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_HPD = NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_CTRL_CLK = NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_C_CTRL_DATA = NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_MLN<3..0> = NC_DP_IG_D_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_MLP<3..0> = NC_DP_IG_D_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_AUXN = NC_DP_IG_D_AUXN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_AUXP = NC_DP_IG_D_AUXP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_HPD = NC_DP_IG_D_HPD MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_CTRL_CLK = NC_DP_IG_D_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE
- TP_DP_IG_D_CTRL_DATA = NC_DP_IG_D_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_TVCLKINN = NC_SDVO_TVCLKINN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_TVCLKINP = NC_SDVO_TVCLKINP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_STALLN = NC_SDVO_STALLN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_STALLP = NC_SDVO_STALLP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_INTN = NC_SDVO_INTN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SDVO_INTP = NC_SDVO_INTP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_L_BKLTCTL = NC_PCH_L_BKLTCTL MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_L_BKLTEN = NC_PCH_L_BKLTEN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_L_VDD_EN = NC_PCH_L_VDD_EN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_CLKOUT_DPN = NC_PCH_CLKOUT_DPN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_CLKOUT_DPP = NC_PCH_CLKOUT_DPP MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED FDI ALIASES

- TP_CPU_FDI_TX_N<7..0> = NC_CPU_FDI_TXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_CPU_FDI_TX_P<7..0> = NC_CPU_FDI_TXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_FDI_RX_N<7..0> = NC_PCH_FDI_RXN<7..0> MAKE_BASE=TRUE NO_TEST=TRUE
- TP_PCH_FDI_RX_P<7..0> = NC_PCH_FDI_RXP<7..0> MAKE_BASE=TRUE NO_TEST=TRUE

NC ON UNUSED SATA ALIASES

- TP_SATA_D_D2RN = NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_D_D2RP = NC_SATA_D_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_D_R2D_CN = NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_D_R2D_CP = NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_E_D2RN = NC_SATA_E_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_E_D2RP = NC_SATA_E_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_E_R2D_CN = NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_E_R2D_CP = NC_SATA_E_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_F_D2RN = NC_SATA_F_D2RN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_F_D2RP = NC_SATA_F_D2RP MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_F_R2D_CN = NC_SATA_F_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE
- TP_SATA_F_R2D_CP = NC_SATA_F_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: UNUSED SIGNAL ALIAS

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DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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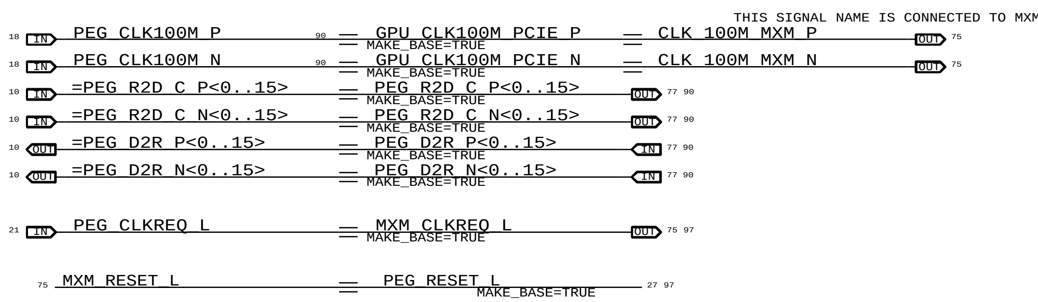
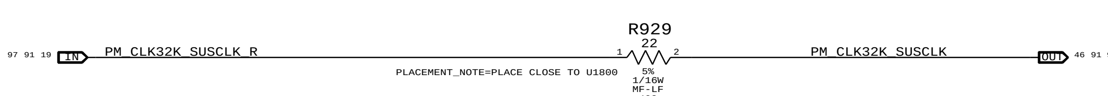
PEG Slot Support





SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

Signal Aliases

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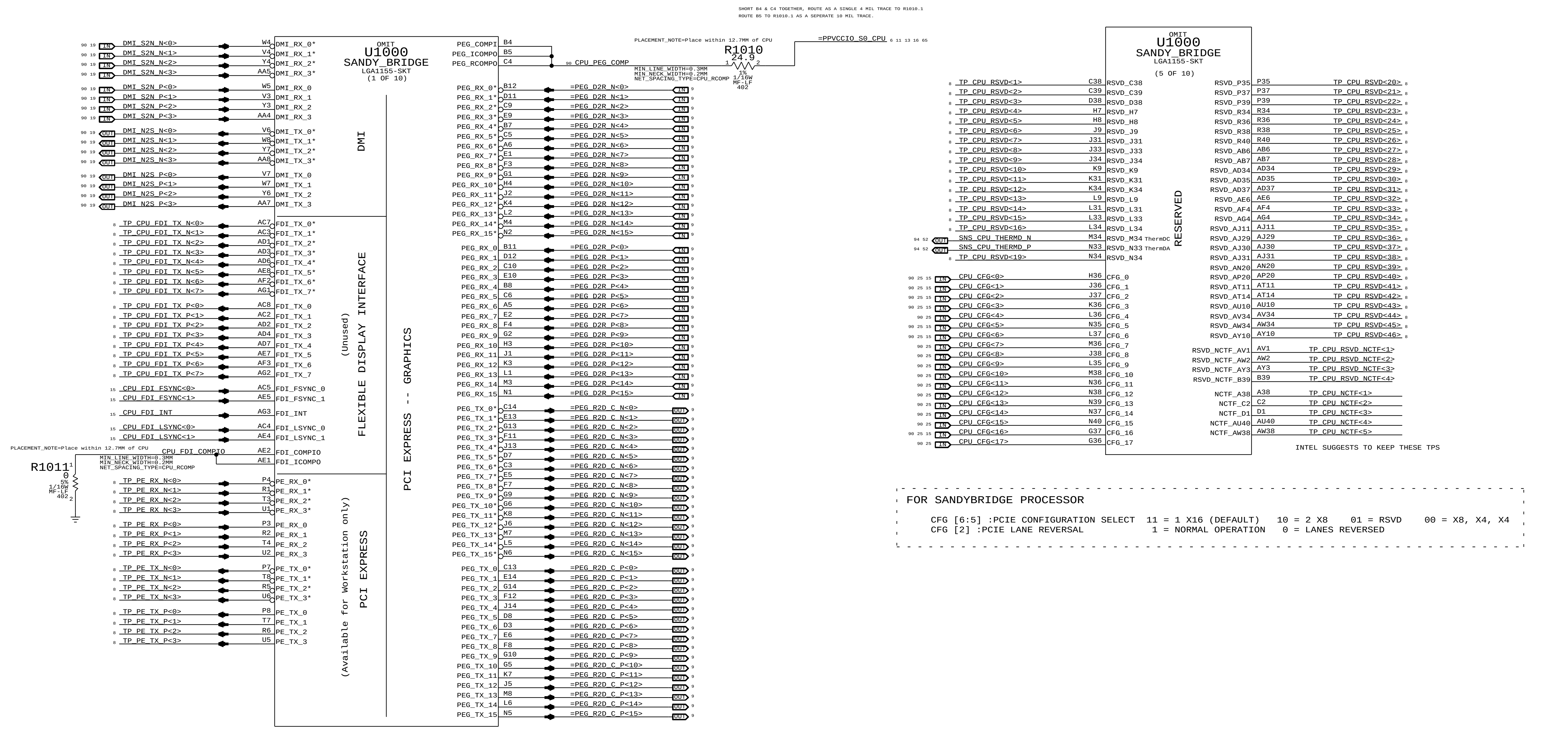

FOR SANDYBRIDGE PROCESSOR

CFG [6:5] :PCIE CONFIGURATION SELECT 11 = 1 X16 (DEFAULT) 10 = 2 X8 01 = RSVD 00 = X8, X4, X4
CFG [2] :PCIE LANE REVERSAL 1 = NORMAL OPERATION 0 = LANES REVERSED

PAGE TITLE
CPU DMI/PEG/FDI/RSVD

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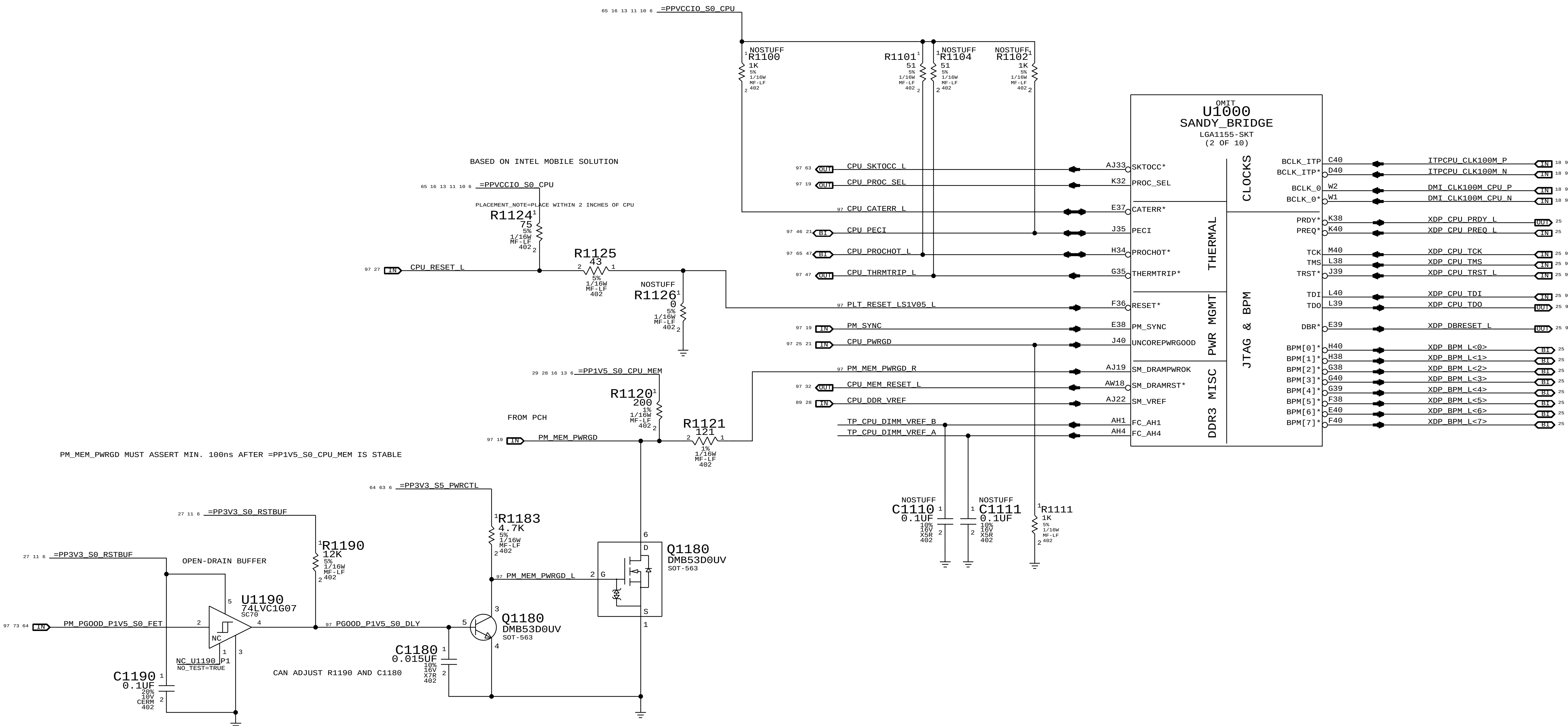

CPU CLOCK/MISC/JTAG

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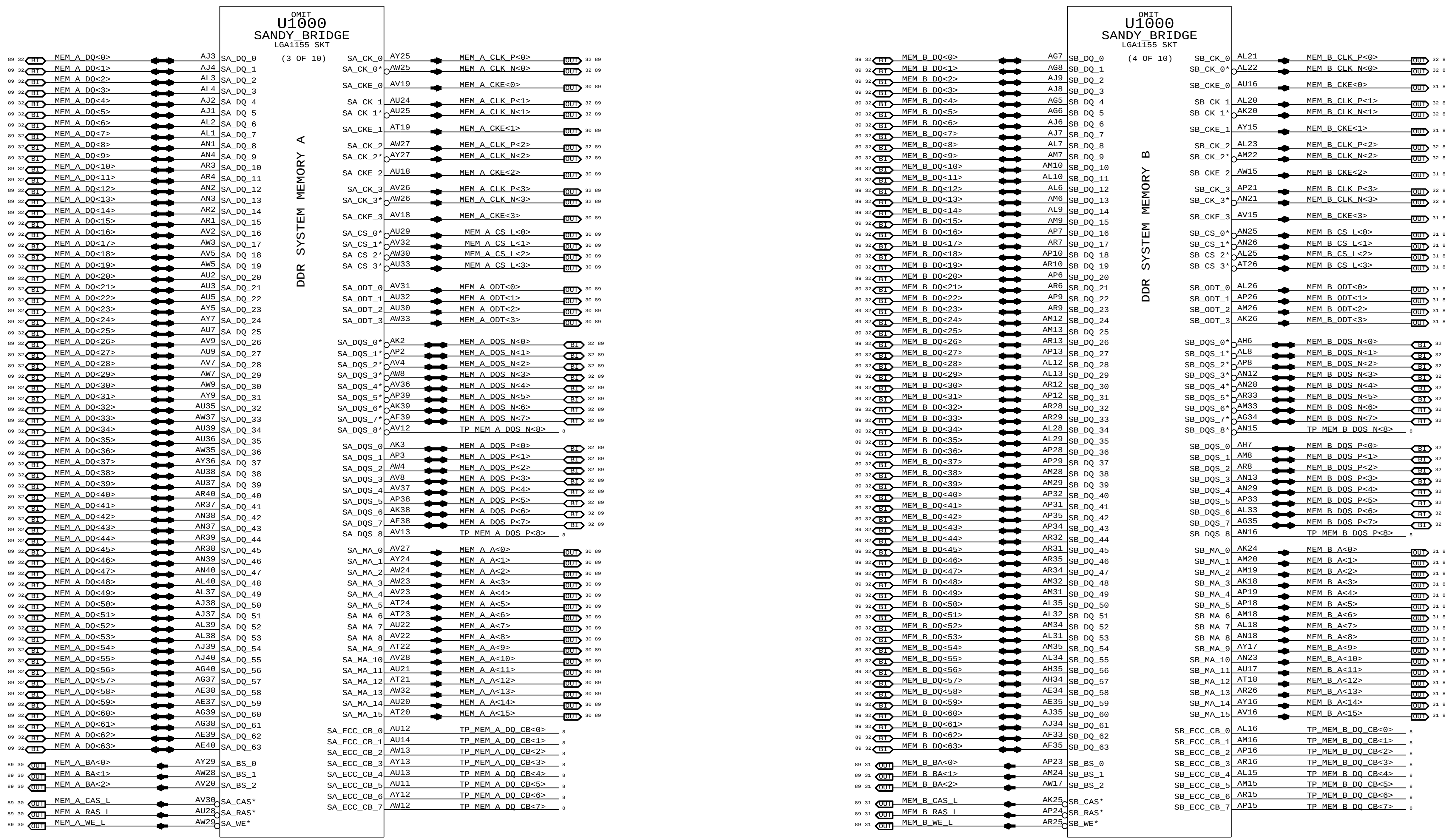

SYNC_DATE=01/06/2011

PAGE TITLE

CPU DDR3 INTERFACES

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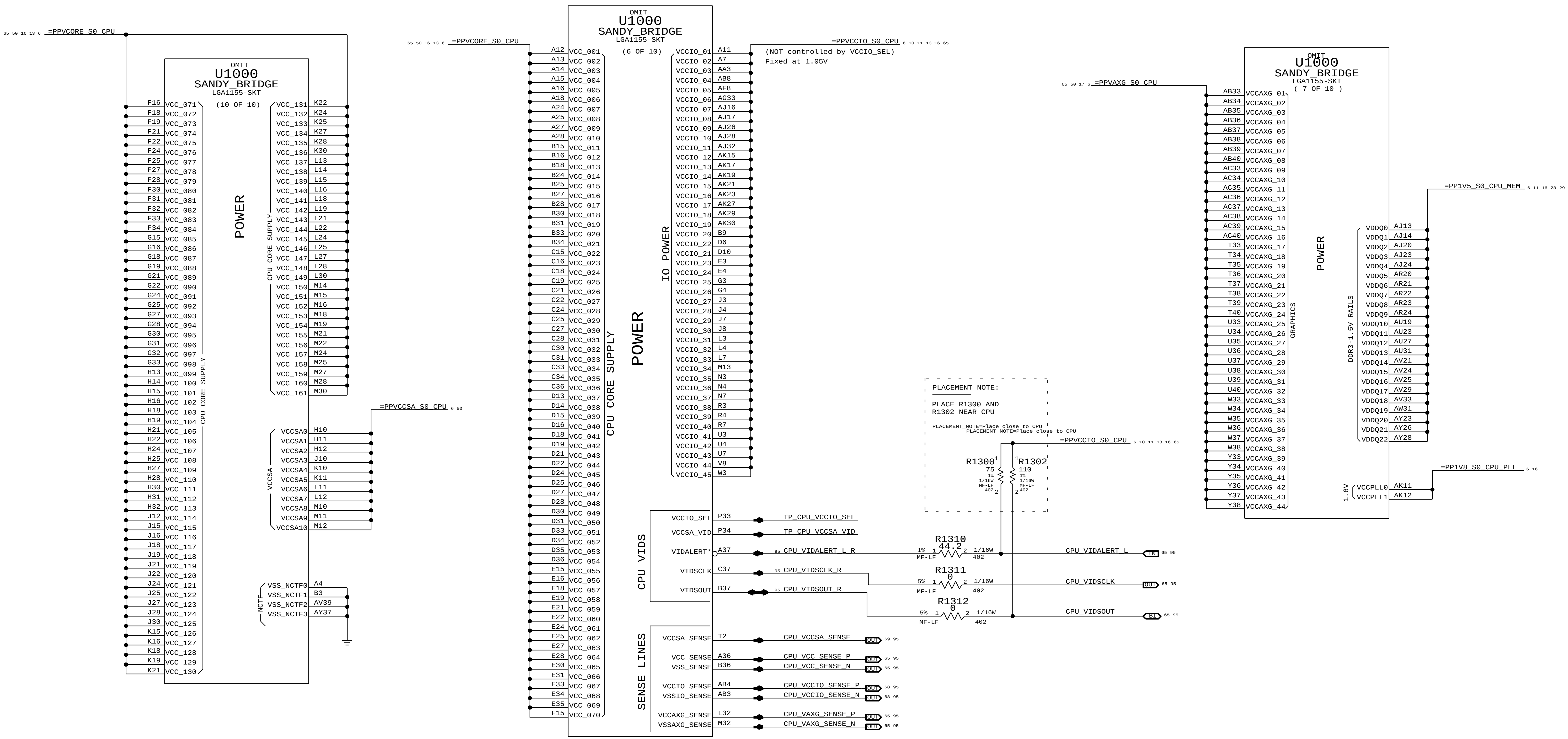

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
CPU POWER

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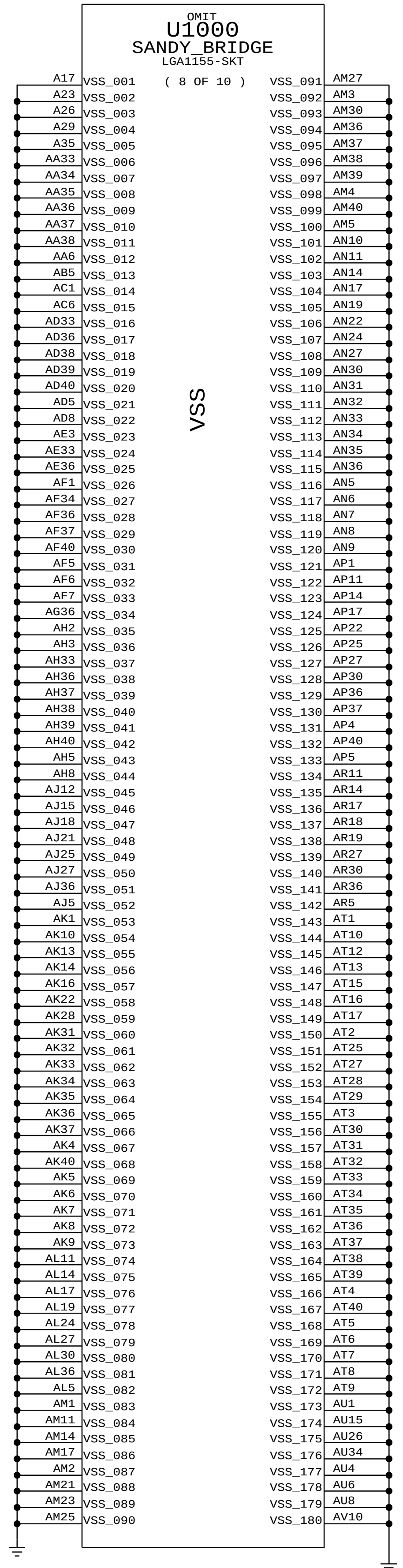

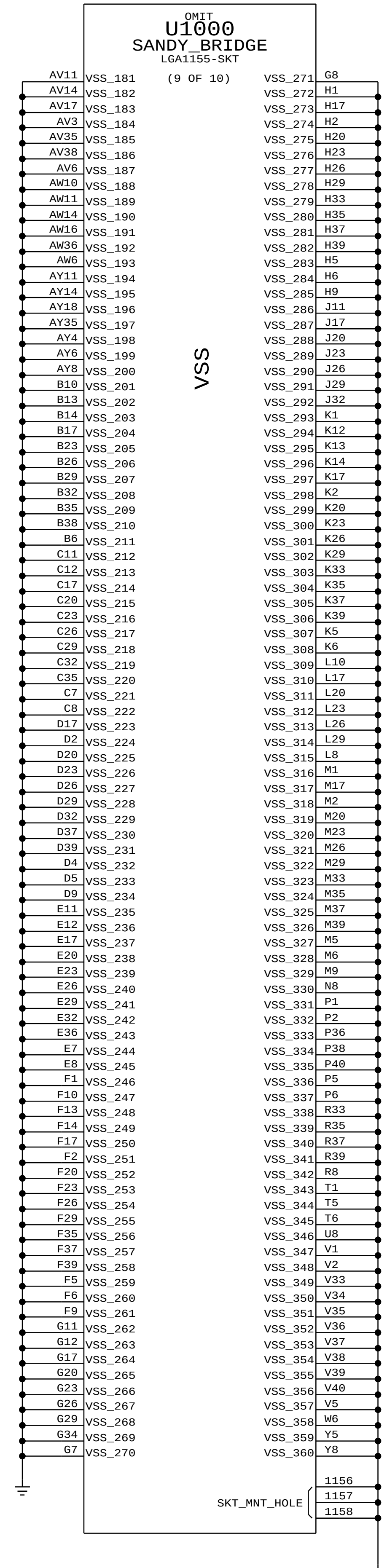

PAGE TITLE
CPU GROUNDS

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DRAWING NUMBER 051-8115 SIZE D

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PAGE 14 OF 110

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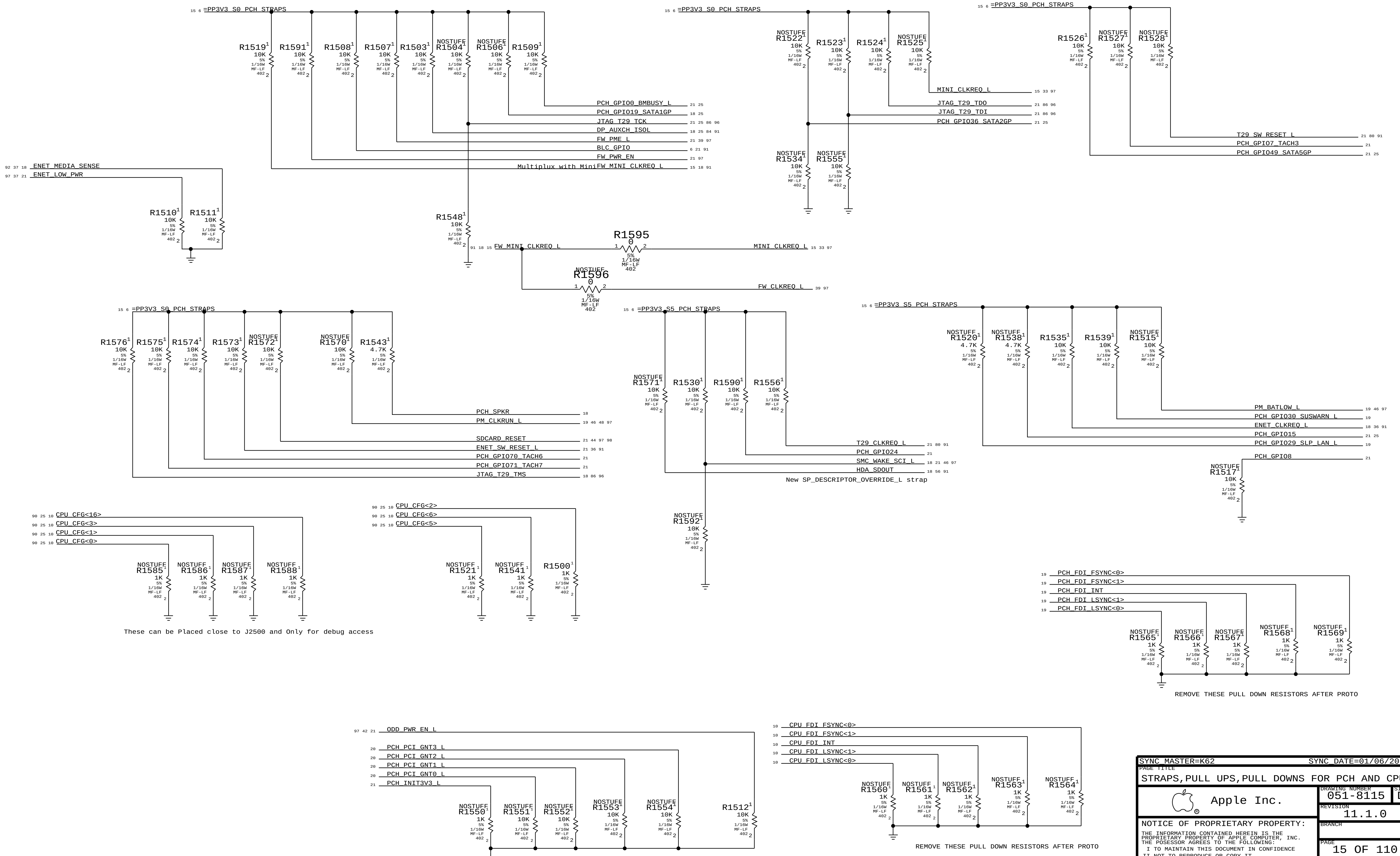

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU

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CPU VCORE DECOUPLING

14x 22UF,0805 INTEL RECOMMENDATION 18X 22UF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613):

Place inside socket cavity

BULK CAPS ON CPU VREG PAGE 72

10x 10UF and 10x 1UF CAPACITORS

Place inside socket cavity

CPU VCCIO DECOUPLING

8X 22UF 0805, 6X 10UF 0805 INTEL RECOMMENDATION 9X22UF 0805,16X 0805 placeholders

PLACEMENT_NOTE (C1650-C1657):

Place under socket cavity on secondary side.

PLACEMENT_NOTE (C1660-C1665):

Place at edge of socket.

BULK CAPS ON CPU VREG PAGE 72

Memory (CPU VCCDDR) DECOUPLING

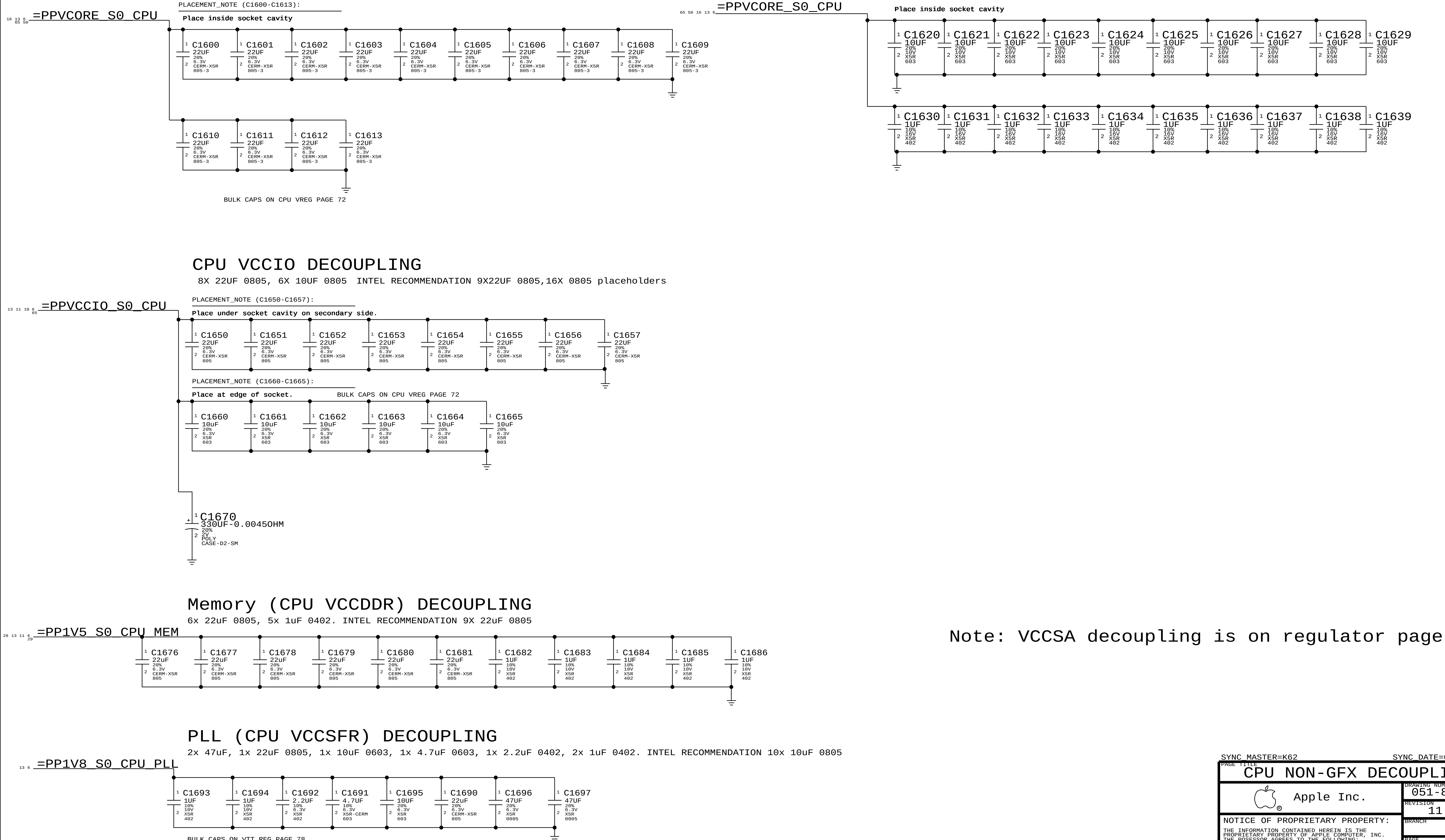
6x 22uF 0805, 5x 1uF 0402. INTEL RECOMMENDATION 9X 22uF 0805

Note: VCCSA decoupling is on regulator page

PLL (CPU VCCSFR) DECOUPLING

2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 10x 10uF 0805

BULK CAPS ON VTT REG PAGE 78



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: CPU NON-GFX DECOUPLING

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REVISION: 11.1.0

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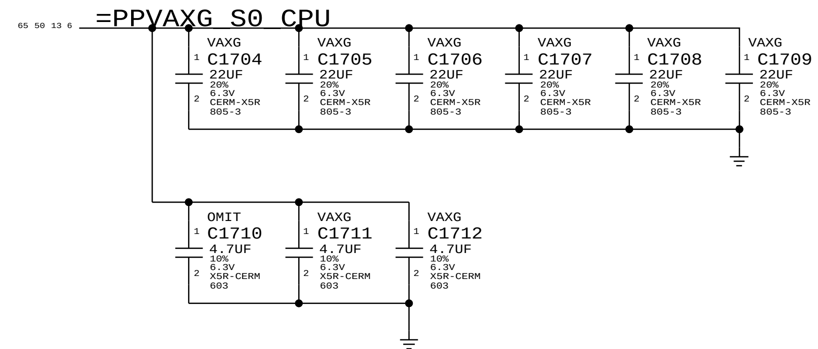
SHEET: 16 OF 98

VAXG DECOUPLING

INTEL RECOMMENDATION 6X22UF 0805,3X 4.7UF

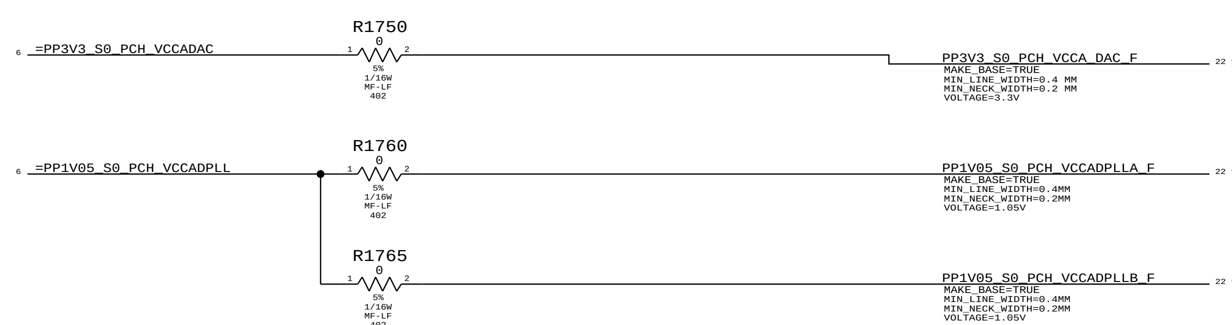
PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



BULK CAPS ON CPU VREG PAGE 73

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
138S0586	1	CAP,4.7UF,10%,6.3V,0603	C1710	VAXG
113S0022	1	RES,0 OHM,5%,0603	C1710	NO_VAXG



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: GFX DECOUPLING & PCH PWR ALIAS

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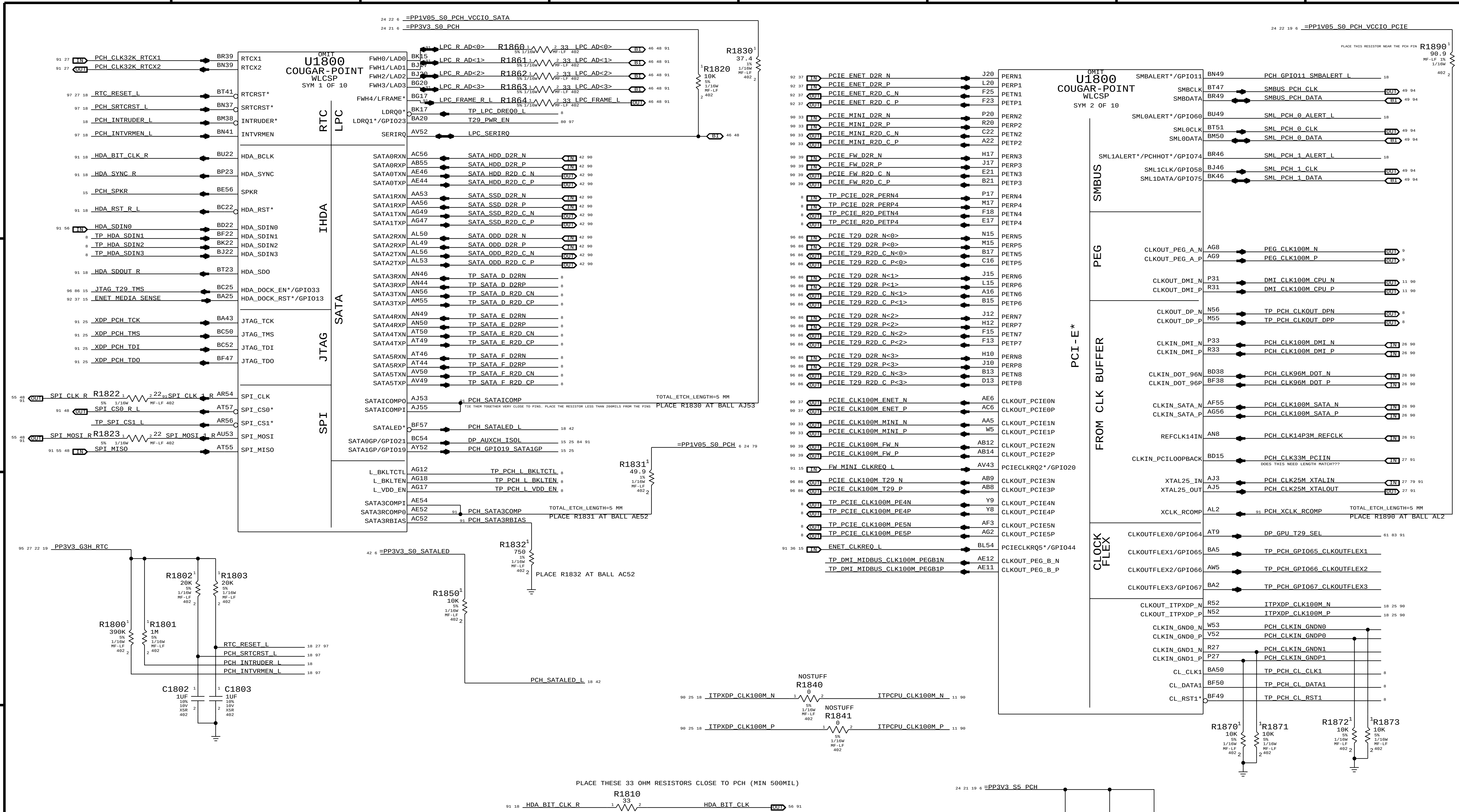

SYNC_MASTER=K62
SYNC_DATE=01/06/2011

PAGE TITLE
PCH SATA/PCIE/CLK/LPC/SPI

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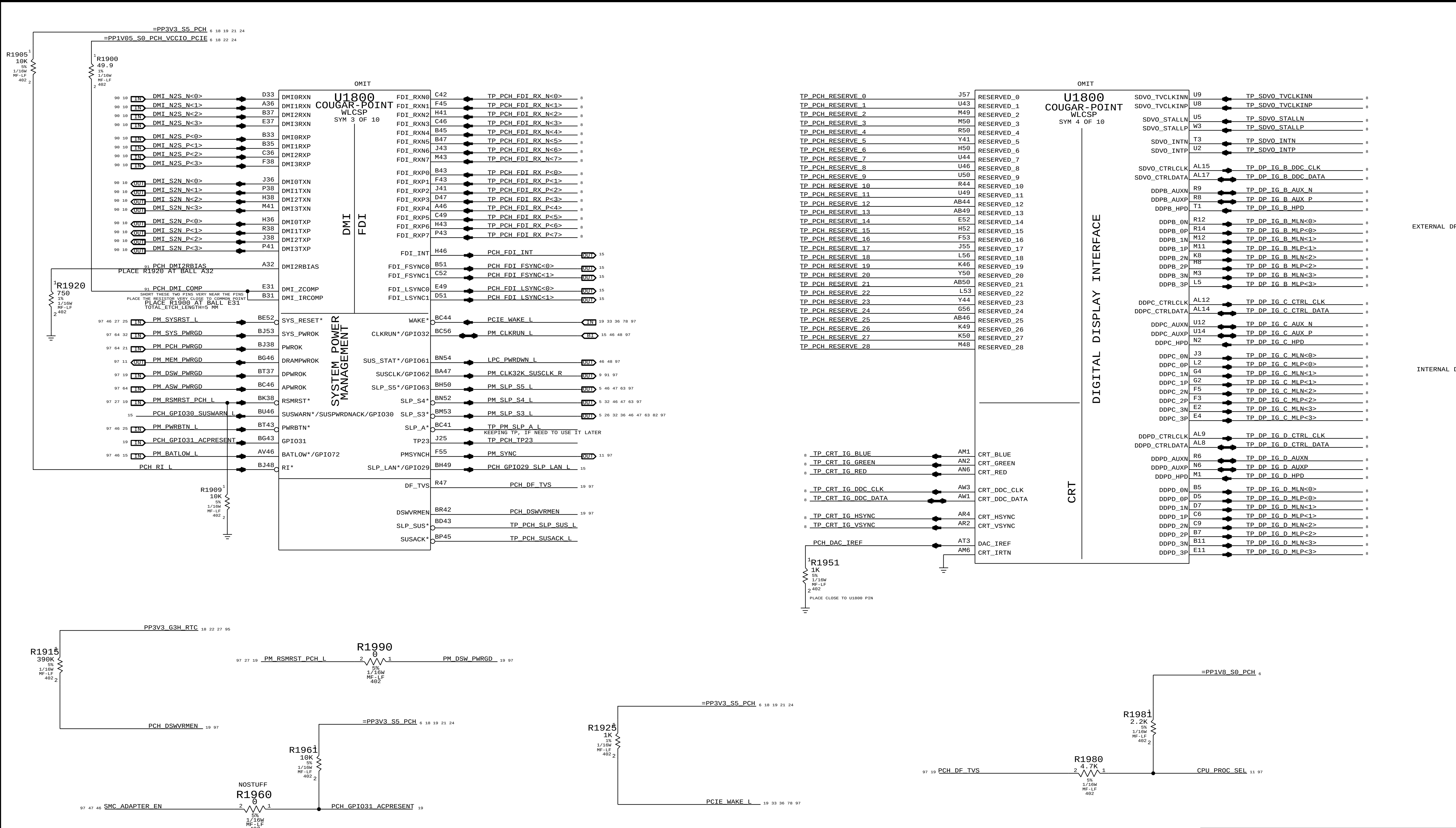
SHEET 18 OF 98



PAGE TITLE

PCH DMI/FDI/GRAPHICS

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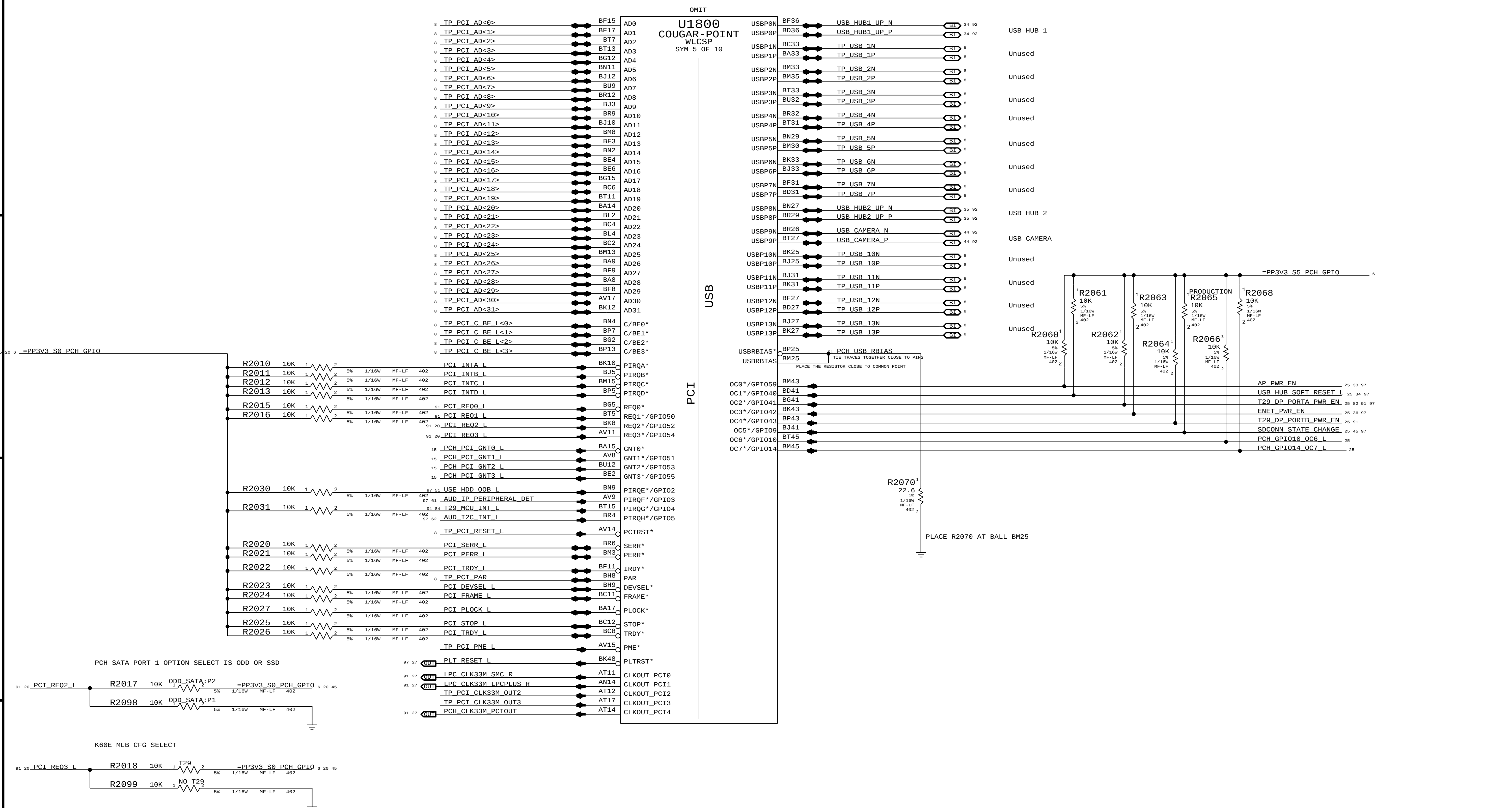

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
PCH PCI/FLASHCACHE/USB

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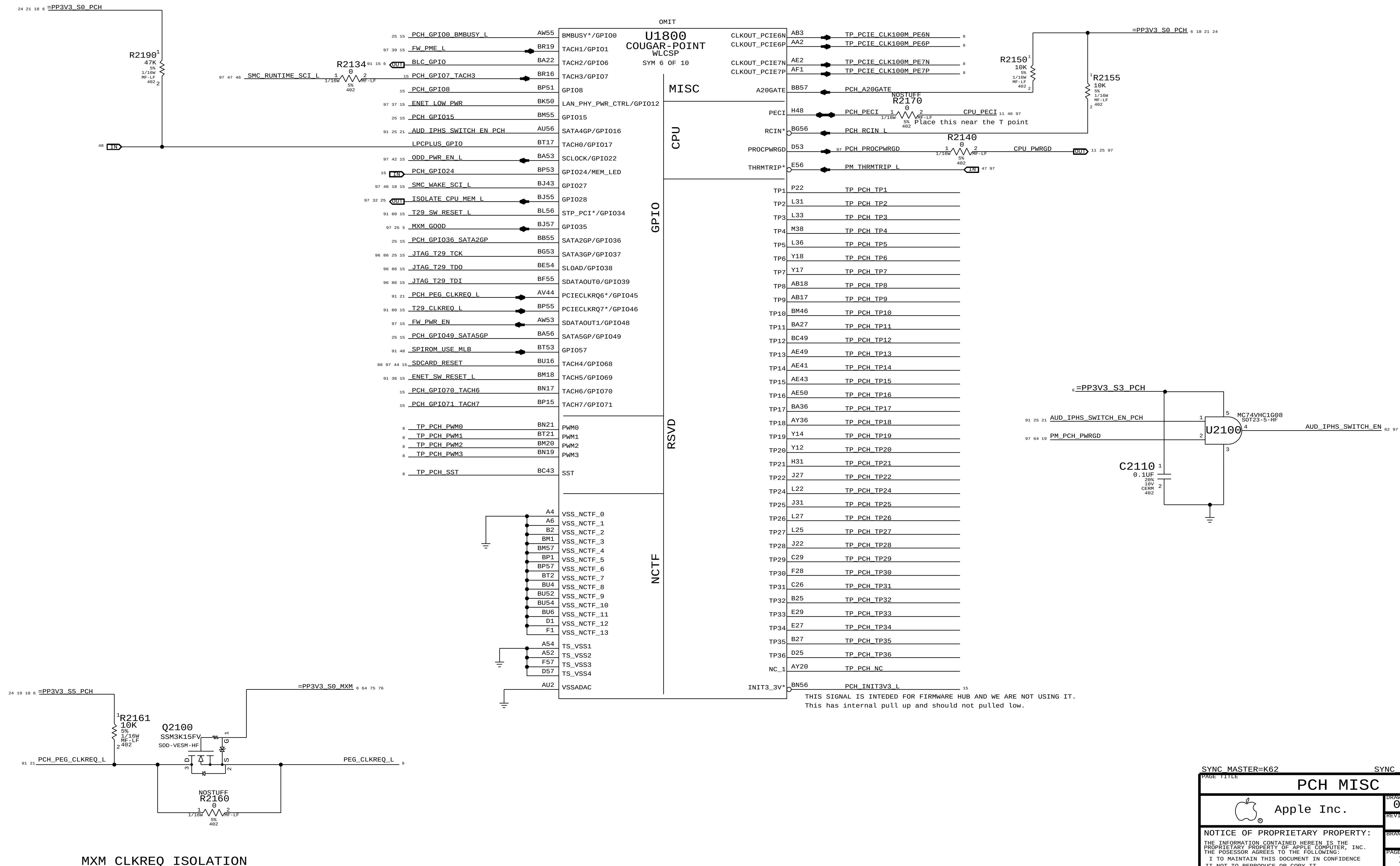

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

PCH MISC

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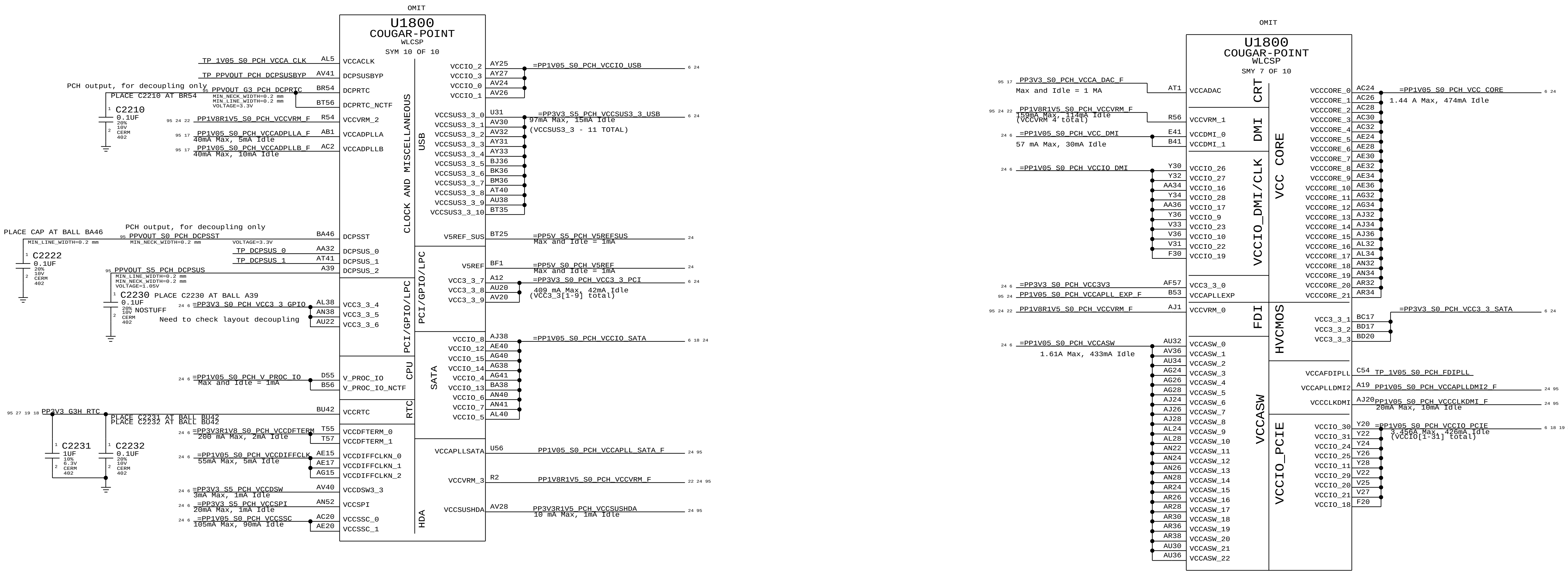
D C B A



PAGE TITLE

PCH POWER

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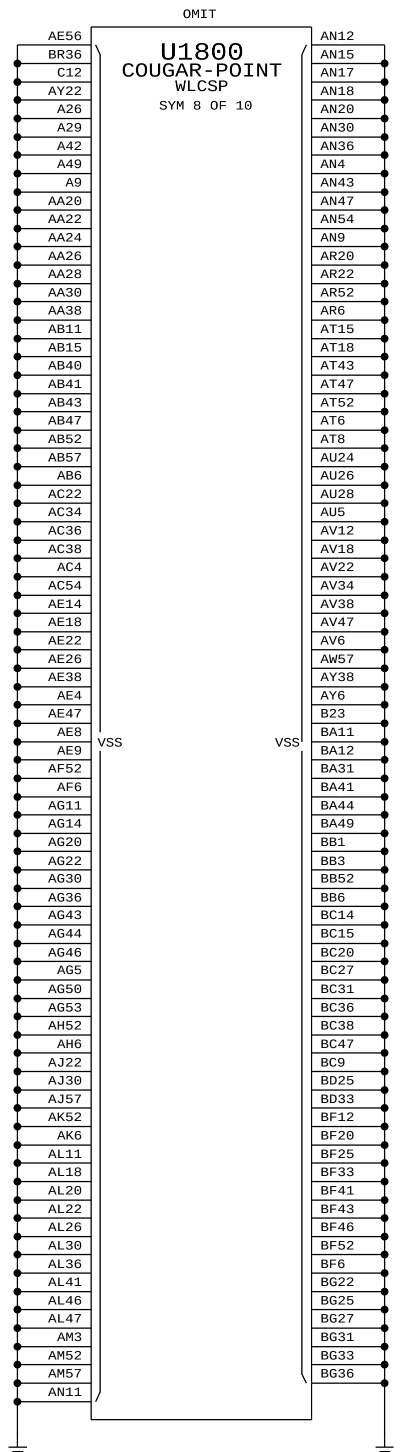

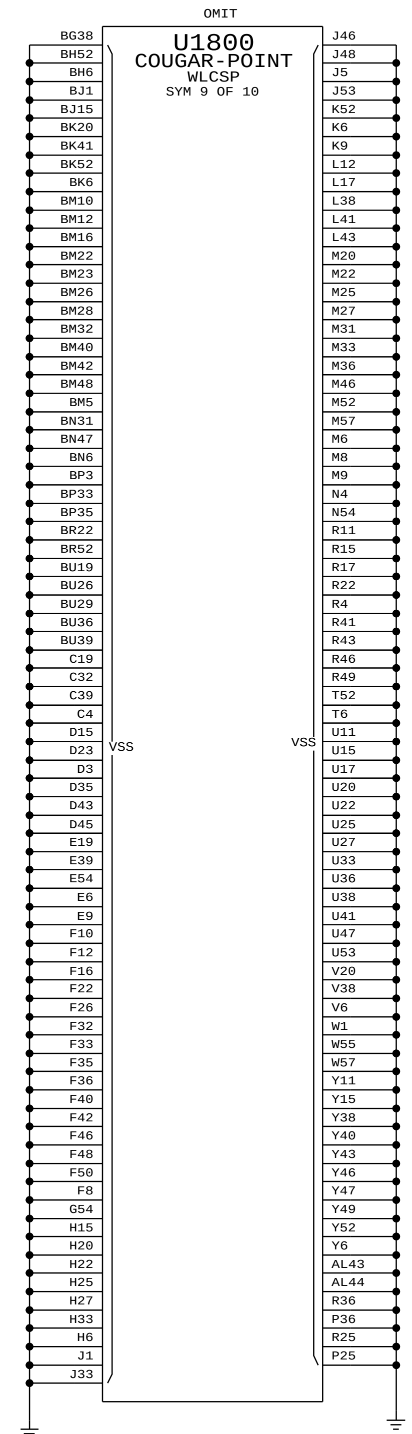

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
PCH GROUNDS

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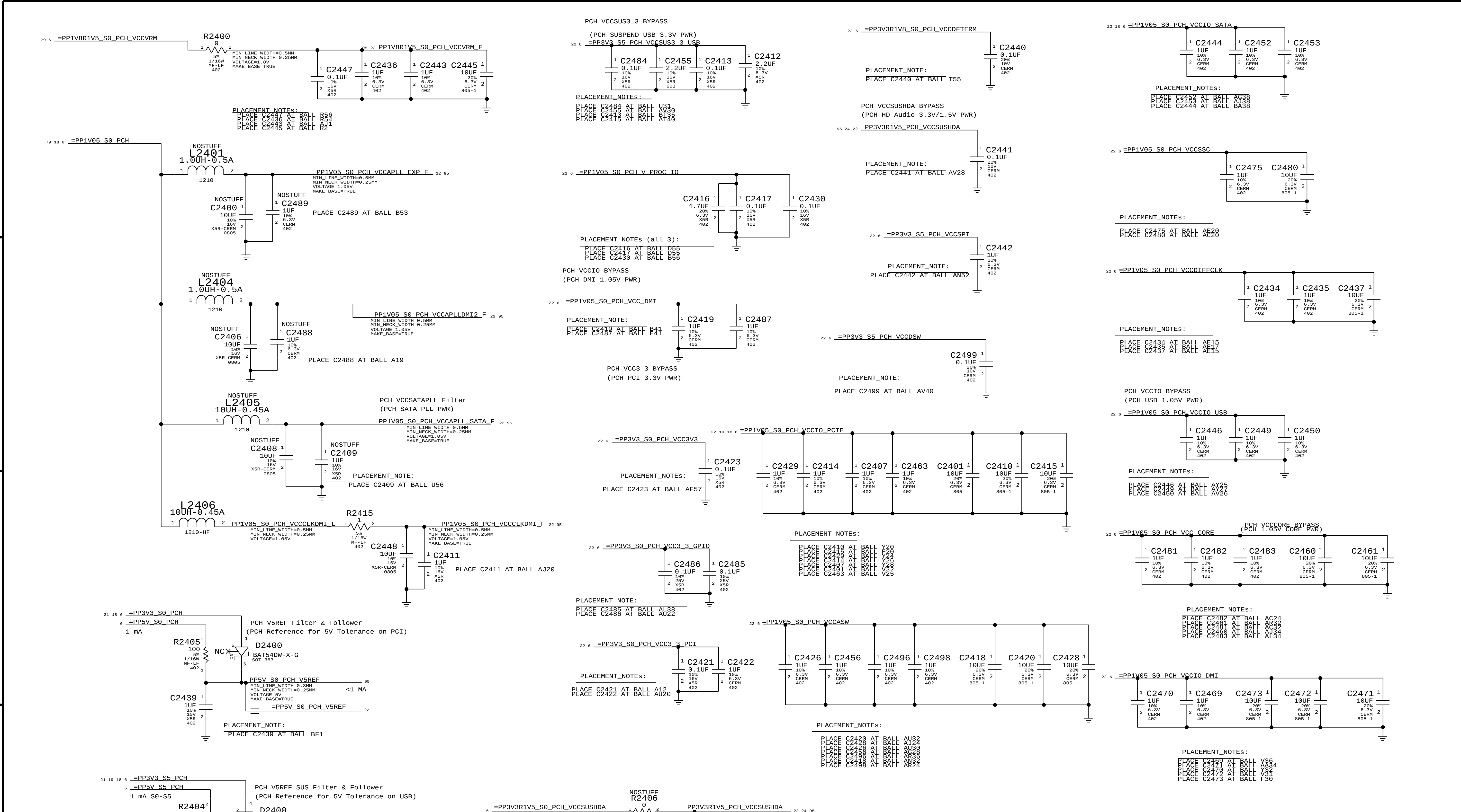

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PAGE TITLE

PCH DECOUPLING

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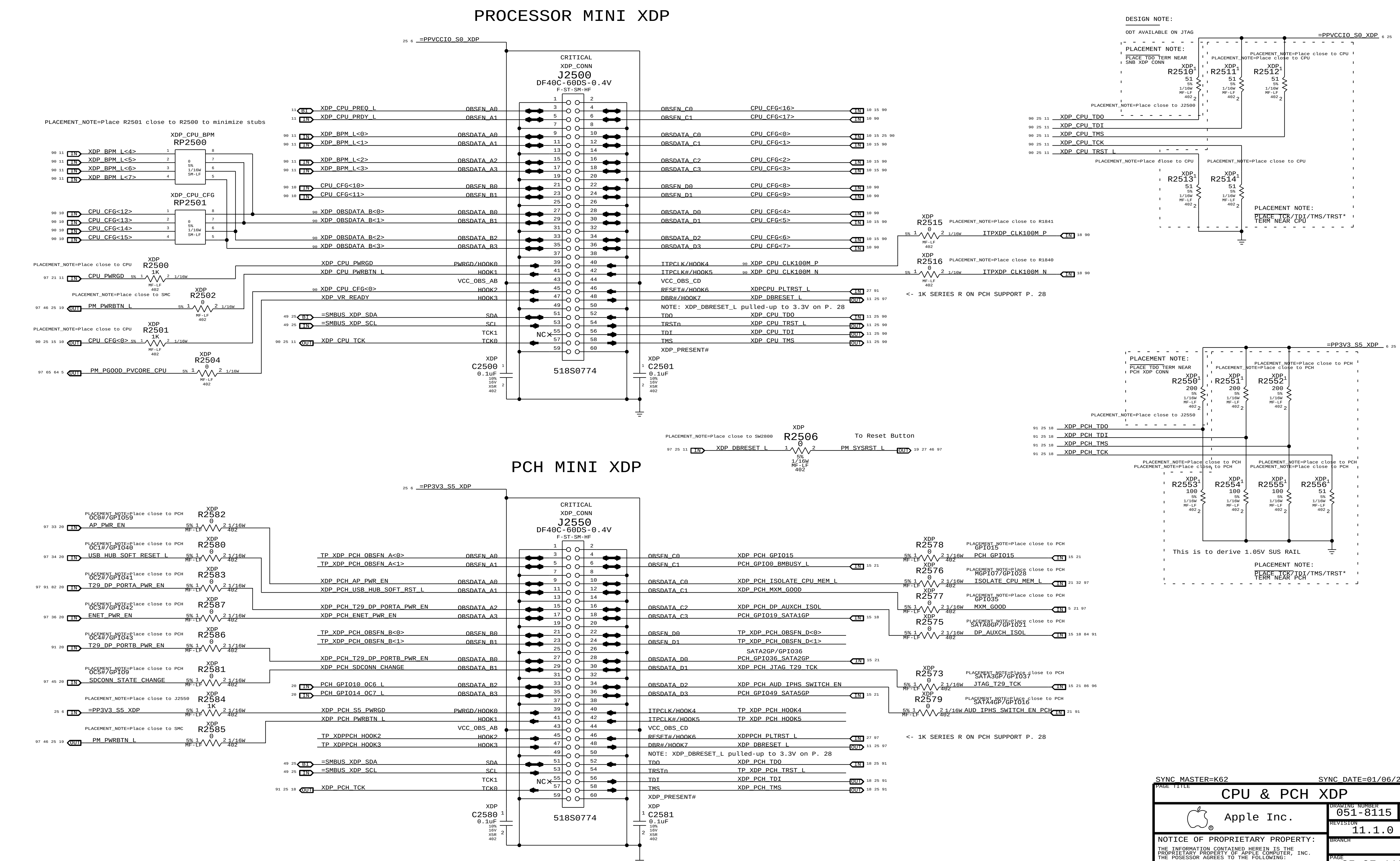

SYNC_MASTER=K62
SYNC_DATE=01/06/2011

PAGE TITLE
CPU & PCH XDP

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SIZE D
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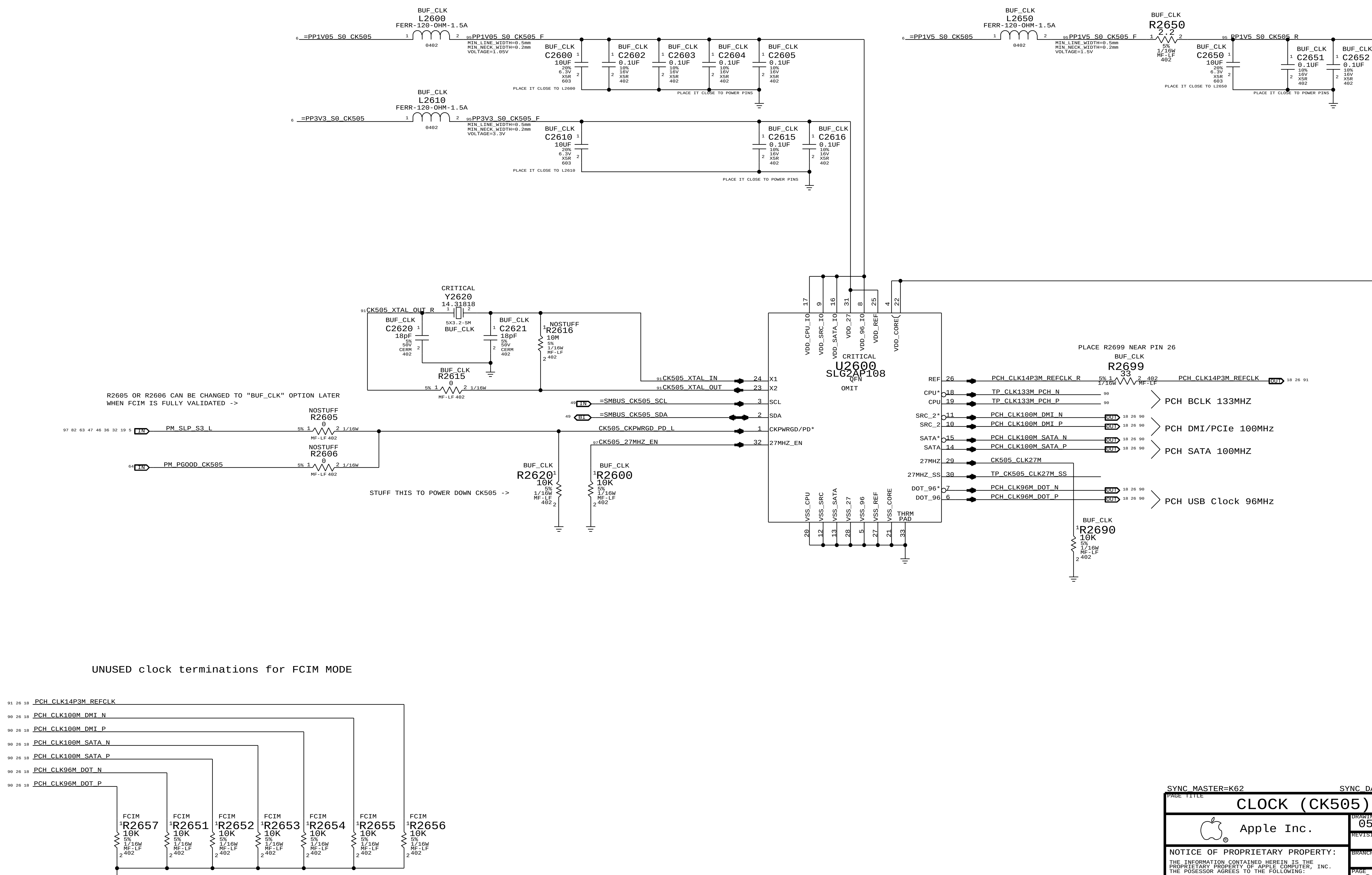
PAGE 25 OF 110
SHEET 25 OF 98



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
CLOCK (CK505)

Apple Inc.

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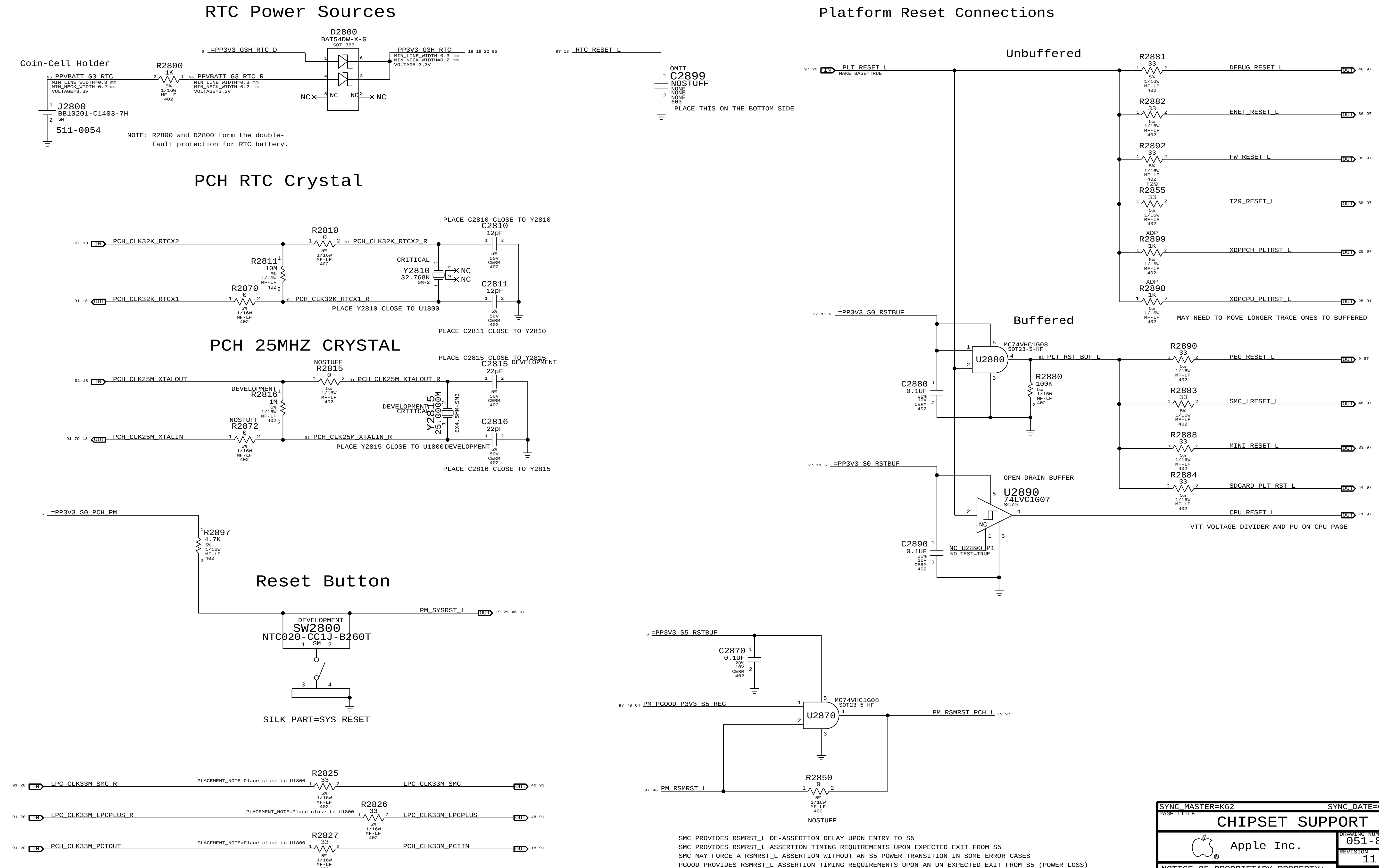

SMC PROVIDES RSMRST_L DE-ASSERTION DELAY UPON ENTRY TO S5
SMC PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON EXPECTED EXIT FROM S5
SMC MAY FORCE A RSMRST_L ASSERTION WITHOUT AN S5 POWER TRANSITION IN SOME ERROR CASES
PGOOD PROVIDES RSMRST_L ASSERTION TIMING REQUIREMENTS UPON AN UN-EXPECTED EXIT FROM S5 (POWER LOSS)

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

CHIPSET SUPPORT

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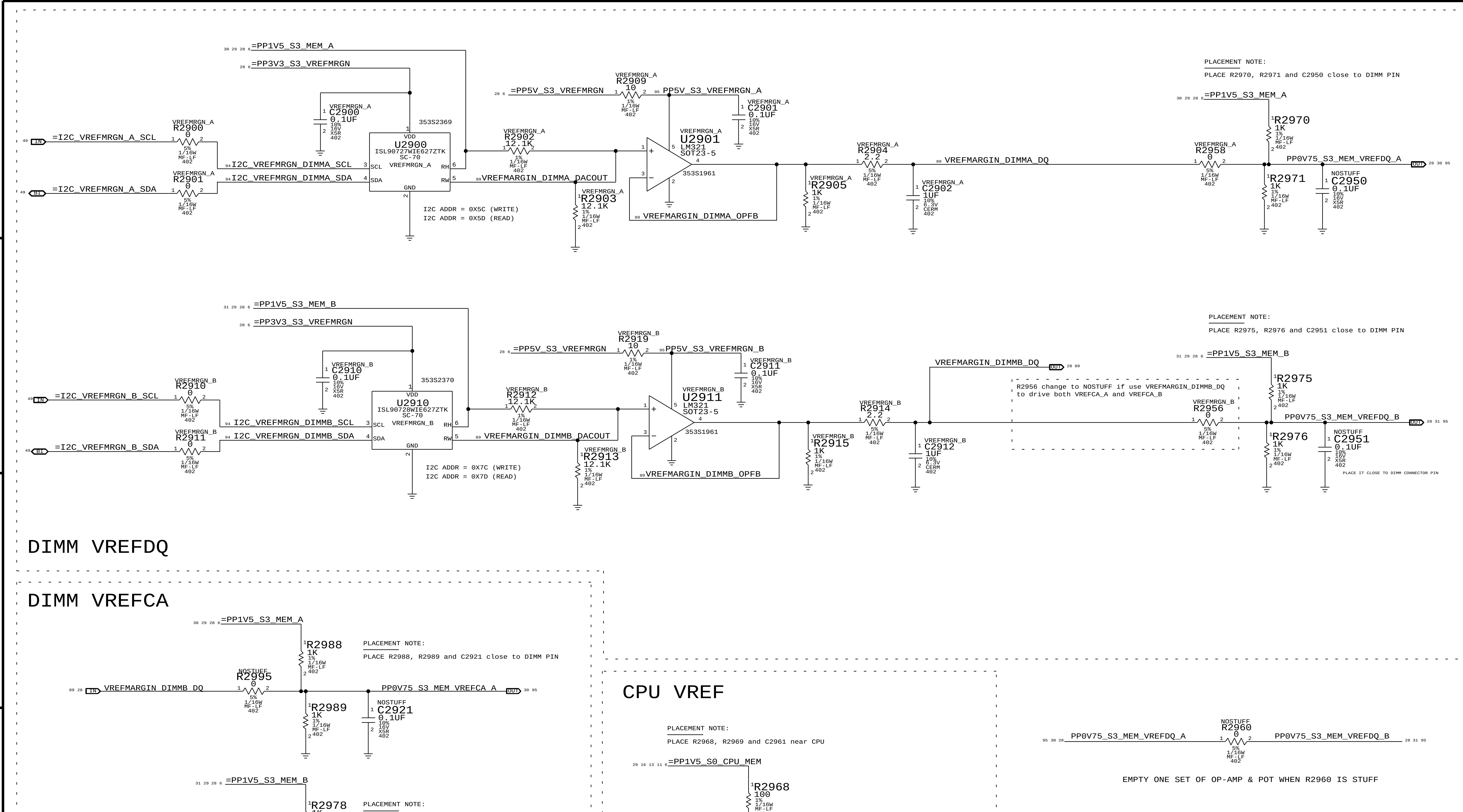

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

DDR3 VREF MARGINING

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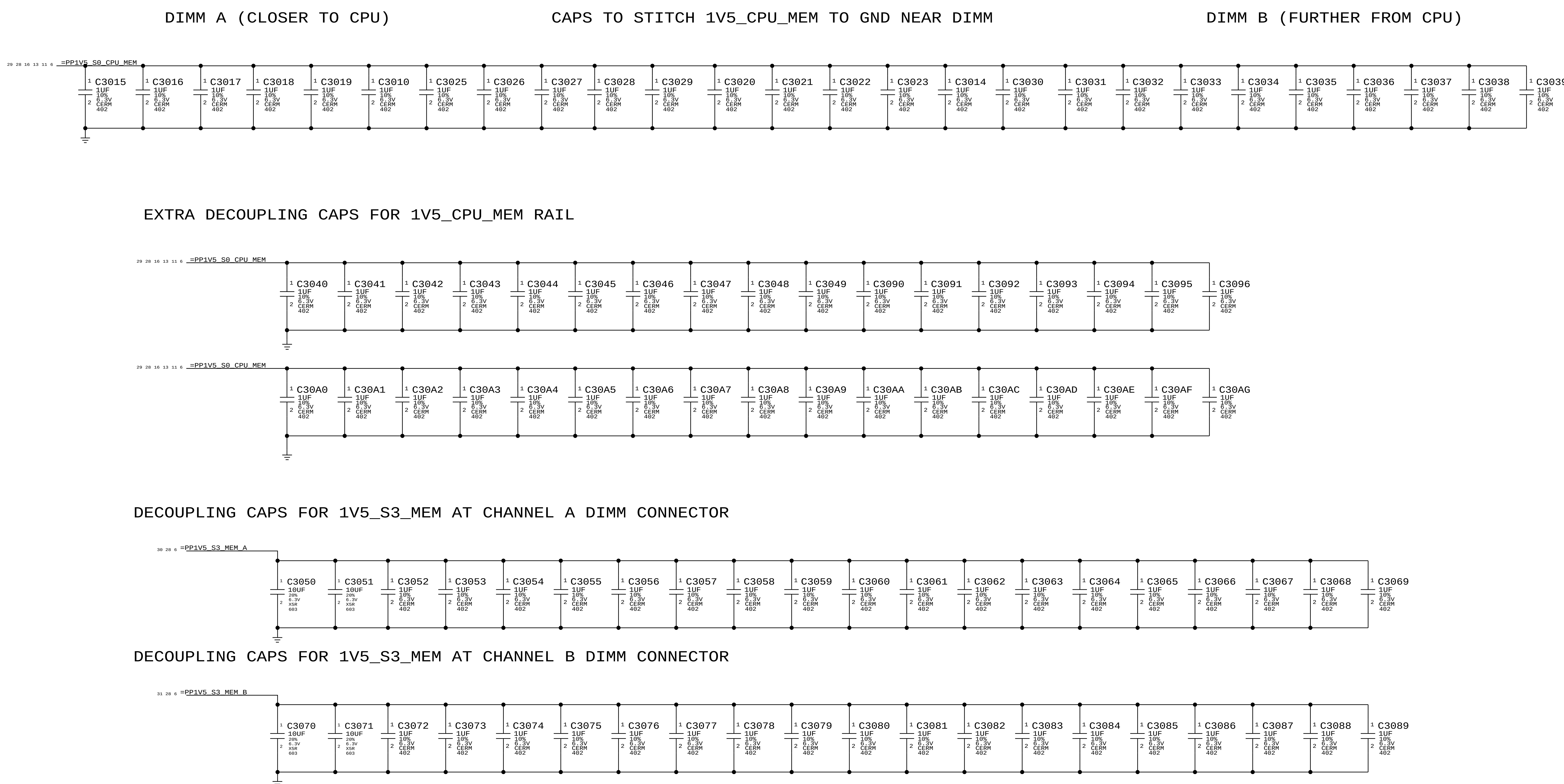

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

MEMORY CAPS

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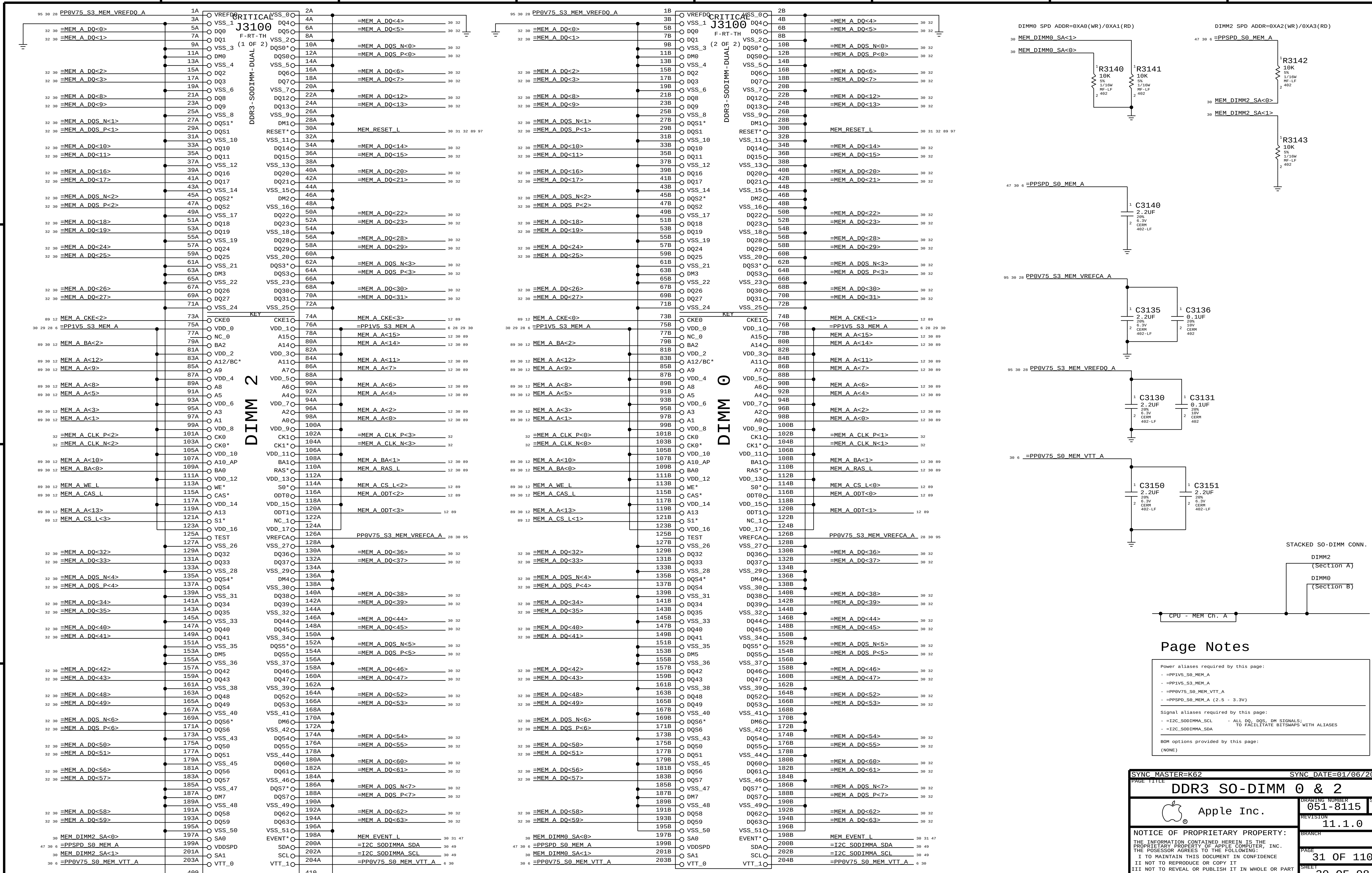

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S0_MEM_VTT_A
- =PPSPD_S0_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

- ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES

BOM options provided by this page:
(NONE)

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: DDR3 SO-DIMM 0 & 2

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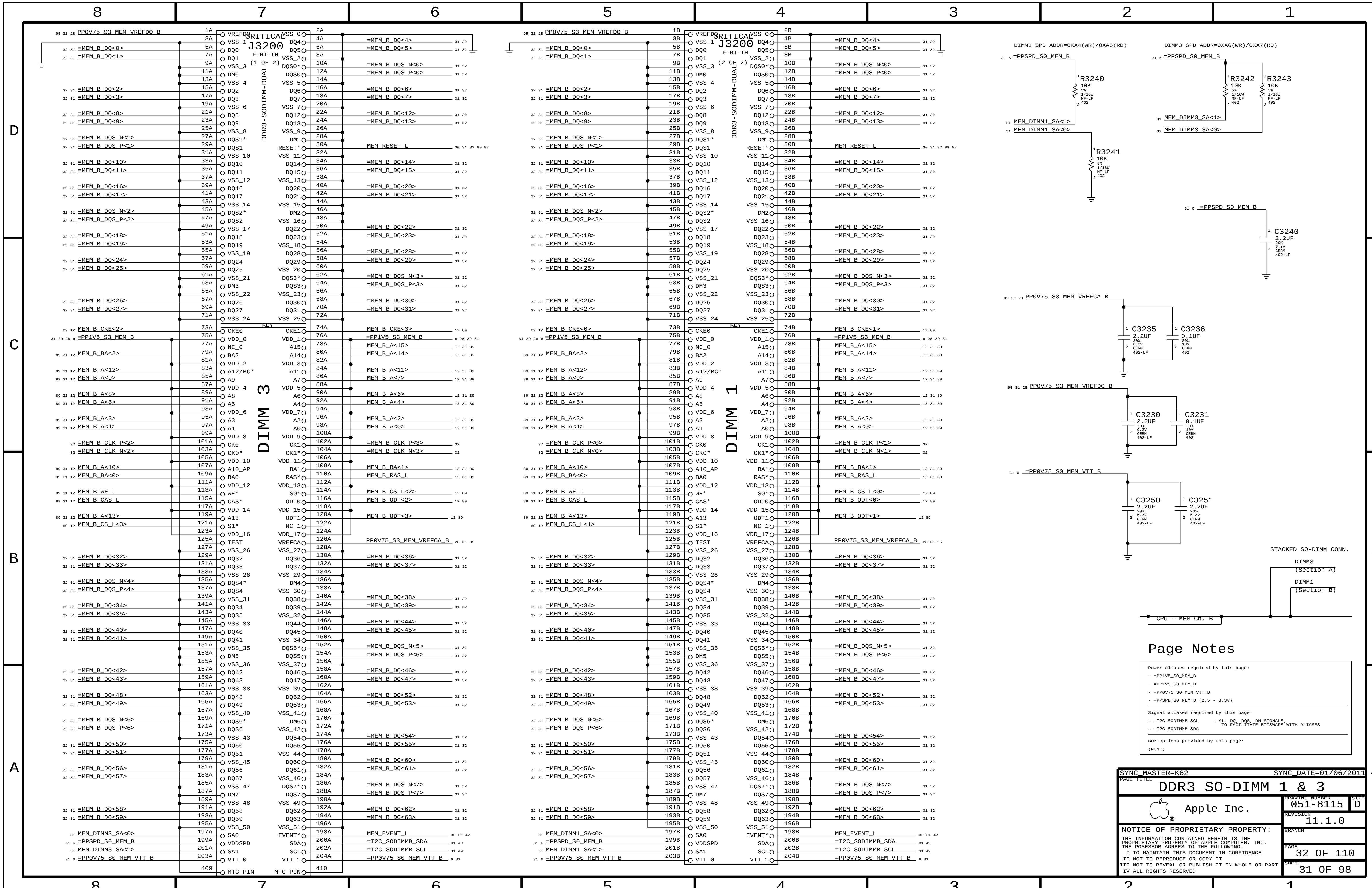

Page Notes

Power aliases required by this page:
- =PP1V5_S0_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S0_MEM_VTT_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMB_SCL — ALL DQ, DQS, DM SIGNALS; TO FACILITATE BITSWAPS WITH ALIASES
- =I2C_SODIMMB_SDA

BOM options provided by this page:
(NONE)

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
DDR3 SO-DIMM 1 & 3

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SHEET 31 OF 98

8	7	6	5	4	3	2	1

CPU CHANNEL A DQS 0 -> DIMM A DQS 7

89 12	MEM_A_DQS_N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<7>	30
89 12	MEM_A_DQS_P<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<7>	30
89 12	MEM_A_DQ<7>	MAKE_BASE=TRUE		=MEM_A_DQ<57>	30
89 12	MEM_A_DQ<6>	MAKE_BASE=TRUE		=MEM_A_DQ<56>	30
89 12	MEM_A_DQ<5>	MAKE_BASE=TRUE		=MEM_A_DQ<58>	30
89 12	MEM_A_DQ<4>	MAKE_BASE=TRUE		=MEM_A_DQ<59>	30
89 12	MEM_A_DQ<3>	MAKE_BASE=TRUE		=MEM_A_DQ<60>	30
89 12	MEM_A_DQ<2>	MAKE_BASE=TRUE		=MEM_A_DQ<61>	30
89 12	MEM_A_DQ<1>	MAKE_BASE=TRUE		=MEM_A_DQ<62>	30
89 12	MEM_A_DQ<0>	MAKE_BASE=TRUE		=MEM_A_DQ<63>	30

CPU CHANNEL A DQS 1 -> DIMM A DQS 6

89 12	MEM_A_DQS_N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<6>	30
89 12	MEM_A_DQS_P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<6>	30
89 12	MEM_A_DQ<15>	MAKE_BASE=TRUE		=MEM_A_DQ<49>	30
89 12	MEM_A_DQ<14>	MAKE_BASE=TRUE		=MEM_A_DQ<52>	30
89 12	MEM_A_DQ<13>	MAKE_BASE=TRUE		=MEM_A_DQ<51>	30
89 12	MEM_A_DQ<12>	MAKE_BASE=TRUE		=MEM_A_DQ<50>	30
89 12	MEM_A_DQ<11>	MAKE_BASE=TRUE		=MEM_A_DQ<48>	30
89 12	MEM_A_DQ<10>	MAKE_BASE=TRUE		=MEM_A_DQ<53>	30
89 12	MEM_A_DQ<9>	MAKE_BASE=TRUE		=MEM_A_DQ<55>	30
89 12	MEM_A_DQ<8>	MAKE_BASE=TRUE		=MEM_A_DQ<54>	30

CPU CHANNEL A DQS 2 -> DIMM A DQS 5

89 12	MEM_A_DQS_N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<5>	30
89 12	MEM_A_DQS_P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<5>	30
89 12	MEM_A_DQ<23>	MAKE_BASE=TRUE		=MEM_A_DQ<40>	30
89 12	MEM_A_DQ<22>	MAKE_BASE=TRUE		=MEM_A_DQ<45>	30
89 12	MEM_A_DQ<21>	MAKE_BASE=TRUE		=MEM_A_DQ<46>	30
89 12	MEM_A_DQ<20>	MAKE_BASE=TRUE		=MEM_A_DQ<47>	30
89 12	MEM_A_DQ<19>	MAKE_BASE=TRUE		=MEM_A_DQ<41>	30
89 12	MEM_A_DQ<18>	MAKE_BASE=TRUE		=MEM_A_DQ<44>	30
89 12	MEM_A_DQ<17>	MAKE_BASE=TRUE		=MEM_A_DQ<43>	30
89 12	MEM_A_DQ<16>	MAKE_BASE=TRUE		=MEM_A_DQ<42>	30

CPU CHANNEL A DQS 3 -> DIMM A DQS 4

89 12	MEM_A_DQS_N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<4>	30
89 12	MEM_A_DQS_P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<4>	30
89 12	MEM_A_DQ<31>	MAKE_BASE=TRUE		=MEM_A_DQ<37>	30
89 12	MEM_A_DQ<30>	MAKE_BASE=TRUE		=MEM_A_DQ<33>	30
89 12	MEM_A_DQ<29>	MAKE_BASE=TRUE		=MEM_A_DQ<35>	30
89 12	MEM_A_DQ<28>	MAKE_BASE=TRUE		=MEM_A_DQ<34>	30
89 12	MEM_A_DQ<27>	MAKE_BASE=TRUE		=MEM_A_DQ<36>	30
89 12	MEM_A_DQ<26>	MAKE_BASE=TRUE		=MEM_A_DQ<32>	30
89 12	MEM_A_DQ<25>	MAKE_BASE=TRUE		=MEM_A_DQ<38>	30
89 12	MEM_A_DQ<24>	MAKE_BASE=TRUE		=MEM_A_DQ<39>	30

CPU CHANNEL A DQS 4 -> DIMM A DQS 3

89 12	MEM_A_DQS_N<4>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<3>	30
89 12	MEM_A_DQS_P<4>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<3>	30
89 12	MEM_A_DQ<39>	MAKE_BASE=TRUE		=MEM_A_DQ<28>	30
89 12	MEM_A_DQ<38>	MAKE_BASE=TRUE		=MEM_A_DQ<29>	30
89 12	MEM_A_DQ<37>	MAKE_BASE=TRUE		=MEM_A_DQ<27>	30
89 12	MEM_A_DQ<36>	MAKE_BASE=TRUE		=MEM_A_DQ<31>	30
89 12	MEM_A_DQ<35>	MAKE_BASE=TRUE		=MEM_A_DQ<25>	30
89 12	MEM_A_DQ<34>	MAKE_BASE=TRUE		=MEM_A_DQ<24>	30
89 12	MEM_A_DQ<33>	MAKE_BASE=TRUE		=MEM_A_DQ<26>	30
89 12	MEM_A_DQ<32>	MAKE_BASE=TRUE		=MEM_A_DQ<30>	30

CPU CHANNEL A DQS 5 -> DIMM A DQS 2

89 12	MEM_A_DQS_N<5>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<2>	30
89 12	MEM_A_DQS_P<5>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<2>	30
89 12	MEM_A_DQ<47>	MAKE_BASE=TRUE		=MEM_A_DQ<17>	30
89 12	MEM_A_DQ<46>	MAKE_BASE=TRUE		=MEM_A_DQ<16>	30
89 12	MEM_A_DQ<45>	MAKE_BASE=TRUE		=MEM_A_DQ<22>	30
89 12	MEM_A_DQ<44>	MAKE_BASE=TRUE		=MEM_A_DQ<23>	30
89 12	MEM_A_DQ<43>	MAKE_BASE=TRUE		=MEM_A_DQ<21>	30
89 12	MEM_A_DQ<42>	MAKE_BASE=TRUE		=MEM_A_DQ<20>	30
89 12	MEM_A_DQ<41>	MAKE_BASE=TRUE		=MEM_A_DQ<18>	30
89 12	MEM_A_DQ<40>	MAKE_BASE=TRUE		=MEM_A_DQ<19>	30

CPU CHANNEL A DQS 6 -> DIMM A DQS 1

89 12	MEM_A_DQS_N<6>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<1>	30
89 12	MEM_A_DQS_P<6>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<1>	30
89 12	MEM_A_DQ<55>	MAKE_BASE=TRUE		=MEM_A_DQ<12>	30
89 12	MEM_A_DQ<54>	MAKE_BASE=TRUE		=MEM_A_DQ<13>	30
89 12	MEM_A_DQ<53>	MAKE_BASE=TRUE		=MEM_A_DQ<10>	30
89 12	MEM_A_DQ<52>	MAKE_BASE=TRUE		=MEM_A_DQ<15>	30
89 12	MEM_A_DQ<51>	MAKE_BASE=TRUE		=MEM_A_DQ<8>	30
89 12	MEM_A_DQ<50>	MAKE_BASE=TRUE		=MEM_A_DQ<9>	30
89 12	MEM_A_DQ<49>	MAKE_BASE=TRUE		=MEM_A_DQ<11>	30
89 12	MEM_A_DQ<48>	MAKE_BASE=TRUE		=MEM_A_DQ<14>	30

CPU CHANNEL A DQS 7 -> DIMM A DQS 0

89 12	MEM_A_DQS_N<7>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_N<0>	30
89 12	MEM_A_DQS_P<7>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_DQS_P<0>	30
89 12	MEM_A_DQ<63>	MAKE_BASE=TRUE		=MEM_A_DQ<5>	30
89 12	MEM_A_DQ<62>	MAKE_BASE=TRUE		=MEM_A_DQ<4>	30
89 12	MEM_A_DQ<61>	MAKE_BASE=TRUE		=MEM_A_DQ<3>	30
89 12	MEM_A_DQ<60>	MAKE_BASE=TRUE		=MEM_A_DQ<2>	30
89 12	MEM_A_DQ<59>	MAKE_BASE=TRUE		=MEM_A_DQ<1>	30
89 12	MEM_A_DQ<58>	MAKE_BASE=TRUE		=MEM_A_DQ<0>	30
89 12	MEM_A_DQ<57>	MAKE_BASE=TRUE		=MEM_A_DQ<7>	30
89 12	MEM_A_DQ<56>	MAKE_BASE=TRUE		=MEM_A_DQ<6>	30

CPU CHANNEL B DQS 0 -> DIMM B DQS 7

89 12	MEM_B_DQS_N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<7>	31
89 12	MEM_B_DQS_P<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<7>	31
89 12	MEM_B_DQ<7>	MAKE_BASE=TRUE		=MEM_B_DQ<61>	31
89 12	MEM_B_DQ<6>	MAKE_BASE=TRUE		=MEM_B_DQ<60>	31
89 12	MEM_B_DQ<5>	MAKE_BASE=TRUE		=MEM_B_DQ<62>	31
89 12	MEM_B_DQ<4>	MAKE_BASE=TRUE		=MEM_B_DQ<63>	31
89 12	MEM_B_DQ<3>	MAKE_BASE=TRUE		=MEM_B_DQ<56>	31
89 12	MEM_B_DQ<2>	MAKE_BASE=TRUE		=MEM_B_DQ<57>	31
89 12	MEM_B_DQ<1>	MAKE_BASE=TRUE		=MEM_B_DQ<58>	31
89 12	MEM_B_DQ<0>	MAKE_BASE=TRUE		=MEM_B_DQ<59>	31

CPU CHANNEL B DQS 1 -> DIMM B DQS 6

89 12	MEM_B_DQS_N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<6>	31
89 12	MEM_B_DQS_P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<6>	31
89 12	MEM_B_DQ<15>	MAKE_BASE=TRUE		=MEM_B_DQ<51>	31
89 12	MEM_B_DQ<14>	MAKE_BASE=TRUE		=MEM_B_DQ<54>	31
89 12	MEM_B_DQ<13>	MAKE_BASE=TRUE		=MEM_B_DQ<53>	31
89 12	MEM_B_DQ<12>	MAKE_BASE=TRUE		=MEM_B_DQ<52>	31
89 12	MEM_B_DQ<11>	MAKE_BASE=TRUE		=MEM_B_DQ<48>	31
89 12	MEM_B_DQ<10>	MAKE_BASE=TRUE		=MEM_B_DQ<49>	31
89 12	MEM_B_DQ<9>	MAKE_BASE=TRUE		=MEM_B_DQ<50>	31
89 12	MEM_B_DQ<8>	MAKE_BASE=TRUE		=MEM_B_DQ<55>	31

CPU CHANNEL B DQS 2 -> DIMM B DQS 5

89 12	MEM_B_DQS_N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<5>	31
89 12	MEM_B_DQS_P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<5>	31
89 12	MEM_B_DQ<23>	MAKE_BASE=TRUE		=MEM_B_DQ<45>	31
89 12	MEM_B_DQ<22>	MAKE_BASE=TRUE		=MEM_B_DQ<40>	31
89 12	MEM_B_DQ<21>	MAKE_BASE=TRUE		=MEM_B_DQ<42>	31
89 12	MEM_B_DQ<20>	MAKE_BASE=TRUE		=MEM_B_DQ<47>	31
89 12	MEM_B_DQ<19>	MAKE_BASE=TRUE		=MEM_B_DQ<44>	31
89 12	MEM_B_DQ<18>	MAKE_BASE=TRUE		=MEM_B_DQ<41>	31
89 12	MEM_B_DQ<17>	MAKE_BASE=TRUE		=MEM_B_DQ<46>	31
89 12	MEM_B_DQ<16>	MAKE_BASE=TRUE		=MEM_B_DQ<43>	31

CPU CHANNEL B DQS 3 -> DIMM B DQS 4

89 12	MEM_B_DQS_N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<4>	31
89 12	MEM_B_DQS_P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<4>	31
89 12	MEM_B_DQ<31>	MAKE_BASE=TRUE		=MEM_B_DQ<32>	31
89 12	MEM_B_DQ<30>	MAKE_BASE=TRUE		=MEM_B_DQ<39>	31
89 12	MEM_B_DQ<29>	MAKE_BASE=TRUE		=MEM_B_DQ<37>	31
89 12	MEM_B_DQ<28>	MAKE_BASE=TRUE		=MEM_B_DQ<38>	31
89 12	MEM_B_DQ<27>	MAKE_BASE=TRUE		=MEM_B_DQ<35>	31
89 12	MEM_B_DQ<26>	MAKE_BASE=TRUE		=MEM_B_DQ<34>	31
89 12	MEM_B_DQ<25>	MAKE_BASE=TRUE		=MEM_B_DQ<36>	31
89 12	MEM_B_DQ<24>	MAKE_BASE=TRUE		=MEM_B_DQ<33>	31

CPU CHANNEL B DQS 4 -> DIMM B DQS 3

89 12	MEM_B_DQS_N<4>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<3>	31
89 12	MEM_B_DQS_P<4>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<3>	31
89 12	MEM_B_DQ<39>	MAKE_BASE=TRUE		=MEM_B_DQ<28>	31
89 12	MEM_B_DQ<38>	MAKE_BASE=TRUE		=MEM_B_DQ<24>	31
89 12	MEM_B_DQ<37>	MAKE_BASE=TRUE		=MEM_B_DQ<31>	31
89 12	MEM_B_DQ<36>	MAKE_BASE=TRUE		=MEM_B_DQ<30>	31
89 12	MEM_B_DQ<35>	MAKE_BASE=TRUE		=MEM_B_DQ<29>	31
89 12	MEM_B_DQ<34>	MAKE_BASE=TRUE		=MEM_B_DQ<25>	31
89 12	MEM_B_DQ<33>	MAKE_BASE=TRUE		=MEM_B_DQ<27>	31
89 12	MEM_B_DQ<32>	MAKE_BASE=TRUE		=MEM_B_DQ<26>	31

CPU CHANNEL B DQS 5 -> DIMM B DQS 2

89 12	MEM_B_DQS_N<5>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<2>	31
89 12	MEM_B_DQS_P<5>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<2>	31
89 12	MEM_B_DQ<47>	MAKE_BASE=TRUE		=MEM_B_DQ<18>	31
89 12	MEM_B_DQ<46>	MAKE_BASE=TRUE		=MEM_B_DQ<20>	31
89 12	MEM_B_DQ<45>	MAKE_BASE=TRUE		=MEM_B_DQ<23>	31
89 12	MEM_B_DQ<44>	MAKE_BASE=TRUE		=MEM_B_DQ<22>	31
89 12	MEM_B_DQ<43>	MAKE_BASE=TRUE		=MEM_B_DQ<19>	31
89 12	MEM_B_DQ<42>	MAKE_BASE=TRUE		=MEM_B_DQ<21>	31
89 12	MEM_B_DQ<41>	MAKE_BASE=TRUE		=MEM_B_DQ<17>	31
89 12	MEM_B_DQ<40>	MAKE_BASE=TRUE		=MEM_B_DQ<16>	31

CPU CHANNEL B DQS 6 -> DIMM B DQS 1

89 12	MEM_B_DQS_N<6>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<1>	31
89 12	MEM_B_DQS_P<6>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<1>	31
89 12	MEM_B_DQ<55>	MAKE_BASE=TRUE		=MEM_B_DQ<8>	31
89 12	MEM_B_DQ<54>	MAKE_BASE=TRUE		=MEM_B_DQ<14>	31
89 12	MEM_B_DQ<53>	MAKE_BASE=TRUE		=MEM_B_DQ<10>	31
89 12	MEM_B_DQ<52>	MAKE_BASE=TRUE		=MEM_B_DQ<11>	31
89 12	MEM_B_DQ<51>	MAKE_BASE=TRUE		=MEM_B_DQ<9>	31
89 12	MEM_B_DQ<50>	MAKE_BASE=TRUE		=MEM_B_DQ<15>	31
89 12	MEM_B_DQ<49>	MAKE_BASE=TRUE		=MEM_B_DQ<12>	31
89 12	MEM_B_DQ<48>	MAKE_BASE=TRUE		=MEM_B_DQ<13>	31

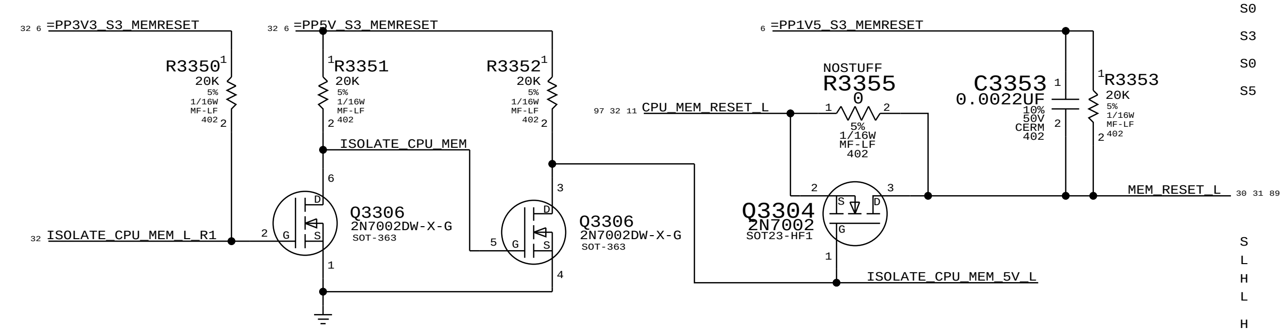
CPU CHANNEL B DQS 7 -> DIMM B DQS 0

89 12	MEM_B_DQS_N<7>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_N<0>	31
89 12	MEM_B_DQS_P<7>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_DQS_P<0>	31
89 12	MEM_B_DQ<63>	MAKE_BASE=TRUE		=MEM_B_DQ<2>	31
89 12	MEM_B_DQ<62>	MAKE_BASE=TRUE		=MEM_B_DQ<1>	31
89 12	MEM_B_DQ<61>	MAKE_BASE=TRUE		=MEM_B_DQ<3>	31
89 12	MEM_B_DQ<60>	MAKE_BASE=TRUE		=MEM_B_DQ<7>	31
89 12	MEM_B_DQ<59>	MAKE_BASE=TRUE		=MEM_B_DQ<4>	31
89 12	MEM_B_DQ<58>	MAKE_BASE=TRUE		=MEM_B_DQ<5>	31
89 12	MEM_B_DQ<57>	MAKE_BASE=TRUE		=MEM_B_DQ<6>	31
89 12	MEM_B_DQ<56>	MAKE_BASE=TRUE		=MEM_B_DQ<0>	31

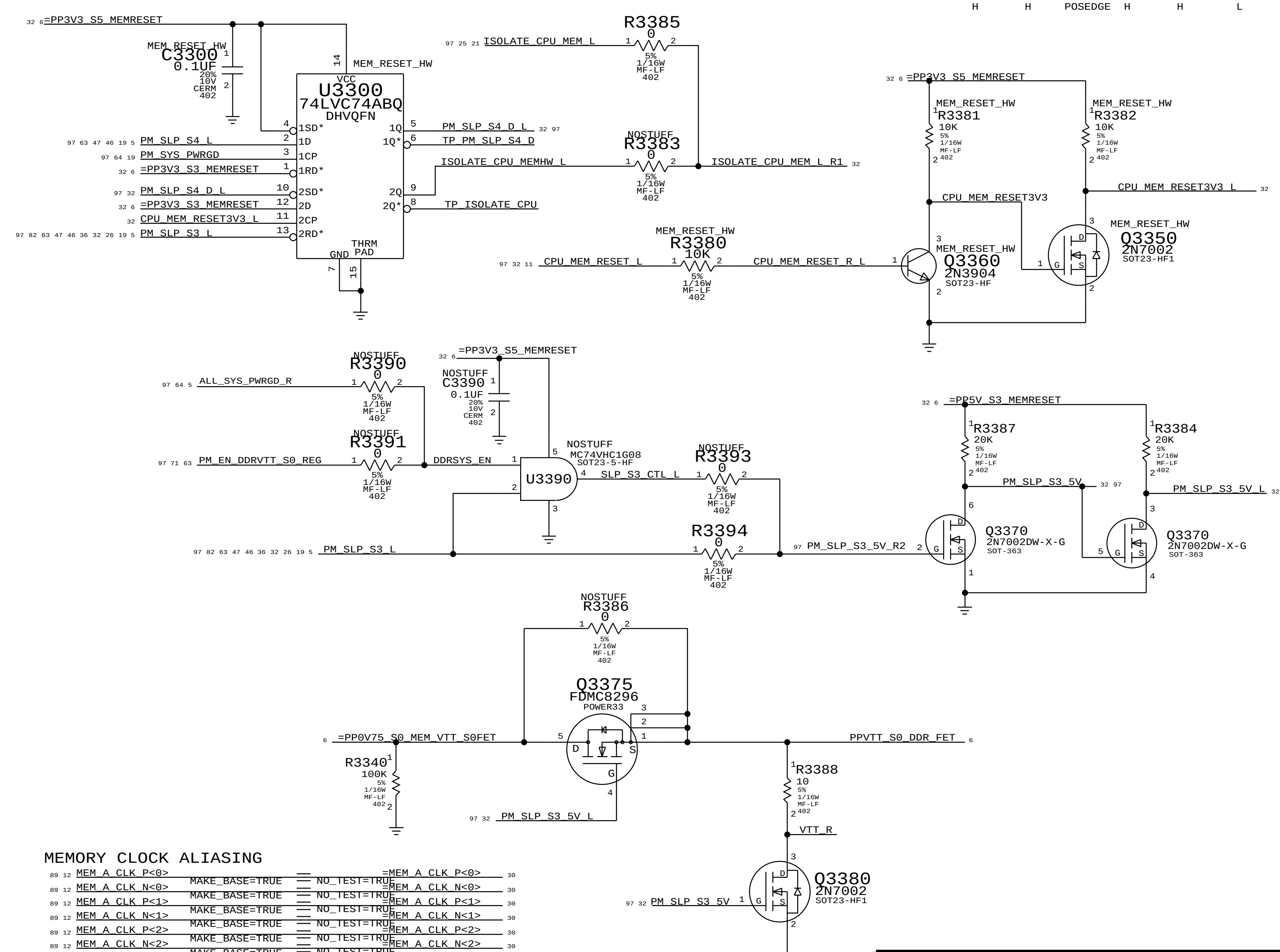
DDR3 RESET SUPPORT

SNB? CANNOT CONTROL THIS SIGNAL DIRECTLY SINCE IT MUST BE HIGH IN SLEEP AND CPU MEM RAILS ARE NOT POWERED IN SLEEP.

	CPU_RESET_L	ISOLATE_L	MEM_RESET_L
S5	0	3.3V	0
S0	0	3.3V	0
S0	1.5V	3.3V	1.5V
S3	0	0	1.5V
S0	1.5V	3.3V	1.5V
S5	0	3.3V	0



S	R	CLK	D	Q	QB
L	H	X	X	H	L
H	L	X	X	L	H
L	L	X	X	H	H
H	H	POSEDGE	L	L	H
H	H	POSEDGE	H	H	L



MEMORY CLOCK ALIASING

89 12	MEM_A_CLK_P<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_P<0>	30
89 12	MEM_A_CLK_N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_N<0>	30
89 12	MEM_A_CLK_P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_P<1>	30
89 12	MEM_A_CLK_N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_N<1>	30
89 12	MEM_A_CLK_P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_P<2>	30
89 12	MEM_A_CLK_N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_N<2>	30
89 12	MEM_A_CLK_P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_P<3>	30
89 12	MEM_A_CLK_N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_A_CLK_N<3>	30
89 12	MEM_B_CLK_P<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_P<0>	31
89 12	MEM_B_CLK_N<0>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_N<0>	31
89 12	MEM_B_CLK_P<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_P<1>	31
89 12	MEM_B_CLK_N<1>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_N<1>	31
89 12	MEM_B_CLK_P<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_P<2>	31
89 12	MEM_B_CLK_N<2>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_N<2>	31
89 12	MEM_B_CLK_P<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_P<3>	31
89 12	MEM_B_CLK_N<3>	MAKE_BASE=TRUE	NO_TEST=TRUE	=MEM_B_CLK_N<3>	31

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
DDR3 SUPPORT AND BITSWAPS

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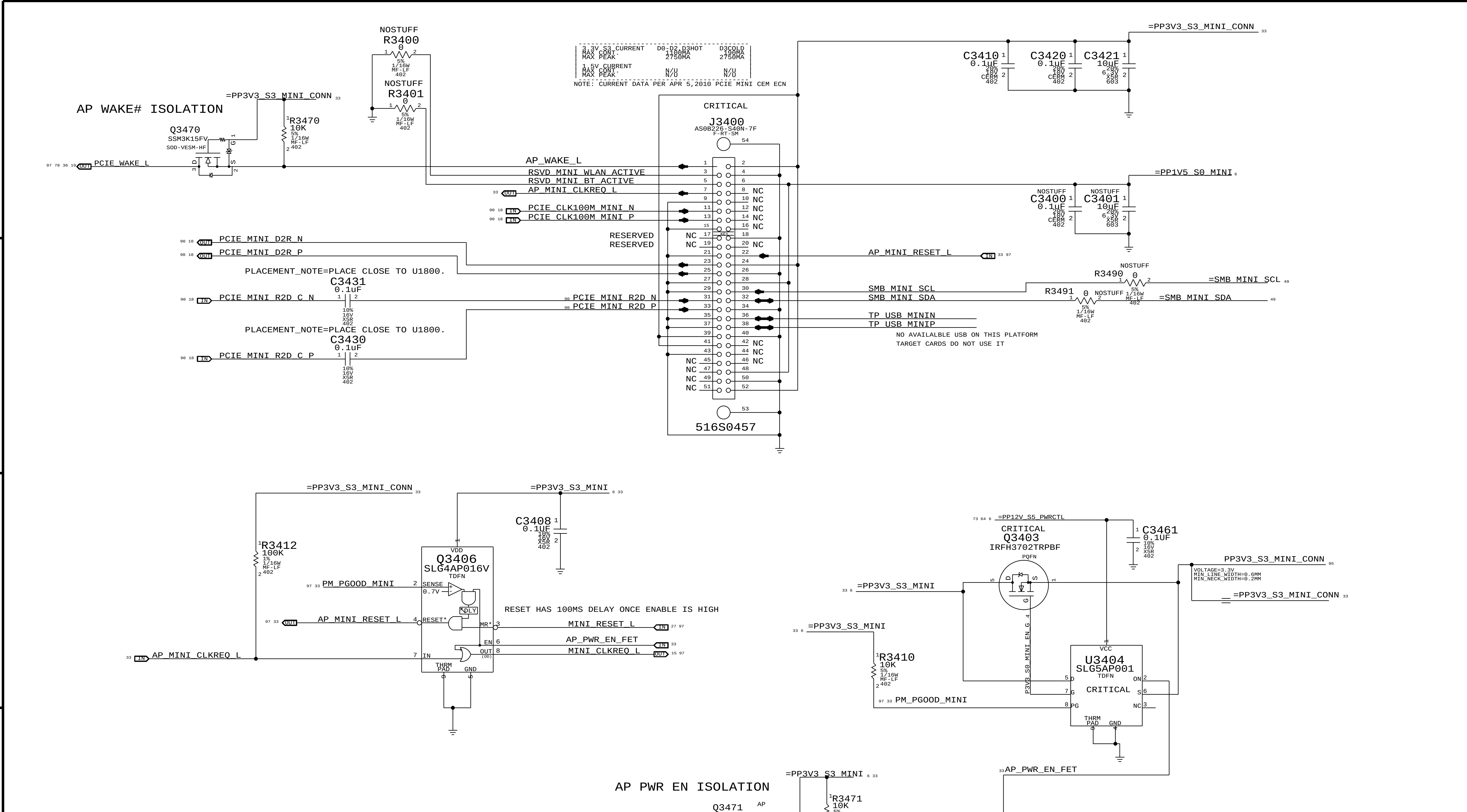
8	7	6	5	4	3	2	1



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PCI-E Wireless Connector

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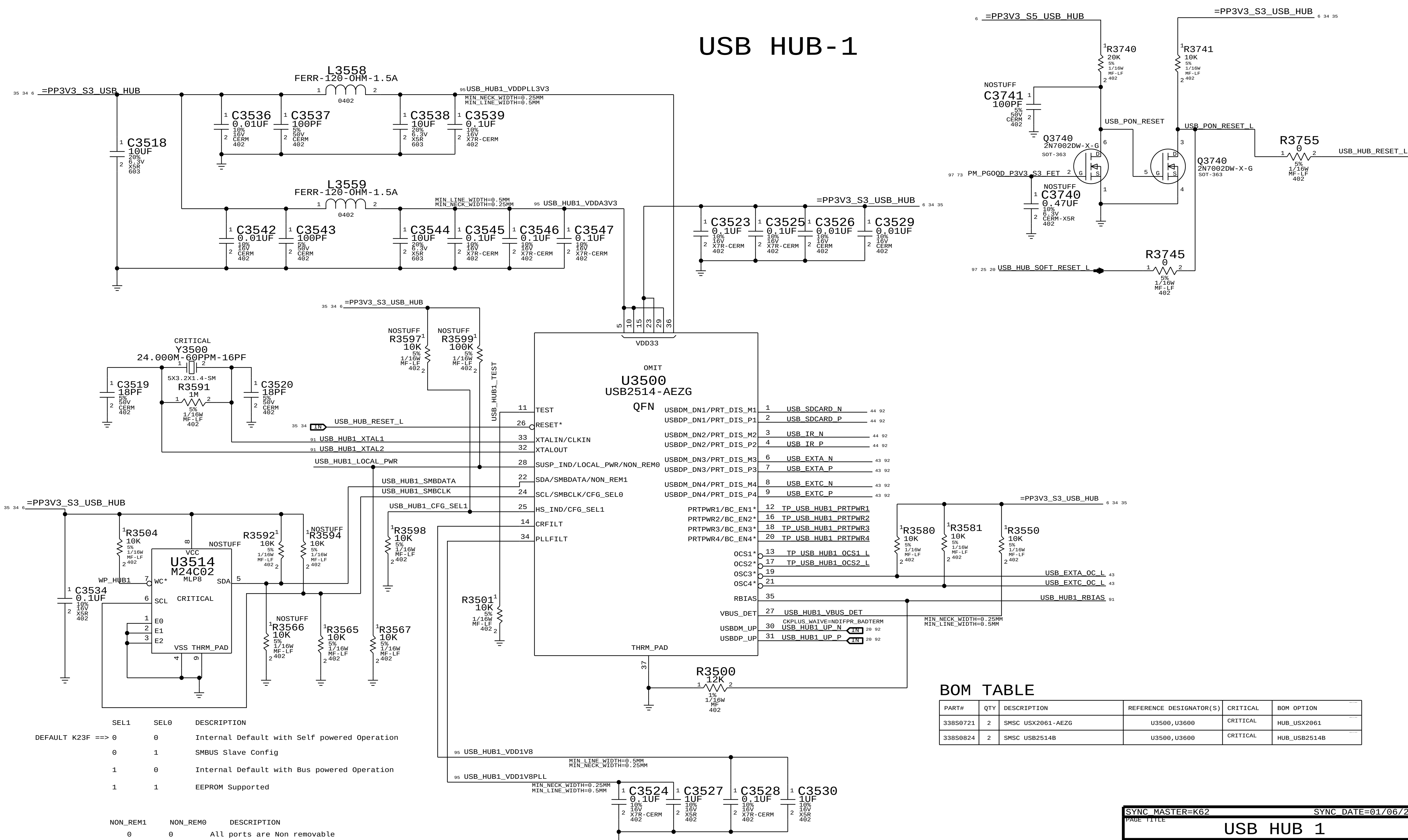
USB HUB-1



SEL1	SEL0	DESCRIPTION
DEFAULT K23F ==> 0	0	Internal Default with Self powered Operation
0	1	SMBUS Slave Config
1	0	Internal Default with Bus powered Operation
1	1	EEPROM Supported

NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are Non removable
0	1	Port1 is non removable
DEFAULT K23F ==> 1	0	Port 1 and 2 are non removable
1	1	Port1,2 and 3 are non Removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
338S0721	2	SMSC USX2061-AEZG	U3500,U3600	CRITICAL	HUB_USX2061
338S0824	2	SMSC USB2514B	U3500,U3600	CRITICAL	HUB_USB2514B

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PAGE TITLE
USB HUB 1

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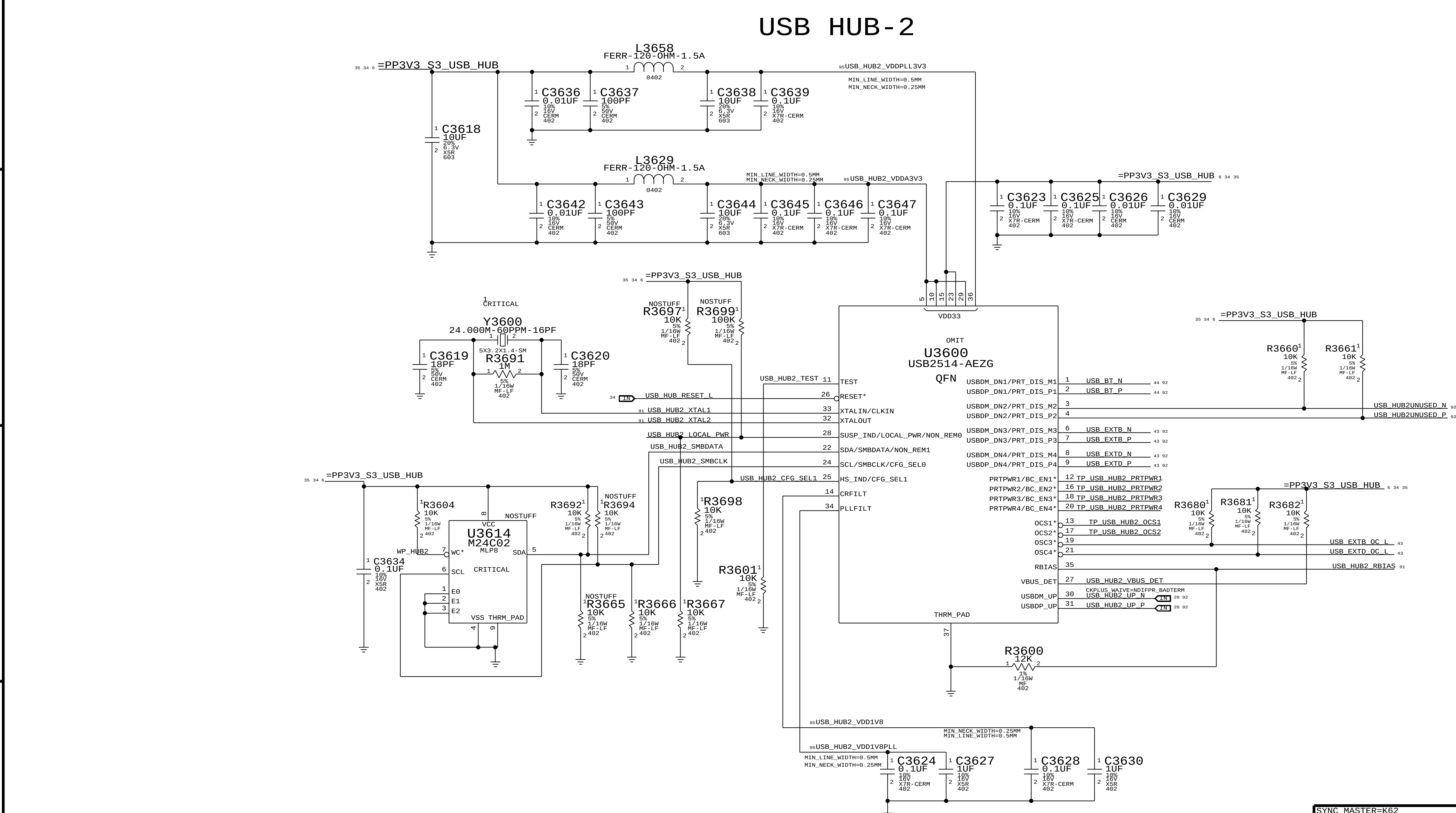

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PAGE TITLE

USB HUB 2

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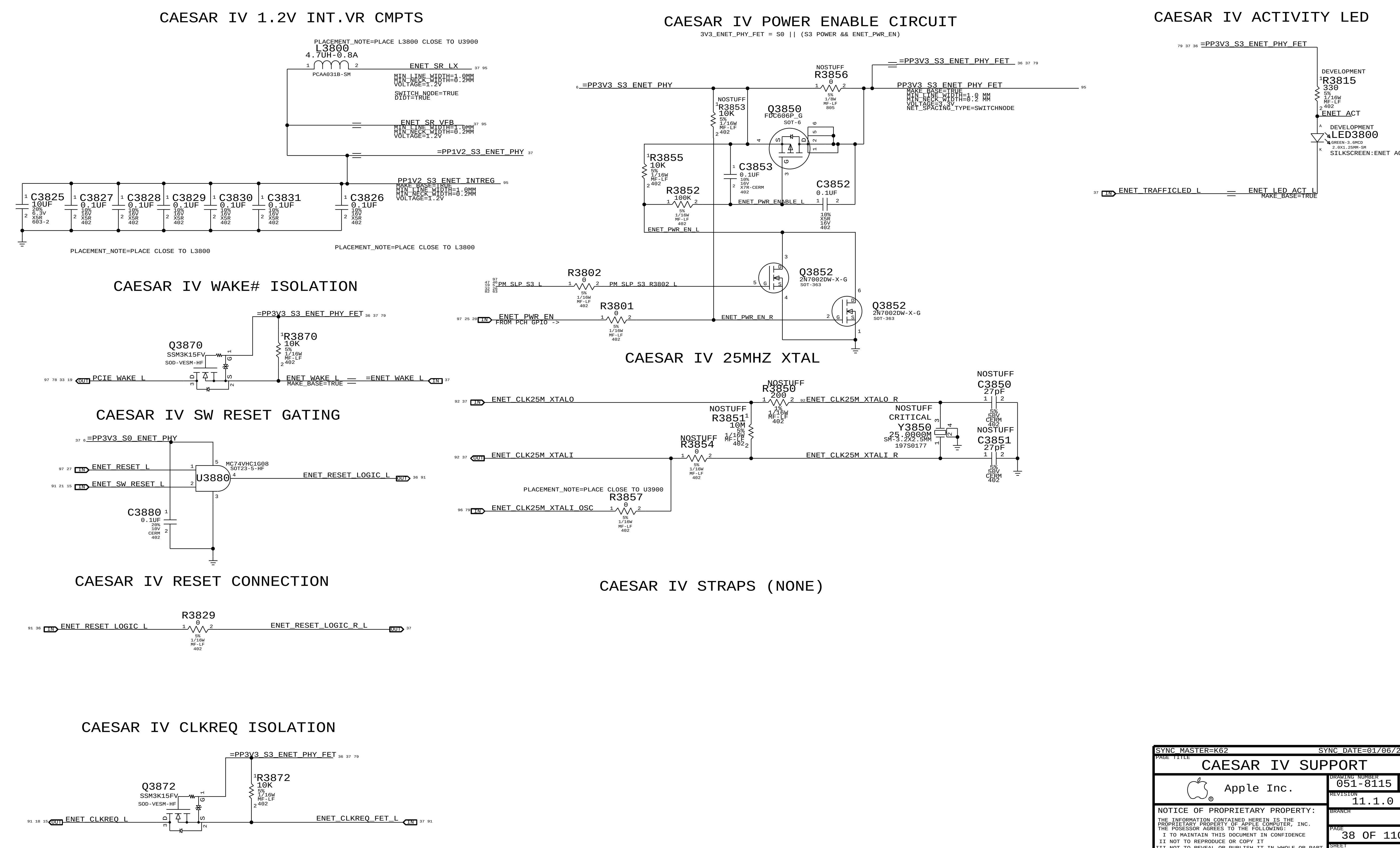

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

CAESAR IV SUPPORT

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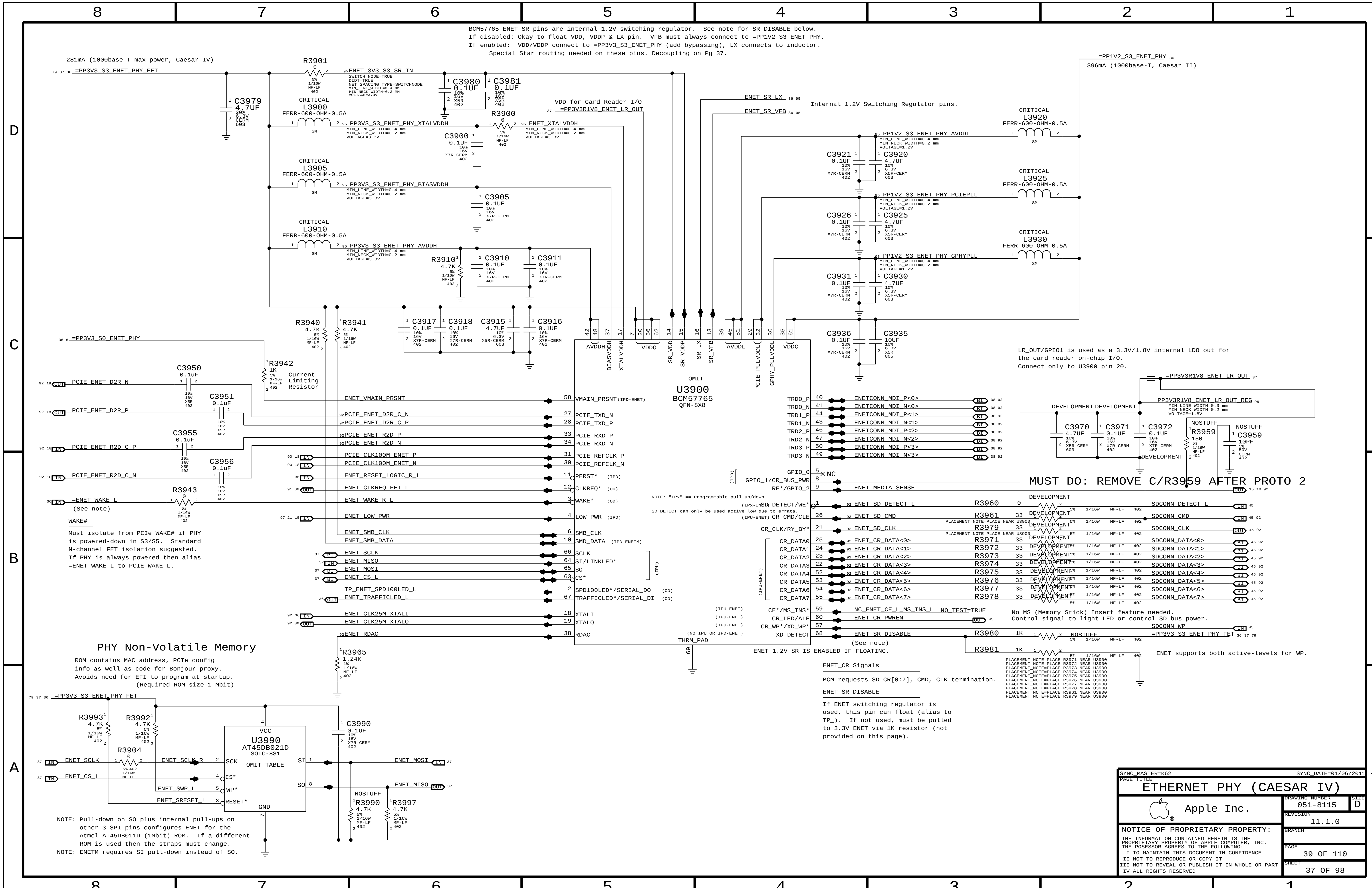

BCM57765 ENET SR pins are internal 1.2V switching regulator. See note for SR_DISABLE below.
If disabled: Okay to float VDD, VDDP & LX pin. VFB must always connect to =PP1V2_S3_ENET_PHY.
If enabled: VDD/VDDP connect to =PP3V3_S3_ENET_PHY (add bypassing), LX connects to inductor.
Special Star routing needed on these pins. Decoupling on Pg 37.
281mA (1000base-T max power, Caesar IV)
=PP3V3_S3_ENET_PHY_FET
R3901
ENET_3V3_S3_SR_IN
C3980
C3981
C3979
CRITICAL
L3900
FERR-600-OHM-0.5A
PP3V3_S3_ENET_PHY_XTALVDDH
C3900
R3900
ENET_XTALVDDH
VDD for Card Reader I/O
=PP3V3R1V8_ENET_LR_OUT
ENET_SR_LX
ENET_SR_VFB
Internal 1.2V Switching Regulator pins.
=PP1V2_S3_ENET_PHY
396mA (1000base-T, Caesar II)
L3905
PP3V3_S3_ENET_PHY_BIASVDDH
C3905
L3910
PP3V3_S3_ENET_PHY_AVDDH
R3910
C3910
C3911
PP1V2_S3_ENET_PHY_AVDDL
L3920
C3921
C3920
PP1V2_S3_ENET_PHY_PCIEPLL
L3925
C3926
C3925
PP1V2_S3_ENET_PHY_GPHYPLL
L3930
C3931
C3930
C3936
C3935
R3940
R3941
C3917
C3918
C3915
C3916
=PP3V3_S0_ENET_PHY
R3942
Current Limiting Resistor
C3950
PCIE_ENET_D2R_N
C3951
PCIE_ENET_D2R_P
C3955
PCIE_ENET_R2D_C_P
C3956
PCIE_ENET_R2D_C_N
R3943
=ENET_WAKE_L
(See note)
WAKE#
Must isolate from PCIe WAKE# if PHY is powered-down in S3/S5. Standard N-channel FET isolation suggested. If PHY is always powered then alias =ENET_WAKE_L to PCIE_WAKE_L.
OMIT
U3900
BCM57765
QFN-8X8
ENET_VMAIN_PRSNT
PCIE_ENET_D2R_C_N
PCIE_ENET_D2R_C_P
PCIE_ENET_R2D_P
PCIE_ENET_R2D_N
PCIE_CLK100M_ENET_P
PCIE_CLK100M_ENET_N
ENET_RESET_LOGIC_R_L
ENET_CLKREQ_FET_L
ENET_WAKE_R_L
ENET_LOW_PWR
ENET_SMB_CLK
ENET_SMB_DATA
ENET_SCLK
ENET_MISO
ENET_MOSI
ENET_CS_L
TP_ENET_SPD100LED_L
ENET_TRAFFICLED_L
ENET_CLK25M_XTALI
ENET_CLK25M_XTALO
ENET_RDAC
ENETCONN_MDI_P<0>
ENETCONN_MDI_N<0>
ENETCONN_MDI_P<1>
ENETCONN_MDI_N<1>
ENETCONN_MDI_P<2>
ENETCONN_MDI_N<2>
ENETCONN_MDI_P<3>
ENETCONN_MDI_N<3>
ENET_MEDIA_SENSE
NOTE: "IPx" == Programmable pull-up/down
SD_DETECT can only be used active low due to errata.
LR_OUT/GPIO1 is used as a 3.3V/1.8V internal LDO out for the card reader on-chip I/O.
Connect only to U3900 pin 20.
=PP3V3R1V8_ENET_LR_OUT
PP3V3R1V8_ENET_LR_OUT_REG
C3970
C3971
C3972
R3959
C3959
MUST DO: REMOVE C/R3959 AFTER PROTO 2
ENET_SD_DETECT_L
ENET_SD_CMD
ENET_SD_CLK
ENET_CR_DATA<0>
ENET_CR_DATA<1>
ENET_CR_DATA<2>
ENET_CR_DATA<3>
ENET_CR_DATA<4>
ENET_CR_DATA<5>
ENET_CR_DATA<6>
ENET_CR_DATA<7>
SDCONN_DETECT_L
SDCONN_CMD
SDCONN_CLK
SDCONN_DATA<0>
SDCONN_DATA<1>
SDCONN_DATA<2>
SDCONN_DATA<3>
SDCONN_DATA<4>
SDCONN_DATA<5>
SDCONN_DATA<6>
SDCONN_DATA<7>
NC_ENET_CE_L_MS_INS_L NO_TEST=TRUE
ENET_CR_PWREN
ENET_SR_DISABLE
(See note)
No MS (Memory Stick) Insert feature needed.
Control signal to light LED or control SD bus power.
SDCONN_WP
=PP3V3_S3_ENET_PHY_FET
ENET 1.2V SR IS ENABLED IF FLOATING.
ENET supports both active-levels for WP.
ENET_CR Signals
BCM requests SD CR[0:7], CMD, CLK termination.
ENET_SR_DISABLE
If ENET switching regulator is used, this pin can float (alias to TP_). If not used, must be pulled to 3.3V ENET via 1K resistor (not provided on this page).
PHY Non-Volatile Memory
ROM contains MAC address, PCIe config info as well as code for Bonjour proxy. Avoids need for EFI to program at startup. (Required ROM size 1 Mbit)
=PP3V3_S3_ENET_PHY_FET
R3993
R3992
R3904
ENET_SCLK
ENET_CS_L
ENET_SCLK_R
ENET_SWP_L
ENET_SRESET_L
U3990
AT45DB021D
SOIC-8S1
OMIT_TABLE
C3990
ENET_MOSI
ENET_MISO
NOSTUFF
R3990
R3997
NOTE: Pull-down on SO plus internal pull-ups on other 3 SPI pins configures ENET for the Atmel AT45DB011D (1Mbit) ROM. If a different ROM is used then the straps must change.
NOTE: ENETM requires SI pull-down instead of SO.
SYNC_MASTER=K62
SYNC_DATE=01/06/2011
ETHERNET PHY (CAESAR IV)
Apple Inc.
051-8115
D
11.1.0
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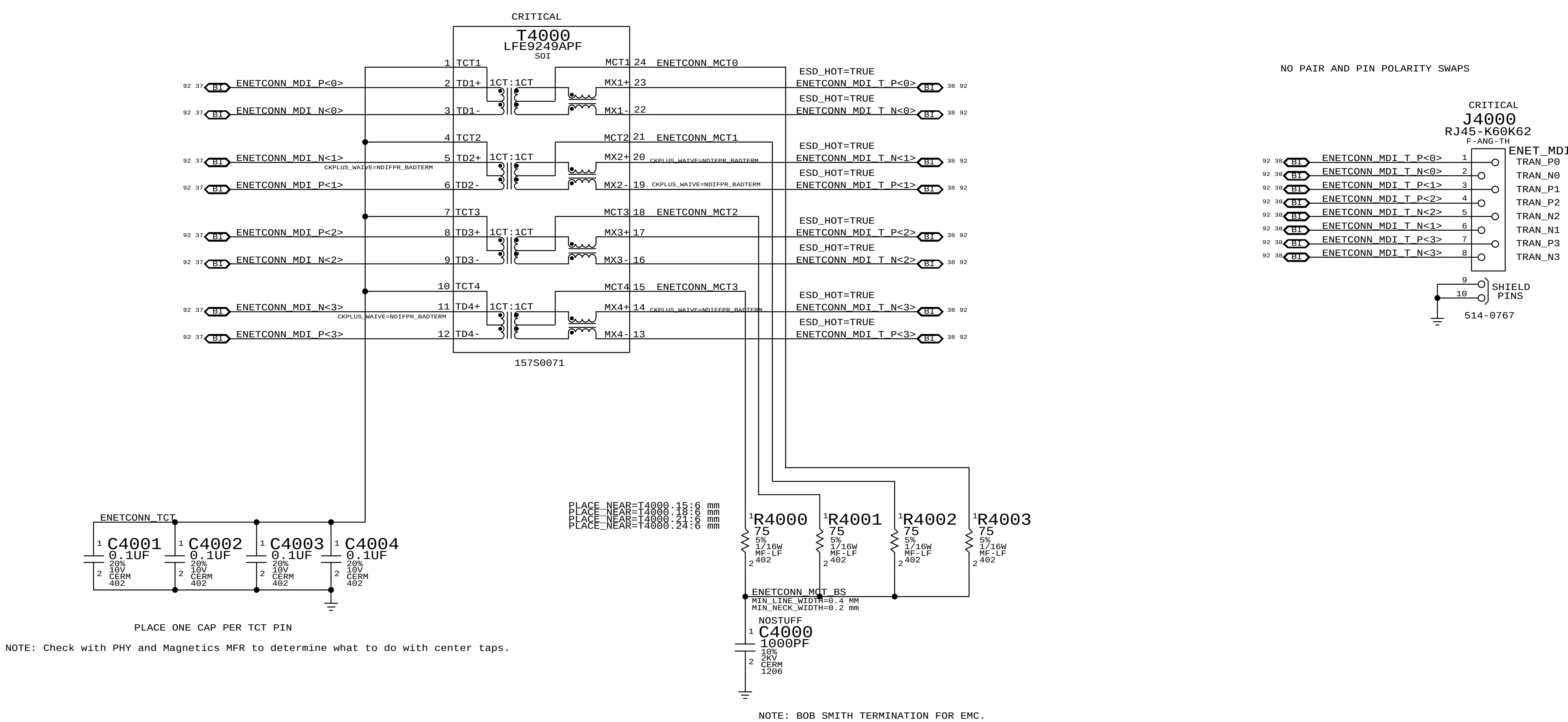
THIS PAGE DIFFERENT BETWEEN K60 and K62.



SYNC_MASTER=K60_MARK SYNC_DATE=01/06/2011

PAGE TITLE
Ethernet Connector

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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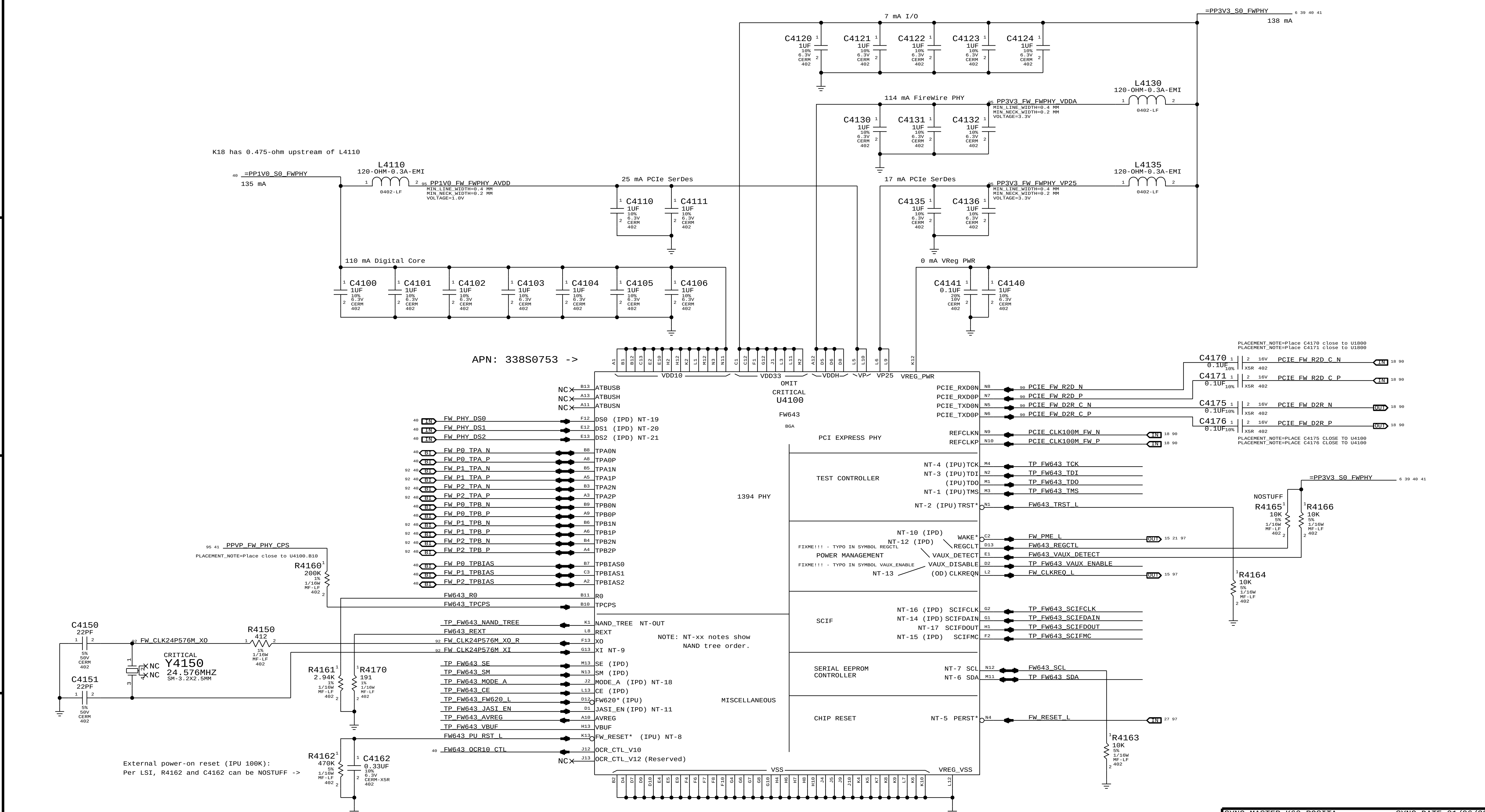
PAGE 40 OF 110

SHEET 38 OF 98



SYNC_MASTER=K60_ROSITA SYNC_DATE=01/06/2011

PAGE TITLE

FireWire LLC/PHY (FW643)

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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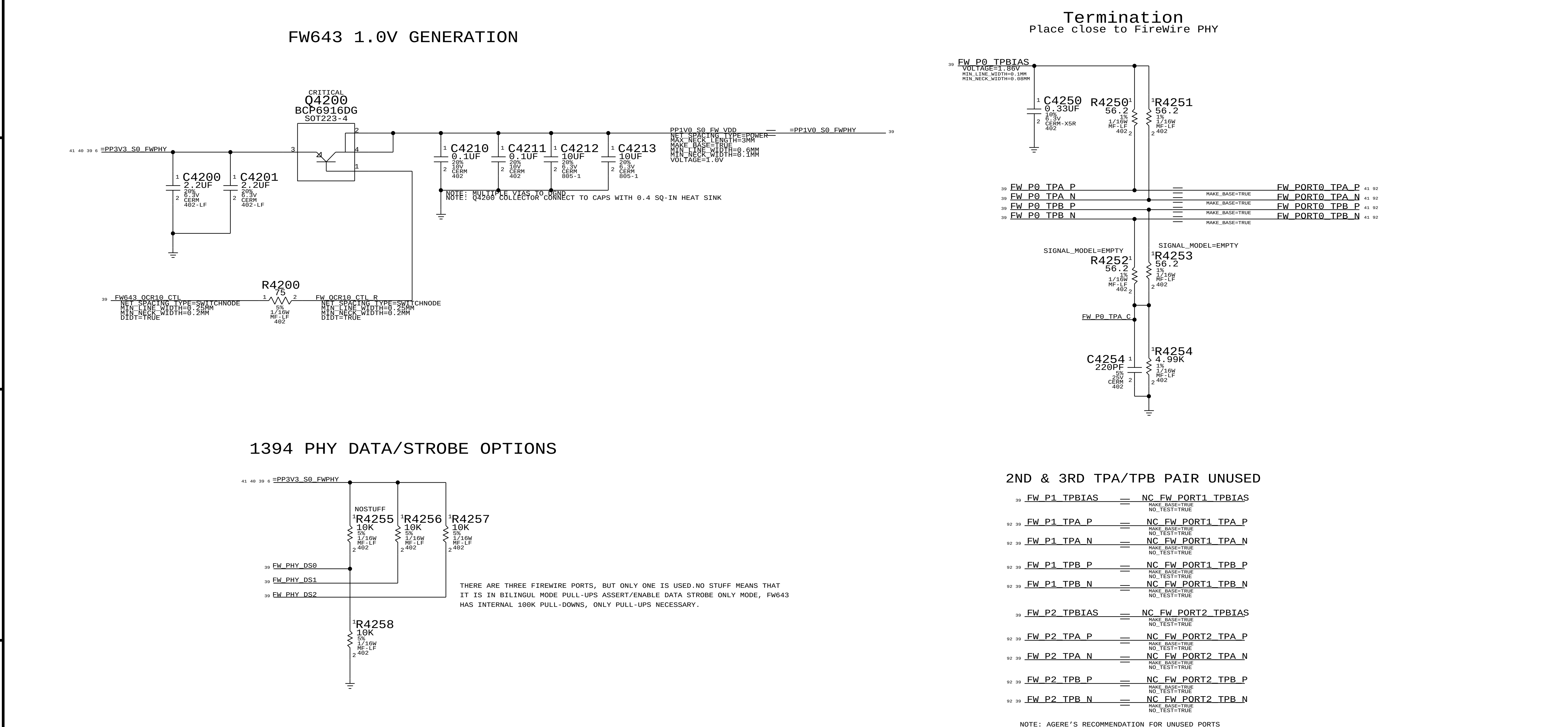

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

FireWire: 1394B MISC

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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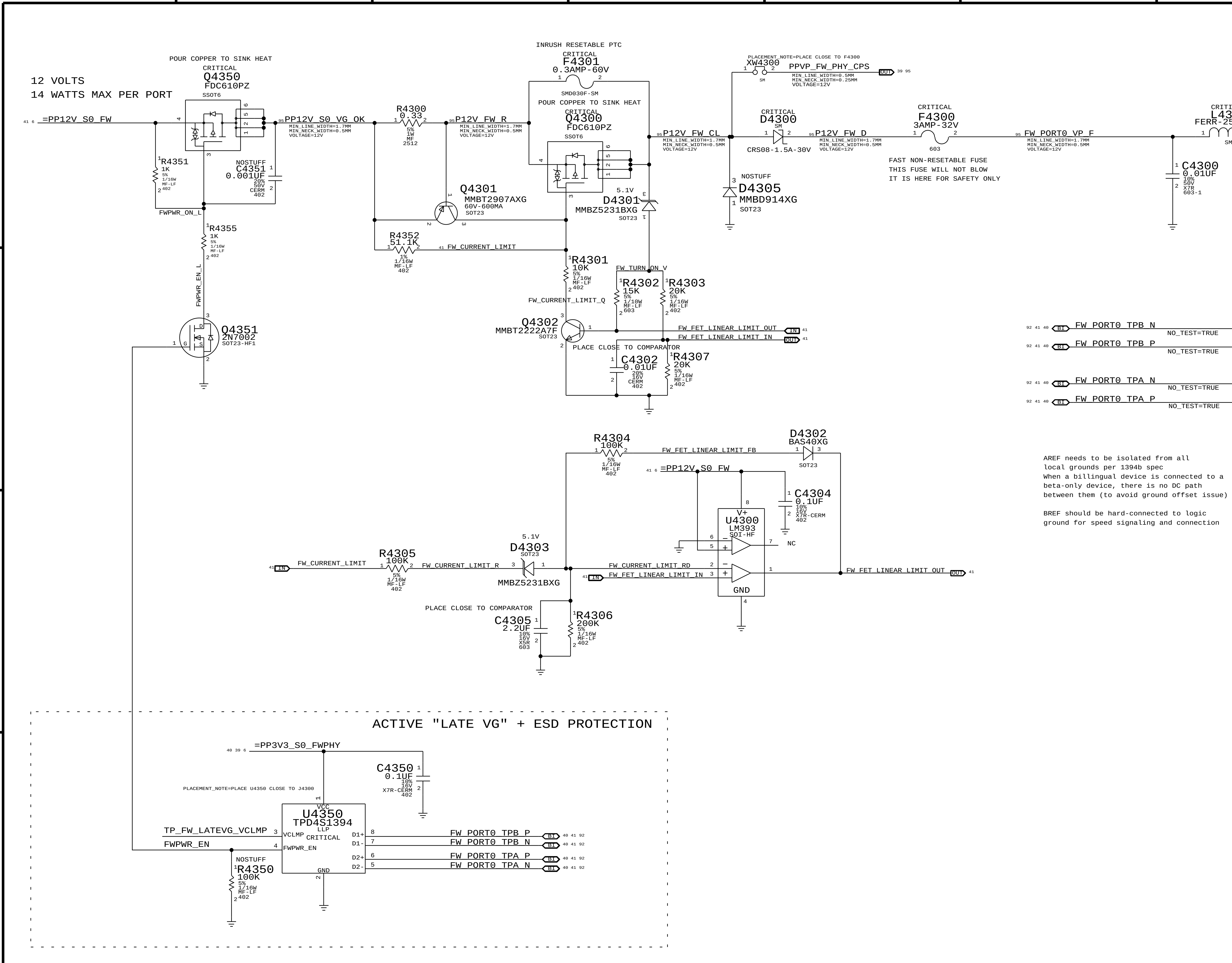

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

FIREWIRE CONNECTOR

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

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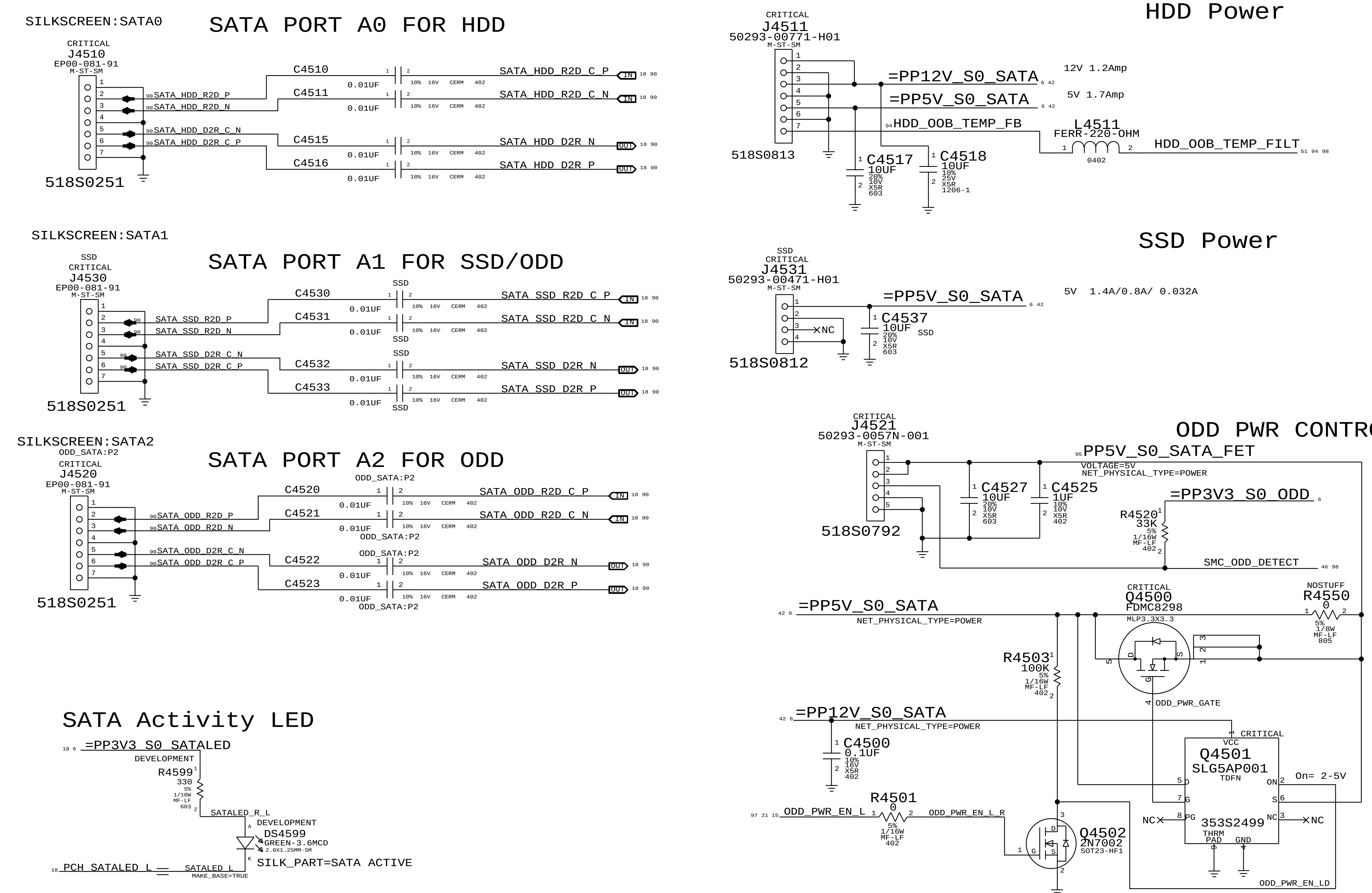
NOTICE OF PROPRIETARY PROPERTY:

BOMOPTION OPTIONS FOR SATA PORT A1 AND A2

A1	A2	ODD_SATA:P1	ODD_SATA:P2
SSD	ODD		X
ODD		X	

USE OF PORT A2 FOR SSD IS NOT INTENDED VIA BOMOPTION THOUGH MLB SUPPORTS IT.
K60E should stuff S_P1_ODD:YES because it has no SSD option.



SYNC_MASTER=K60_JERRY SYNC_DATE=01/06/2011

PAGE TITLE: SATA Connectors

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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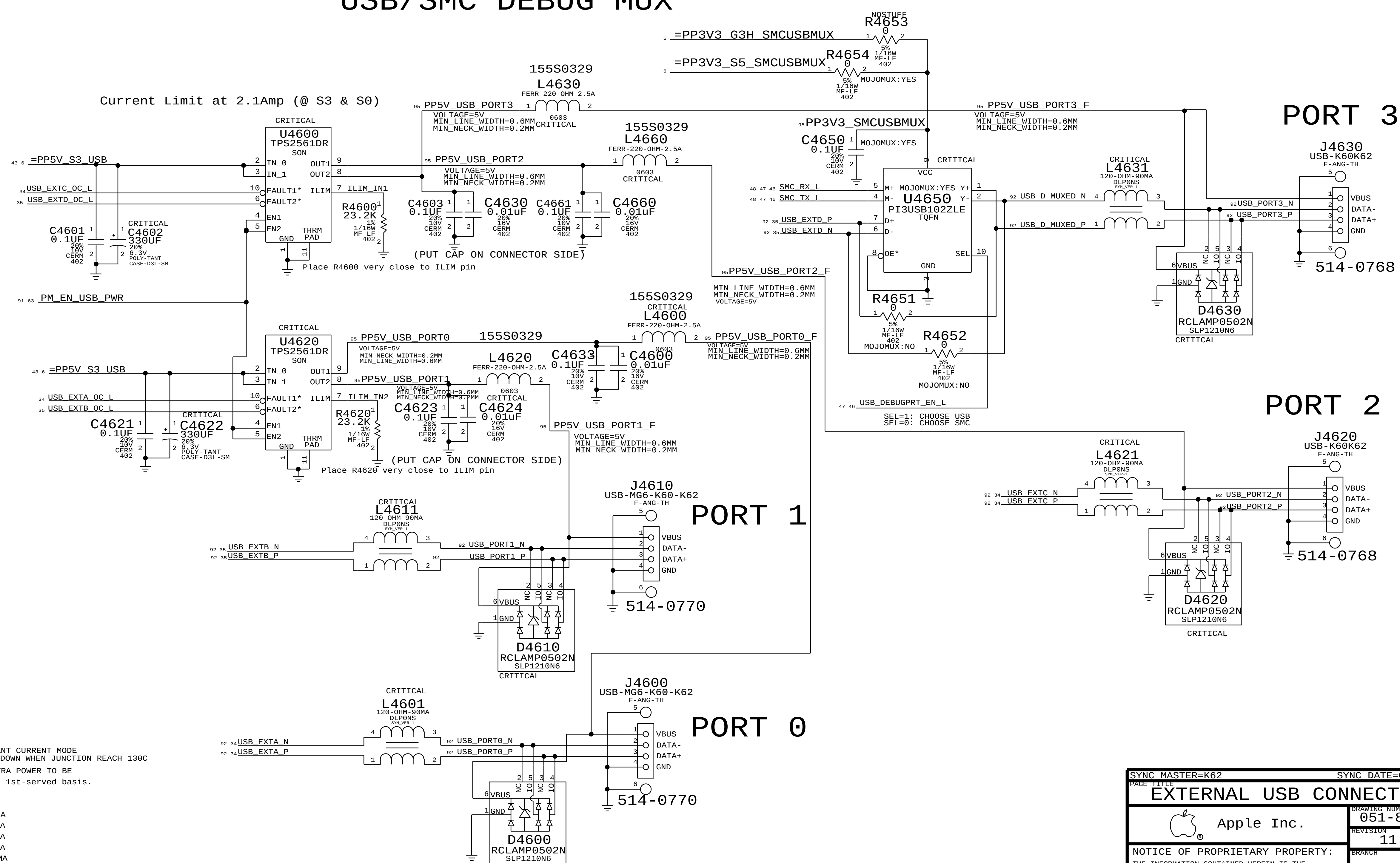
USB/SMC DEBUG MUX



USB PORT POWER:

EACH PORT IS HARDWARE Capable of :
STATE   MAX   MIN (WITHIN THE TOLERANCE)
S0, S3   2.7A   2.1A   -- PER PORT
WHEN CURRENT HITS LIMIT, TPS2561 BECOME CONSTANT CURRENT MODE
AND STAY AT THE LIMIT LEVEL UNTIL THERMAL SHUTDOWN WHEN JUNCTION REACH 130C
SOFTWARE WILL ALOW 500MA/PORT, PLUS 2700MA EXTRA POWER TO BE
distributed to approved devices on a 1st-come, 1st-served basis.

EXAMPLE: Port 1 - iPad fast charging = 2100mA
Port 2 - Wired Keyboard = 1100mA
Port 3 - iPhone fast charging = 1000mA
PORT 4 - USB 2.0 500MA = 500MA
TOTAL: 4700MA

SYNC_MASTER=K62 SYNC_DATE=01/06/2011
PAGE TITLE
EXTERNAL USB CONNECTORS
Apple Inc.
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PAGE 46 OF 110
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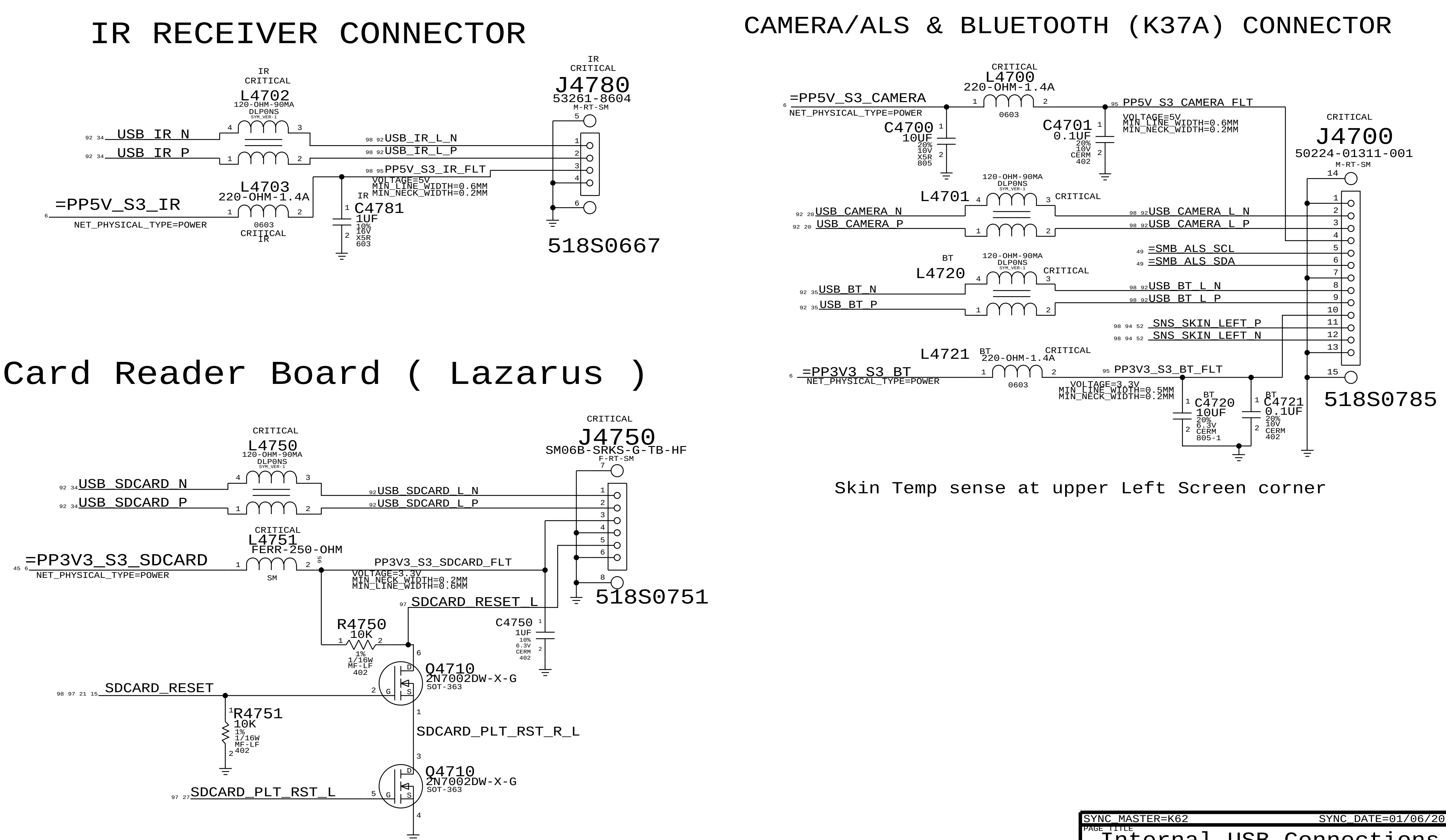

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: Internal USB Connections

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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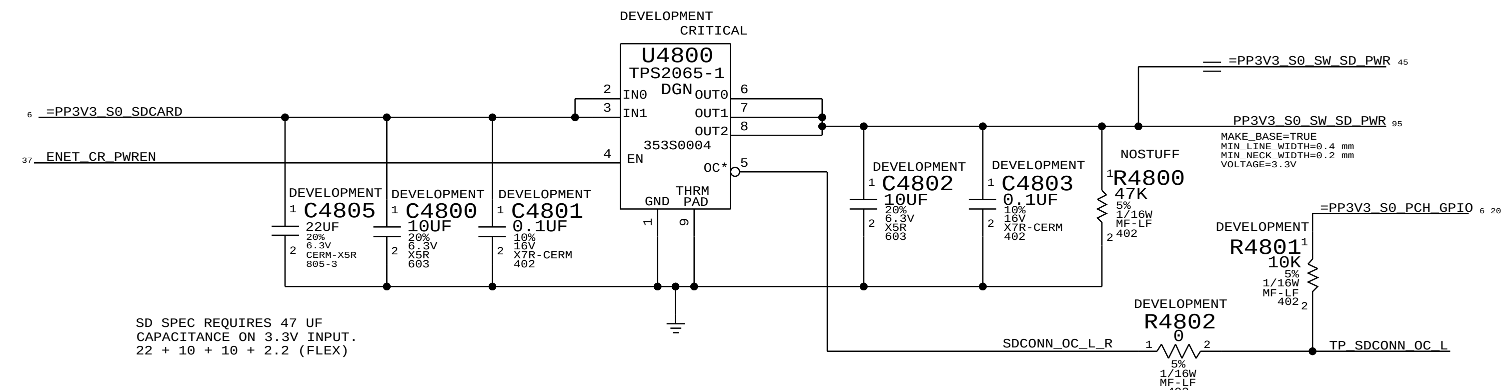
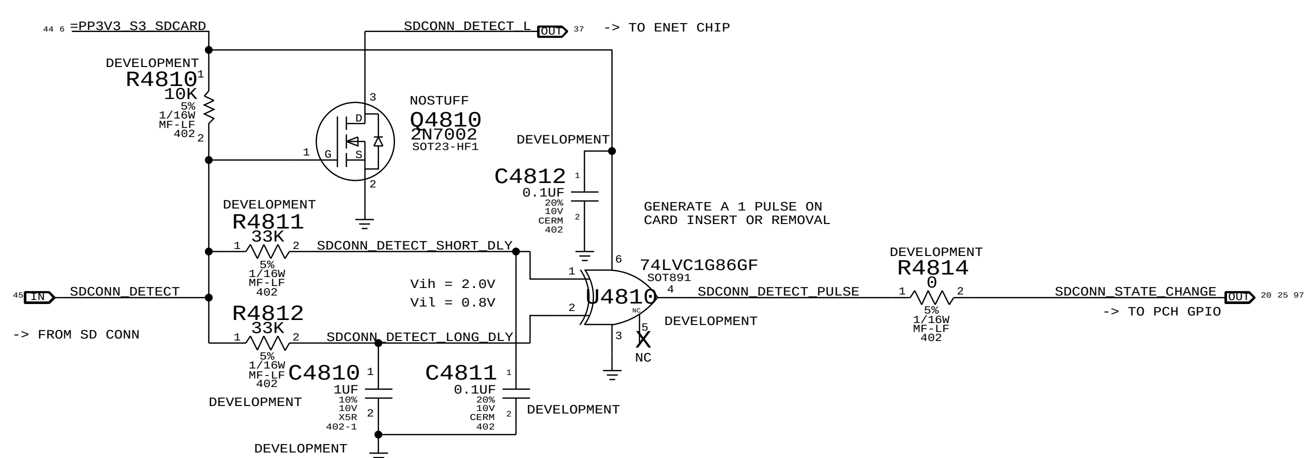
SD CARD 3.3V OVERCURRENT PROTECTION CHIP WITH ACTIVE LOAD DISCHARGE

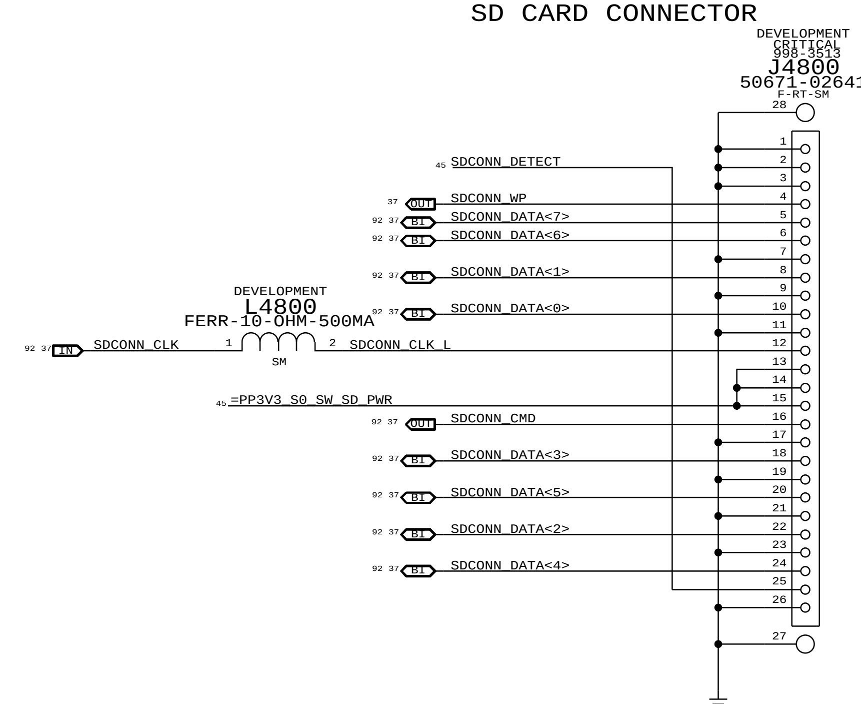
TPS2065-1 (1.0A LIMIT) HAS ACTIVE LOAD DISCHARGE SO R4800 IS NOSTUFF.



SDCONN DETECT DEBOUNCE, INVERSION, AND DETECT-CHANGED PCH GPIO CIRCUIT



SD CARD CONNECTOR



(CARD INSERTED = OPEN)
CAESAR-IV CARD DETECT IS PROGRAMMABLE, BUT A SILICON BUG
MAKES THE ACTIVE-HIGH CASE UNUSABLE.

SYNC_MASTER=K62	SYNC_DATE=01/06/2011
PAGE TITLE: SD READER CONNECTOR	
Apple Inc.	DRAWING NUMBER 051-8115, SIZE D
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	PAGE 48 OF 110
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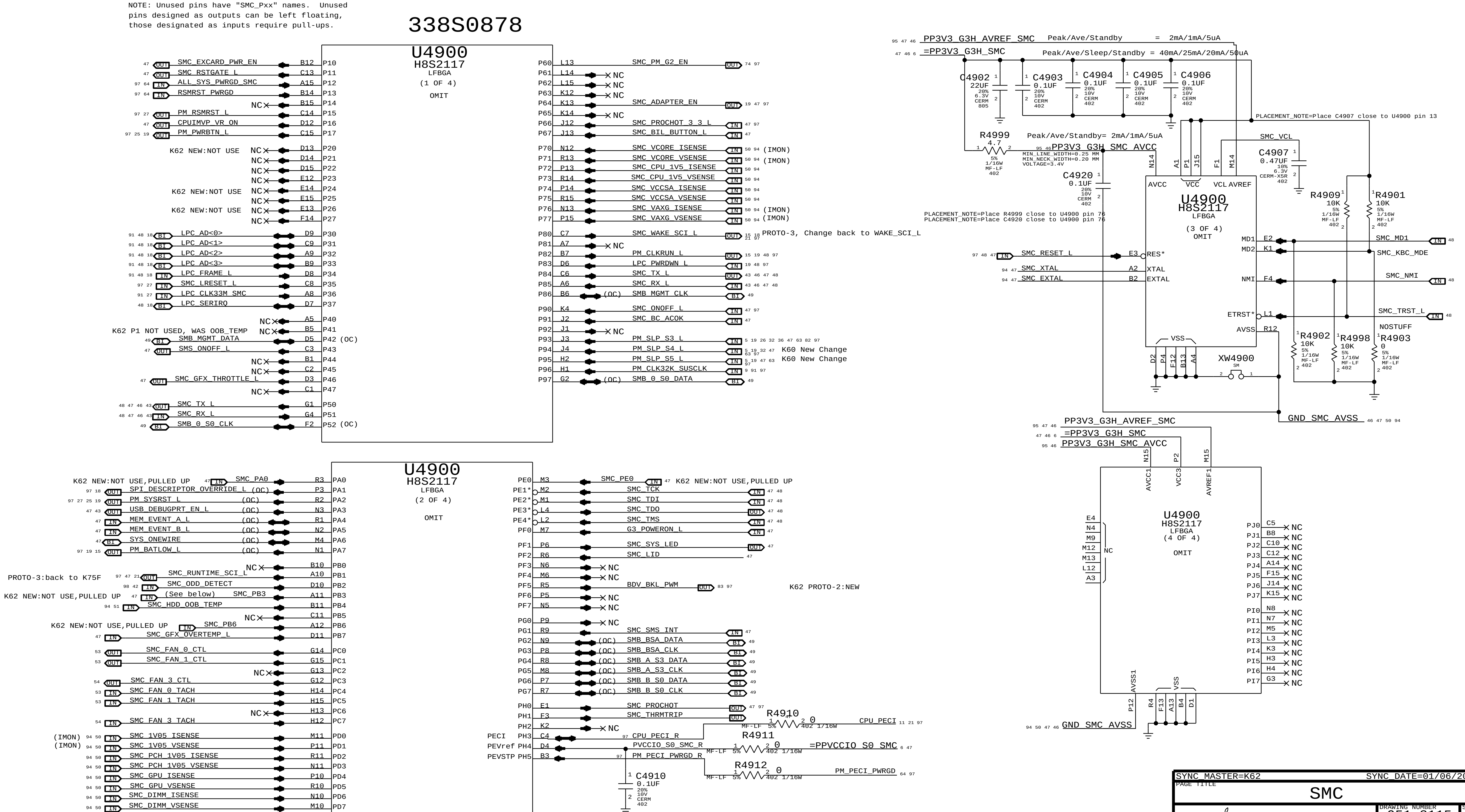
NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.



SMC PB3: SMC_IG_THROTTLE_L for MG systems.
Otherwise, TP/NC okay (was ISENSE_CAL_EN)

SMC PG1: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: SMC

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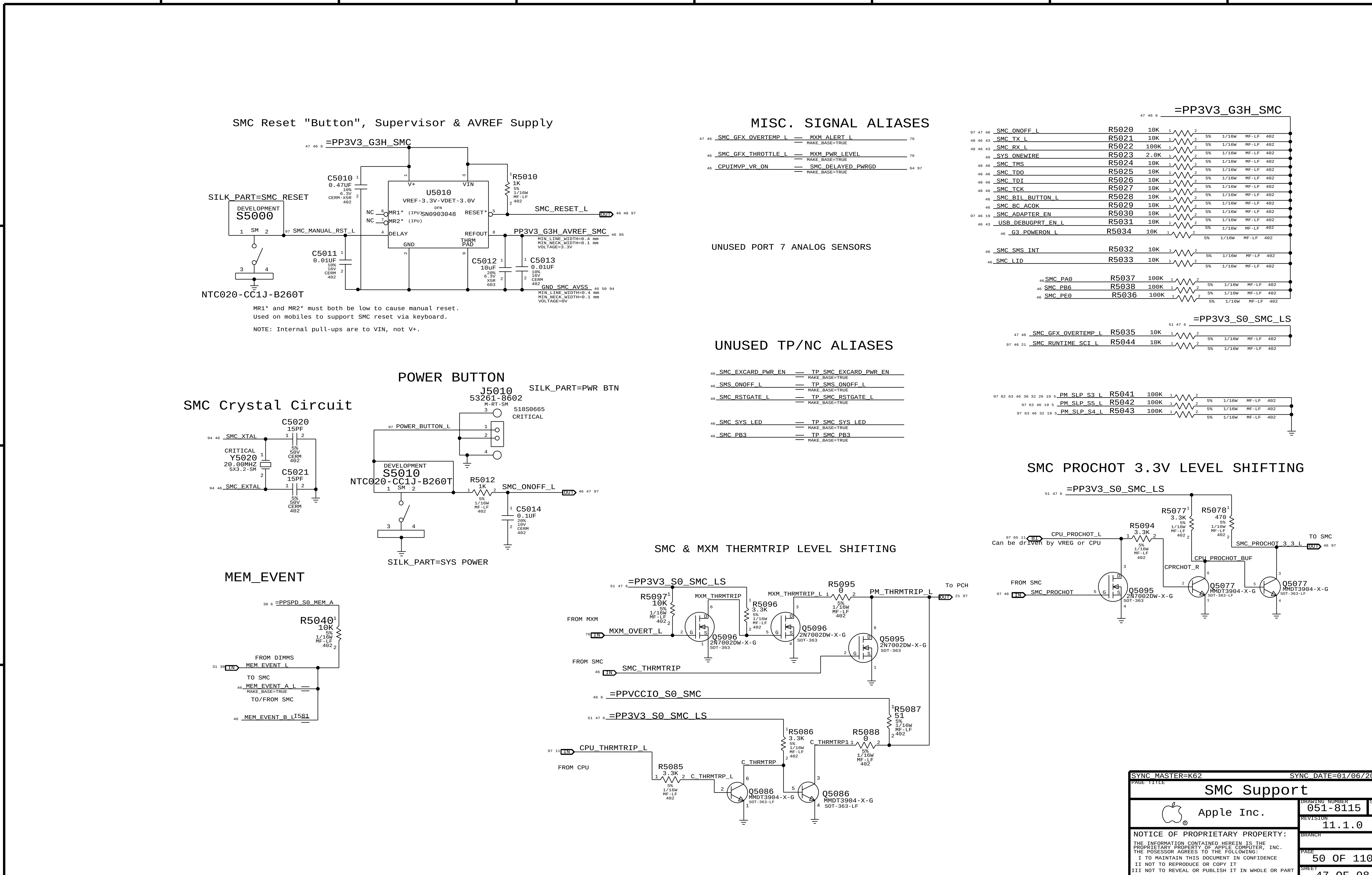

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

SMC Support

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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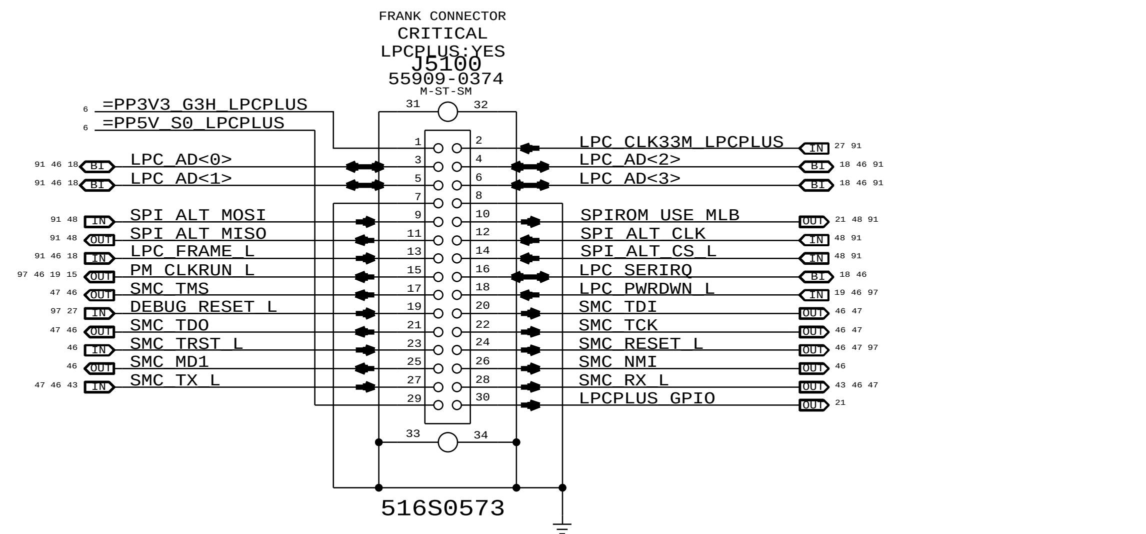
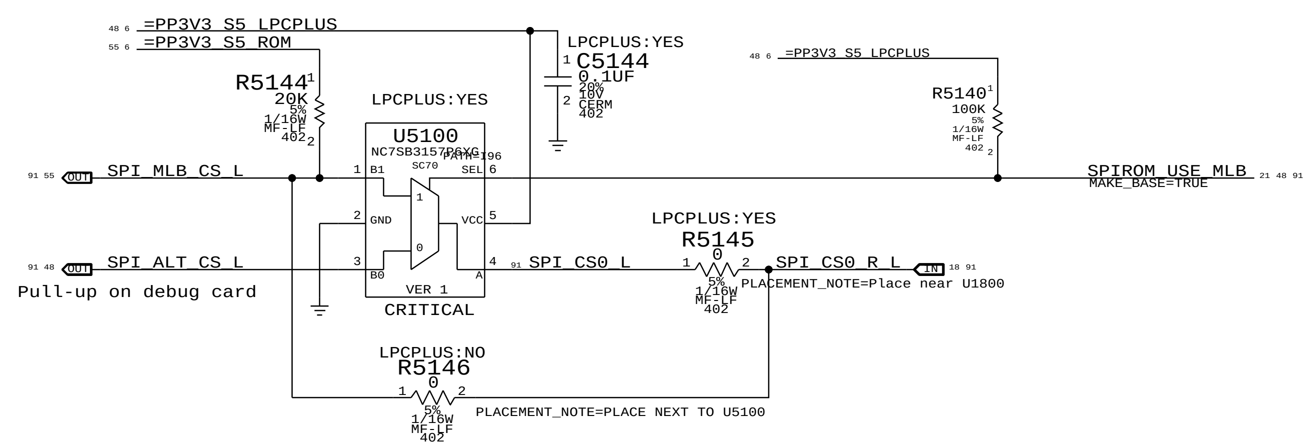
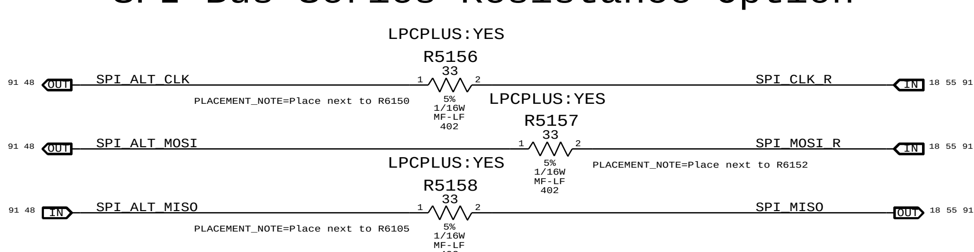
LPC+SPI Connector



Alternate SPI ROM Support



SPI Bus Series Resistance Option



SYNC_MASTER=K62_AARON SYNC_DATE=11/30/2009

PAGE TITLE
LPC+SPI Debug Connector

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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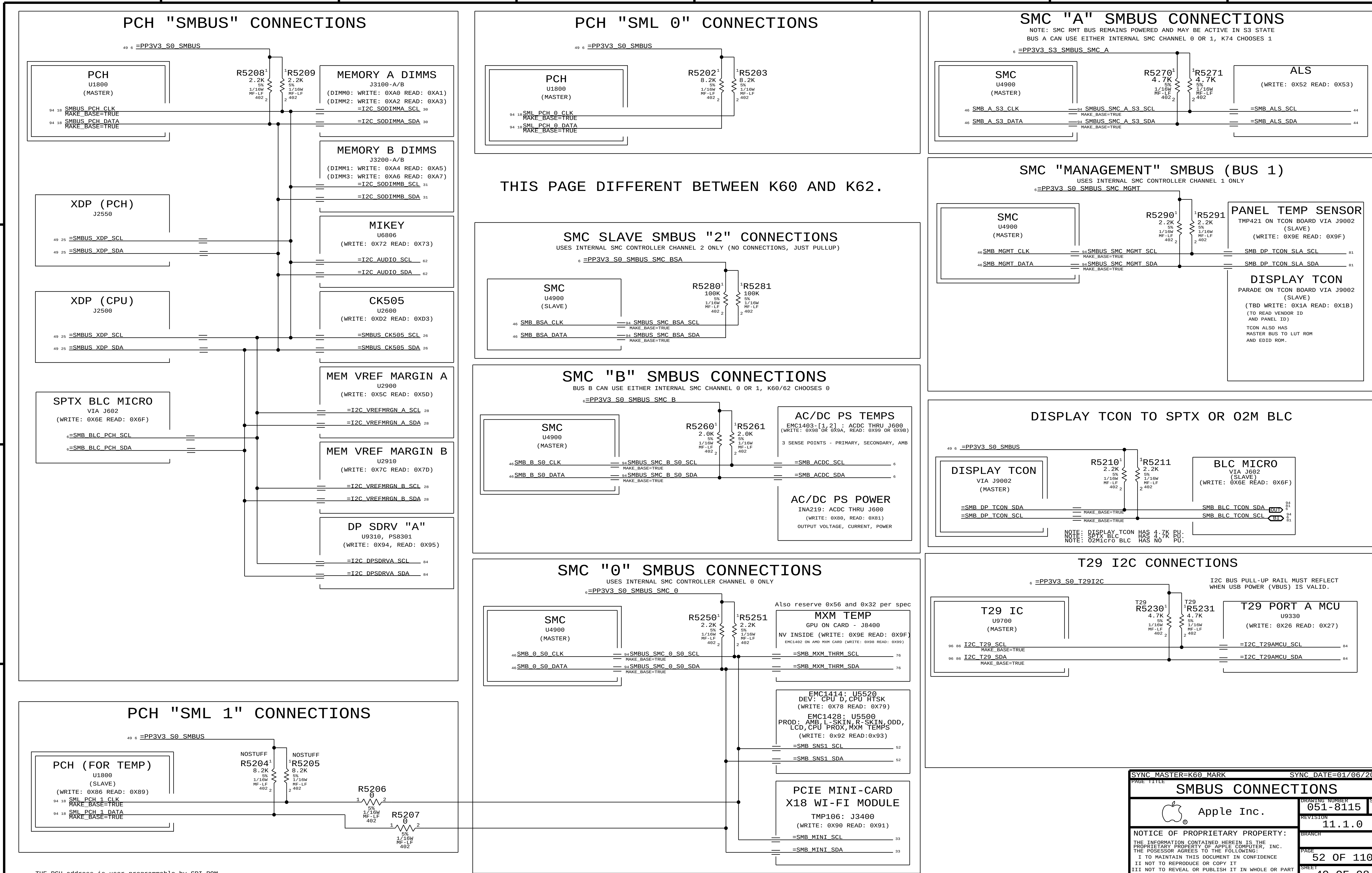

SYNC_MASTER=K60_MARK SYNC_DATE=01/06/2011

PAGE TITLE: SMBUS CONNECTIONS

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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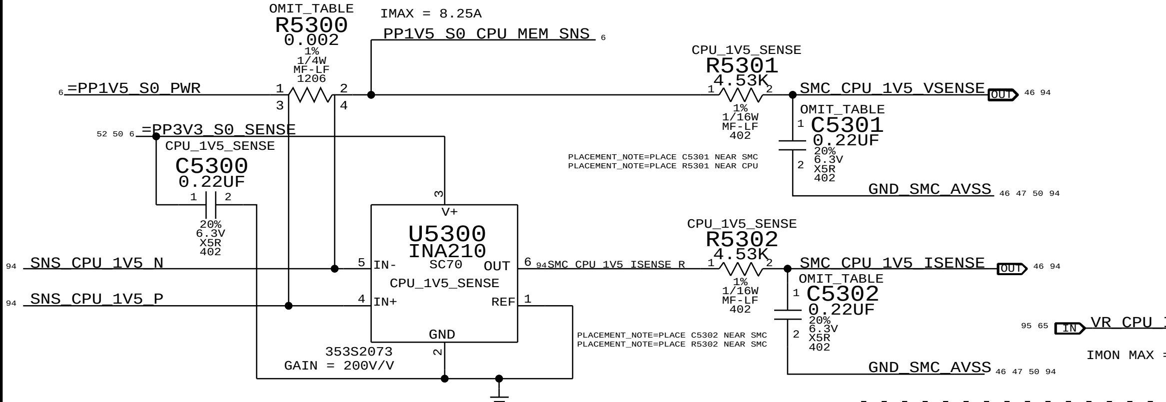
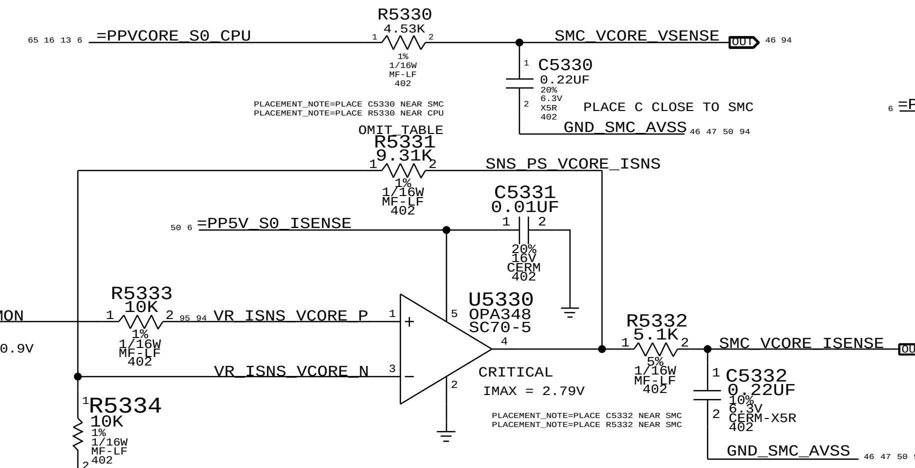
CPU VDD 1.5V
SENSE RESISTOR CURRENT (IC5R) AND VOLTAGE (VC5R) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5300	CPU_1V5_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5300	NO_CPU_1V5_SENSE
132S0080	2	CAP,0.22UF,402	C5301,C5302	CPU_1V5_SENSE
116S0004	2	RES,0 OHM,402	C5301,C5302	NO_CPU_1V5_SENSE

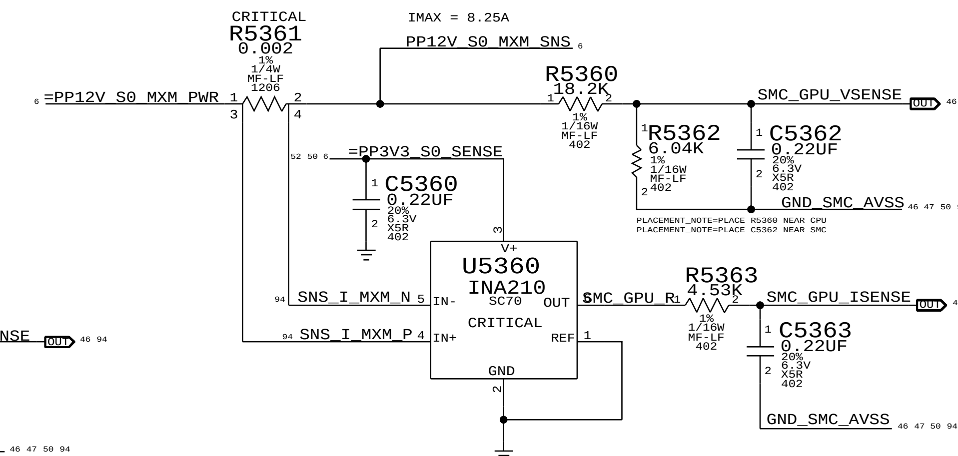
THE NO_1V05_PCH_SENSE, NO_CPU_VCCSA_SENSE, AND NO_CPU_1V5_SENSE BOMOPTIONS SHOULD IDEALLY NEVER BE USED AS TOTAL CPU POWER SENSING REQUIRES ALL 3 SENSORS.

CPU VCC (VCORE)
IMON CURRENT (IC0C) AND VOLTAGE (VC0C) SENSE

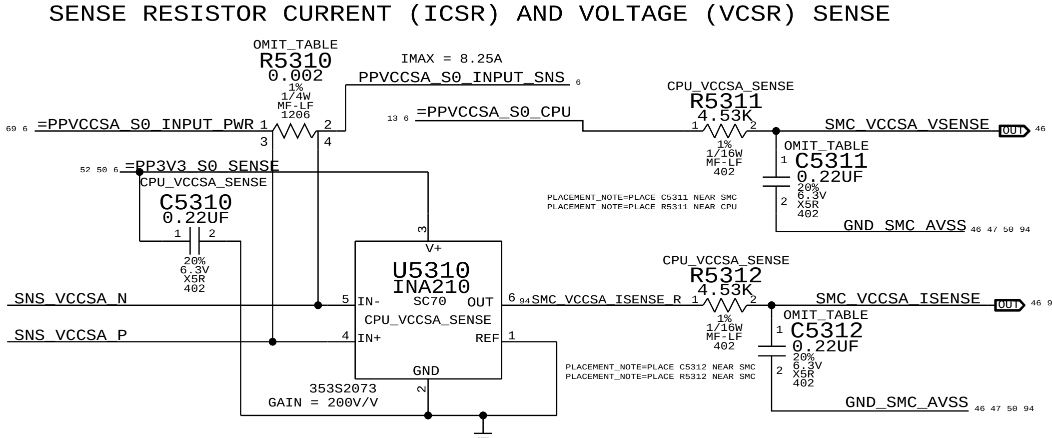


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0312	1	RES,MTL FILM,1/16W,9.31K,0402	R5331	CPUVCORE-3PH
114S0345	1	RES,MTL FILM,1/16W,21K,0402	R5331	CPUVCORE-4PH

GPU MXM
SENSE RESISTOR CURRENT (IG0R) AND VOLTAGE (VG0R) SENSE

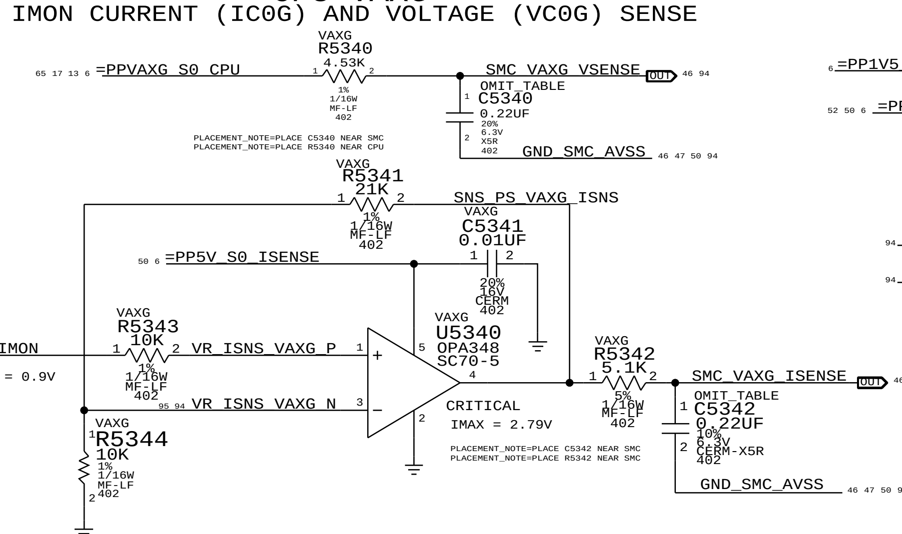


CPU VCCSA
SENSE RESISTOR CURRENT (ICSR) AND VOLTAGE (VCSR) SENSE



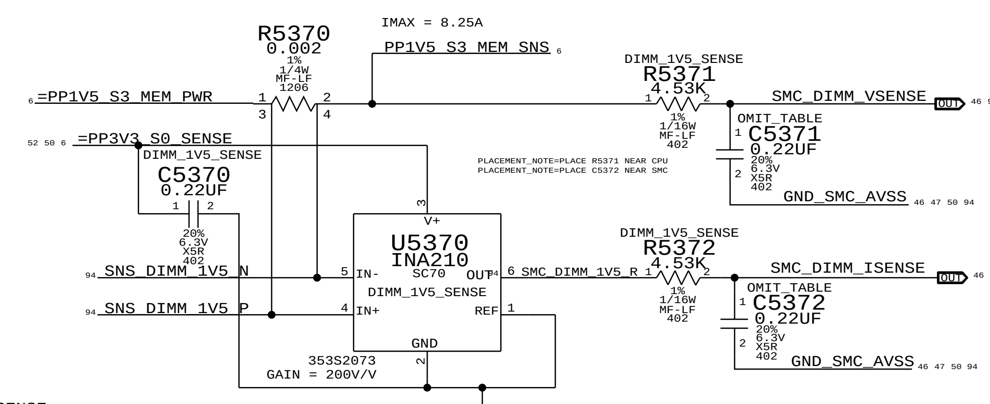
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5310	CPU_VCCSA_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5310	NO_CPU_VCCSA_SENSE
132S0080	2	CAP,0.22UF,402	C5311,C5312	CPU_VCCSA_SENSE
116S0004	2	RES,0 OHM,402	C5311,C5312	NO_CPU_VCCSA_SENSE

CPU VAXG
IMON CURRENT (IC0G) AND VOLTAGE (VC0G) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	2	CAP,0.22UF,402	C5340,C5342	VAXG
116S0004	2	RES,0 OHM,402	C5340,C5342	NO_VAXG

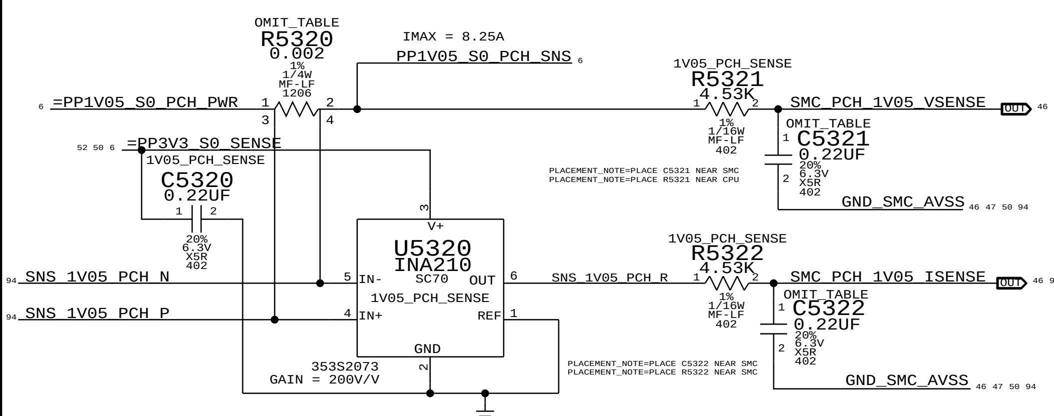
DIMM VDD 1.5V (LIKELY DEVELOPMENT ONLY)
SENSE RESISTOR CURRENT (IM0R) AND VOLTAGE (VM0R) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	2	CAP,0.22UF,402	C5371,C5372	DIMM_1V5_SENSE
116S0004	2	RES,0 OHM,402	C5371,C5372	PRODUCTION

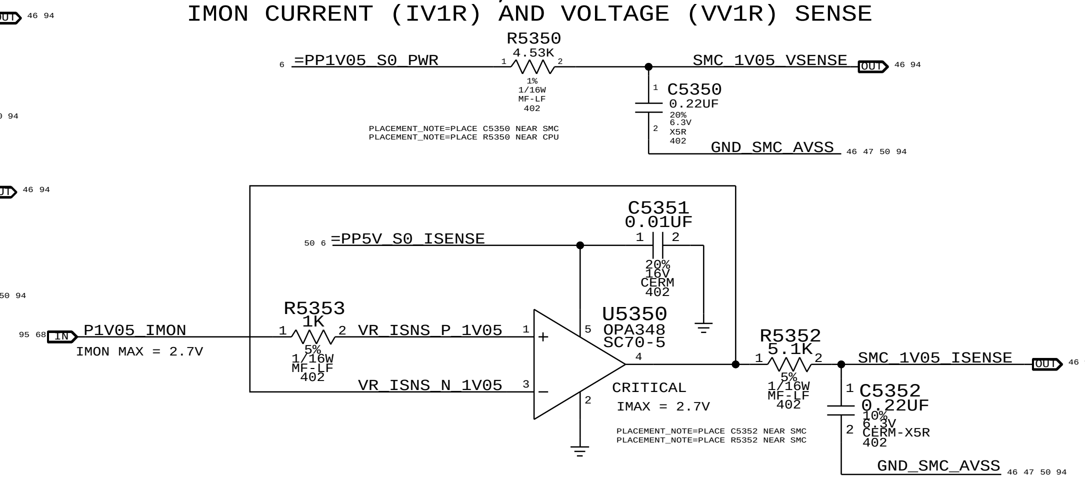
NOTE: TOTAL CPU POWER = VC0C*IC0C + VC5R*IC5R + VCSR*ICSR + VC0G*IC0G + VV1R*IC1R
where
IC1R = IV1R - IN1R - ICSR

PCH 1.05V
SENSE RESISTOR CURRENT (IN1R) AND VOLTAGE (VN1R) SENSE



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
104S0018	1	RES,2 MILLIOHM,1206	R5320	1V05_PCH_SENSE
101S0414	1	RES,0 OHM,1206,20 MILLIOHM MAX	R5320	NO_1V05_PCH_SENSE
132S0080	2	CAP,0.22UF,402	C5321,C5322	1V05_PCH_SENSE
116S0004	2	RES,0 OHM,402	C5321,C5322	NO_1V05_PCH_SENSE

1.05V FOR CPU VCCIO, CPU VCCSA & PCH 1.05V
IMON CURRENT (IV1R) AND VOLTAGE (VV1R) SENSE



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
CPU/PCH/GPU POWER SENSE

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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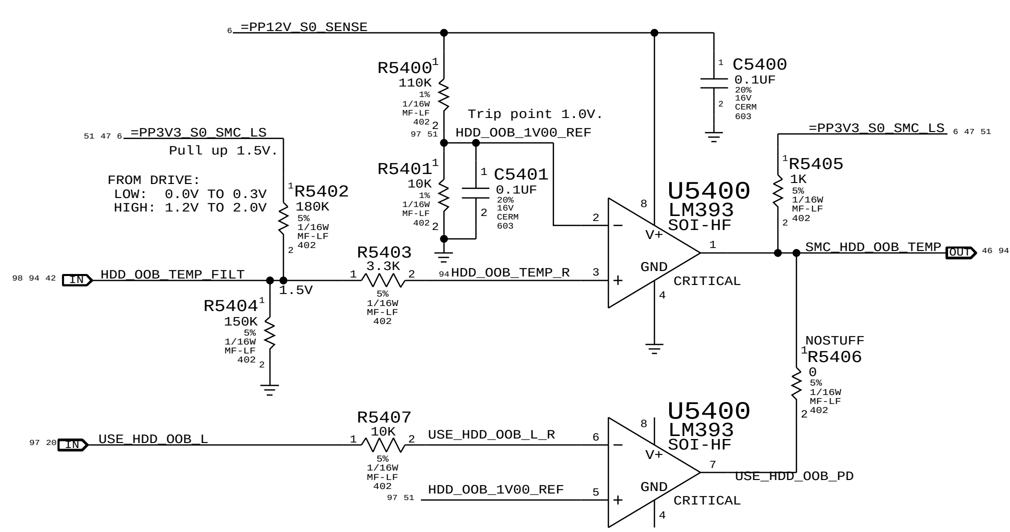
HDD OOB TEMPERATURE SENSING



DRIVE ACTIVE = VALID SIGNAL PROTOCOL BETWEEN 0-2.0V.
DRIVE ASLEEP = HDD DRIVES HDD_OOB_TEMP LOW
DRIVE ABSENT = OOB IS PULLED HIGH UNLESS PCH DETERMINES SSD PRESENT AND DRIVES USE_HDD_OOB_L LOW WHICH THEN PULLS HDD_OOB_TEMP LOW.

NOTE: WILL BE CONNECTED TO SATA PWR CONNECTOR PIN 11
THIS PIN IS ORIGINALLY INTENDED FOR HDD LED OUTPUT,
AND ALSO FOR HDD STAGGERED PIN UP (FLOATING) OR IMMEDIATE SPIN-UP (GROUND).
BOTH FUNCTIONS NOT USED.

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
HDD OOB SENSE

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DRAWING NUMBER 051-8115 SIZE D
REVISION 11.1.0
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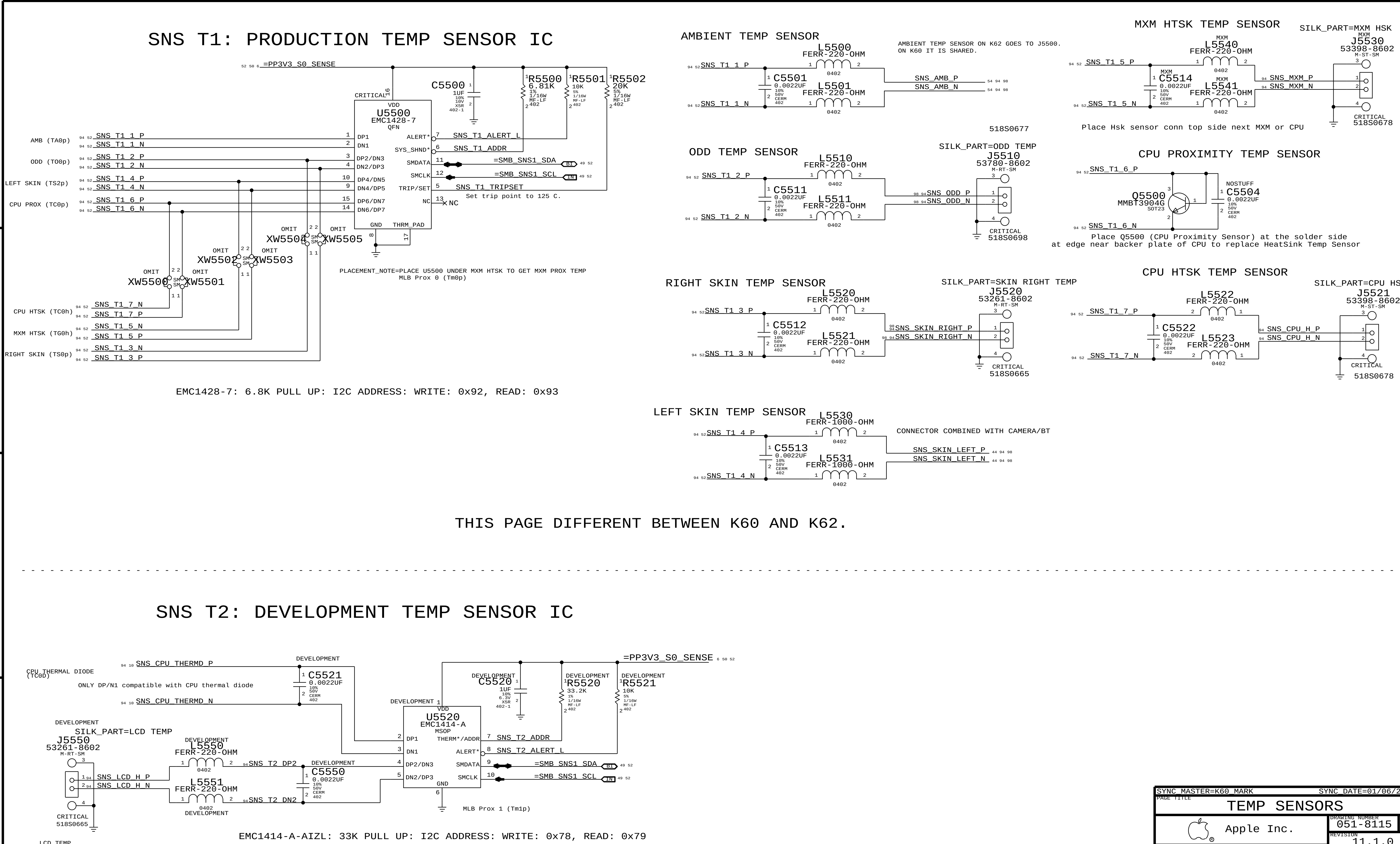

SYNC_MASTER=K60_MARK SYNC_DATE=01/06/2011

PAGE TITLE: TEMP SENSORS

Apple Inc.

DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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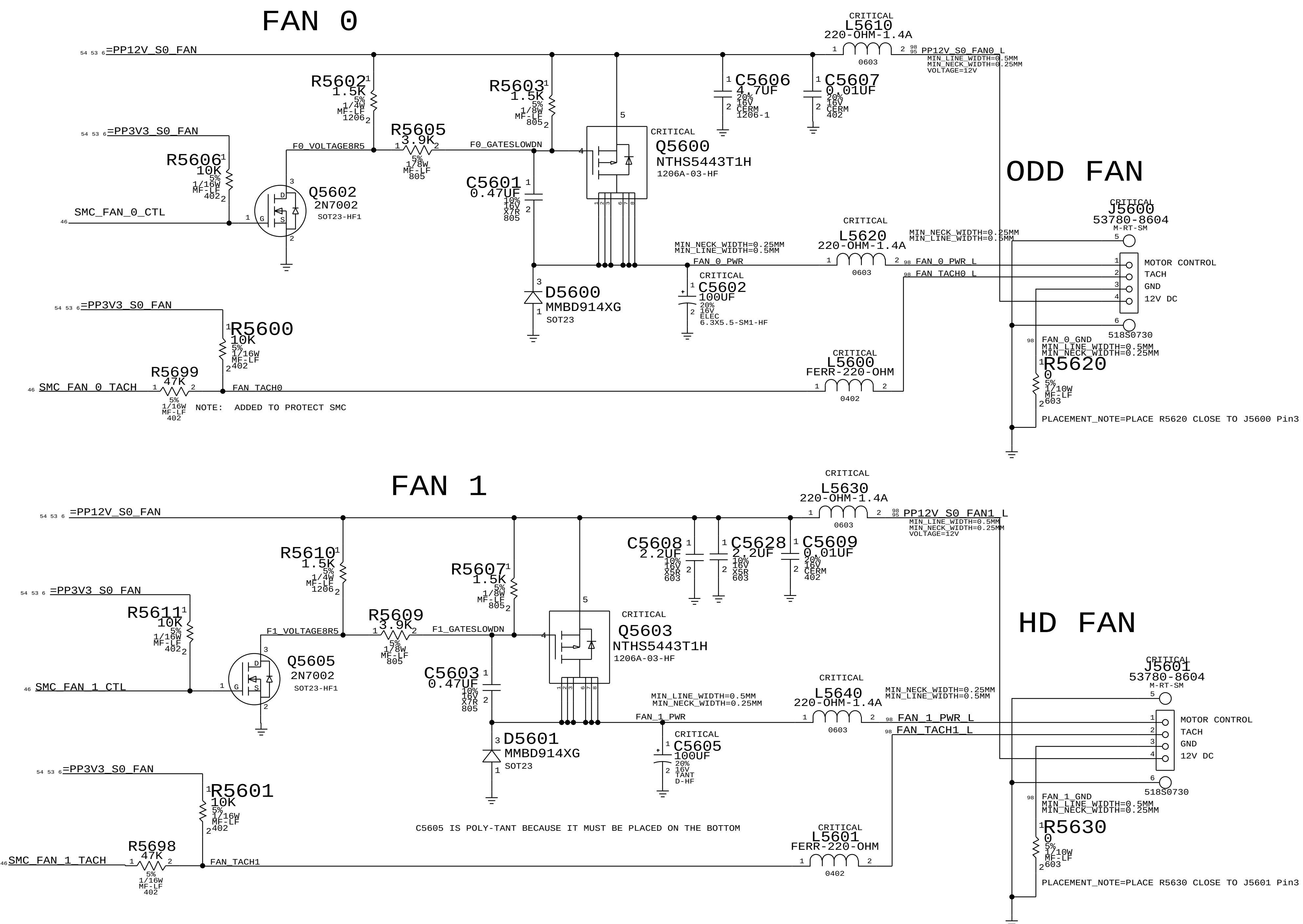

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

HD AND OD FAN

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

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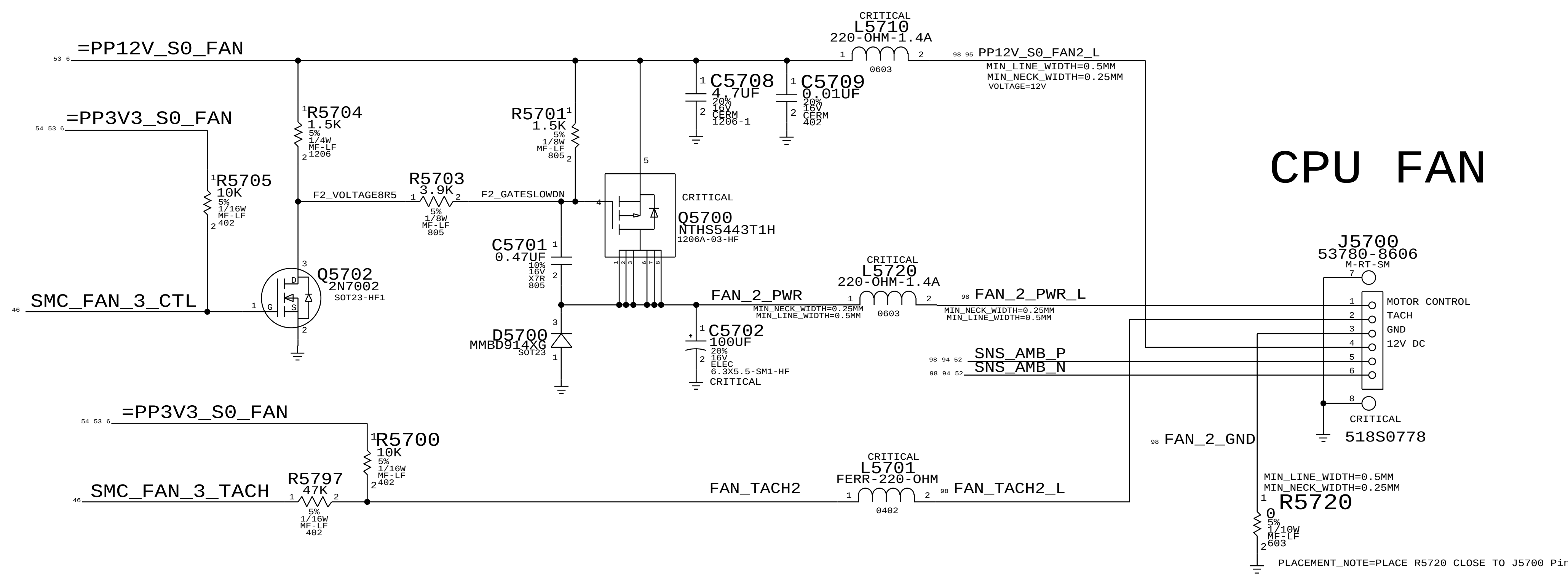
SMC'S FAN3 OUTPUT CONTROL FAN 2



FAN 3 SMC CONTROL (UNUSED)

SYNC_MASTER=K60_JERRY SYNC_DATE=01/06/2011

PAGE TITLE: CPU FAN

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DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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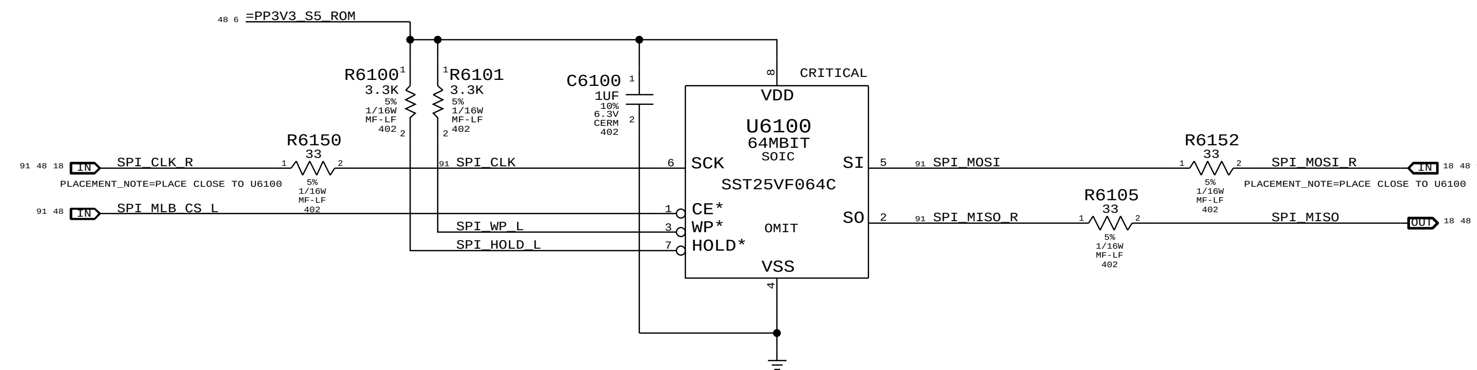

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

SPI ROM

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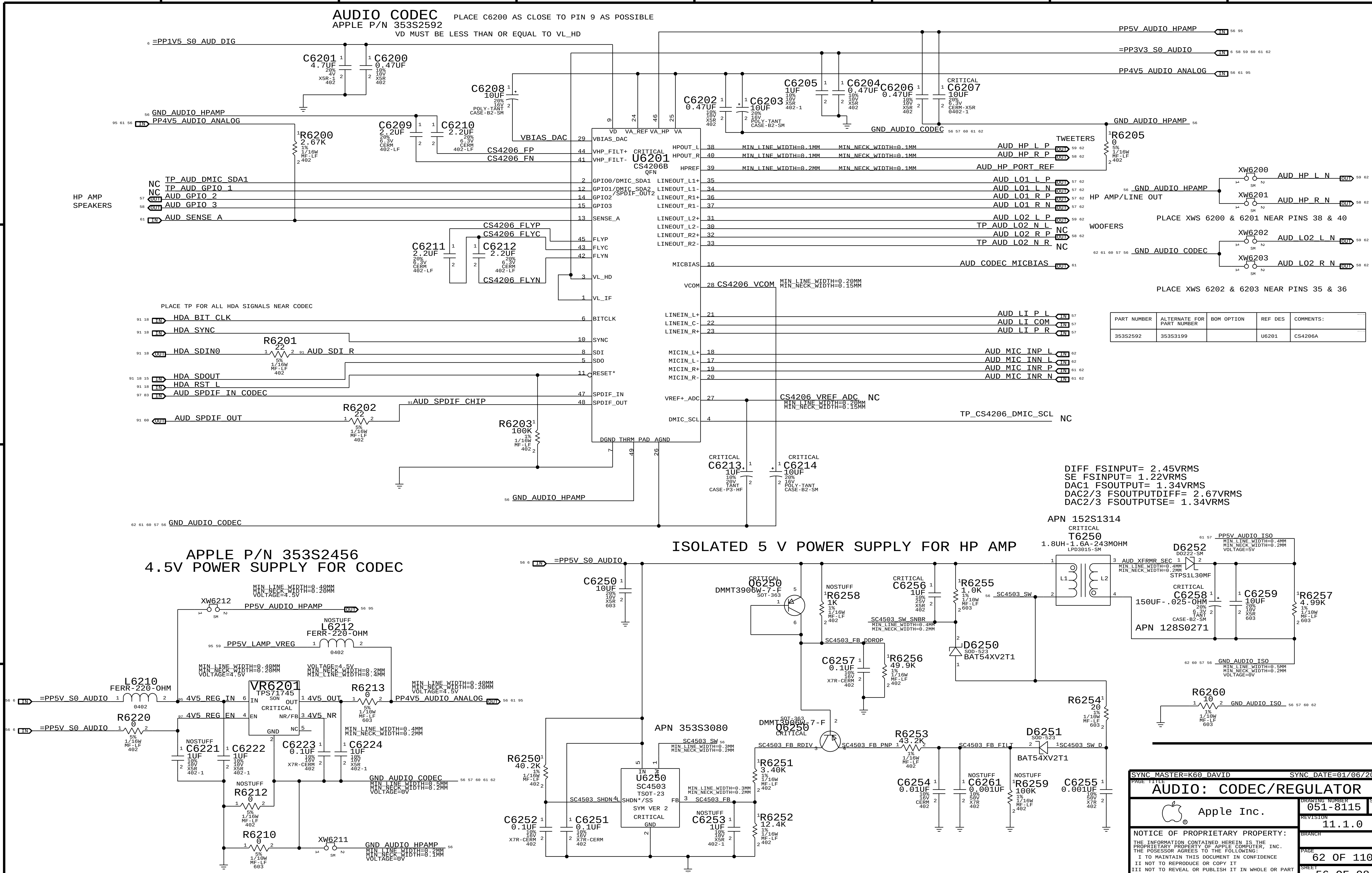

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S2592	353S3199		U6201	CS4206A

SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE: AUDIO: CODEC/REGULATOR

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DRAWING NUMBER: 051-8115 SIZE: D

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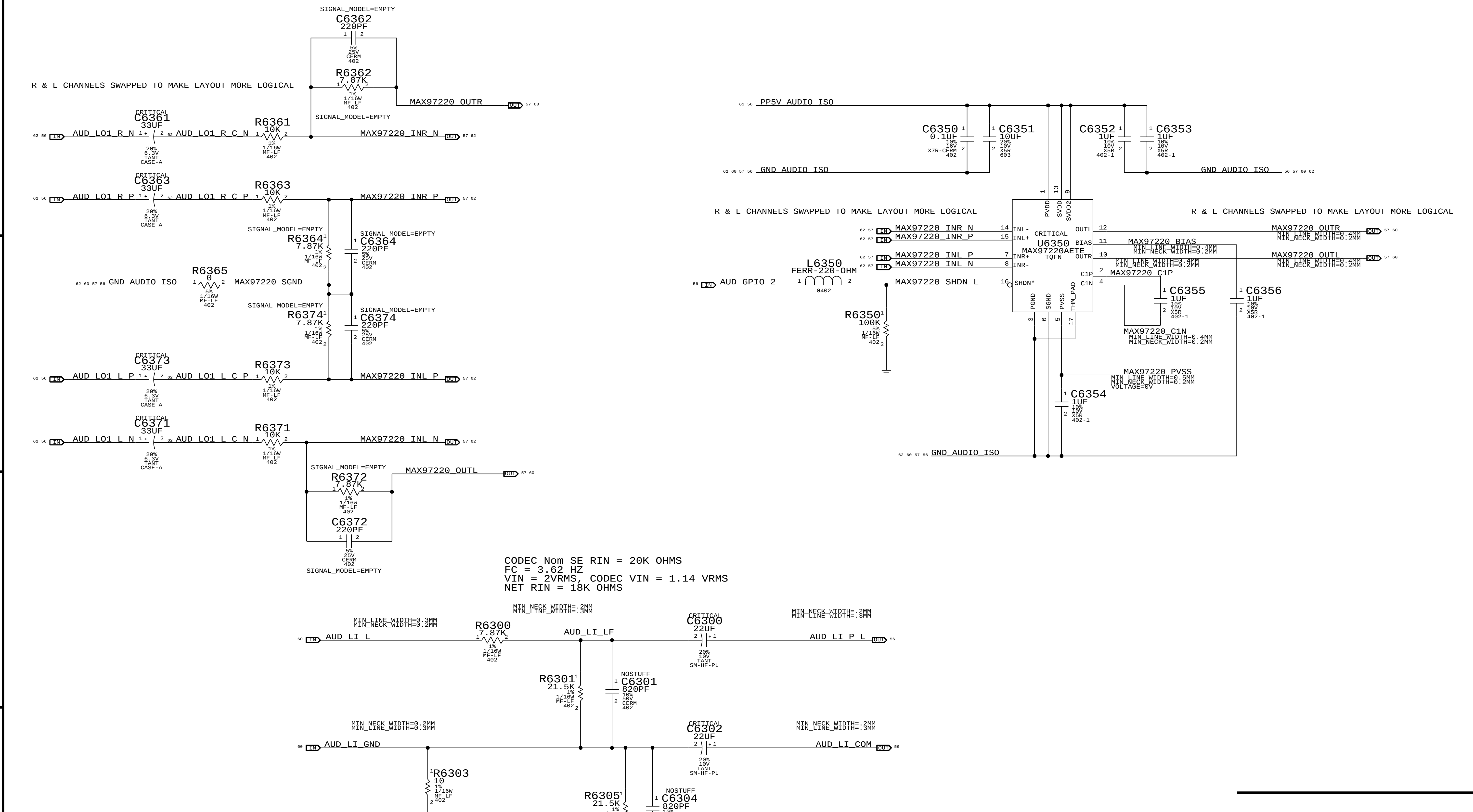

SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE
AUDIO: FILTER/BUFFER

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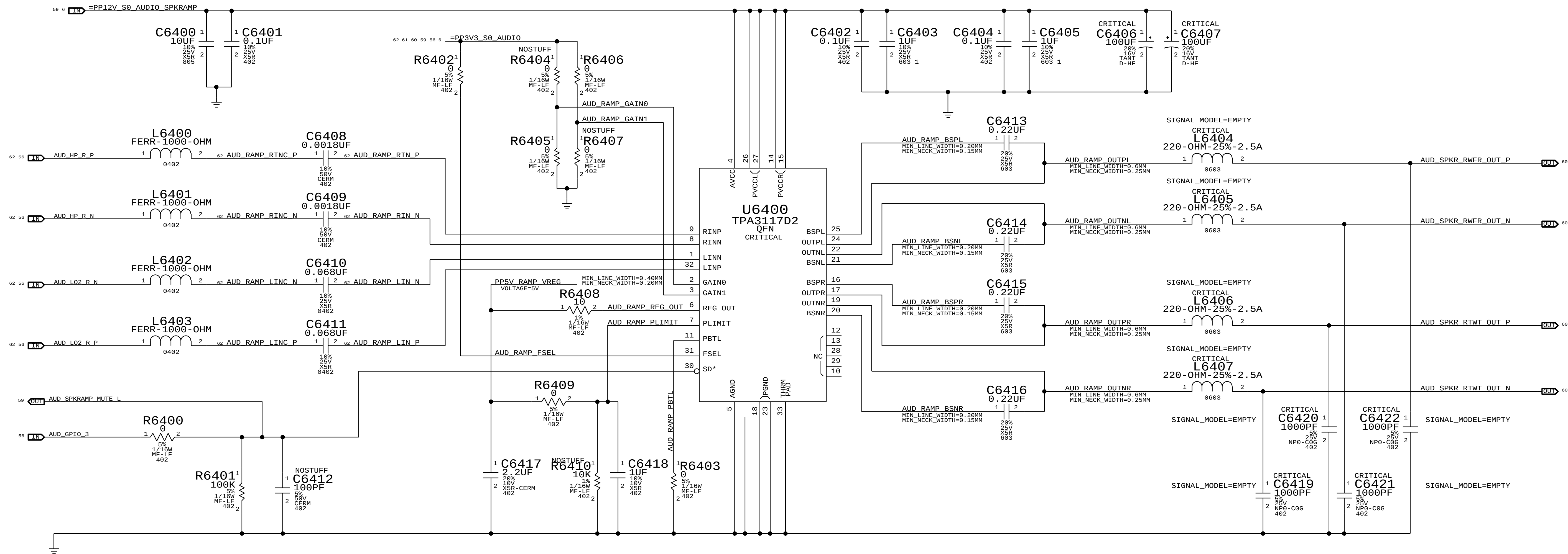
RIGHT CH SPEAKER AMP

APPLE P/N 353S3069

R6402 = HIGH = 400 KHZ
SPEAKER AMP GAIN = +15.2 DB
SPEAKER AMP RIN = 104K NOMINAL W/ +15.2 DB GAIN
FC_HPF, TWEETERS = ~850 HZ (0.0018 UF)
FC_HPF, WOOFERS = ~22.5 HZ (0.068 UF)



SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE

AUDIO: SPEAKER AMP_1

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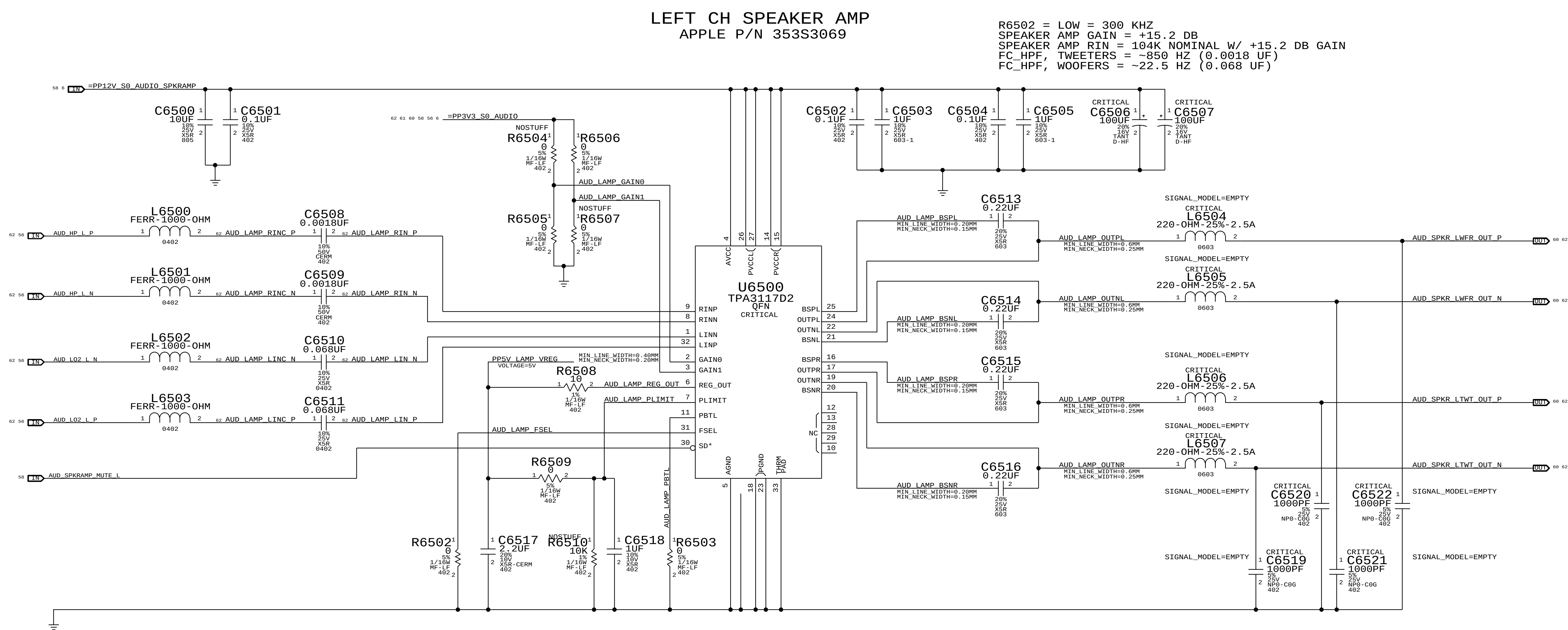

SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE

AUDIO: SPEAKER AMP

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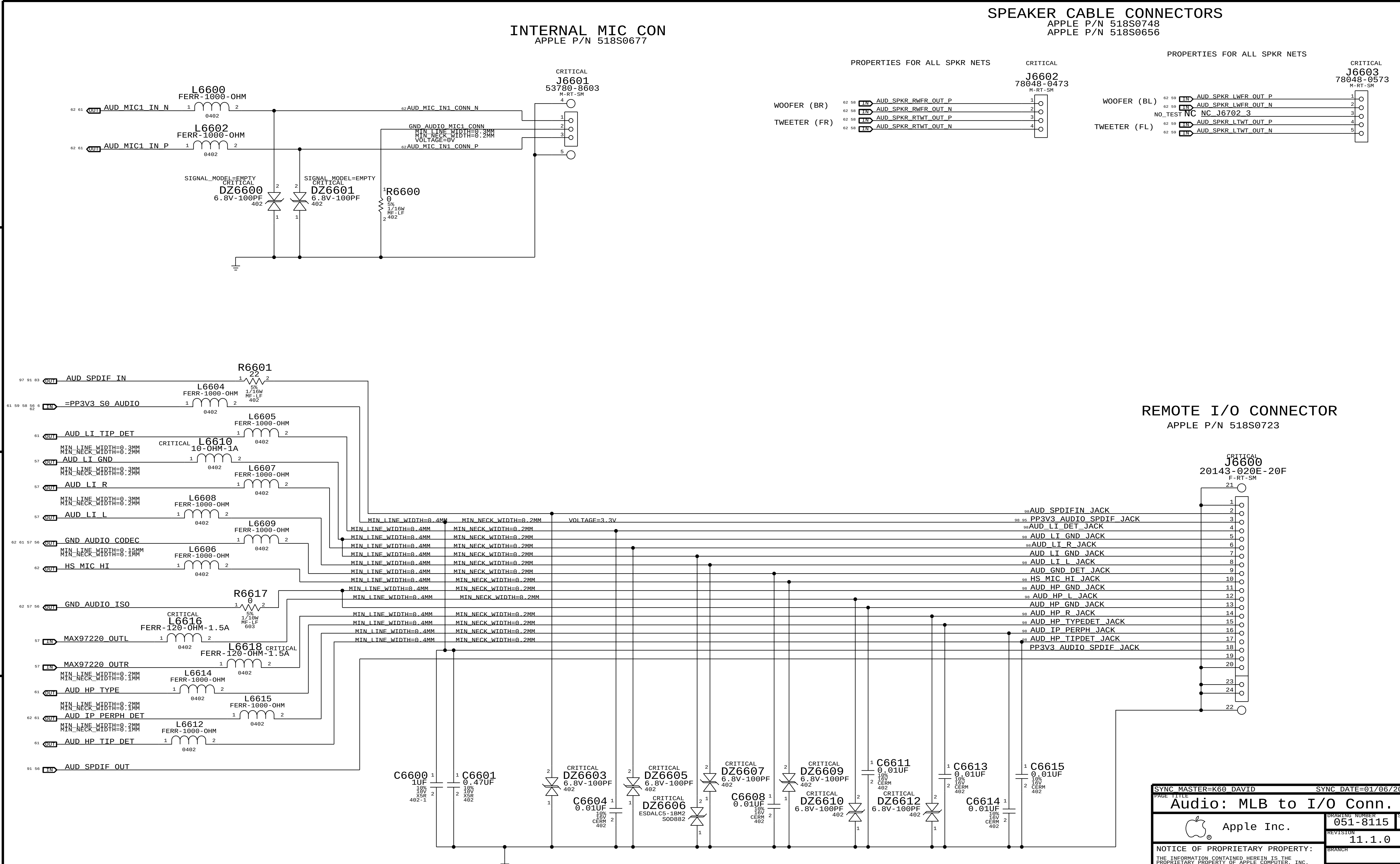

SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE

Audio: MLB to I/O Conn.

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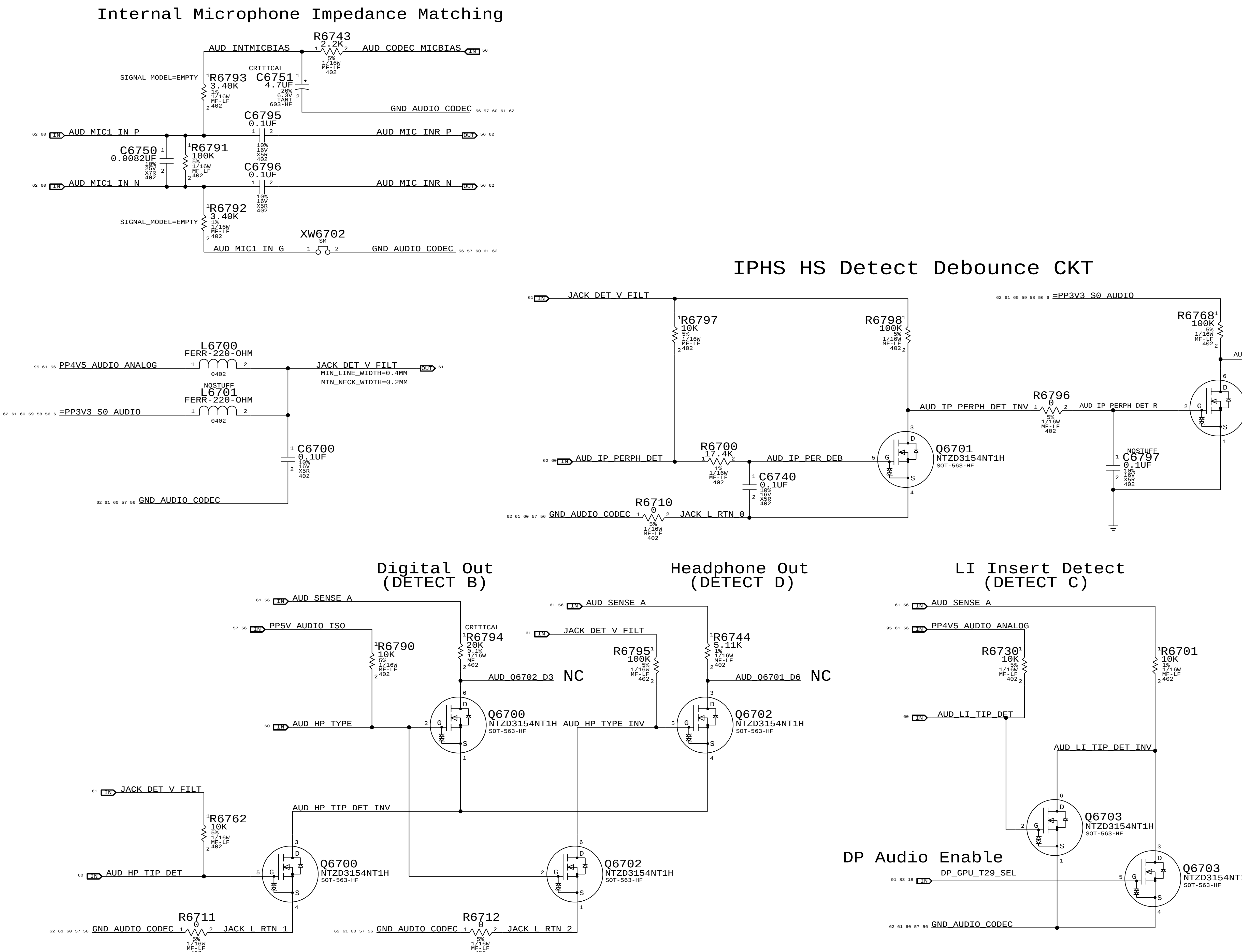
NOTICE OF PROPRIETARY PROPERTY:



SYNC_MASTER=K60_DAVID SYNC_DATE=11/24/2010

PAGE TITLE

AUDIO: Detects/Grounding

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NOTICE OF PROPRIETARY PROPERTY:

CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME/MUTE	CONVERTER	PIN COMPLEX	SHDN	DET ASSIGNMENT
HP/LINE OUT	0X03 (3)	0X03 (3)	0X0A (10,D)	GPIO_2	0X0A (D)
PRIMARY SPKRS (WFR)	0X04 (4)	0X04 (4)	0X0B (11)	GPIO_3	N/A
SECONDARY SPKRS (TWT)	0X02 (2)	0X02 (2)	0X09 (09)	GPIO_3	N/A
SPDIF OUT	N/A	0X08 (8)	0x10 (16)	N/A	0X0D (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	ENABLE/CONTROL	DET ASSIGNMENT
LINE IN	0X05 (5)	0X12 (12,C)	N/A	0X12 (C)
SPDIF IN	0X07 (7)	0x0F (15)	N/A	N/A
INTERNAL MIC	0X06 (6)	0X0E (14,LEFT & RIGHT)	N/A	N/A
EXTERNAL MIC	0X06 (6)	0X0D (13,V22,B,LEFT)	COUGAR POINT GPIO 16	COUGAR POINT GPIO 5 (RCVR INT) COUGAR POINT GPIO 3 (PERIPH DET)

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
AUDIO	*	0.1 MM	?
SPKROUT	*	0.2 MM	?

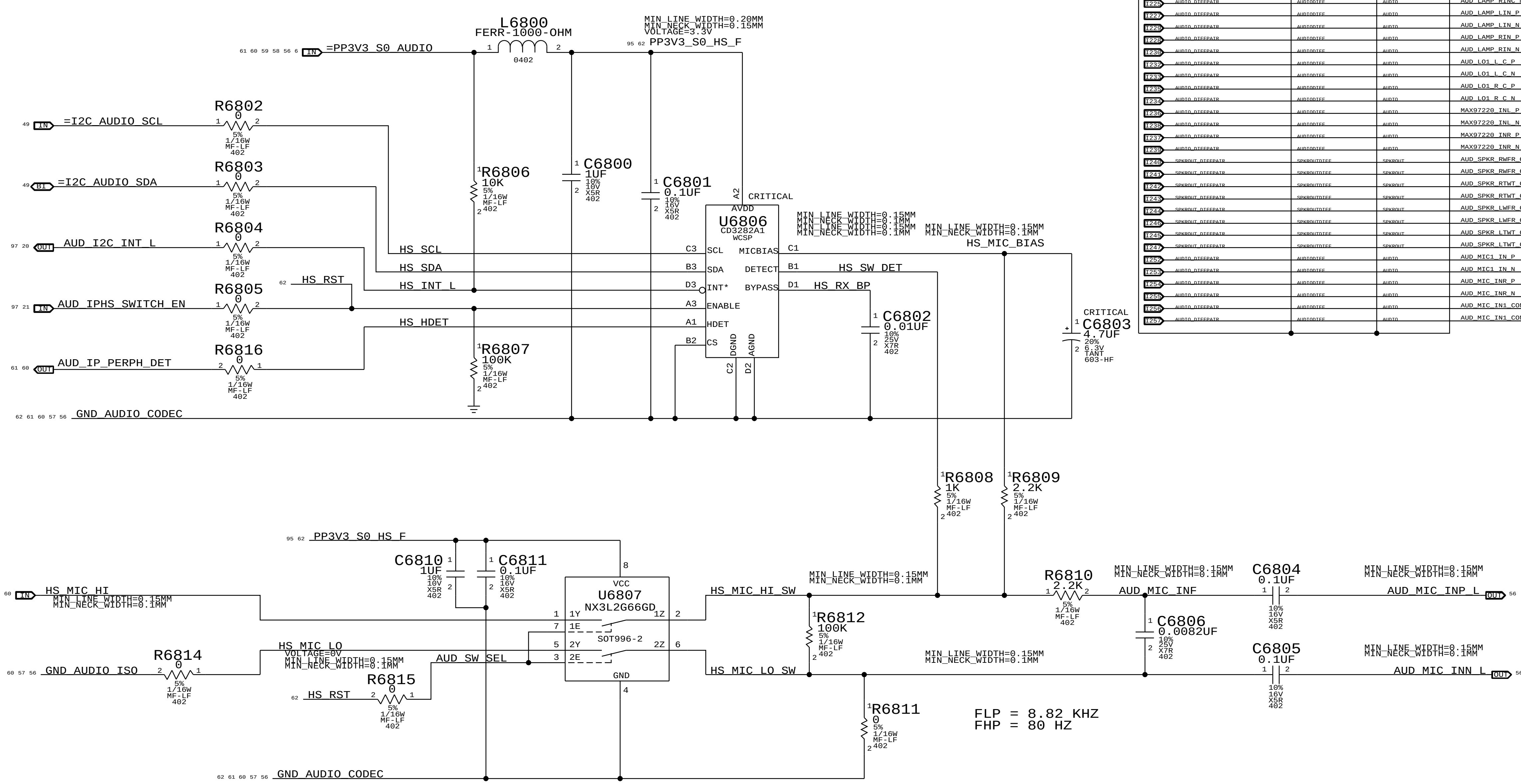
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
AUDIODIFF	*	AUDIODIFF
SPKROUTDIFF	*	SPKROUTDIFF

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
AUDIODIFF	*	Y	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
SPKROUTDIFF	*	Y	0.6 MM	0.25 MM	10 MM	0.2 MM	0.2 MM

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
I213 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_L_P	56 59
I214 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_L_N	56 59
I215 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_R_P	56 58
I216 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_HP_R_N	56 58
I217 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_P	56 57
I218 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_N	56 57
I211 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_P	56 57
I212 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_N	56 57
I210 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_P	56 59
I209 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_L_N	56 59
I208 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_P	56 58
I206 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO2_R_N	56 58
I207 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_P	58
I204 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LINC_N	58
I205 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_P	58
I203 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RINC_N	58
I220 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_P	58
I219 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_LIN_N	58
I222 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_P	58
I221 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_RAMP_RIN_N	58
I224 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_P	59
I223 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LINC_N	59
I226 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_P	59
I225 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RINC_N	59
I227 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_P	59
I229 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_LIN_N	59
I228 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_P	59
I230 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LAMP_RIN_N	59
I232 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_P	57
I233 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_L_C_N	57
I235 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_P	57
I234 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_LO1_R_C_N	57
I236 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_P	57
I238 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INL_N	57
I237 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_P	57
I239 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	MAX97220_INR_N	57
I240 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_P	58 60
I241 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RWFR_OUT_N	58 60
I242 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_P	58 60
I243 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_RTWT_OUT_N	58 60
I244 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_P	59 60
I246 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LWFR_OUT_N	59 60
I245 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_P	59 60
I247 SPKROUT_DIFFPAIR	SPKROUTDIFF	SPKROUT	AUD_SPKR_LTWT_OUT_N	59 60
I252 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC1_IN_P	60 61
I253 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC1_IN_N	60 61
I254 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INR_P	56 61
I255 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_INR_N	56 61
I256 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_IN1_CONN_P	60
I257 AUDIO_DIFFPAIR	AUDIODIFF	AUDIO	AUD_MIC_IN1_CONN_N	60

MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640



SYNC_MASTER=K60_DAVID SYNC_DATE=01/06/2011

PAGE TITLE

AUDIO: Mikey

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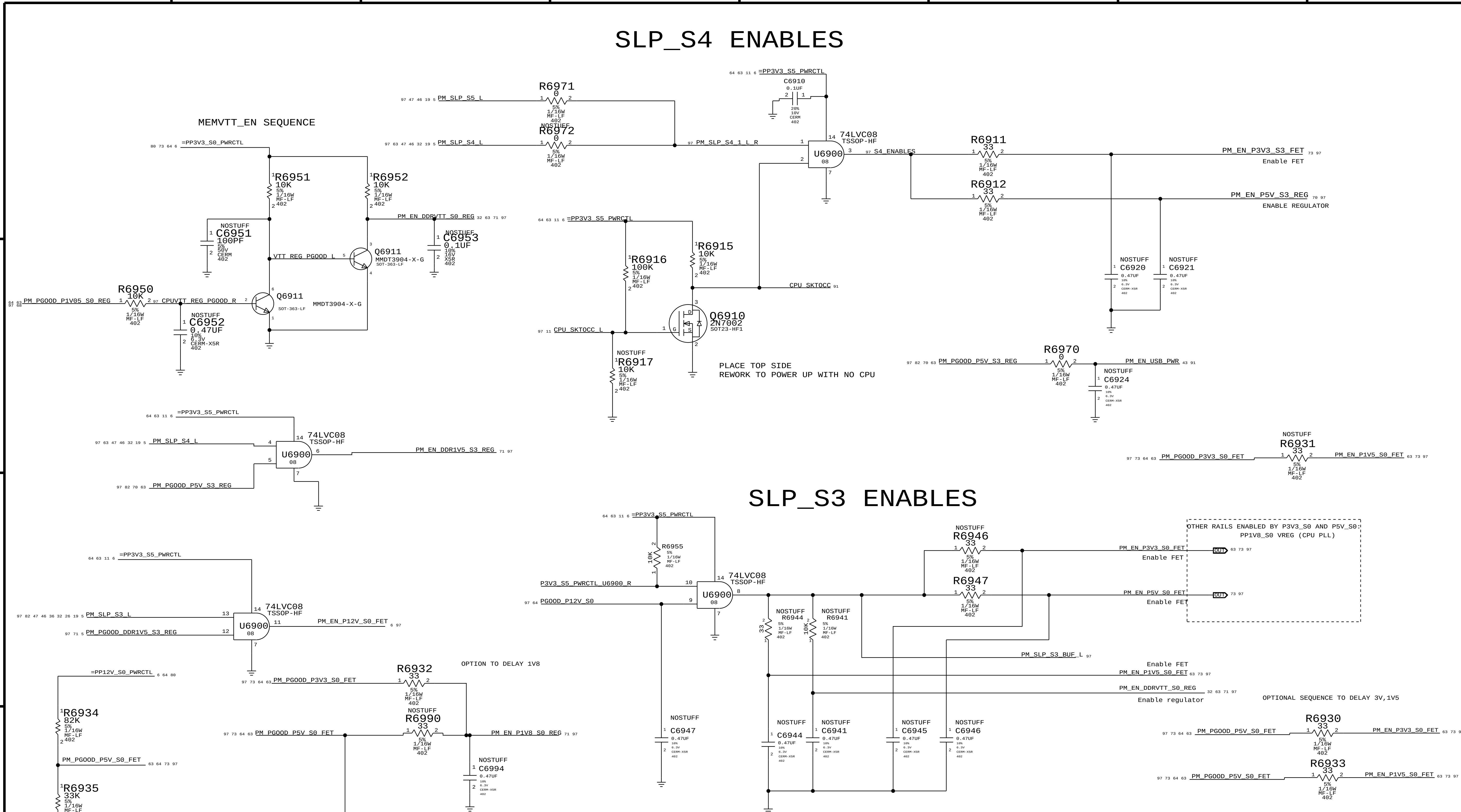

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

POWER SEQUENCING ENABLES

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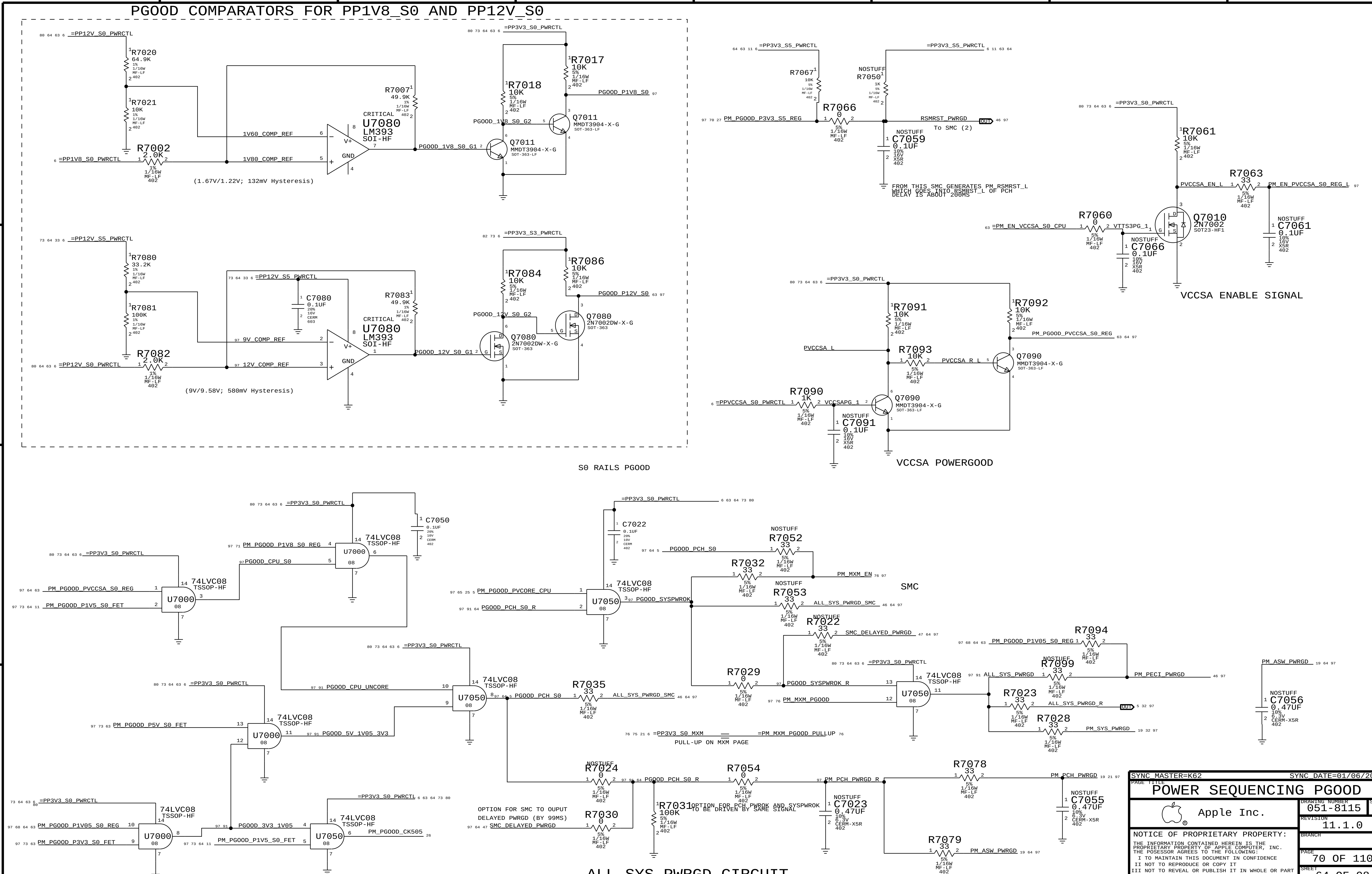

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

POWER SEQUENCING PGOOD

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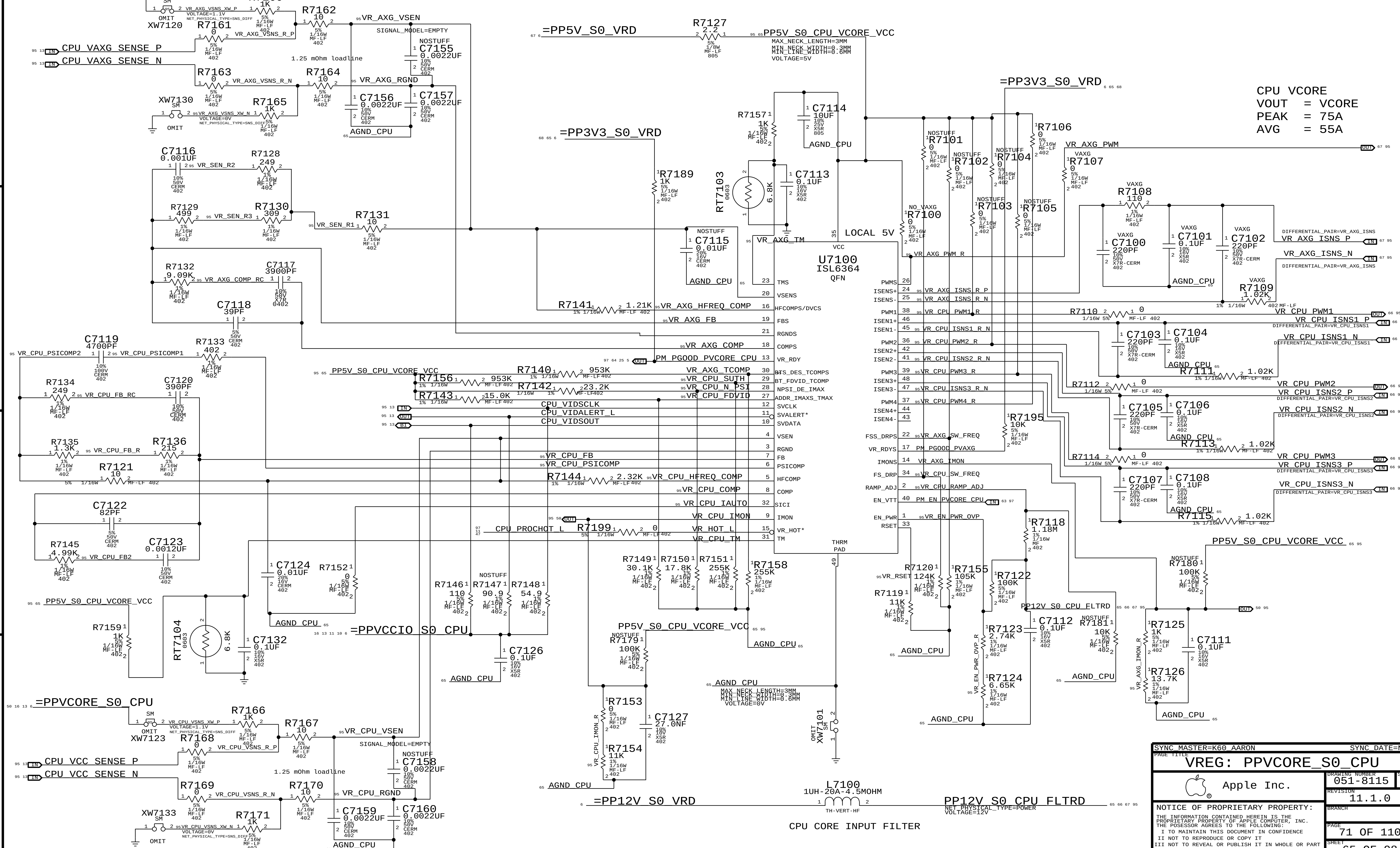
SHEET 64 OF 98

CPU CORE/AXG REG 1.1V/75A O/P= PPVCORE_S0_CPU_REG



SYNC_MASTER=K60_AARON SYNC_DATE=N/A

PAGE TITLE: VREG: PPVCORE_S0_CPU

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DRAWING NUMBER: 051-8115 SIZE: D

REVISION: 11.1.0

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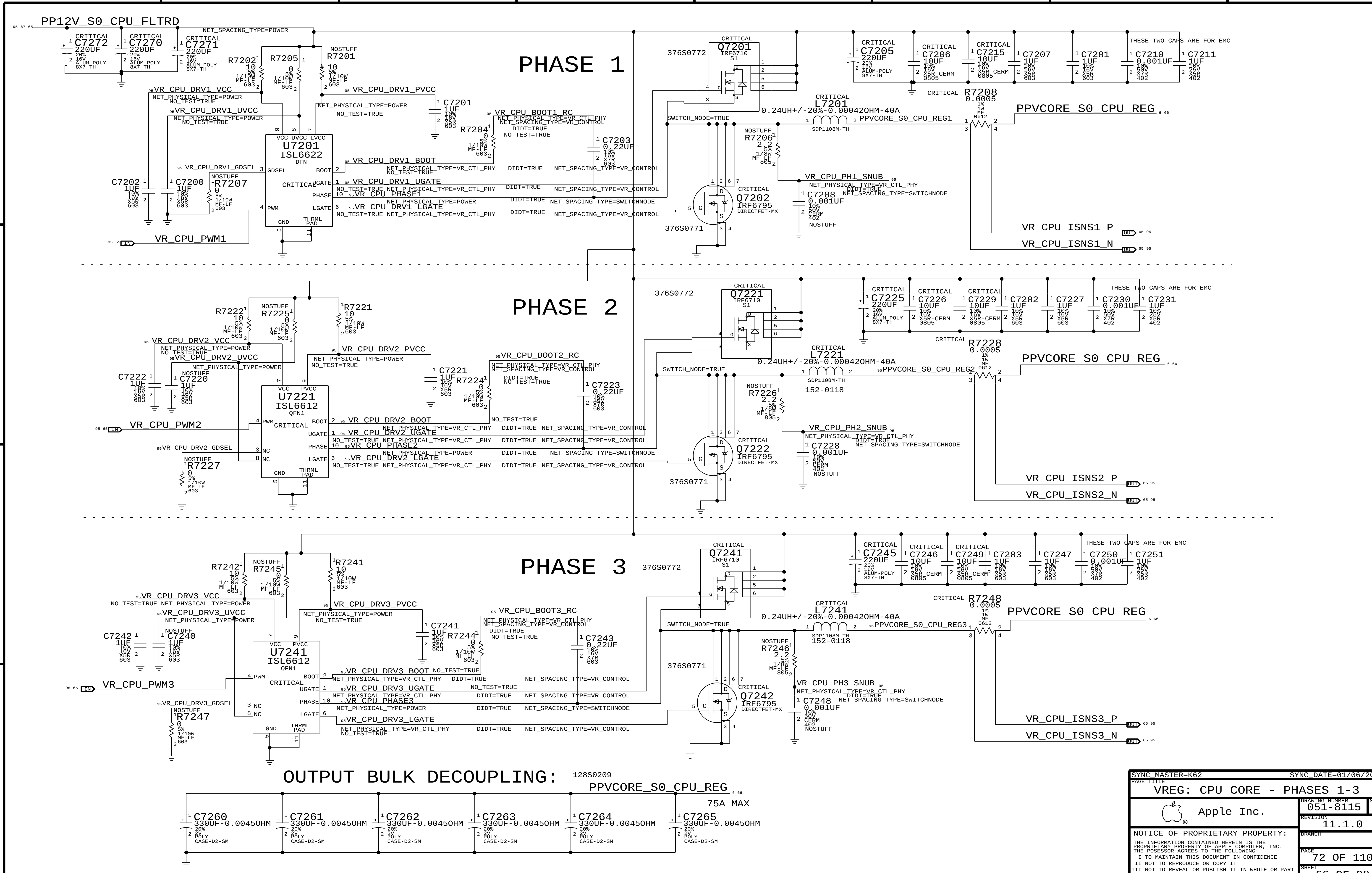
PAGE: 71 OF 110

SHEET: 65 OF 98



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: VREG: CPU CORE - PHASES 1-3

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REVISION: 11.1.0

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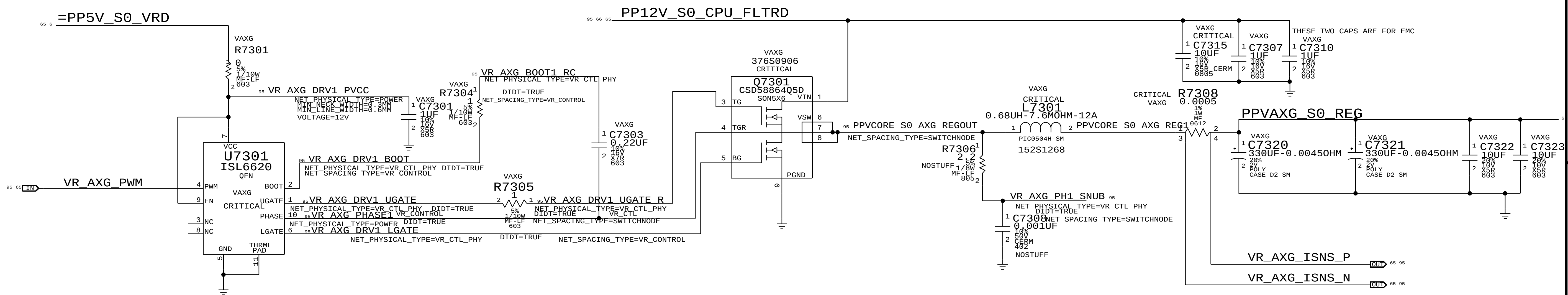
AXG PHASE (MAX 15A)



SYNC_MASTER=K60_AARON SYNC_DATE=N/A

PAGE TITLE
VREG:AXG PHASE/CORE - CAPS

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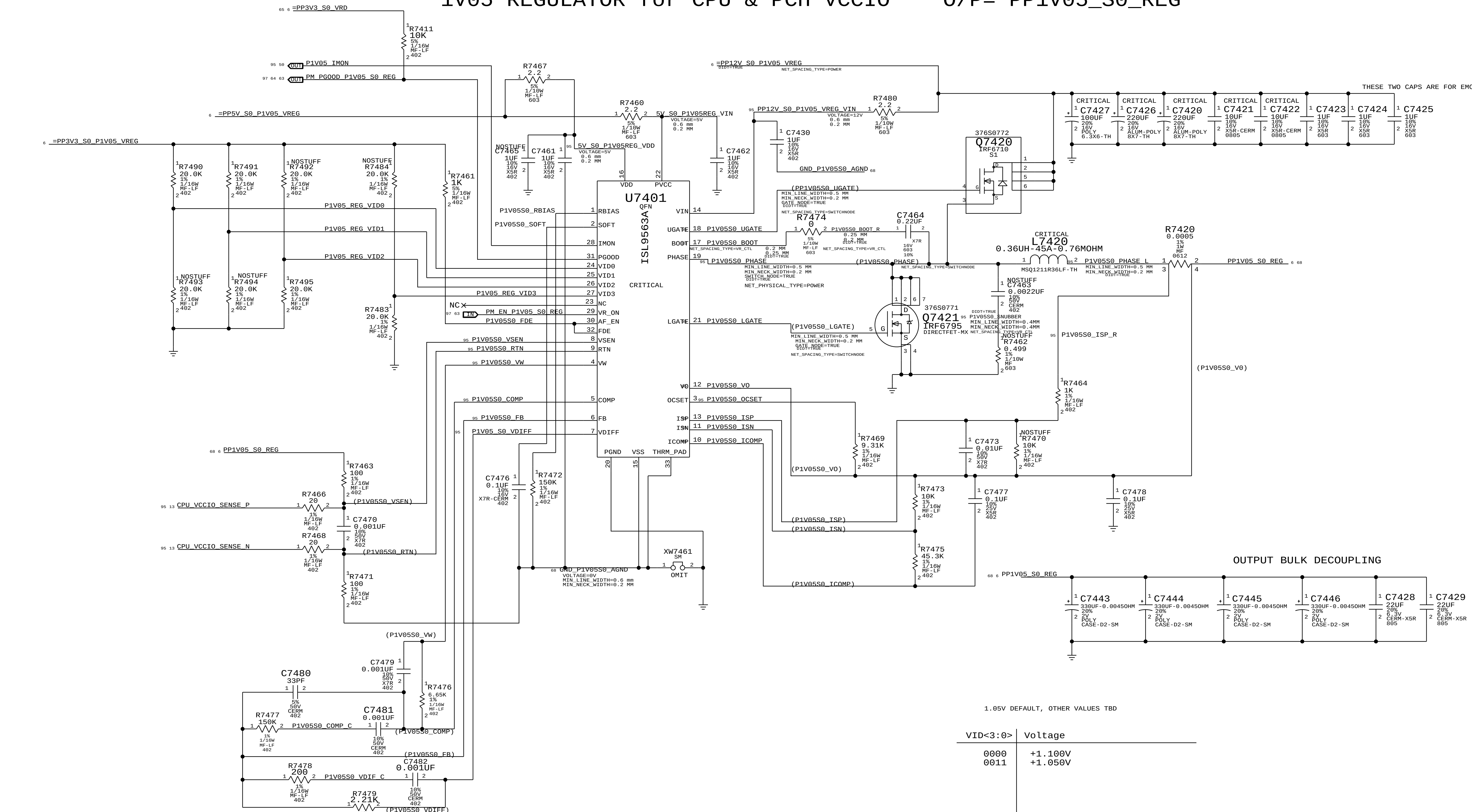
PAGE 73 OF 110

1V05 REGULATOR for CPU & PCH VCCIO   O/P= PP1V05_S0_REG



1.05V DEFAULT, OTHER VALUES TBD

VID<3:0>	Voltage
0000	+1.100V
0011	+1.050V

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
1V05 REGULATOR

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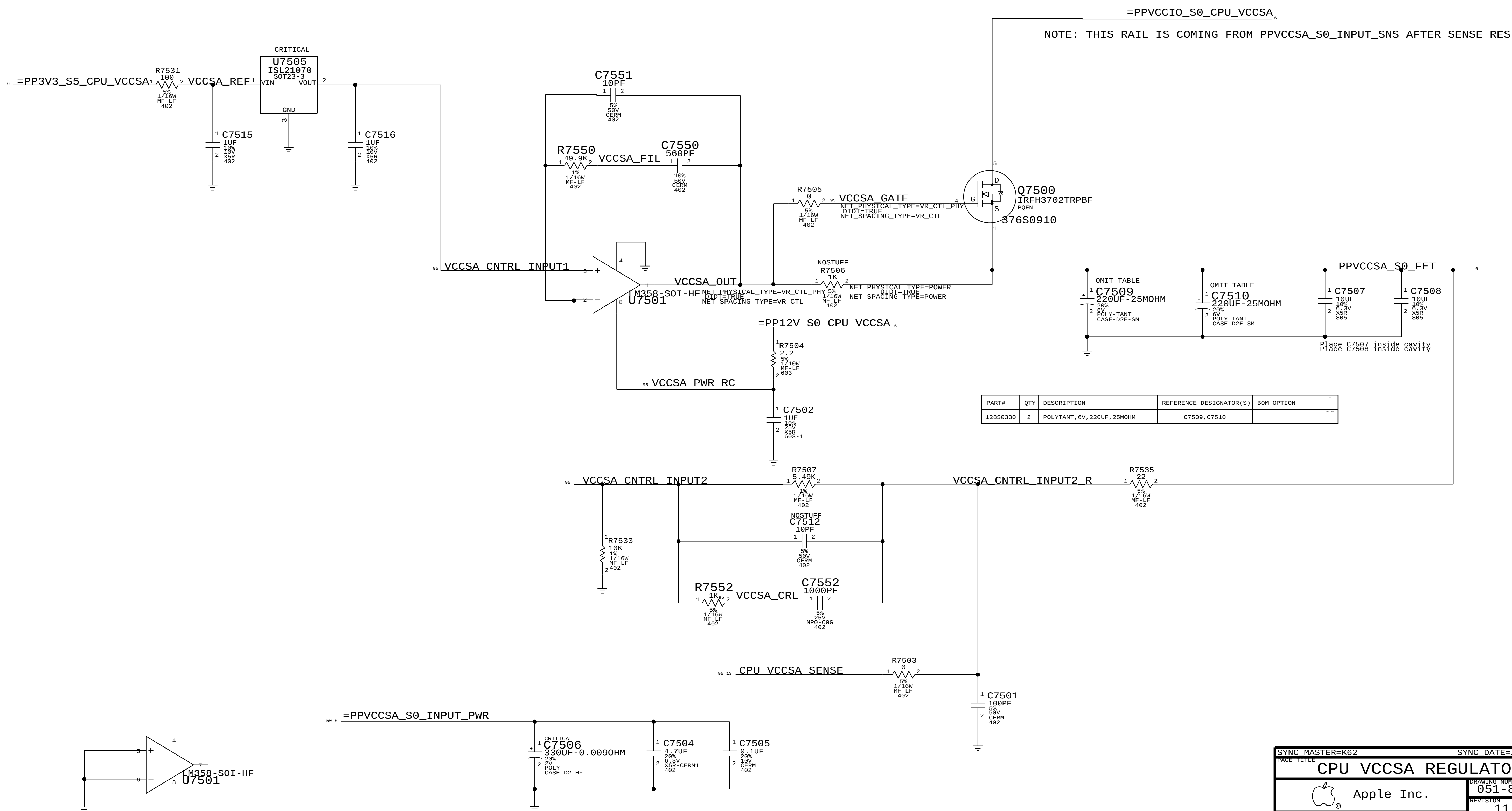
CPU VCCSA 0.925V (8.8A MAX)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
128S0330	2	POLYTANT,6V,220UF,25MOHM	C7509,C7510	

NOTE: THIS POWER RAIL IS BEFORE THE SENSE RES R5310

SYNC_MASTER=K62 SYNC_DATE=11/15/2010

PAGE TITLE
CPU VCCSA REGULATOR

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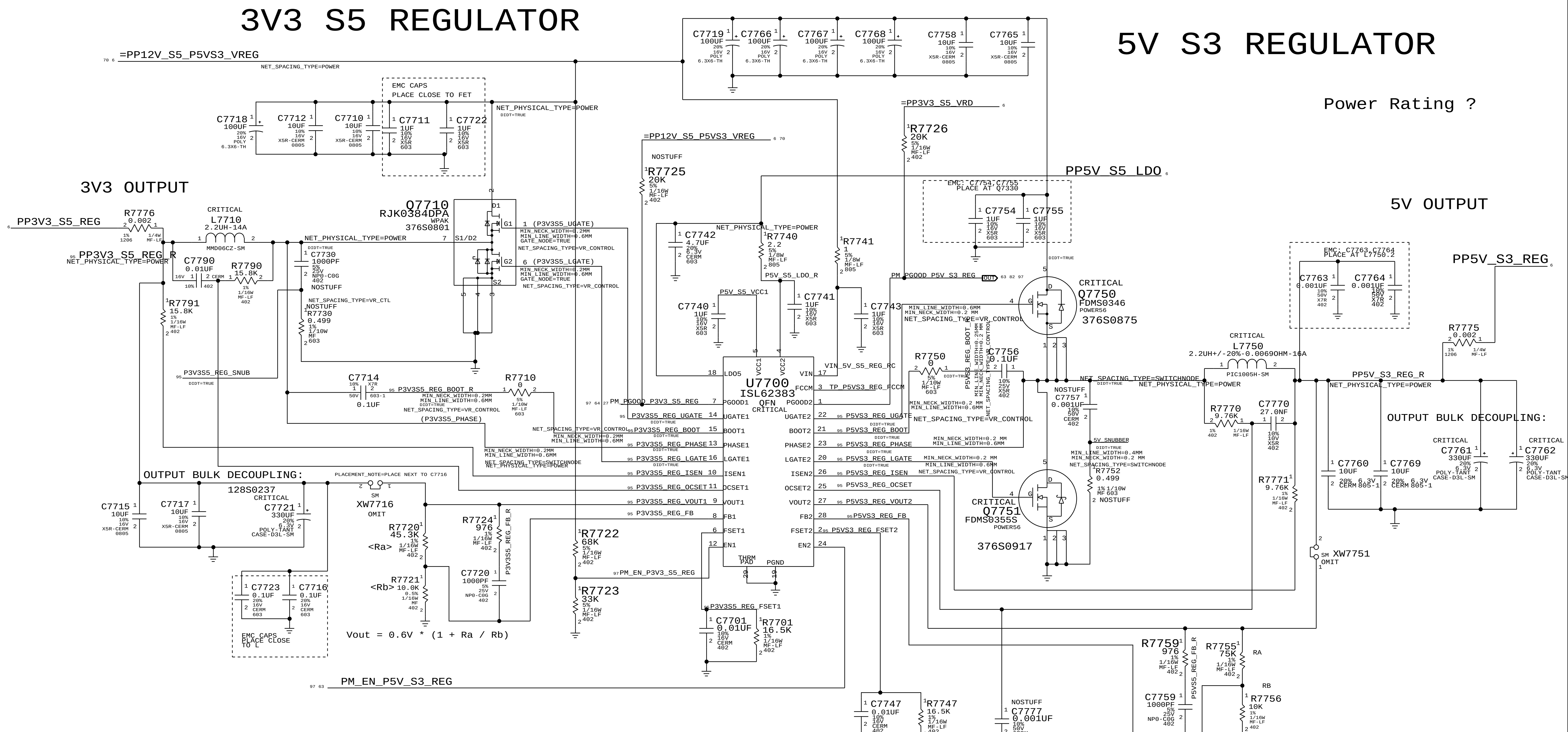

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

5V_S3 / 3V3_S5 VREGS

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PAGE 77 OF 110

SHEET 70 OF 98

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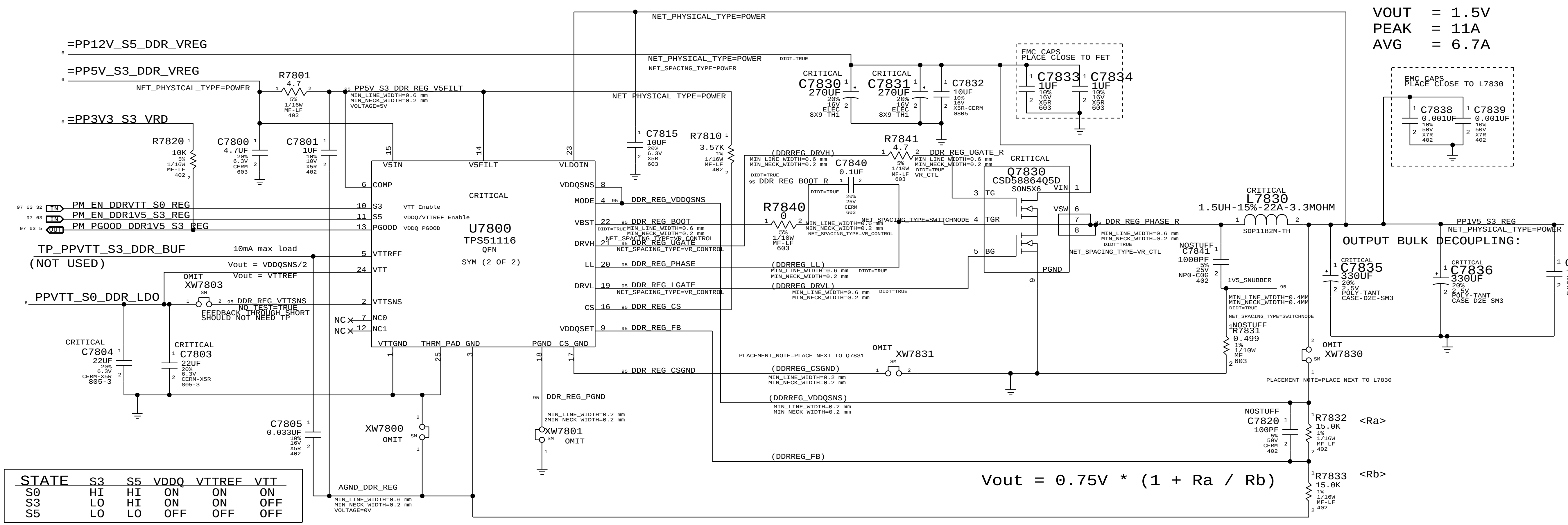
1.5 V DDR SUPPLY

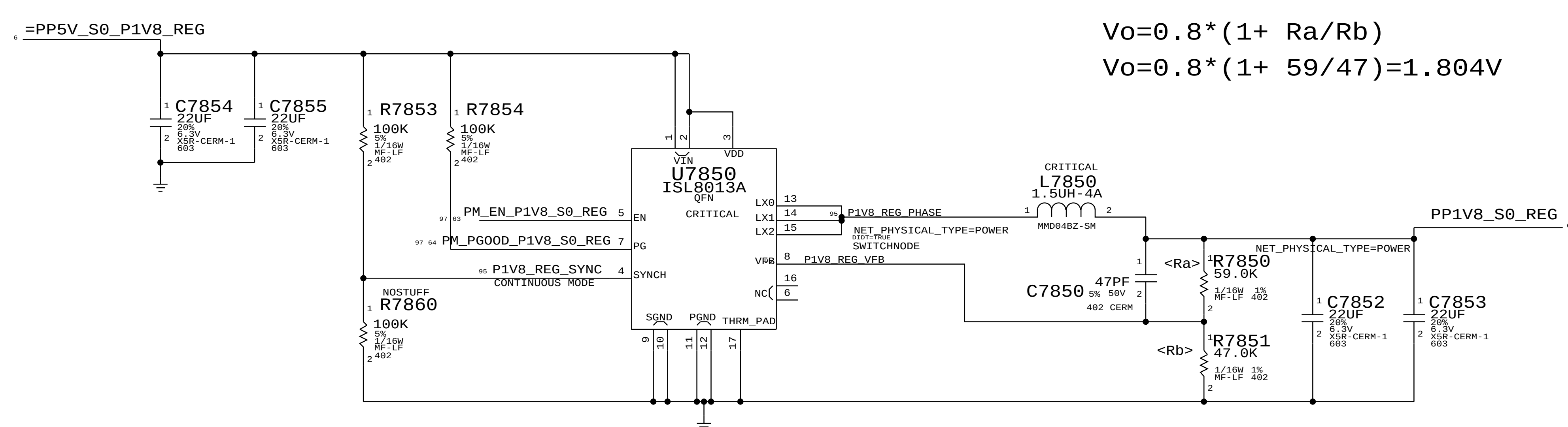


PPDDR_S3_REG
VOUT = 1.5V
PEAK = 11A
AVG = 6.7A

STATE	S3	S5	VDDQ	VTTREF	VTT
S0	HI	HI	ON	ON	ON
S3	LO	HI	ON	ON	OFF
S5	LO	LO	OFF	OFF	OFF

Vout = 0.75V * (1 + Ra / Rb)

1.8 V SUPPLY



1A Average current

Vo=0.8*(1+ Ra/Rb)

Vo=0.8*(1+ 59/47)=1.804V

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: 1.5V / 1.8V VREGS

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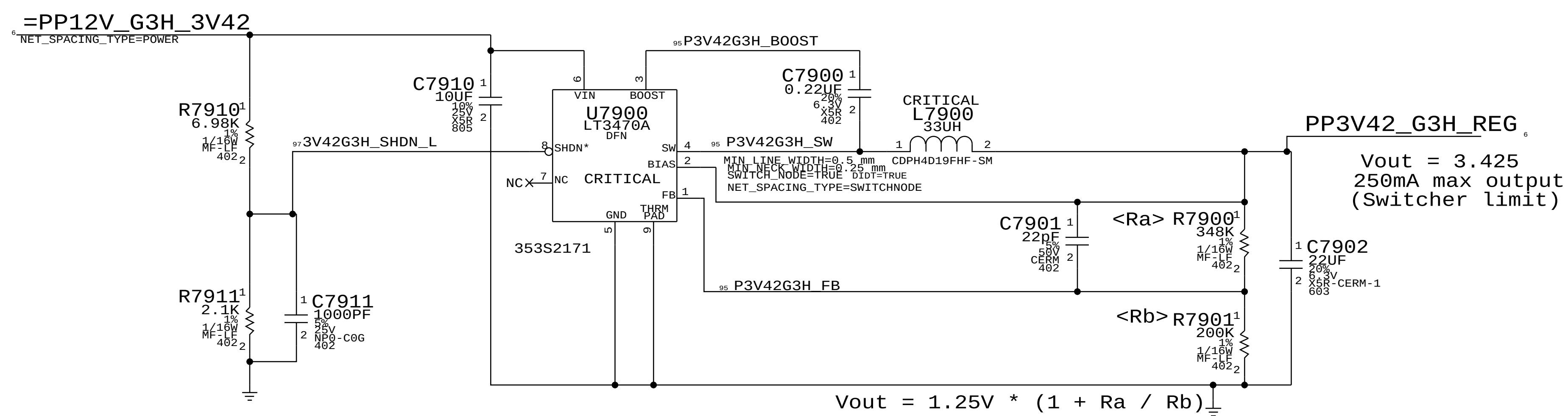
REVISION: 11.1.0

3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC VRef generator



SYNC_MASTER=K62		SYNC_DATE=01/06/2011
PAGE TITLE: 3.42 G3HOT SUPPLY		
Apple Inc.	DRAWING NUMBER: 051-8115	SIZE: D
	REVISION: 11.1.0	
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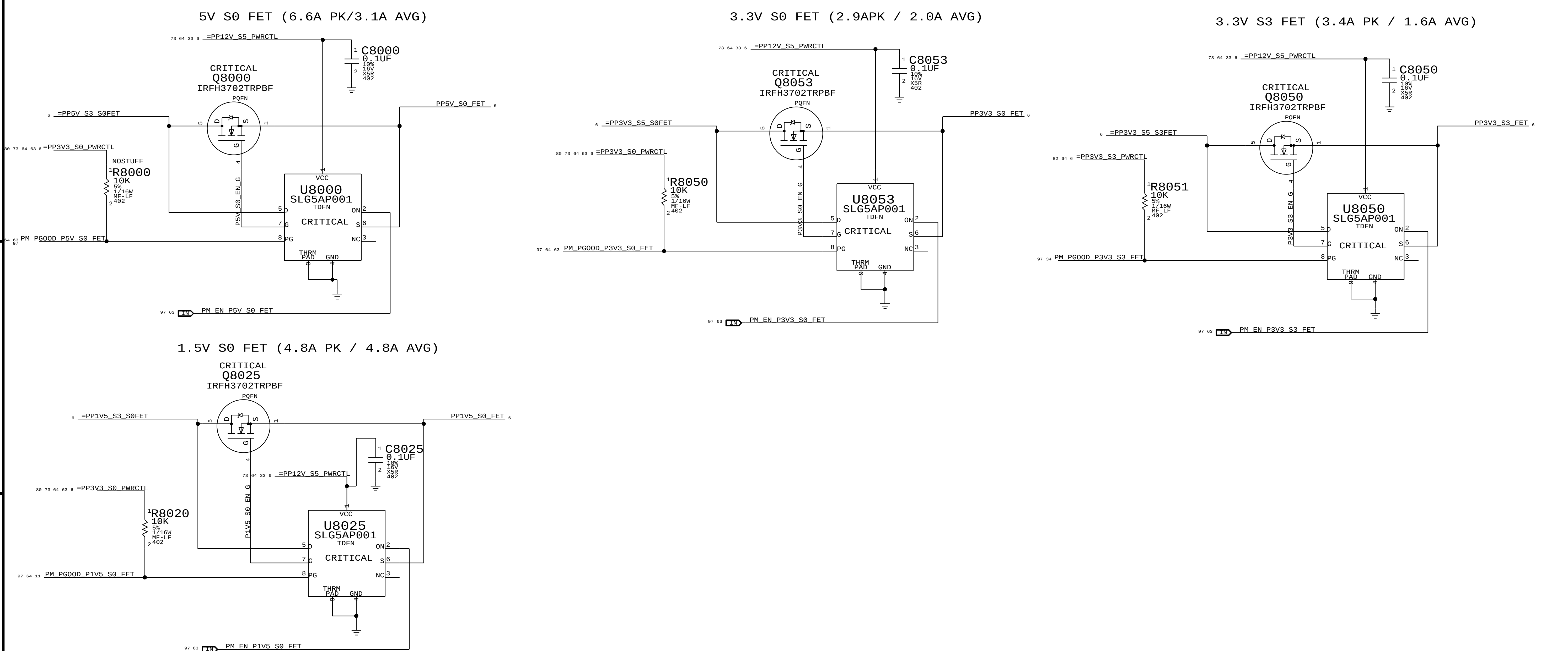
	PAGE: 79 OF 110	
	SHEET: 72 OF 98	



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

S3+S0 FETS

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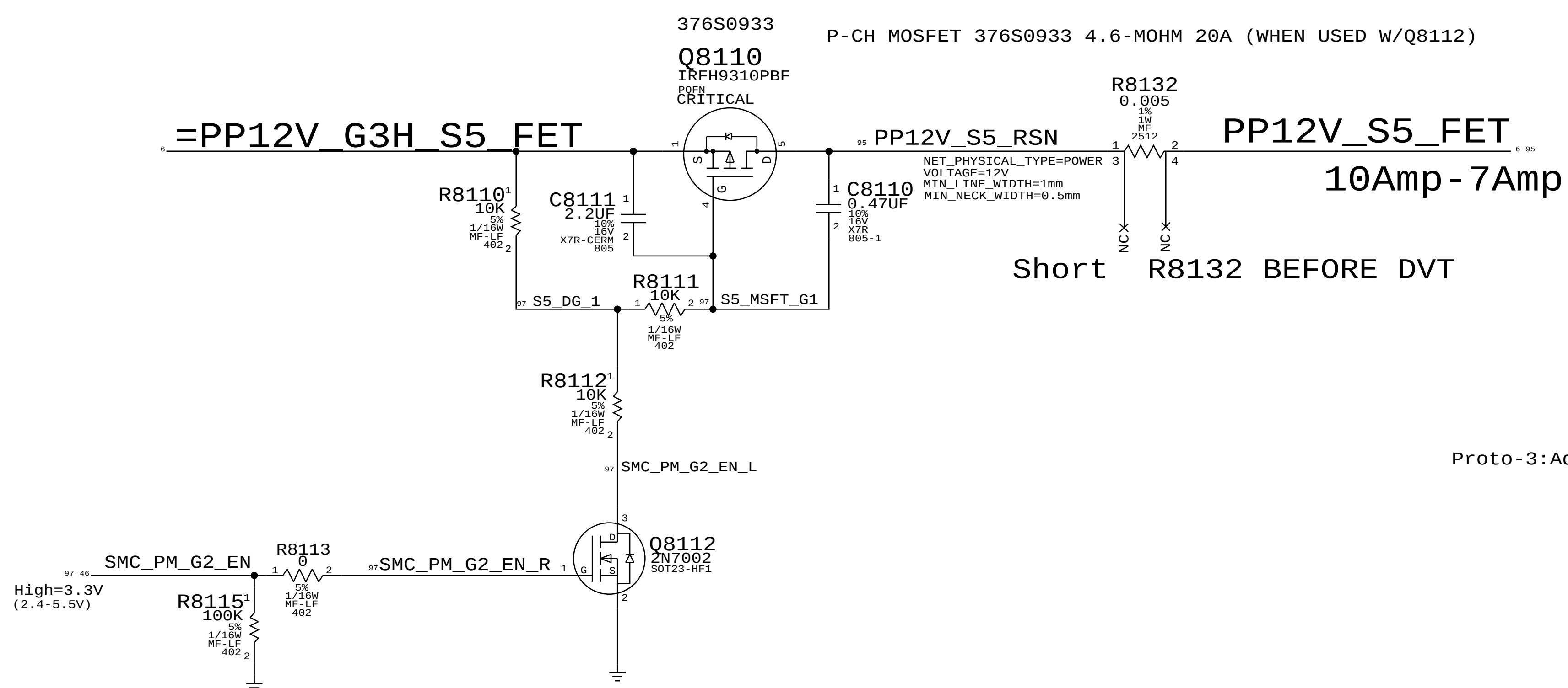

Proto-3:Add C8111

SYNC_MASTER=K60_JERRY SYNC_DATE=01/06/2011

PAGE TITLE

12V_S0 & 12V_S5 switch

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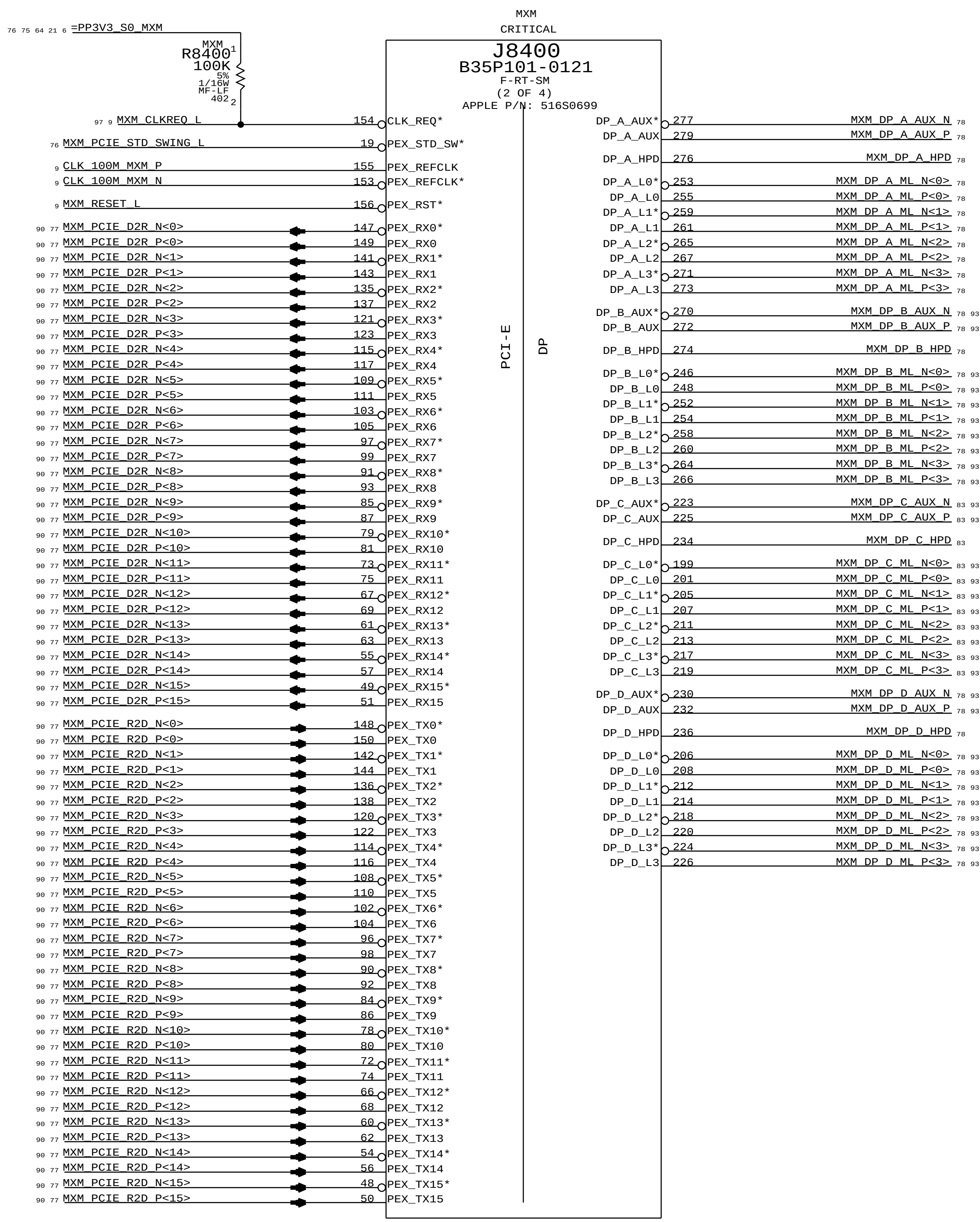
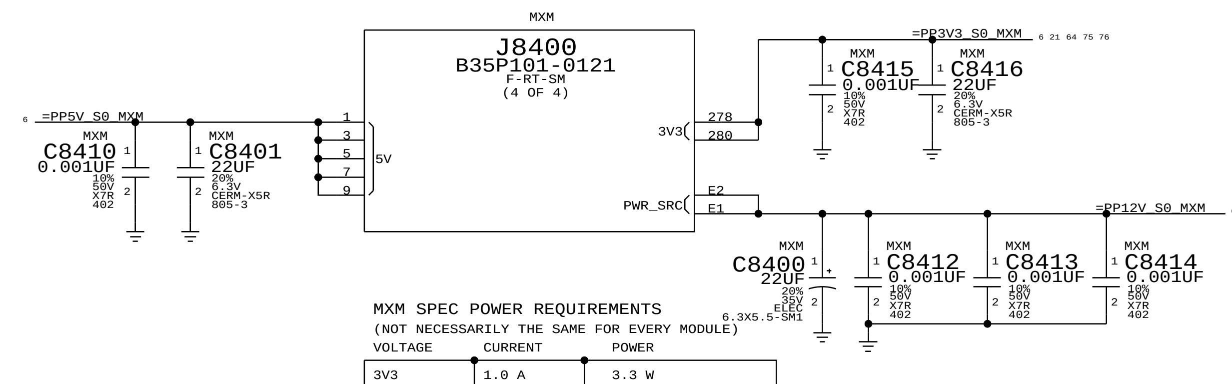
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM
- =PP5V_S0_MXM
- =PPV_S0_MXM_PWRSRC

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
- MXM





MXM SPEC POWER REQUIREMENTS
(NOT NECESSARILY THE SAME FOR EVERY MODULE)

VOLTAGE	CURRENT	POWER
3V3	1.0 A	3.3 W
5V	2.5 A	12.5 W
PWR (7-20V)	UP TO 10 A	PLATFORM DEPENDENT

MXM DP PORT ROUTING

	K62	K60
DP A	EXT DP1	EXT DP1
DP B	T29 DP2	T29 DP2
DP C	INT DP	INT DP
DP D	T29 DP1	T29 DP1
DP E	EXT DP2	

SYNC_MASTER=K62 SYNC_DATE=01/06/2011
PAGE TITLE
MXM PCIe, DP & Power
Apple Inc.
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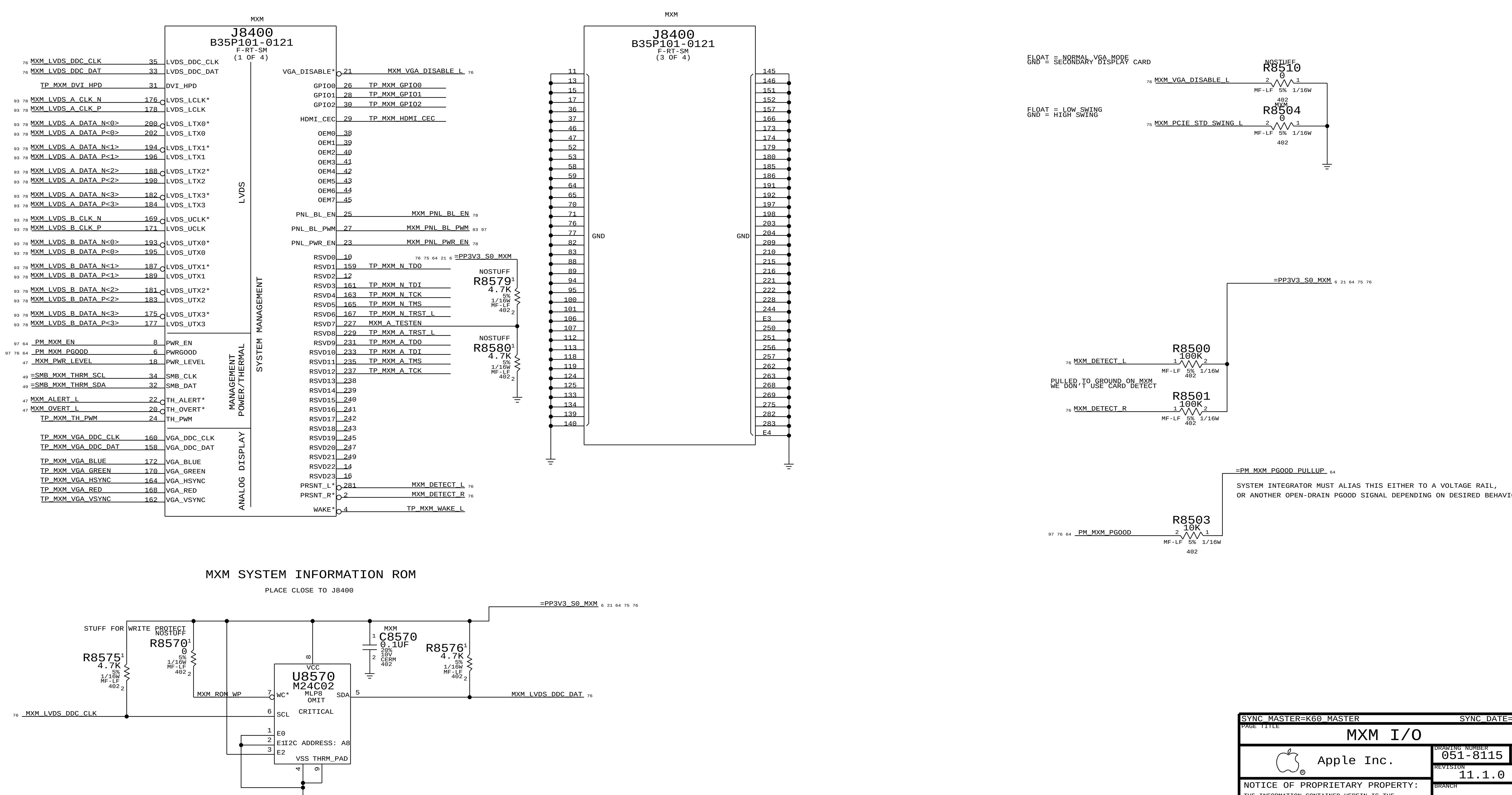
Page Notes

Power aliases required by this page:
- =PP3V3_S0_MXM

Signal aliases required by this page:
- =SMB_MXM_THRM_DATA   - =PM_MXM_PGOOD_PULLUP
- =SMB_MXM_THRM_CLK

BOM options provided by this page:

PULLUPS & PULLDOWNS AT MXM CONNECTOR



SYNC_MASTER=K60_MASTER SYNC_DATE=N/A

PAGE TITLE: MXM I/O

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MXM TX CAPS

MXM RX CAPS

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

MXM PCIE CAPS

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Page Notes

Power aliases required by this page:

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

MXM ALIAS

75	MXM_DP_A_ML_P<0..3>	DP_EXTA_ML_C_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	84 93
75	MXM_DP_A_ML_N<0..3>	DP_EXTA_ML_C_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	84 93
75	MXM_DP_A_AUX_P	DP_EXTA_AUXCH_C_P MAKE_BASE=TRUE NO_TEST=TRUE	78 84 93
75	MXM_DP_A_AUX_N	DP_EXTA_AUXCH_C_N MAKE_BASE=TRUE NO_TEST=TRUE	78 84 93
75	MXM_DP_A_HPD	DP_EXTA_HPD MAKE_BASE=TRUE NO_TEST=TRUE	84

T29 CONN POWER CONTROL ALIAS

95 84	PP3V3_SW_DPAPWR	=PP3V3_SW_DPAPWR	6
97 36 33 19	PCIE_WAKE_L	=T29_WAKE_L	84

T29 MXM DP ALIAS

93 75	MXM_DP_D_ML_P<0..3>	DP_T29SNK0_ML_C_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_D_ML_N<0..3>	DP_T29SNK0_ML_C_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_D_AUX_P	DP_T29SNK0_AUXCH_C_P MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_D_AUX_N	DP_T29SNK0_AUXCH_C_N MAKE_BASE=TRUE NO_TEST=TRUE	86 96
75	MXM_DP_D_HPD	DP_T29SNK0_HPD MAKE_BASE=TRUE NO_TEST=TRUE	86
93 75	MXM_DP_B_ML_P<0..3>	DP_T29SNK1_ML_C_P<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_B_ML_N<0..3>	DP_T29SNK1_ML_C_N<0..3> MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_B_AUX_P	DP_T29SNK1_AUXCH_C_P MAKE_BASE=TRUE NO_TEST=TRUE	86 96
93 75	MXM_DP_B_AUX_N	DP_T29SNK1_AUXCH_C_N MAKE_BASE=TRUE NO_TEST=TRUE	86 96
75	MXM_DP_B_HPD	DP_T29SNK1_HPD MAKE_BASE=TRUE NO_TEST=TRUE	86

Unused MXM Interfaces

93 76	MXM_LVDS_A_CLK_N	NC_MXM_LVDS_A_CLK_N MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_A_CLK_P	NC_MXM_LVDS_A_CLK_P MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_A_DATA_N<3..0>	NC_MXM_LVDS_A_DATA_N<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_A_DATA_P<3..0>	NC_MXM_LVDS_A_DATA_P<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_B_CLK_N	NC_MXM_LVDS_B_CLK_N MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_B_CLK_P	NC_MXM_LVDS_B_CLK_P MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_B_DATA_N<3..0>	NC_MXM_LVDS_B_DATA_N<3..0> MAKE_BASE=TRUE NO_TEST=TRUE
93 76	MXM_LVDS_B_DATA_P<3..0>	NC_MXM_LVDS_B_DATA_P<3..0> MAKE_BASE=TRUE NO_TEST=TRUE

Unused T29 Interfaces

96 86	T29_D2R_N<2..3>	NC_T29_D2R_N<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		
96 86	T29_D2R_P<2..3>	NC_T29_D2R_P<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		
96 86	T29_R2D_C_N<2..3>	NC_T29_R2D_C_N<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		
96 86	T29_R2D_C_P<2..3>	NC_T29_R2D_C_P<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		
86	T29_LSEO<3>	T29_LSEO_LSOE3 MAKE_BASE=TRUE NO_TEST=TRUE	T29_LSOE<3>	86
86	T29_LSEO<2>	T29_LSEO_LSEO2 MAKE_BASE=TRUE NO_TEST=TRUE	T29_LSOE<2>	86

UNUSED MXM CONTROL SIGNALS

76	MXM_PNL_BL_EN	NC_MXM_PNL_BL_EN MAKE_BASE=TRUE NO_TEST=TRUE
76	MXM_PNL_PWR_EN	NC_MXM_PNL_PWR_EN MAKE_BASE=TRUE NO_TEST=TRUE

DDC/AUX ALIAS

93 84 78	DP_EXTA_AUXCH_C_P MAKE_BASE=TRUE	DP_EXTA_DDC_CLK	84
93 84 78	DP_EXTA_AUXCH_C_N MAKE_BASE=TRUE	DP_EXTA_DDC_DATA	84

System RTC Power Source & 32kHz / 25MHz Clock Generator

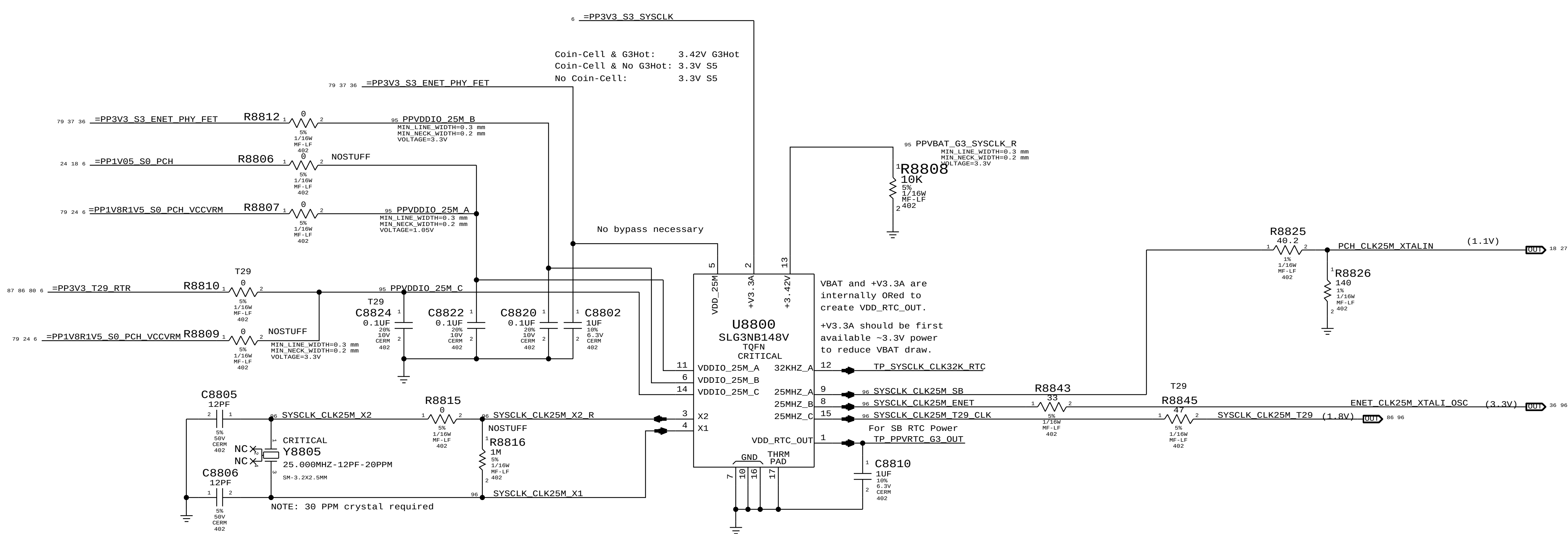
GreenCLK Implementation Notes:

VBAT: Alias as appropriate (see note below & Desktop Example)
+V3.3A: Alias as appropriate (see note below)
VDD_25M: 3.3V matching 'highest' VDDIO power state (ENET)

VDDIO_25M_A: SB power rail for XTAL circuit.
VDDIO_25M_B: Ethernet power rail for XTAL circuit.
VDDIO_25M_C: T29 power rail for XTAL circuit.

NOTE: VDD_25M must be powered if any VDDIO_25M_x is powered.

For Cougar Point Desktop: VDDIO = VCCVRM (1.8V), Vclk = 1.1V Max, Divider: 604 / 1000
For Cougar Point Mobile: VDDIO = VCCVRM (1.5V), Vclk = 1.1V Max, Divider: 332 / 1000
For Caesar-IV (BCM57765): VDDIO = XTALVDDH (3.3V), Vclk = 3.3V Max. No Divider Necessary



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: GREEN CLOCK

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PAGE: 88 OF 110

SHEET: 79 OF 98

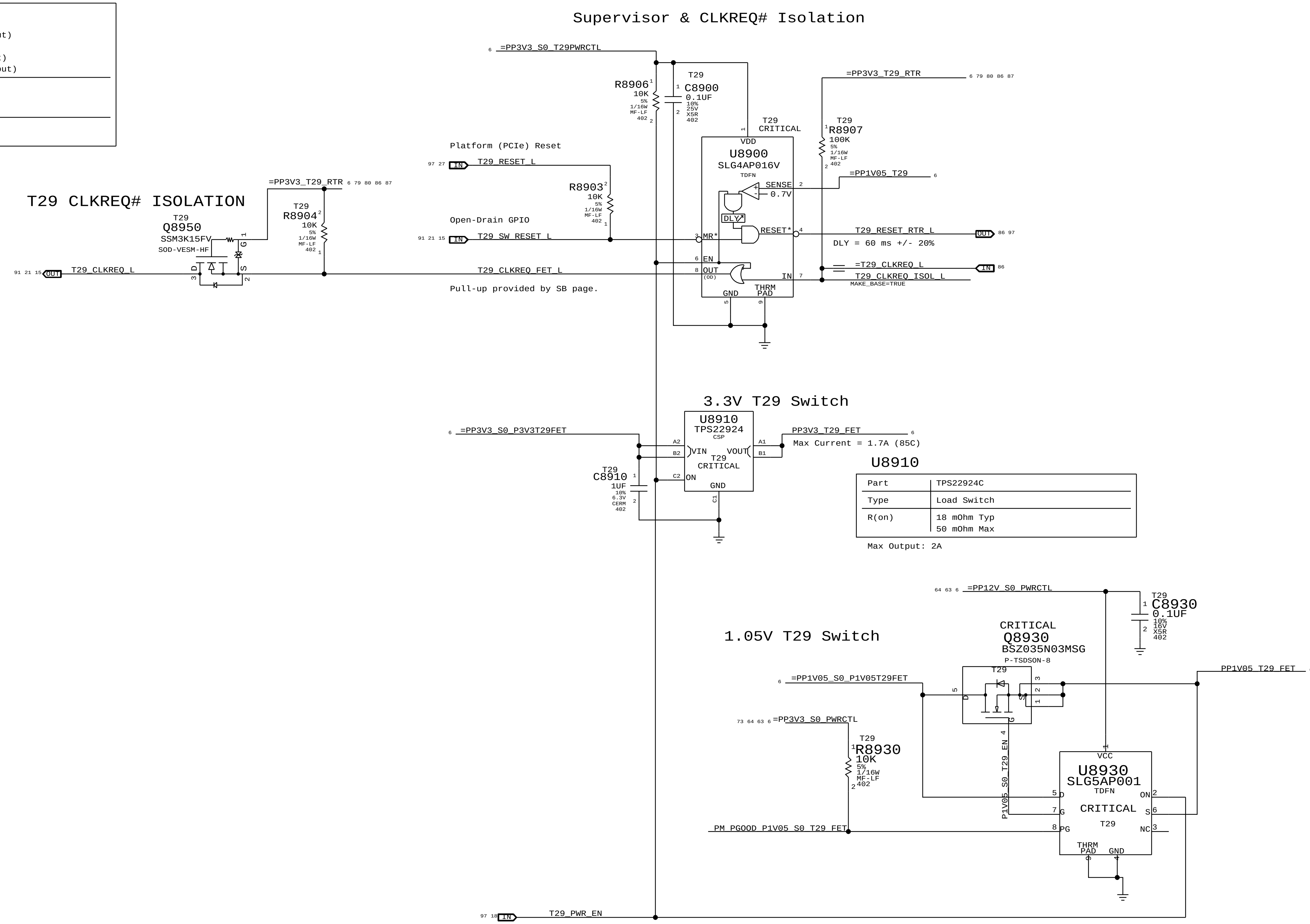
D C B A

Page Notes

Power aliases required by this page:
- =PP3V3_T29_P3V3T29FET (3.3V FET Input)
- =PP3V3_T29_FET (3.3V FET Output)
- =PP3V3_S0_T29PWRCTL
- =PP1V05_T29_P1V05T29FET (1.05V FET Input)
- =PP1V05_T29_FET (1.05V FET Output)

Signal aliases required by this page:
- =T29_CLKREQ_L
- T29_RESET_L

BOM options provided by this page:
(NONE)



Page Notes

Power aliases required by this page:

- =PP12V_S0_LCD
- =PP3V3_S0_VIDEO

Signal aliases required by this page:
(NONE)

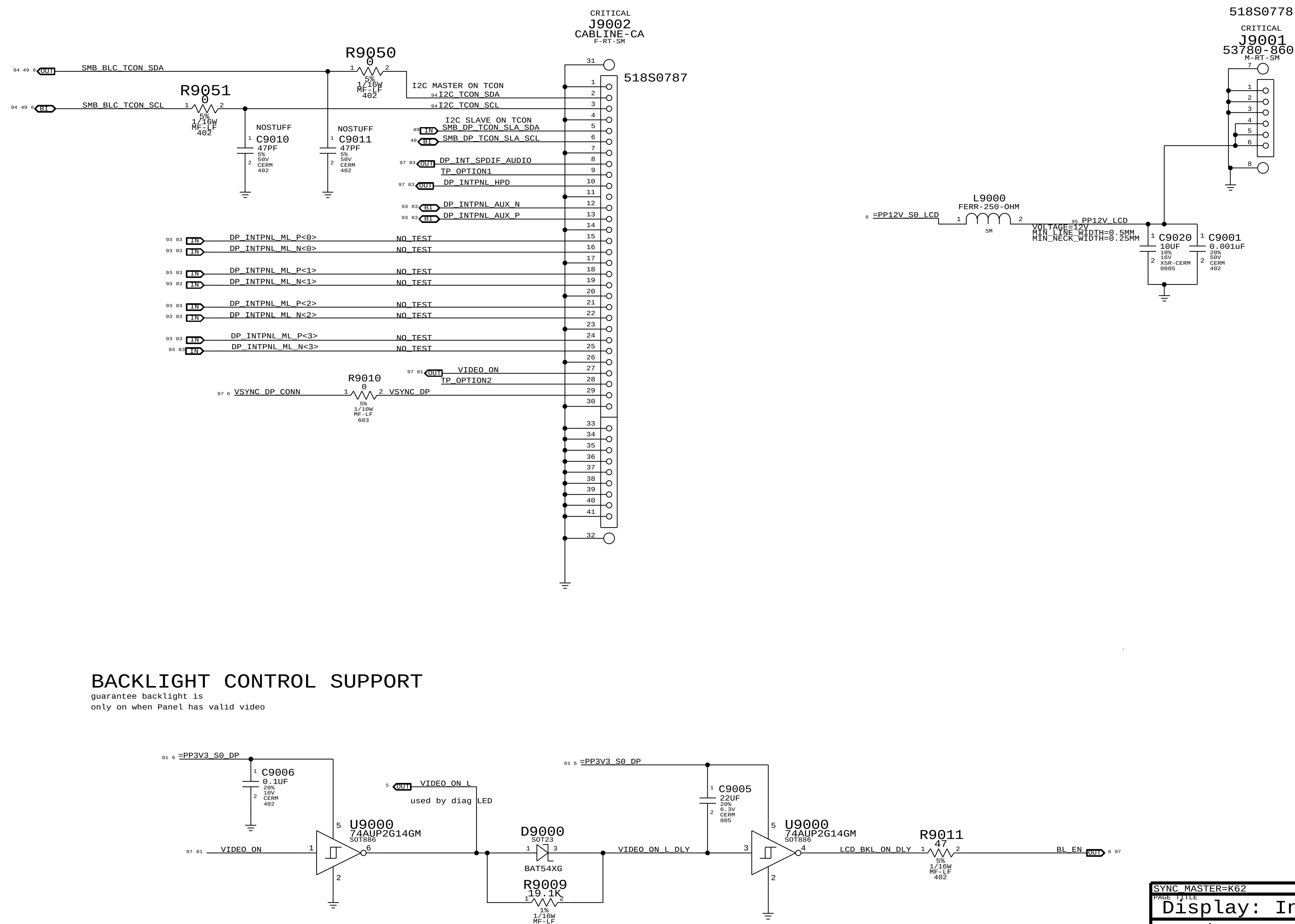
BOM options provided by this page:
IG, MXM, MLB_PNL_PWR, LCD_PNL_PWR

INTERNAL DP INTERFACE

INTERNAL DP POWER

BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video



SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
Display: Int DP Connector

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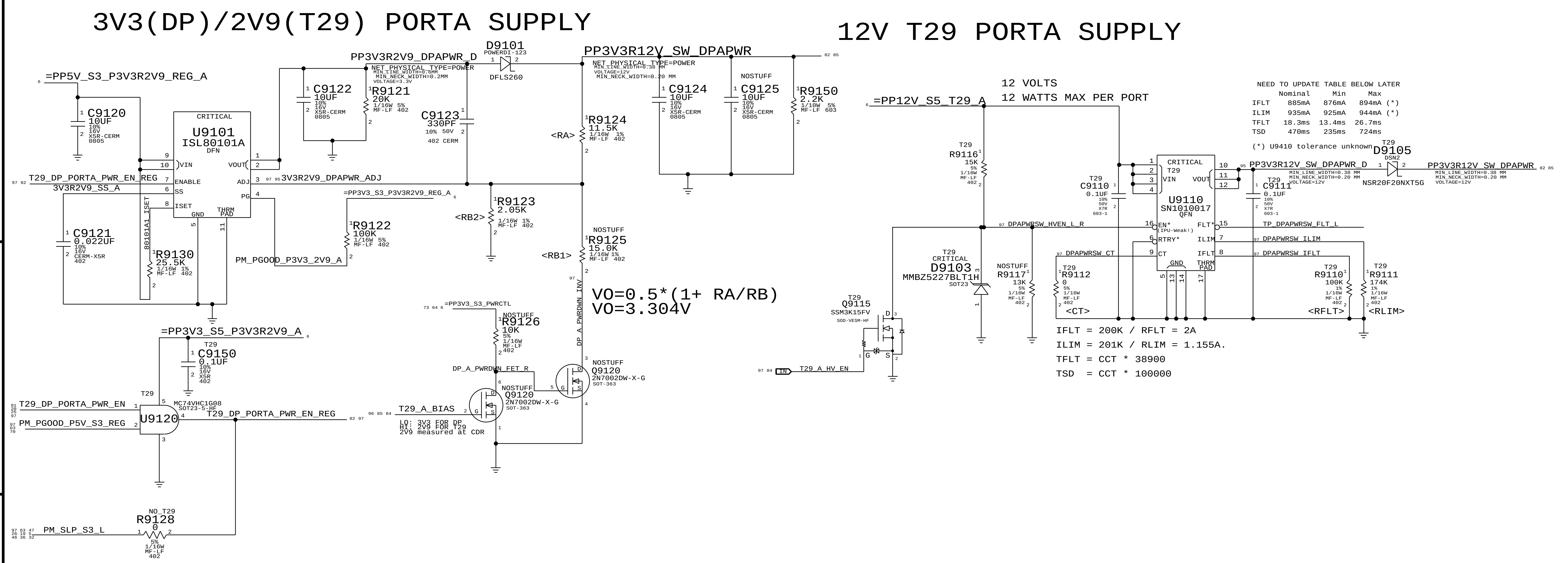

SYNC_MASTER=K62 SYNC_DATE=11/14/2010

PAGE TITLE

2V9/3V3/12V POWER SWITCH

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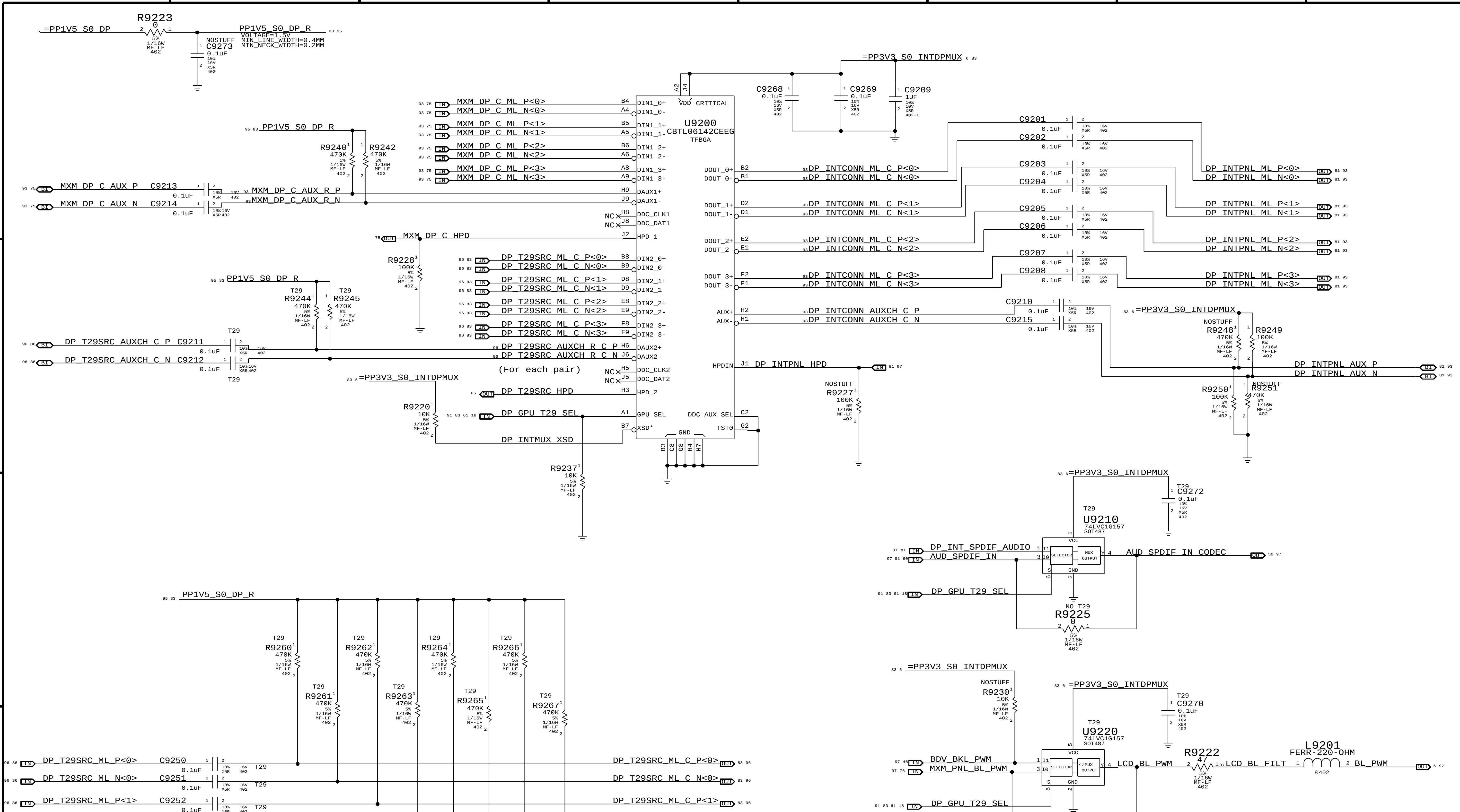

SYNC_MASTER=K62_AARON SYNC_DATE=N/A

PAGE TITLE

Internal DP MUXing

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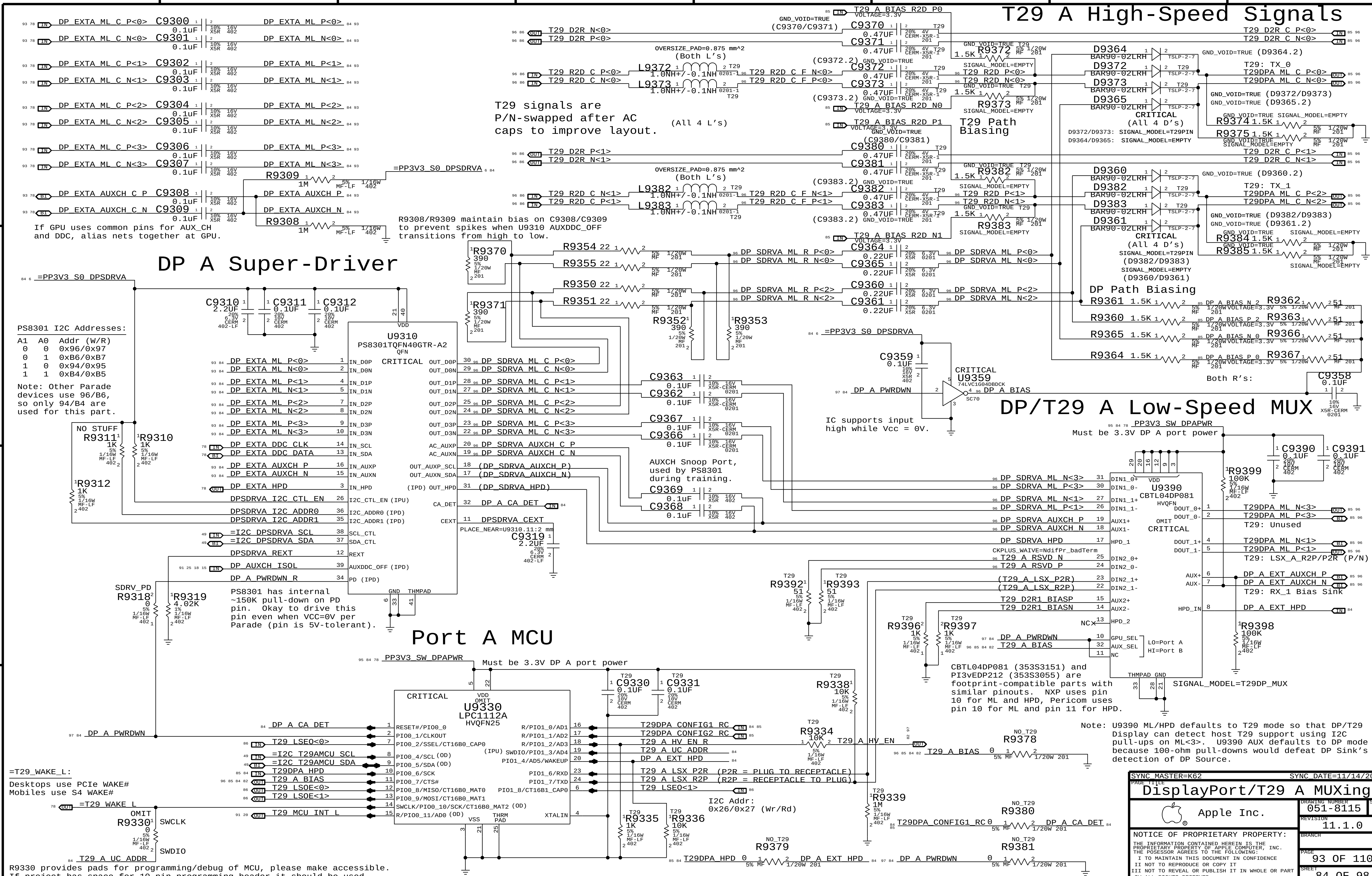
T29 A High-Speed Signals



SYNC_MASTER=K62 SYNC_DATE=11/14/2010

PAGE TITLE: DisplayPort/T29 A MUXing

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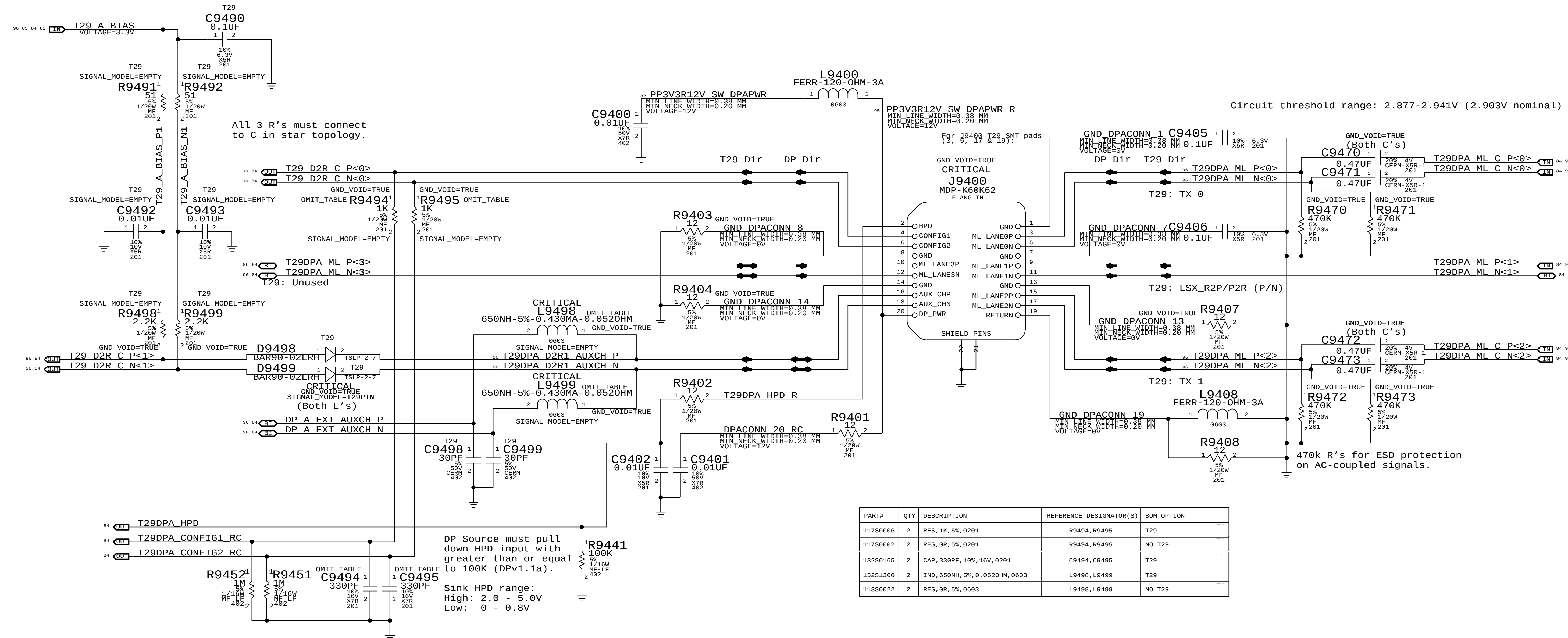
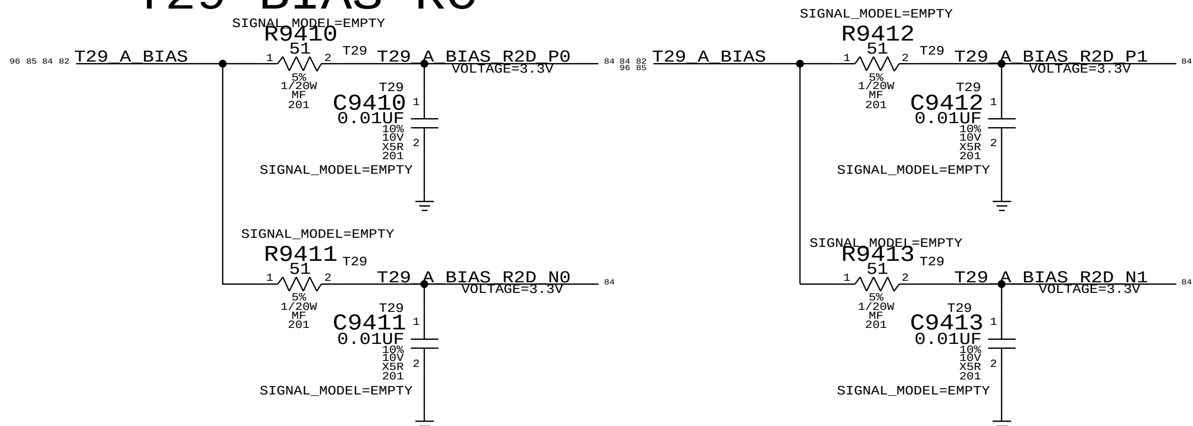
DisplayPort/T29 A Connector

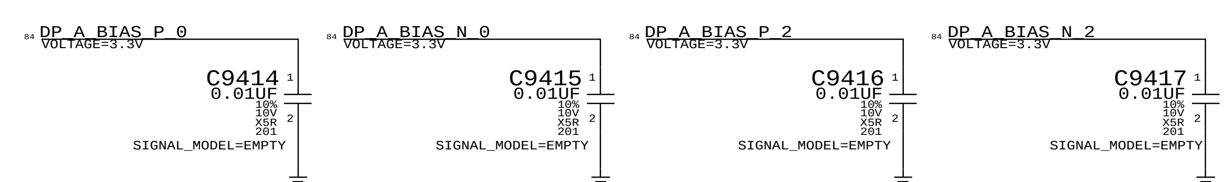


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
117S0006	2	RES,1K,5%,0201	R9494,R9495	T29
117S0002	2	RES,0R,5%,0201	R9494,R9495	NO_T29
132S0165	2	CAP,330PF,10%,16V,0201	C9494,C9495	T29
152S1300	2	IND,650NH,5%,0.052OHM,0603	L9498,L9499	T29
113S0022	2	RES,0R,5%,0603	L9498,L9499	NO_T29

T29 BIAS RC



DP BIAS CAPS



SYNC_MASTER=(MASTER) SYNC_DATE=(MASTER)

PAGE TITLE: DisplayPort/T29 A Connector

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DRAWING NUMBER: 051-8115 SIZE: D

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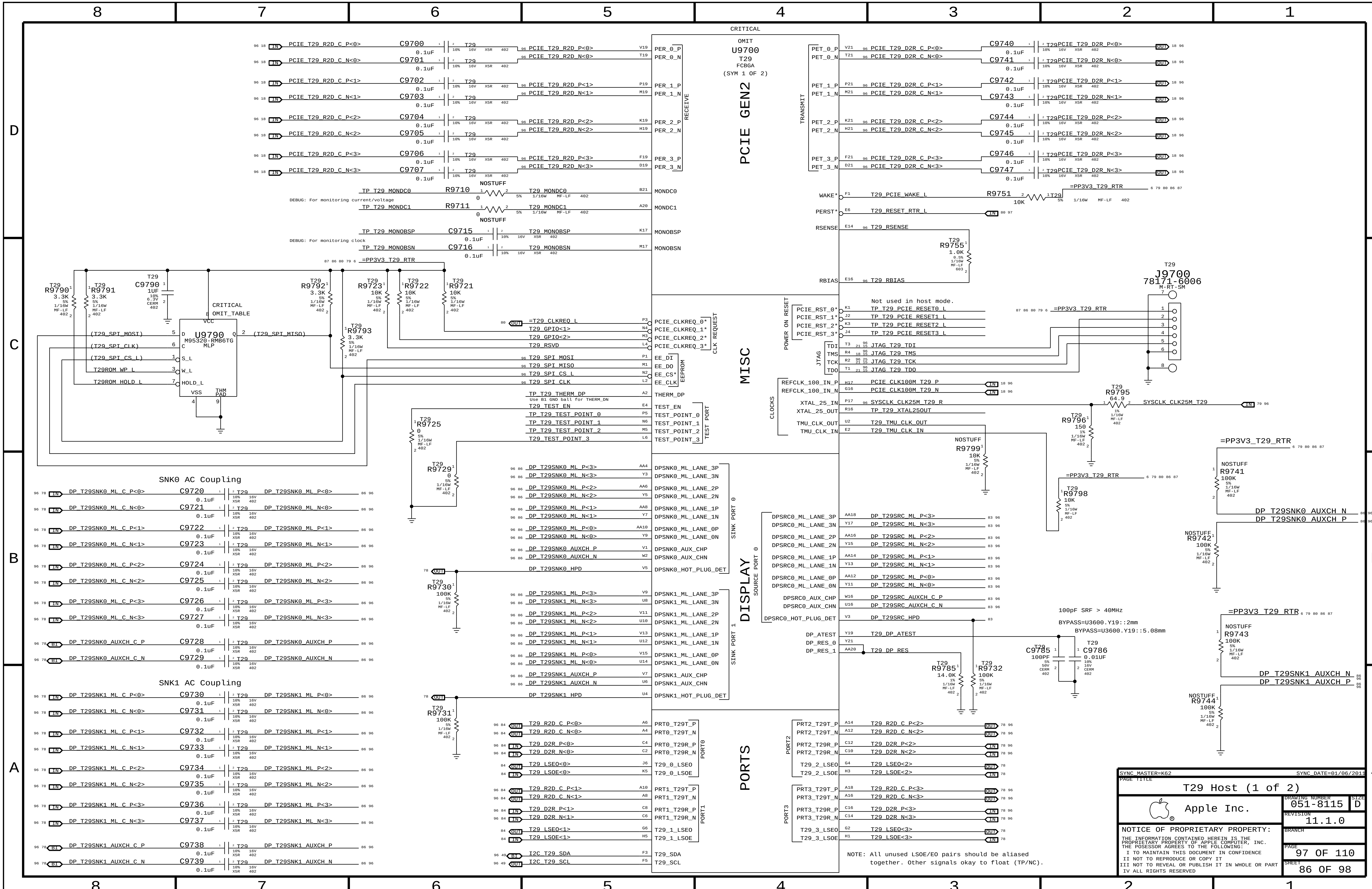

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE: T29 Host (1 of 2)

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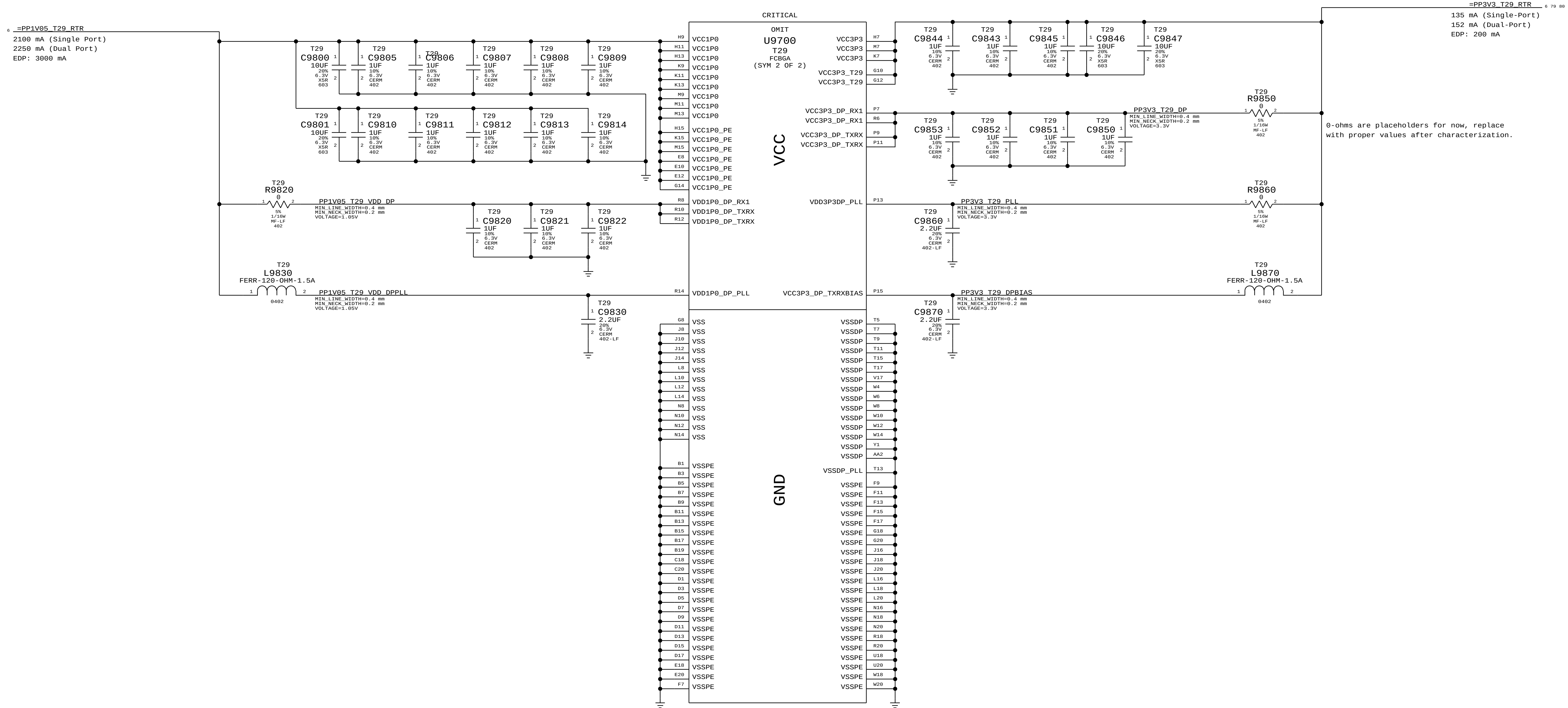

Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

T29 Host (2 of 2)

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K60/62 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,BOTTOM	NO_TYPE,BGA,BGA_P1MM	MM	15.5.1

PHYSICAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	=50_OHM_SE	100 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
34_OHM_SE	TOP,BOTTOM	Y	0.21 MM	0.090 MM	=STANDARD		
34_OHM_SE	*	Y	0.19 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
39_OHM_SE	TOP,BOTTOM	Y	0.175 MM	0.090 MM	=STANDARD		
39_OHM_SE	*	Y	0.159 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
42_OHM_SE	TOP,BOTTOM	Y	0.151 MM	0.090 MM	=STANDARD		
42_OHM_SE	*	Y	0.135 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.1 MM	0.085 MM	15 MM		
50_OHM_SE	*	Y	0.1 MM	0.085 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP,BOTTOM	Y	0.085 MM	0.085 MM	=STANDARD		
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
68_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
68_OHM_DIFF	ISL3,ISL6	Y	0.16 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM
68_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.090 MM	=STANDARD	0.13 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL6	Y	0.115 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL6	Y	0.099 MM	0.085 MM	12 MM	0.2 MM	0.1 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.110 MM	0.085 MM	=STANDARD	0.2 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3,ISL6	Y	0.081 MM	0.081 MM	=STANDARD	0.25 MM	0.1 MM
100_OHM_DIFF	TOP,BOTTOM	Y	0.091 MM	0.085 MM	=STANDARD	0.25 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3,ISL6	Y	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	TOP,BOTTOM	Y	0.075 MM	0.075 MM	=STANDARD	0.320 MM	0.15 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM
POWER_WIDTH	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_CTL	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING RULE SET

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
3.5:1_SPACING	*	0.35 MM	?
4:1_SPACING	*	0.4 MM	?
5:1_SPACING	*	0.5 MM	?
6:1_SPACING	*	0.6 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	*	0.155 MM	?
2X_DIELECTRIC	TOP,BOTTOM	0.145 MM	?
3X_DIELECTRIC	*	0.230 MM	?
3X_DIELECTRIC	TOP,BOTTOM	0.215 MM	?
4X_DIELECTRIC	*	0.305 MM	?
4X_DIELECTRIC	TOP,BOTTOM	0.285 MM	?
5X_DIELECTRIC	*	0.380 MM	?
5X_DIELECTRIC	TOP,BOTTOM	0.355 MM	?
7X_DIELECTRIC	*	0.532 MM	?
7X_DIELECTRIC	TOP,BOTTOM	0.497 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	*	*	STANDARD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?
GND_P2MM	*	0.2 MM	1000
PWR_P2MM	*	0.2 MM	1000
SWITCHNODE	*	0.8 MM	1000

CONSTRAINTS FOR BGA AREA

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
BGA_P1MM	*	=DEFAULT	?
BGA_P1P5MM	*	0.15 MM	?
BGA_P2MM	*	0.2 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
*	*	BGA	BGA_P1MM
MEM_CLK	*	BGA	BGA_P1P5MM
CLK_PCIE	*	BGA	BGA_P1MM
CLK_LPC	*	BGA	BGA_P1MM
CLK_PCI	*	BGA	BGA_P1MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CONTROL	*	*	SWITCHNODE
VR_CONTROL	VR_CONTROL	*	STANDARD
VR_CONTROL	SWITCHNODE	*	STANDARD
VR_CONTROL	GND	*	STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER	BGA_P1MM	POWER_CTL
POWER	*	POWER_WIDTH
VR_CTL_PHY	BGA_P1MM	DEFAULT
VR_CTL_PHY	*	POWER_CTL

BOARD STACK-UP

TOP	HALF OZ	SIGNAL
	0.071	PREPREG
2	TWO OZ	GND
	0.076	PREPREG
3	ONE OZ	SIGNAL
	0.370	PREPREG
4	TWO OZ	POWER
	0.101	CORE
5	TWO OZ	POWER
	0.370	PREPREG
6	ONE OZ	SIGNAL
	0.076	PREPREG
7	TWO OZ	GND
	0.071	PREPREG
BOTTOM	HALF OZ	SIGNAL

BOARD THICKNESS = 62 MIL (1.5748 mm)

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
K60/K62 RULE DEFINITIONS

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
MEM_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
MEM_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
MEM_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF
MEM_42S_D	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2.5:1_SPACING	?
MEM_CTRL2MEM	*	=3:1_SPACING	?
MEM_CMD2CMD	*	=2:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DQ_SAMEBYTE	*	=3:1_SPACING	?
MEM_DQ_DIFFBYTE	*	=5:1_SPACING	?
MEM_DATA2MEM	*	=4:1_SPACING	?
MEM_DQS2MEM	*	=4:1_SPACING	?
MEM_2OTHER	*	=5:1_SPACING	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE0	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE1	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE2	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE3	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE4	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE5	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE6	*	MEM_DQS2MEM
MEM_DQS	MEM_DQ_BYTE7	*	MEM_DQS2MEM
MEM_DQS	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE5	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE5	MEM_DQ_BYTE5	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE5	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE6	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE6	MEM_DQ_BYTE6	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE6	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE6	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE7	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE7	MEM_DQ_BYTE7	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE7	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE0	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE0	MEM_DQ_BYTE0	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE1	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE0	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE1	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE1	MEM_DQ_BYTE1	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE2	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE1	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE2	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE2	MEM_DQ_BYTE2	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE3	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE2	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE3	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE3	MEM_DQ_BYTE3	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE4	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE3	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQ_BYTE4	MEM_CTRL	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_CMD	*	MEM_DATA2MEM
MEM_DQ_BYTE4	MEM_DQ_BYTE4	*	MEM_DQ_SAMEBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE5	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE6	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	MEM_DQ_BYTE7	*	MEM_DQ_DIFFBYTE
MEM_DQ_BYTE4	*	*	MEM_2OTHER

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_68D	MEM_CLK	MEM_A_CLK_P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM_A_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_A_CKE<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_CS_L<3..0>	12 30
	MEM_39S	MEM_CTRL	MEM_A_ODT<3..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_A<15..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_BA<2..0>	12 30
	MEM_34S	MEM_CMD	MEM_A_RAS_L	12 30
	MEM_34S	MEM_CMD	MEM_A_CAS_L	12 30
	MEM_34S	MEM_CMD	MEM_A_WE_L	12 30
	MEM_42S	MEM_DQ_BYTE0	MEM_A_DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM_A_DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM_A_DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM_A_DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM_A_DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM_A_DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM_A_DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM_A_DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM_A_DQS_N<7>	12 32
I177	MEM_50S	PM	MEM_RESET_L	30 31 32 97

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	MEM_68D	MEM_CLK	MEM_B_CLK_P<3..0>	12 32
	MEM_68D	MEM_CLK	MEM_B_CLK_N<3..0>	12 32
	MEM_39S	MEM_CTRL	MEM_B_CKE<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_CS_L<3..0>	12 31
	MEM_39S	MEM_CTRL	MEM_B_ODT<3..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_A<15..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_BA<2..0>	12 31
	MEM_34S	MEM_CMD	MEM_B_RAS_L	12 31
	MEM_34S	MEM_CMD	MEM_B_CAS_L	12 31
	MEM_34S	MEM_CMD	MEM_B_WE_L	12 31
	MEM_42S	MEM_DQ_BYTE0	MEM_B_DQ<7..0>	12 32
	MEM_42S	MEM_DQ_BYTE1	MEM_B_DQ<15..8>	12 32
	MEM_42S	MEM_DQ_BYTE2	MEM_B_DQ<23..16>	12 32
	MEM_42S	MEM_DQ_BYTE3	MEM_B_DQ<31..24>	12 32
	MEM_42S	MEM_DQ_BYTE4	MEM_B_DQ<39..32>	12 32
	MEM_42S	MEM_DQ_BYTE5	MEM_B_DQ<47..40>	12 32
	MEM_42S	MEM_DQ_BYTE6	MEM_B_DQ<55..48>	12 32
	MEM_42S	MEM_DQ_BYTE7	MEM_B_DQ<63..56>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<0>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<1>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<2>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<3>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<4>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<5>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<6>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_P<7>	12 32
	MEM_42S_D	MEM_DQS	MEM_B_DQS_N<7>	12 32

MEMORY MISC PROPERTIES

VOLTAGE	NET_TYPE PHYSICAL	NET_TYPE SPACING	VOLTAGE	NET	PAGES
	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_A	
	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_B	
I162	MEM_POWER_PHY	MEM_POWER		CPU_DDR_VREF	11 28
I168	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_DACOUT	28
I169	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_OPFB	28
I170	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMA_DQ	28
I174	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_A_SW	
I171	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_DACOUT	28
I173	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_OPFB	28
I172	MEM_POWER_PHY	MEM_POWER		VREFMARGIN_DIMMB_DQ	28
I175	MEM_POWER_PHY	MEM_POWER		CPU_DIMM_VREF_B_SW	

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_POWER_WIDTH	*	Y	0.500 MM	0.250 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_POWER_PHY	*	MEM_POWER_WIDTH

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_POWER	*	=3:1_SPACING	?

PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	*	=4X_DIELECTRIC	?
CLK_PCIE	*	0.5 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE	TOP,BOTTOM	=4:1_SPACING	?

CPU

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_RCOMP_PHY	*	Y	0.254 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ITP	*	0.2 MM	?
CPU_RCOMP	*	0.2 MM	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	*	=6:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA	TOP,BOTTOM	=6:1_SPACING	?

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
SATA				
I300	SATA_90D	SATA	SATA_SSD_R2D_C_P	18 42
I299	SATA_90D	SATA	SATA_SSD_R2D_C_N	18 42
I301	SATA_90D	SATA	SATA_SSD_R2D_P	42
I302	SATA_90D	SATA	SATA_SSD_R2D_N	42
I303	SATA_90D	SATA	SATA_SSD_D2R_P	18 42
I304	SATA_90D	SATA	SATA_SSD_D2R_N	18 42
I305	SATA_90D	SATA	SATA_SSD_D2R_C_P	42
I306	SATA_90D	SATA	SATA_SSD_D2R_C_N	42
PCIE				
I319	PCIE_85D	PCIE	PCIE_USB3_1_R2D_P	
I320	PCIE_85D	PCIE	PCIE_USB3_1_R2D_N	
I321	PCIE_85D	PCIE	PCIE_USB3_1_R2D_C_P	
I322	PCIE_85D	PCIE	PCIE_USB3_1_R2D_C_N	
I323	PCIE_85D	PCIE	PCIE_USB3_1_D2R_P	
I324	PCIE_85D	PCIE	PCIE_USB3_1_D2R_N	
I325	PCIE_85D	PCIE	PCIE_USB3_1_D2R_C_P	
I326	PCIE_85D	PCIE	PCIE_USB3_1_D2R_C_N	
I327	PCIE_85D	PCIE	PCIE_USB3_2_R2D_P	
I328	PCIE_85D	PCIE	PCIE_USB3_2_R2D_N	
I330	PCIE_85D	PCIE	PCIE_USB3_2_R2D_C_P	
I329	PCIE_85D	PCIE	PCIE_USB3_2_R2D_C_N	
I332	PCIE_85D	PCIE	PCIE_USB3_2_D2R_P	
I331	PCIE_85D	PCIE	PCIE_USB3_2_D2R_N	
I333	PCIE_85D	PCIE	PCIE_USB3_2_D2R_C_P	
I334	PCIE_85D	PCIE	PCIE_USB3_2_D2R_C_N	
CPU ITP				
	CPU_50S	CPU_ITP	XDP_BPM_L<7..0>	11 25
	CPU_50S	CPU_ITP	CPU_CFG<17..0>	10 15 25
I344	CPU_50S	CPU_ITP	XDP_OBSDATA_B<3..0>	25
I343	CPU_50S	CPU_ITP	XDP_CPU_CFG<0>	25
I345	CPU_50S	CPU_ITP	XDP_CPU_TDO	11 25
I347	CPU_50S	CPU_ITP	XDP_CPU_TDI	11 25
I348	CPU_50S	CPU_ITP	XDP_CPU_TMS	11 25
I349	CPU_50S	CPU_ITP	XDP_CPU_TCK	11 25
I350	CPU_50S	CPU_ITP	XDP_CPU_TRST_L	11 25
CPU_MISC				
	CPU_RCOMP_PHY	CPU_RCOMP	CPU_PEG_COMP	10

ANY OTHER LYNNFIELD CONSTRAINTS NOT COVERED ON PAGES 101 AND 107 SHOULD GO ON THIS PAGE TOO

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
PCIE GRAPHICS				
	PCIE_85D	PCIE	PEG_R2D_C_P<15..0>	9 77
	PCIE_85D	PCIE	PEG_R2D_C_N<15..0>	9 77
	PCIE_85D	PCIE	PEG_D2R_P<15..0>	9 77
	PCIE_85D	PCIE	PEG_D2R_N<15..0>	9 77
	PCIE_85D	PCIE	MXM_PCIE_R2D_P<7..0>	75 77
	PCIE_85D	PCIE	MXM_PCIE_R2D_N<7..0>	75 77
	PCIE_85D	PCIE	MXM_PCIE_D2R_P<7..0>	75 77
	PCIE_85D	PCIE	MXM_PCIE_D2R_N<7..0>	75 77
PCIE I/O				
	PCIE_85D	PCIE	PCIE_MINI_R2D_P	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_N	33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_R2D_C_N	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_P	18 33
	PCIE_85D	PCIE	PCIE_MINI_D2R_N	18 33
	PCIE_85D	PCIE	PCIE_FW_R2D_P	39
	PCIE_85D	PCIE	PCIE_FW_R2D_N	39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_P	18 39
	PCIE_85D	PCIE	PCIE_FW_R2D_C_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_P	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_N	18 39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_P	39
	PCIE_85D	PCIE	PCIE_FW_D2R_C_N	39
DMI				
I315	CLK_PCIE_90D	CLK_PCIE	DMI_MIDBUS_CLK100M_N	
I318	CLK_PCIE_90D	CLK_PCIE	DMI_MIDBUS_CLK100M_P	
I316	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_N	11 18
I317	CLK_PCIE_90D	CLK_PCIE	DMI_CLK100M_CPU_P	11 18
	PCIE_85D	PCIE	DMI_S2N_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_S2N_N<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_P<3..0>	10 19
	PCIE_85D	PCIE	DMI_N2S_N<3..0>	10 19
PCIE REF CLOCKS				
	CLK_PCIE_90D	CLK_PCIE	GPU_CLK100M_PCIE_P	9
	CLK_PCIE_90D	CLK_PCIE	GPU_CLK100M_PCIE_N	9
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_MINI_P	18 33
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_MINI_N	18 33
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_P	18 39
	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_FW_N	18 39
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_P	18 37
	ENET_100D	ENET_MII	PCIE_CLK100M_ENET_N	18 37
SATA				
	SATA_90D	SATA	SATA_HDD_R2D_C_P	18 42
	SATA_90D	SATA	SATA_HDD_R2D_C_N	18 42
	SATA_90D	SATA	SATA_HDD_R2D_P	42
	SATA_90D	SATA	SATA_HDD_R2D_N	42
	SATA_90D	SATA	SATA_HDD_D2R_P	18 42
	SATA_90D	SATA	SATA_HDD_D2R_N	18 42
	SATA_90D	SATA	SATA_HDD_D2R_C_P	42
	SATA_90D	SATA	SATA_HDD_D2R_C_N	42
	SATA_90D	SATA	SATA_ODD_R2D_C_P	18 42
	SATA_90D	SATA	SATA_ODD_R2D_C_N	18 42
	SATA_90D	SATA	SATA_ODD_R2D_P	42
	SATA_90D	SATA	SATA_ODD_R2D_N	42
	SATA_90D	SATA	SATA_ODD_D2R_P	18 42
	SATA_90D	SATA	SATA_ODD_D2R_N	18 42
	SATA_90D	SATA	SATA_ODD_D2R_C_P	42
	SATA_90D	SATA	SATA_ODD_D2R_C_N	42
CLOCKS				
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_DMI_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_DMI_N	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK96M_DOT_N	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_P	18 26
	CLK_PCIE_90D	CLK_PCIE	PCH_CLK100M_SATA_N	18 26
	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_N	18 25
	CLK_PCIE_90D	CLK_PCIE	ITPXDP_CLK100M_P	18 25
	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_N	11 18
	CLK_PCIE_90D	CLK_PCIE	ITPCPU_CLK100M_P	11 18
I337	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_P	25
I338	CLK_PCIE_90D	CLK_PCIE	XDP_CPU_CLK100M_N	25
UNUSED CLOCKS				
I335	CLK_PCIE_90D	CLK_PCIE	TP_CLK133M_PCH_N	26
I336	CLK_PCIE_90D	CLK_PCIE	TP_CLK133M_PCH_P	26
UNUSED PCIE				
I339	PCIE_85D	PCIE	MXM_PCIE_R2D_P<8..15>	75 77
I340	PCIE_85D	PCIE	MXM_PCIE_R2D_N<8..15>	75 77
I342	PCIE_85D	PCIE	MXM_PCIE_D2R_P<8..15>	75 77
I341	PCIE_85D	PCIE	MXM_PCIE_D2R_N<8..15>	75 77

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

PCIE/DMI/FDI/SATA CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

BRANCH

PCH CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	0.2 MM	?
ITP_PCH	*	0.2 MM	?

PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	0.2 MM	?

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	0.15 MM	?
CLK_LPC	*	0.2 MM	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	0.2 MM	?

XTAL Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_XTAL	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

NET_TYPE

	PHYSICAL	SPACING		
I329		PM	T29_CLKREQ_L	15 21 80
I330		PM	FW_MINI_CLKREQ_L	15 18
I331		PM	BLC_GPIO	6 15 21
I332		PM	T29_SW_RESET_L	15 21 80
I334		PM	ENET_CLKREQ_L	15 18 36
I337		PM	DP_GPU_T29_SEL	18 61 83
I338		PM	T29_MCU_INT_L	20 84
I339		PM	T29_DP_PORTA_PWR_EN	20 25 82 97
I340		PM	T29_DP_PORTB_PWR_EN	20 25
I341		PM	DP_AUXCH_ISOL	15 18 25 84
I348		PM	PLT_RST_BUF_L	27
I349		PM	XDPCPU_PLTRST_L	25 27
I352		PM	PCH_PEG_CLKREQ_L	21
I357		PM	ENET_SW_RESET_L	15 21 36
I354		PM	CPU_SKTOCC	63
I355		PM	PM_EN_USB_PWR	43 63

NET_TYPE

	PHYSICAL	SPACING		
I372		PM	ENET_RESET_LOGIC_L	36
I373		PM	ENET_RESET_FET_L	
I375		PM	ENET_CLKREQ_FET_L	36 37
I374		PM	PGOOD_5V_1V05_3V3	64 97
I376		PM	PGOOD_CPU_UNCORE	64 97
I377		PM	ALL_SYS_PWRGD	64 97
I378		PM	PGOOD_3V3_1V05	64 97
I379		PM	PGOOD_PCH_S0_R	64 97
I380		PM	AUD_IPHS_SWITCH_EN_PCH	21 25

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
	PCI_55S	PCI	PCI_REQ0_L	20
	PCI_55S	PCI	PCI_REQ1_L	20
I381	PCI_55S	PCI	PCI_REQ2_L	20
I382	PCI_55S	PCI	PCI_REQ3_L	20
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIOUT	20 27
	CLK_PCI_55S	CLK_PCI	PCH_CLK33M_PCIIN	18 27
	LPC_55S	LPC	LPC_AD<3..0>	18 46 48
	LPC_55S	LPC	LPC_FRAME_L	18 46 48
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R	20 27
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC	27 46
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS	27 48
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK_R	9 19 97
	CLK_LPC_55S	PM	PM_CLK32K_SUSCLK	9 46 97
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS_R	20 27
I325	LPC_55S	LPC	LPC_R_AD<3..0>	18
I326	LPC_55S	LPC	LPC_FRAME_R_L	18
I335	SPI_55S	SPI	SPI_CLK_1_R	18
I336	SPI_55S	SPI	SPI_MOSI_1_R	18
I350	CLK_XTAL	XTAL	USB_HUB2_XTAL1	35
I351	CLK_XTAL	XTAL	USB_HUB2_XTAL2	35
I344	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1_R	27
I345	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R	27
I342	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 91
I343	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 91
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1	18 27 91
	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2	18 27 91
	CLK_XTAL	XTAL	CK505_XTAL_IN	26
	CLK_XTAL	XTAL	CK505_XTAL_OUT	26
I324	CLK_XTAL	XTAL	CK505_XTAL_OUT_R	26
	CLK_PCH_55S	CLK_PCH	PCH_CLK14P3M_REFCLK	18 26

NET_TYPE

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING		
	SPI_55S	SPI	SPI_CLK_R	18 48 55
	SPI_55S	SPI	SPI_CLK	55
	SPI_55S	SPI	SPI_MOSI_R	18 48 55
	SPI_55S	SPI	SPI_MOSI	55
	SPI_55S	SPI	SPI_MISO	18 48 55
	SPI_55S	SPI	SPI_MISO_R	55
	SPI_55S	SPI	SPI_CS0_R_L	18 48
	SPI_55S	SPI	SPI_CS0_L	48
	SPI_55S	SPI	SPI_MLB_CS_L	48 55
	SPI_55S	SPI	SPI_ALT_CS_L	48
	SPI_55S	SPI	SPIROM_USE_MLB	21 48
	SPI_55S	SPI	SPI_ALT_MOSI	48
	SPI_55S	SPI	SPI_ALT_MISO	48
	SPI_55S	SPI	SPI_ALT_CLK	48
	HDA_55S	HDA	HDA_BIT_CLK	18 56
	HDA_55S	HDA	HDA_BIT_CLK_R	18
	HDA_55S	HDA	HDA_RST_L	18 56
	HDA_55S	HDA	HDA_RST_R_L	18
	HDA_55S	HDA	HDA_SDOUT	15 18 56
	HDA_55S	HDA	HDA_SDOUT_R	18
	HDA_55S	HDA	HDA_SYNC	18 56
	HDA_55S	HDA	HDA_SYNC_R	18
	HDA_55S	HDA	HDA_SDIN0	18 56
	HDA_55S	HDA	AUD_SDI_R	56
		PM	AUD_SPDIF_IN	60 83 97
		HDA	AUD_SPDIF_OUT	56 60
		HDA	AUD_SPDIF_CHIP	56
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_NOUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO1L_POUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_NOUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO1R_POUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_NOUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO2L_POUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_NOUT	98
	HDA_55S	HDA	AUD_SPKR_OUTLO2R_POUT	98
I346	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT_R	27
I347	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN_R	27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALOUT	18 27
	CLK_XTAL	XTAL	PCH_CLK25M_XTALIN	18 27 79
	PCH_55S	COMP_PCH	PCH_USB_RBIAS	20
	PCH_55S	COMP_PCH	PCH_SATA3COMP	18
	PCH_55S	COMP_PCH	PCH_XCLK_RCOMP	18
	PCH_55S	COMP_PCH	PCH_DMI_COMP	19
I322	PCH_55S	COMP_PCH	PCH_SATAICOMP	18
	CLK_XTAL	XTAL	USB_HUB1_XTAL1	34
	CLK_XTAL	XTAL	USB_HUB1_XTAL2	34
	PCH_55S	COMP_PCH	USB_HUB1_RBIAS	34
	PCH_55S	ITP_PCH	XDP_PCH_TCK	18 25
	PCH_55S	ITP_PCH	XDP_PCH_TMS	18 25
	PCH_55S	ITP_PCH	XDP_PCH_TDI	18 25
I323	PCH_55S	ITP_PCH	XDP_PCH_TDO	18 25
I327	PCH_55S	COMP_PCH	PCH_DMI2RBIAS	19
I328	PCH_55S	COMP_PCH	PCH_SATA3RBIAS	18
I356	PCH_55S	COMP_PCH	USB_HUB2_RBIAS	35

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE

IBEX PEAK CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

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CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MII	*	0.3 MM	?
ENET_SE	*	=STANDARD	?

SOURCE:BROADCOM 5764M-DS04-RDS. PAGE 38

CAESAR II (ETHERNET) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF	*	0.6 MM	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_2OTHER	*	=50MIL_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
50MIL_SPACING	*	1.27 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_DIFF_T	*	*	ENET_2OTHER

CAESAR IV (SD) CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD	*	=3:1_SPACING	?

FireWire Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FW_110D	*	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF	=110_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FW_TP	*	=3:1_SPACING	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=3:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=4x_DIELECTRIC	?

NET_TYPE (Ethernet / SD)

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
	ENET_50S	ENET_SE	ENET_RDAC	37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALI	36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALO	36 37
	CLK_PCH_55S	XTAL	ENET_CLK25M_XTALO_R	36
	ENET_100D	ENET_DIFF	ENETCONN_MDI_P<3..0>	37 38
	ENET_100D	ENET_DIFF	ENETCONN_MDI_N<3..0>	37 38
	ENET_100D	ENET_DIFF_T	ENETCONN_MDI_T_P<3..0>	38
	ENET_100D	ENET_DIFF_T	ENETCONN_MDI_T_N<3..0>	38
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_P	37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_N	37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_P	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_N	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_C_P	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_R2D_C_N	18 37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_C_P	37
	PCIE_85D	ENET_MII	PCIE_ENET_D2R_C_N	37
	SD_50S	SD	ENET_SD_CMD	37
	SD_50S	SD	SDCONN_CMD	37 45
	SD_50S	SD	SDCONN_CLK	37 45
	SD_50S	SD	ENET_SD_CLK	37
	SD_50S	SD	SDCONN_DATA<7..0>	37 45
	SD_50S	SD	ENET_CR_DATA<7..0>	37
	SD_50S	PM	ENET_MEDIA_SENSE	15 18 37
	SD_50S	SD	ENET_MEDIA_SENSE_R	
	SD_50S	SD	ENET_SD_DETECT_L	37
	SD_50S	PM	SDCONN_DETECT_BUF_L	97

NET_TYPE (USB)

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
I252	USB_90D	USB	USB_EXTA_P	34 43
I253	USB_90D	USB	USB_EXTA_N	34 43
I254	USB_90D	USB	USB_PORT0_P	43
I256	USB_90D	USB	USB_PORT0_N	43
I255	USB_90D	USB	USB_EXTB_P	35 43
I257	USB_90D	USB	USB_EXTB_N	35 43
I258	USB_90D	USB	USB_PORT1_P	43
I259	USB_90D	USB	USB_PORT1_N	43
I260	USB_90D	USB	USB_EXTC_P	34 43
I261	USB_90D	USB	USB_EXTC_N	34 43
I262	USB_90D	USB	USB_PORT2_P	43
I264	USB_90D	USB	USB_PORT2_N	43
I263	USB_90D	USB	USB_EXTD_P	35 43
I265	USB_90D	USB	USB_EXTD_N	35 43
I266	USB_90D	USB	USB_D_MUXED_P	43
I267	USB_90D	USB	USB_D_MUXED_N	43
I268	USB_90D	USB	USB_PORT3_P	43
I269	USB_90D	USB	USB_PORT3_N	43
I270	USB_90D	USB	USB_CAMERA_P	20 44
I271	USB_90D	USB	USB_CAMERA_N	20 44
I272	USB_90D	USB	USB_CAMERA_L_P	44 98
I274	USB_90D	USB	USB_CAMERA_L_N	44 98
I273	USB_90D	USB	USB_BT_P	35 44
I275	USB_90D	USB	USB_BT_N	35 44
I276	USB_90D	USB	USB_BT_L_P	44 98
I277	USB_90D	USB	USB_BT_L_N	44 98
I278	USB_90D	USB	USB_IR_P	34 44
I279	USB_90D	USB	USB_IR_N	34 44
I280	USB_90D	USB	USB_IR_L_P	44 98
I281	USB_90D	USB	USB_IR_L_N	44 98
I286	USB_90D	USB	USB_SDCARD_P	34 44
I287	USB_90D	USB	USB_SDCARD_N	34 44
I288	USB_90D	USB	USB_SDCARD_L_P	44
I289	USB_90D	USB	USB_SDCARD_L_N	44
I282	USB_90D	USB	USB_HUB1_UP_P	20 34
I283	USB_90D	USB	USB_HUB1_UP_N	20 34
I284	USB_90D	USB	USB_HUB2_UP_P	20 35
I285	USB_90D	USB	USB_HUB2_UP_N	20 35
I290	USB_90D	USB	USB_HUB2UNUSED_P	35
I291	USB_90D	USB	USB_HUB2UNUSED_N	35

FireWire Net Properties

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
I251	CLK_PCH_55S	XTAL	FW_CLK24P576M_XO	39
I249	CLK_PCH_55S	XTAL	FW_CLK24P576M_XO_R	39
I250	CLK_PCH_55S	XTAL	FW_CLK24P576M_XI	39
	FW_110D	FW_TP	FW_PORT0_TPA_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPA_N	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_P	40 41
	FW_110D	FW_TP	FW_PORT0_TPB_N	40 41

UNUSED FW NETS PHYSICAL PROPERTIES

ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	NET	PAGES
I243	FW_110D	FW_TP	FW_P1_TPA_P	39 40
I245	FW_110D	FW_TP	FW_P1_TPA_N	39 40
I244	FW_110D	FW_TP	FW_P2_TPA_P	39 40
I246	FW_110D	FW_TP	FW_P2_TPA_N	39 40
	FW_110D	FW_TP	FW_P1_TPB_P	39 40
	FW_110D	FW_TP	FW_P1_TPB_N	39 40
	FW_110D	FW_TP	FW_P2_TPB_P	39 40
	FW_110D	FW_TP	FW_P2_TPB_N	39 40

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

PAGE TITLE
USB/ENET/SD/FW/AUD CONSTRAINTS

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

BRANCH

GRAPHICS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

USE 5X_DIELECTRIC IN K62

PAIRS SHOULD BE WITHIN 100 MILS OF CLOCK LENGTH.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DIsplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.

ELECTRICAL_CONSTRAINT_SET
ASSINGED IN CONT. MGR. NET_TYPE

	PHYSICAL	SPACING		
I169	DP_85D	DISPLAYPORT	DP_INTCONN_ML_C_P<3..0>	83
I170	DP_85D	DISPLAYPORT	DP_INTCONN_ML_C_N<3..0>	83
	DP_85D	DISPLAYPORT	DP_INTCONN_AUXCH_C_P	83
	DP_85D	DISPLAYPORT	DP_INTCONN_AUXCH_C_N	83
I153	DP_85D	DISPLAYPORT	DP_INTPNL_ML_P<3..0>	81 83
I154	DP_85D	DISPLAYPORT	DP_INTPNL_ML_N<3..0>	81 83
	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_P	81 83
	DP_85D	DISPLAYPORT	DP_INTPNL_AUX_N	81 83
I185	DP_85D	DISPLAYPORT	DP_EXTA_ML_P<3..0>	84
I186	DP_85D	DISPLAYPORT	DP_EXTA_ML_N<3..0>	84
I188	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_P	84
I187	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_N	84
I171	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_P<3..0>	78 84
I174	DP_85D	DISPLAYPORT	DP_EXTA_ML_C_N<3..0>	78 84
I172	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_P	78 84
I173	DP_85D	DISPLAYPORT	DP_EXTA_AUXCH_C_N	78 84
I191	DP_85D	DISPLAYPORT	DP_EXTB_ML_P<3..0>	
I192	DP_85D	DISPLAYPORT	DP_EXTB_ML_N<3..0>	
I193	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_P	
I194	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_N	
I195	DP_85D	DISPLAYPORT	DP_EXTB_ML_C_P<3..0>	
I196	DP_85D	DISPLAYPORT	DP_EXTB_ML_C_N<3..0>	
I198	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_C_P	
I197	DP_85D	DISPLAYPORT	DP_EXTB_AUXCH_C_N	
I199	DP_85D	DISPLAYPORT	MXM_DP_B_ML_P<3..0>	75 78
I200	DP_85D	DISPLAYPORT	MXM_DP_B_ML_N<3..0>	75 78
I201	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_P	75 78
I202	DP_85D	DISPLAYPORT	MXM_DP_B_AUX_N	75 78
I203	DP_85D	DISPLAYPORT	MXM_DP_C_ML_P<3..0>	75 83
I204	DP_85D	DISPLAYPORT	MXM_DP_C_ML_N<3..0>	75 83
I205	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_P	75 83
I206	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_N	75 83
I211	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_R_P	83
I212	DP_85D	DISPLAYPORT	MXM_DP_C_AUX_R_N	83
I207	DP_85D	DISPLAYPORT	MXM_DP_D_ML_P<3..0>	75 78
I208	DP_85D	DISPLAYPORT	MXM_DP_D_ML_N<3..0>	75 78
I209	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_P	75 78
I210	DP_85D	DISPLAYPORT	MXM_DP_D_AUX_N	75 78

UNUSED VIDEO NET PHYSICAL CONSTRAINTS

DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_P	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_A_CLK_N	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_P	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_B_CLK_N	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_P<3..0>	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_A_DATA_N<3..0>	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_P<3..0>	76 78
DP_85D	DISPLAYPORT	MXM_LVDS_B_DATA_N<3..0>	76 78

SYNC_MASTER=K62 SYNC_DATE=01/06/2011
PAGE TITLE
GRAPHICS CONSTRAINTS

Apple Inc.

DRAWING NUMBER	051-8115
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SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
THERMAL	*	*	4:1_SPACING
THERMAL	POWER	*	PWR_P2MM
THERMAL	GND	*	GND_P2MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
THERM_DIFF	*	1:1_DIFFPAIR
SNS_DIFF	*	1:1_DIFFPAIR

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	SMB_55S	SMB	SMBUS_SMC_A_S3_SCL	49
	SMB_55S	SMB	SMBUS_SMC_A_S3_SDA	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_B_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SCL	49
	SMB_55S	SMB	SMBUS_SMC_0_S0_SDA	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SCL	49
	SMB_55S	SMB	SMBUS_SMC_BSA_SDA	49
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 94
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 94
	SMB_55S	SMB	SMBUS_SMC_MGMT_SCL	49 94
	SMB_55S	SMB	SMBUS_SMC_MGMT_SDA	49 94
	SMB_55S	SMB	SMBUS_PCH_CLK	18 49
	SMB_55S	SMB	SMBUS_PCH_DATA	18 49
	SMB_55S	SMB	SML_PCH_0_CLK	18 49
	SMB_55S	SMB	SML_PCH_0_DATA	18 49
	SMB_55S	SMB	SML_PCH_1_CLK	18 49
	SMB_55S	SMB	SML_PCH_1_DATA	18 49
	CLK_XTAL	XTAL	SMC_EXTAL	46 47
	CLK_XTAL	XTAL	SMC_XTAL	46 47
I254	SMB_55S	SMB	I2C_VREFMRGN_DIMMA_SCL	28
I253	SMB_55S	SMB	I2C_VREFMRGN_DIMMA_SDA	28
I255	SMB_55S	SMB	I2C_VREFMRGN_DIMMB_SCL	28
I256	SMB_55S	SMB	I2C_VREFMRGN_DIMMB_SDA	28
I265	SMB_55S	SMB	SMB_BLC_TCON_SCL	6 49 81
I264	SMB_55S	SMB	SMB_BLC_TCON_SDA	6 49 81
I268	SMB_55S	SMB	I2C_TCON_SCL	81
I269	SMB_55S	SMB	I2C_TCON_SDA	81
I266	SMB_55S	SMB	SMB_BLC_PCH_SCL_R	6
I267	SMB_55S	SMB	SMB_BLC_PCH_SDA_R	6

SMC THERMAL NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	THERM_DIFF	THERMAL	SNS_T1_1_P	52
	THERM_DIFF	THERMAL	SNS_T1_1_N	52
	THERM_DIFF	THERMAL	SNS_T2_DP2	52
	THERM_DIFF	THERMAL	SNS_T2_DN2	52
I206	THERM_DIFF	THERMAL	SNS_T1_2_P	52
I208	THERM_DIFF	THERMAL	SNS_T1_2_N	52
I207	THERM_DIFF	THERMAL	SNS_T1_3_P	52
I209	THERM_DIFF	THERMAL	SNS_T1_3_N	52
I210	THERM_DIFF	THERMAL	SNS_T1_4_P	52
I212	THERM_DIFF	THERMAL	SNS_T1_4_N	52
I211	THERM_DIFF	THERMAL	SNS_T1_5_P	52
I213	THERM_DIFF	THERMAL	SNS_T1_5_N	52
I214	THERM_DIFF	THERMAL	SNS_T1_6_P	52
I216	THERM_DIFF	THERMAL	SNS_T1_6_N	52
I215	THERM_DIFF	THERMAL	SNS_T1_7_P	52
I217	THERM_DIFF	THERMAL	SNS_T1_7_N	52
I221	THERM_DIFF	THERMAL	SNS_CPU_THERMD_P	10 52
I220	THERM_DIFF	THERMAL	SNS_CPU_THERMD_N	10 52
I219	THERM_DIFF	THERMAL	SNS_LCD_H_P	52
I218	THERM_DIFF	THERMAL	SNS_LCD_H_N	52
	THERM_DIFF	THERMAL	SNS_ODD_P	52 98
	THERM_DIFF	THERMAL	SNS_ODD_N	52 98
	THERM_DIFF	THERMAL	SNS_CPU_H_P	52
	THERM_DIFF	THERMAL	SNS_CPU_H_N	52
I222	THERM_DIFF	THERMAL	SNS_SKIN_RIGHT_P	52 98
I223	THERM_DIFF	THERMAL	SNS_SKIN_RIGHT_N	52 98
	THERM_DIFF	THERMAL	SNS_SKIN_LEFT_P	44 52 98
	THERM_DIFF	THERMAL	SNS_SKIN_LEFT_N	44 52 98
	THERM_DIFF	THERMAL	SNS_AMB_P	52 54 98
	THERM_DIFF	THERMAL	SNS_AMB_N	52 54 98
	THERM_DIFF	THERMAL	SNS_MXM_P	52
	THERM_DIFF	THERMAL	SNS_MXM_N	52
		THERMAL	HDD_OOB_TEMP_FILT	42 51 98
		THERMAL	HDD_OOB_TEMP_FB	42
		THERMAL	HDD_OOB_TEMP_R	51
		THERMAL	SMC_HDD_OOB_TEMP	46 51

SMC VOLTAGE/CURRENT NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	THERM_DIFF	THERMAL	SNS_I_MXM_P	50
	THERM_DIFF	THERMAL	SNS_I_MXM_N	50
	THERM_DIFF	THERMAL	SNS_DIMM_1V5_P	50
	THERM_DIFF	THERMAL	SNS_DIMM_1V5_N	50
	SNS_DIFF	THERMAL	VR_ISNS_VCORE_P	50 95
	SNS_DIFF	THERMAL	VR_ISNS_VCORE_N	50 95
	SNS_DIFF	THERMAL	VR_ISNS_VAXG_P	50 95
	SNS_DIFF	THERMAL	VR_ISNS_VAXG_N	50 95
I243	SNS_DIFF	THERMAL	VR_ISNS_1V05_P	95
I244	SNS_DIFF	THERMAL	VR_ISNS_1V05_N	95
I245	THERM_DIFF	THERMAL	SNS_CPU_1V5_P	50
I246	THERM_DIFF	THERMAL	SNS_CPU_1V5_N	50
I247	THERM_DIFF	THERMAL	SNS_VCCSA_P	50
I248	THERM_DIFF	THERMAL	SNS_VCCSA_N	50
I251	THERM_DIFF	THERMAL	SNS_1V05_PCH_P	50
I252	THERM_DIFF	THERMAL	SNS_1V05_PCH_N	50
I239		THERMAL	GND_SMC_AVSS	46 47 50 94
I241		THERMAL	SMC_CPU_1V5_ISENSE	46 50
I240		THERMAL	SMC_CPU_1V5_ISENSE_R	50
I242		THERMAL	SMC_CPU_1V5_VSENSE	46 50
I229		THERMAL	GND_SMC_AVSS	46 47 50 94
I230		THERMAL	SMC_DIMM_ISENSE	46 50
I228		THERMAL	SMC_DIMM_1V5_R	50
I231		THERMAL	SMC_DIMM_VSENSE	46 50
I225		THERMAL	GND_SMC_AVSS	46 47 50 94
I226		THERMAL	SMC_VCCSA_ISENSE	46 50
I224		THERMAL	SMC_VCCSA_ISENSE_R	50
I227		THERMAL	SMC_VCCSA_VSENSE	46 50
		THERMAL	GND_SMC_AVSS	46 47 50 94
		THERMAL	SMC_PCH_1V05_ISENSE	46 50
		THERMAL	SMC_VAXG_VSENSE	46 50
		THERMAL	SMC_PCH_1V05_VSENSE	46 50
I233		THERMAL	GND_SMC_AVSS	46 47 50 94
I234		THERMAL	SMC_1V05_ISENSE	46 50
I232		THERMAL	SMC_VAXG_ISENSE	46 50
I235		THERMAL	SMC_1V05_VSENSE	46 50
I236		THERMAL	SMC_GPU_ISENSE	46 50
		THERMAL	SMC_GPU_VSENSE	46 50
		THERMAL	SMC_VCORE_ISENSE	46 50
I237		THERMAL	SMC_VCORE_VSENSE	46 50
I238		THERMAL	SMC_CPU_VSENSE	

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PAGE TITLE

SMC Constraints

Apple Inc.

DRAWING NUMBER 051-8115 SIZE D

REVISION 11.1.0

BRANCH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SWITCHNODE	SWITCHNODE	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	BGA_P1MM	BGA_P2MM
SWITCHNODE	GND	BGA_P1MM	BGA_P2MM
SWITCHNODE	*	BGA_P1MM	BGA_P2MM
SWITCHNODE	POWER	*	6:1_SPACING
SWITCHNODE	GND	*	6:1_SPACING
SWITCHNODE	*	*	SWITCHNODE

POWER NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	VOLTAGE		
I128	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE1	66
I129	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE2	66
I130	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE3	66
I777	POWER	SWITCHNODE	1.5V	VR_CPU_PHASE4	
I671	POWER	SWITCHNODE	1.5V	VR_AXG_PHASE1	67
I132	POWER	SWITCHNODE	3.3V	P3V3S5_REG_PHASE	70
I133	POWER	SWITCHNODE	5V	P5VS3_REG_PHASE	70
I134	POWER	SWITCHNODE	1.05V	P1V05S0_PHASE	68
I742	POWER	SWITCHNODE	1.05V	P1V05_REG_PHASE	
I136	POWER	SWITCHNODE	3.4V	P3V42G3H_SW	72
I137	POWER	SWITCHNODE	3.3V	ENET_3V3_S3_SR_IN	37
I666	POWER	SWITCHNODE	1.2V	ENET_SR_LX	36 37
I139	POWER	SWITCHNODE	1.5V	DDR_REG_PHASE	71
I140	POWER	SWITCHNODE	1.8V	P1V8_REG_PHASE	71
I141	POWER	POWER	0.75V	PP0V75_S3_MEM_VREFCA_A	28 30
I142	POWER	POWER	0.75V	PP0V75_S3_MEM_VREFCA_B	28 31
I143	POWER	POWER	0.75V	PP0V75_S3_MEM_VREFDQ_A	28 30
I144	POWER	POWER	0.75V	PP0V75_S3_MEM_VREFDQ_B	28 31
I148	POWER	POWER	12V	PP12V_S0	6
I149	POWER	POWER	12V	PP12V_S0_CPU_FLTRD	65 66 67
I175	POWER	POWER	12V	PP12V_S0_FAN0_L	53 98
I174	POWER	POWER	12V	PP12V_S0_FAN1_L	53 98
I173	POWER	POWER	12V	PP12V_S0_FAN2_L	54 98
I672	POWER	POWER	12V	VR_CPU_DRV1_PVCC	66
I673	POWER	POWER	12V	VR_CPU_DRV1_UVCC	66
I675	POWER	POWER	12V	VR_CPU_DRV2_PVCC	66
I676	POWER	POWER	12V	VR_CPU_DRV2_UVCC	66
I674	POWER	POWER	12V	VR_CPU_DRV3_PVCC	66
I678	POWER	POWER	12V	VR_CPU_DRV3_UVCC	66
I778	POWER	POWER	12V	VR_CPU_DRV4_PVCC	
I779	POWER	POWER	12V	VR_CPU_DRV4_UVCC	
I679	POWER	POWER	12V	VR_AXG_DRV1_PVCC	67
I677	POWER	POWER	12V	VR_AXG_DRV1_UVCC	
I697	POWER	POWER	12V	VR_CPU_DRV1_VCC	66
I698	POWER	POWER	12V	VR_CPU_DRV2_VCC	66
I699	POWER	POWER	12V	VR_CPU_DRV3_VCC	66
I700	POWER	POWER	12V	VR_CPU_DRV4_VCC	
I654	POWER	POWER	12V	PP12V_S0_VG_OK	41
I653	POWER	POWER	12V	P12V_FW_R	41
I651	POWER	POWER	12V	P12V_FW_CL	41
I652	POWER	POWER	12V	P12V_FW_D	41
I267	POWER	POWER	12V	FW_PORT0_VP	41
I270	POWER	POWER	12V	FW_PORT0_VP_F	41
I544	POWER	POWER	12V	PPVP_FW_PHY_CPS	39 41
I680	POWER	POWER	12V	PP12v_S0_P1V05_VREG_VIN	68
I764	POWER	POWER	12V	VCCSA_PWR_RC	69
I774	POWER	POWER	12V	PP12V_S0_BL_R	
I775	POWER	POWER	12V	PP12V_S5_FET	6 74
I787	POWER	POWER	12V	PP12V_LCD	81
I604	POWER	POWER	12V	PP12V_G3H	6
I178	POWER	POWER	12V	PP12V_S5	6
I732	POWER	POWER	12V	PP12V_S5_RSN	74
I268	POWER	POWER	0.75V	PPVTT_S0_DDR	6
I613	POWER	POWER	0.75V	PP0V75_S0	6
I703	POWER	POWER	0.925V	PPVCCSA_S0_CPU	6
I658	POWER	POWER	1.0V	PP1V0_FW_FWPHY_AVDD	39
I657	POWER	POWER	1.0V	PP1V0_S0_FW_VDD	40
I825	POWER	POWER	1.05V	PP1V05_S0_T29	6
I746	POWER	POWER	1.05V	PP1V05_S0_INPUT_VCCSA	6
I717	POWER	POWER	1.05V	PPVAXG_S0	6
I177	POWER	POWER	1.05V	PP1V05_S0	6
I745	POWER	POWER	1.05V	PP1V05_S0_PCH	6
I179	POWER	POWER	1.05V	PP1V05_S0_CK505_F	26
I715	POWER	POWER	1.05V	PPVOUT_S5_PCH_DCPSUS	22
I182	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLA_F	17 22
I185	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCADPLLB_F	17 22
I184	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_EXP_F	22 24
I737	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLLDMI2_F	22 24
I188	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCAPLL_SATA_F	22 24
I736	POWER	POWER	1.05V	PP1V05_S0_PCH_VCCCLKDMI_F	22 24
I753	POWER	POWER	1.1V	PPVCORE_S0_AXG_REG1	67
I752	POWER	POWER	1.1V	PPVCORE_S0_AXG_REGOUT	67
I259	POWER	POWER	1.1V	PPVCORE_S0_CPU	6
I258	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG1	66
I563	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG2	66
I564	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG3	66
I786	POWER	POWER	1.1V	PPVCORE_S0_CPU_REG4	
I659	VR_CTL_PHY	VR_CTL	1.2V	PP1V2_S3_ENET_INTREG	36
I660	POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_AVDDL	37
I661	POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_PCIEPLL	37
I662	POWER	POWER	1.2V	PP1V2_S3_ENET_PHY_GPHYPLL	37
I194	POWER	POWER	1.5V	PP1V5_S0	6
I193	POWER	POWER	1.5V	PP1V5_S0_CK505_F	26
I587	POWER	POWER	1.5V	PP1V5_S0_CK505_R	26
I575	POWER	POWER	1.5V	PP1V5_S0_CPU_MEM	6
I754	POWER	POWER	1.5V	PP1V5_S0_CPU_SNS	
I198	POWER	POWER	1.5V	PP1V5_S3	6
I704	POWER	POWER	1.5V	PP1V5_S3_MEM	6

POWER NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING	VOLTAGE		
I756	POWER	POWER	12V	PP12V_S0_MXM	6
I757	POWER	POWER	12V	PP12V_SLG1	
I807	POWER	POWER	12V	PP3V3R12V_SW_DPAPWR_1	
I806	POWER	POWER	12V	PP3V3R12V_SW_DPBPWR_1	
I797	POWER	POWER	1.8V	USB_HUB2_VDD1V8	35
I798	POWER	POWER	1.8V	USB_HUB2_VDD1V8PLL	35
I743	POWER	POWER	1.8V	PP1V8_S0	6
I744	POWER	POWER	1.8V	PP1V8R1V5_S0_PCH_VCCVRM_F	22 24
I793	POWER	POWER	1.8V	USB_HUB1_VDD1V8	34
I794	POWER	POWER	1.8V	USB_HUB1_VDD1V8PLL	34
I822	POWER	POWER	1.8V	PP3V3R1V8_ENET_LR_OUT_REG	37
I795	POWER	POWER	3.3V	PPVBAT_G3_SYSCLK_R	79
I800	POWER	POWER	3.3V	USB_HUB1_VDDPLL3V3	34
I796	POWER	POWER	3.3V	USB_HUB1_VDDA3V3	34
I799	POWER	POWER	3.3V	USB_HUB2_VDDPLL3V3	35
I790	POWER	POWER	3.3V	USB_HUB2_VDDA3V3	35
I216	POWER	POWER	3.3V	PP3V3R1V5_PCH_VCCSUSHDA	22 24
I218	POWER	POWER	3.3V	PP3V3_S0_CK505_F	26
I221	POWER	POWER	3.3V	PP3V3_S0_HS_F	62
I225	POWER	POWER	3.3V	PP3V3_S0_SW_SD_PWR	45
I223	POWER	POWER	3.3V	PP3V3_S0_PCH_VCCA_DAC_F	17 22
I839	POWER	POWER	3.3V	PP3V3_S0	6
I826	POWER	POWER	3.3V	PP3V3_SMCUSBMUX	43
I224	POWER	POWER	3.3V	PP3V3_S3	6 98
I669	POWER	POWER	3.3V	PP3V3_S3_BT_FLT	44
I557	POWER	POWER	3.3V	PP3V3_S3_MINI_CONN	33
I620	POWER	POWER	3.3V	PP3V3_S3_SDCARD_FLT	44
I748	POWER	SWITCHNODE	3.3V	PP3V3_S3_ENET_PHY_FET	36
I663	POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_XTALVDDH	37
I664	POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_BIASVDDH	37
I665	POWER	POWER	3.3V	PP3V3_S3_ENET_PHY_AVDDH	37
I747	POWER	POWER	3.3V	ENET_XTALVDDH	37
I235	POWER	POWER	3.3V	PP3V3_S5	6
I758	POWER	POWER	3.3V	PP3V3_S5_REG_R	70
I276	POWER	POWER	3.3V	PPVOUT_S0_PCH_DCPSST	22
I266	POWER	POWER	3.3V	PPVOUT_PCH_DCPSUSBYP	22
I262	POWER	POWER	3.3V	PPVOUT_G3_PCH_DCPRTC	22
I257	POWER	POWER	3.3V	PPVBATT_G3_RTC	27
I258	POWER	POWER	3.3V	PPVBATT_G3_RTC_R	27
I207	POWER	POWER	3.3V	PP3V3_AUDIO_SPDIF_JACK	60 98
I209	POWER	POWER	3.3V	PP3V3_FW_FWPHY_VDDA	39
I211	POWER	POWER	3.3V	PP3V3_FW_FWPHY_VP25	39
I819	POWER	POWER	3.3V	PP3V3_S0_T29	6
I611	POWER	POWER	3.3V	PP3V3_SW_DPAPWR	78 84
I612	POWER	POWER	3.3V	PP3V3_SW_DPBPWR	78
I622	POWER	POWER	3.3V	PP3V3_G3H_RTC	18 19 22 27
I233	POWER	POWER	3.3V	PP3V3_G3H_AVREF_SMC	46 47
I238	POWER	POWER	3.4V	PP3V3_G3H_SMC_AVCC	46
I606	POWER	POWER	3.42V	PP3V42_G3H	6
I577	POWER	POWER	4.5V	4V5_REG_IN	56
I231	POWER	POWER	4.5V	PP4V5_AUDIO_ANALOG	56 61
I670	POWER	POWER	4.5V	PP5V_AUDIO_HPAMP	56
I880	POWER	POWER	5V	PP5V_LAMP_VREG	56 59
I881	POWER	POWER	5V	PP5V_RAMP_VREG	58
I237	POWER	POWER	5V	PP5V_S0	6 98
I239	POWER	POWER	5V	PP5V_S0_CPU_VCORE_VCC	65
I240	POWER	POWER	5V	PP5V_S0_PCH_V5REF	24
I656	POWER	POWER	5V	PP5V_S0_SATA_FET	44
I681	POWER	POWER	5V	5V_S0_P1V05REG_VIN	68
I763	POWER	POWER	5V	5V_S0_P1V05REG_VDD	68
I236	POWER	POWER	5V	PP5V_S3	6 98
I730	POWER	POWER	5V	PP5V_S3_VREFMRGN_A	28
I729	POWER	POWER	5V	PP5V_S3_VREFMRGN_B	28
I242	POWER	POWER	5V	PP5V_S3_DDR_REG_V5FILT	71
I243	POWER	POWER	5V	PP5V_S3_CAMERA_FLT	44
I633	POWER	POWER	5V	PP5V_S3_REG_R	70
I786	POWER	POWER	5V	PP5V_S3_IR_FLT	44 98
I234	POWER	POWER	5V	PP5V_S5	6
I245	POWER	POWER	5V	PP5V_S5_PCH_V5REFSUS	24
I247	POWER	POWER	5V	PP5V_USB_PORT0	43
I246	POWER	POWER	5V	PP5V_USB_PORT0_F	43
I250	POWER	POWER	5V	PP5V_USB_PORT1	43
I248	POWER	POWER	5V	PP5V_USB_PORT1_F	43
I251	POWER	POWER	5V	PP5V_USB_PORT2	43
I249	POWER	POWER	5V	PP5V_USB_PORT2_F	43
I253	POWER	POWER	5V	PP5V_USB_PORT3	43
I252	POWER	POWER	5V	PP5V_USB_PORT3_F	43
I548	POWER	POWER		DDR_REG_PGND	71
I546	POWER	POWER		DDR_REG_CSGND	71
I833	POWER	POWER	3.3V	PPVDDIO_25M_B	79
I834	POWER	POWER	1.05V	PPVDDIO_25M_A	79
I835	POWER	POWER	3.3V	PPVDDIO_25M_C	79
I836	POWER	POWER	12V	PP3V3R12V_SW_DPAPWR_R	85
I837	POWER	POWER	12V	PP3V3R12V_SW_DPAPWR_D	82
I838	POWER	POWER	1.5V	PP5V_S0_DP_R	83

SENSING NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING		
I155	SNS_DIFF	THERMAL	VR_CPU_ISNS1_P	65 66
I156	SNS_DIFF	THERMAL	VR_CPU_ISNS1_N	65 66
I158	SNS_DIFF	THERMAL	VR_CPU_ISNS1_R_N	65
I160	SNS_DIFF	THERMAL	VR_CPU_ISNS2_P	65 66
I159	SNS_DIFF	THERMAL	VR_CPU_ISNS2_N	65 66
I163	SNS_DIFF	THERMAL	VR_CPU_ISNS2_R_N	65
I162	SNS_DIFF	THERMAL	VR_CPU_ISNS3_P	65 66
I164	SNS_DIFF	THERMAL	VR_CPU_ISNS3_N	65 66
I166	SNS_DIFF	THERMAL	VR_CPU_ISNS3_R_N	65
I788	SNS_DIFF	THERMAL	VR_CPU_ISNS4_P	
I789	SNS_DIFF	THERMAL	VR_CPU_ISNS4_N	
I790	SNS_DIFF	THERMAL	VR_CPU_ISNS4_R_N	
I683	SNS_DIFF	THERMAL	P1V05S0_ISN	68
I684	SNS_DIFF	THERMAL	P1V05S0_ISP	68
I621	SNS_DIFF	THERMAL	VR_AXG_ISNS_R_P	65
I622	SNS_DIFF	THERMAL	VR_AXG_ISNS_R_N	65
I623	SNS_DIFF	THERMAL	VR_AXG_ISNS_N	65 67
I625	SNS_DIFF	THERMAL	VR_AXG_ISNS_P	65 67
I685	SNS_DIFF	THERMAL	VR_ISNS_1V05_N	94
I686	SNS_DIFF	THERMAL	VR_ISNS_1V05_P	94
I707	SNS_DIFF	THERMAL	VR_ISNS_VAXG_N	50 94
I708	SNS_DIFF	THERMAL	VR_ISNS_VAXG_P	50 94
I710	SNS_DIFF	THERMAL	VR_ISNS_VCORE_N	50 94
I709	SNS_DIFF	THERMAL	VR_ISNS_VCORE_P	50 94
I172	SNS_DIFF		CPU_VAXG_SENSE_P	13 65
I171	SNS_DIFF		CPU_VAXG_SENSE_N	13 65
I687	SNS_DIFF		VR_AXG_VSNS_R_N	65
I688	SNS_DIFF		VR_AXG_VSNS_R_P	65
I58	SNS_DIFF		VR_AXG_VSEN	65
I59	SNS_DIFF		VR_AXG_RGND	65
I713	SNS_DIFF		VR_AXG_VSNS_XW_N	65
I714	SNS_DIFF		VR_AXG_VSNS_XW_P	65
I615	SNS_DIFF		CPU_VCC_SENSE_N	13 65
I616	SNS_DIFF		CPU_VCC_SENSE_P	13 65
I711	SNS_DIFF		VR_CPU_VSEN	65
I712	SNS_DIFF		VR_CPU_RGND	65
I151	SNS_DIFF		VR_CPU_VSNS_R_N	65
I152	SNS_DIFF		VR_CPU_VSNS_R_P	65
I153	SNS_DIFF		VR_CPU_VSNS_XW_P	65
I154	SNS_DIFF		VR_CPU_VSNS_XW_N	65
I56	SNS_DIFF		CPU_VCCIO_SENSE_P	13 68
I57	SNS_DIFF		CPU_VCCIO_SENSE_N	13 68
I419	SNS_DIFF		P1V05S0_VSEN	68
I413	SNS_DIFF		P1V05S0_RTN	68

VR CTRL NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING		
I278	VR_CTL_PHY	SWITCHNODE	VR_CPU_PH1_SNUB	66
I338	VR_CTL_PHY	SWITCHNODE	VR_CPU_PH2_SNUB	66
I339	VR_CTL_PHY	SWITCHNODE	VR_CPU_PH3_SNUB	66
I784	VR_CTL_PHY	SWITCHNODE	VR_CPU_PH4_SNUB	
I695	VR_CTL_PHY	VR_CTL	VR_CPU_COMP	65
I694	VR_CTL_PHY	VR_CTL	VR_AXG_TM	65
I624	VR_CTL_PHY	VR_CTL	VR_AXG_PWM_R	65
I619	VR_CTL_PHY	VR_CTL	VR_AXG_PWM	65 67
I620	VR_CTL_PHY	VR_CTL	VR_CPU_PWM2	65 66
I626	VR_CTL_PHY	VR_CTL	VR_CPU_PWM1	65 66
I341	VR_CTL_PHY	VR_CTL	VR_CPU_PWM1_R	65
I343	VR_CTL_PHY	VR_CTL	VR_CPU_PWM2_R	65
I344	VR_CTL_PHY	VR_CTL	VR_CPU_PWM3	65 66
I345	VR_CTL_PHY	VR_CTL	VR_CPU_PWM3_R	65
I785	VR_CTL_PHY	VR_CTL	VR_CPU_PWM4	
I347	VR_CTL_PHY	VR_CTL	VR_CPU_PWM4_R	65
I348	VR_CTL_PHY	VR_CTL	VR_AXG_SW_FREQ	65
I349	VR_CTL_PHY	VR_CTL	VR_EN_PWR_OVP	65
I350	VR_CTL_PHY	VR_CTL	VR_RSET	65
I353	VR_CTL_PHY	VR_CTL	VR_EN_PWR_OVP_R	65
I352	VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT1_RC	66
I351	VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT2_RC	66
I354	VR_CTL_PHY	VR_CONTROL	VR_CPU_BOOT3_RC	66
I696	VR_CTL_PHY	VR_CONTROL	VR_AXG_BOOT1_RC	67
I357	VR_CTL_PHY	VR_CTL	VR_CPU_BOOT4_RC	
I359	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_BOOT	66
I360	VR_CTL_PHY	VR_CTL	VR_CPU_DRV1_GDSEL	66
I363	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_LGATE	66
I361	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV1_UGATE	66
I365	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_BOOT	66
I367	VR_CTL_PHY	VR_CTL	VR_CPU_DRV2_GDSEL	66
I368	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_LGATE	66
I370	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV2_UGATE	66
I371	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_BOOT	66
I374	VR_CTL_PHY	VR_CTL	VR_CPU_DRV3_GDSEL	66
I376	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_LGATE	66
I377	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV3_UGATE	66
I780	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_BOOT	
I781	VR_CTL_PHY	VR_CTL	VR_CPU_DRV4_GDSEL	
I783	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_LGATE	
I782	VR_CTL_PHY	VR_CONTROL	VR_CPU_DRV4_UGATE	
I690	VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_BOOT	67
I691	VR_CTL_PHY	VR_CTL	VR_AXG_DRV1_UGATE_R	67
I692	VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_LGATE	67
I693	VR_CTL_PHY	VR_CONTROL	VR_AXG_DRV1_UGATE	67
I387	VR_CTL_PHY	VR_CTL	VR_CPU_SW_FREQ	65
I388	VR_CTL_PHY	VR_CTL	VR_CPU_FB	65
I389	VR_CTL_PHY	VR_CTL	VR_CPU_FB_R	65
I390	VR_CTL_PHY	VR_CTL	VR_CPU_RAMP_ADJ	65
I393	VR_CTL_PHY	VR_CTL	VR_CPU_IMON	50 65
I635	VR_CTL_PHY	VR_CTL	VR_CPU_SUTH	65
I637	VR_CTL_PHY	VR_CTL	VR_CPU_N_PSI	65
I638	VR_CTL_PHY	VR_CTL	VR_CPU_FDVID	65
I641	VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP	65
I639	VR_CTL_PHY	VR_CTL	VR_CPU_IAUTO	65
I642	VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP1	65
I645	VR_CTL_PHY	VR_CTL	VR_CPU_PSICOMP2	65
I644	VR_CTL_PHY	VR_CTL	VR_CPU_FB_RC	65
I647	VR_CTL_PHY	VR_CTL	VR_CPU_FB2	65
I646	VR_CTL_PHY	VR_CTL	VR_CPU_IMON_R	65
I649	VR_CTL_PHY	VR_CTL	VR_CPU_HFREQ_COMP	65
I689	VR_CTL_PHY	SWITCHNODE	VR_AXG_PH1_SNUB	67
I627	VR_CTL_PHY	VR_CTL	VR_AXG_IMON	50 65
I650	VR_CTL_PHY	VR_CTL	VR_AXG_IMON_R	65
I628	VR_CTL_PHY	VR_CTL	VR_AXG_HFREQ_COMP	65
I630	VR_CTL_PHY	VR_CTL	VR_AXG_FB	65
I629	VR_CTL_PHY	VR_CTL	VR_SEN_R1	65
I631	VR_CTL_PHY	VR_CTL	VR_SEN_R2	65
I634	VR_CTL_PHY	VR_CTL	VR_SEN_R3	65
I632	VR_CTL_PHY	VR_CTL	VR_AXG_COMP_RC	65
I633	VR_CTL_PHY	VR_CTL	VR_AXG_COMP	65
I636	VR_CTL_PHY	VR_CTL	VR_AXG_TCOMP	65
I394	VR_CTL_PHY	SWITCHNODE	P1V05S0_UGATE	68
I396	VR_CTL_PHY	SWITCHNODE	P1V05S0_LGATE	68
I395	VR_CTL_PHY	VR_CTL	P1V05S0_BOOT	68
I682	VR_CTL_PHY	VR_CTL	P1V05S0_BOOT_R	68
I716	VR_CTL_PHY	VR_CTL	P1V05_IMON	50 68
I398	VR_CTL_PHY	VR_CTL	P1V05S0_FB	68
I399	VR_CTL_PHY	VR_CTL	P1V05S0_PHASE_L	68
I594	VR_CTL_PHY	VR_CTL	P1V05S0_VW	68
I403	VR_CTL_PHY	VR_CTL	P1V05S0_COMP	68
I402	VR_CTL_PHY	VR_CTL	P1V05S0_ICOMP	68
I401	VR_CTL_PHY	VR_CTL	P1V05_S0_VDIFF	68
I404	VR_CTL_PHY	VR_CTL	P1V05S0_COMP_C	68
I406	VR_CTL_PHY	VR_CTL	P1V05S0_VDIF_C	68
I405	VR_CTL_PHY	VR_CTL	P1V05S0_VO	68
I408	VR_CTL_PHY	VR_CTL	P1V05S0_OCSET	68
I412	VR_CTL_PHY	VR_CTL	P1V05S0_SNUBBER	68
I730	VR_CTL_PHY	VR_CTL	P1V05_REG_VFB	
I760	VR_CTL_PHY	VR_CTL	ENET_SR_VFB	36 37
I762	VR_CTL_PHY	VR_CTL	P1V05S0_ISP_R	68
I805	VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2_R	69 95
I808	VR_CTL_PHY	VR_CTL	T29BST_FBX	
I607	VR_CTL_PHY	VR_CTL	P3V42G3H_BOOST	72
I608	VR_CTL_PHY	VR_CTL	P3V42G3H_FB	72

VR CTRL NET PROPERTIES

	NET_TYPE PHYSICAL	SPACING		
I437	VR_CTL_PHY	VR_CTL	DDR_REG_CS	71
I595	VR_CTL_PHY	VR_CTL	DDR_REG_FB	71
I439	VR_CTL_PHY	VR_CONTROL	DDR_REG_LGATE	71
I596	VR_CTL_PHY	VR_CONTROL	DDR_REG_UGATE	71
I443	VR_CTL_PHY	VR_CONTROL	DDR_REG_BOOT	71
I442	VR_CTL_PHY	VR_CONTROL	DDR_REG_BOOT_R	71
I441	VR_CTL_PHY	VR_CTL	DDR_REG_VDDQSNS	71
I444	VR_CTL_PHY	VR_CTL	DDR_REG_VTTSNS	71
I728	VR_CTL_PHY	VR_CTL	DDR_REG_PHASE_R	71
I776	VR_CTL_PHY	VR_CTL	DDR_REG_UGATE_R	71
I729	VR_CTL_PHY	SWITCHNODE	1V5_SNUBBER	71
I726	VR_CTL_PHY	VR_CTL	P1V8_REG_VFB	71
I727	VR_CTL_PHY	VR_CTL	P1V8_REG_SYNC	71
I450	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_BOOT	70
I452	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_BOOT_R	70
I451	VR_CTL_PHY	VR_CTL	P3V3S5_REG_FB	70
I453	VR_CTL_PHY	VR_CTL	P3V3S5_REG_ISEN	70
I455	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_LGATE	70
I454	VR_CTL_PHY	VR_CTL	P3V3S5_REG_OCSET	70
I768	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_VOUT1	70
I770	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_FSET1	70
I458	VR_CTL_PHY	VR_CONTROL	P3V3S5_REG_UGATE	70
I457	VR_CTL_PHY	VR_CTL	P3V3S5_REG_SNUB	70
I459	VR_CTL_PHY	VR_CONTROL	P5VS3_REG_BOOT	70
I461	VR_CTL_PHY	VR_CTL	P5VS3_REG_FB	70
I462	VR_CTL_PHY	VR_CTL	P5VS3_REG_ISEN	70
I463	VR_CTL_PHY	VR_CONTROL	P5VS3_REG_LGATE	70
I464	VR_CTL_PHY	VR_CTL	P5VS3_REG_OCSET	70
I769	VR_CTL_PHY	VR_CTL	P5VS3_REG_FSET2	70
I466	VR_CTL_PHY	VR_CONTROL	P5VS3_REG_UGATE	70
I767	VR_CTL_PHY	VR_CONTROL	P5VS3_REG_VOUT2	70
I718	VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT1	69
I719	VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2	69
I755	VR_CTL_PHY	VR_CTL	VCCSA_CNTRL_INPUT2_R	69 95
I720	VR_CTL_PHY	VR_CTL	VCCSA_OUT	69
I721	VR_CTL_PHY	VR_CTL	VCCSA_GATE	69
I830	VR_CTL_PHY	VR_CTL	VCCSA_REF	69
I831	VR_CTL_PHY	VR_CTL	VCCSA_CRL	69
I723	VR_CTL_PHY	VR_CTL	CPU_VCCSA_SENSE	13 69
I791	VR_CTL_PHY	VR_CTL	VR_CPU_TM	65
I832	VR_CTL_PHY	PM	3V3R2V9_DPAPWR_ADJ	82 97

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VID_PHY	*	50_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL	*	0.35MM	?

VR VID NET PROPERTIES

PULL-UP STUB < 1-INCH
VID LENGTH SKEW < 1-INCH
VID LENGTH RANGE< 1 TO 15-INCH

	NET_TYPE PHYSICAL	SPACING		
I771	VID_PHY	VR_CTL_VIDS	CPU_VIDSCLK_R	13
I772	VID_PHY	VR_CTL_VIDS	CPU_VIDALERT_L_R	13
I773	VID_PHY	VR_CTL_VIDS	CPU_VIDSOUT_R	13
I491	VID_PHY	VR_CTL_VIDS	CPU_VIDSCLK	13 65
I492	VID_PHY	VR_CTL_VIDS	CPU_VIDALERT_L	13 65
I521	VID_PHY	VR_CTL_VIDS	CPU_VIDSOUT	13 65

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_VIDS	*	=4X_DIELECTRIC	?

SYNC_MASTER=K62 SYNC_DATE=01/06/2011

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POWER CONSTRAINTS

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T29 ELECTRICAL ROUTES

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29	*	=5X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29	TOP,BOTTOM	=7X_DIELECTRIC	?

T29 PCI-EXPRESS (SAME RULE AS PCIE)

T29 SPI INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SPI	*	0.2 MM	?

T29 XTAL CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_XTAL_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_XTAL	*	=4X_DIELECTRIC	?

T29 SMBUS INTERFACE CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_SMB	*	=2x_DIELECTRIC	?

GREEN CLOCK CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

T29 BIAS CONSTRAINTS

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
T29_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
T29_COMP	*	0.2 MM	?

T29 NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_P<3..0>	78 84 86
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_N<3..0>	78 84 86
NO_TEST=TRUE	T29_90D	T29	T29_D2R_C_P<3..0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29_D2R_C_N<3..0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29_R2D_P<3..0>	84
NO_TEST=TRUE	T29_90D	T29	T29_R2D_N<3..0>	84
NO_TEST=TRUE	T29_90D	T29	T29_D2R_P<3..0>	78 84 86
NO_TEST=TRUE	T29_90D	T29	T29_D2R_N<3..0>	78 84 86
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_F_P<3..0>	84
NO_TEST=TRUE	T29_90D	T29	T29_R2D_C_F_N<3..0>	84
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_P<3..0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_N<3..0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_P<3..0>	
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_N<3..0>	
NO_TEST=TRUE	T29_90D	T29	DP_A_EXT_AUXCH_P	84 85
NO_TEST=TRUE	T29_90D	T29	DP_A_EXT_AUXCH_N	84 85
NO_TEST=TRUE	T29_90D	T29	DP_SDRVA_AUXCH_C_P	84
NO_TEST=TRUE	T29_90D	T29	DP_SDRVA_AUXCH_C_N	84
NO_TEST=TRUE	T29_90D	T29	DP_SDRVB_AUXCH_C_P	
NO_TEST=TRUE	T29_90D	T29	DP_SDRVB_AUXCH_C_N	
NO_TEST=TRUE	T29_90D	T29	DP_B_EXT_AUXCH_P	
NO_TEST=TRUE	T29_90D	T29	DP_B_EXT_AUXCH_N	
NO_TEST=TRUE	T29_90D	T29	T29DPA_D2R1_AUXCH_P	85
NO_TEST=TRUE	T29_90D	T29	T29DPA_D2R1_AUXCH_N	85
NO_TEST=TRUE	T29_90D	T29	T29DPB_D2R3_AUXCH_P	
NO_TEST=TRUE	T29_90D	T29	T29DPB_D2R3_AUXCH_N	
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_N<0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_P<0>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_N<2>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPA_ML_C_P<2>	84 85
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_N<0>	
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_P<0>	
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_N<2>	
NO_TEST=TRUE	T29_90D	T29	T29DPB_ML_C_P<2>	
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_P<3..0>	86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_N<3..0>	86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_P<3..0>	18 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_R2D_C_N<3..0>	18 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_P<3..0>	18 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_N<3..0>	18 86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_C_P<3..0>	86
NO_TEST=TRUE	PCIE_85D	PCIE	PCIE_T29_D2R_C_N<3..0>	86
NO_TEST=TRUE	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_P	18 86
NO_TEST=TRUE	CLK_PCIE_90D	CLK_PCIE	PCIE_CLK100M_T29_N	18 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_P<3..0>	83
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_C_N<3..0>	83
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_P<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_ML_N<3..0>	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_P<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_N<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_P	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_N	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_P<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_ML_C_N<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_P	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK0_AUXCH_C_N	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_P<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_N<3..0>	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_P	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_N	86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_P<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_ML_C_N<3..0>	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_P	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SNK1_AUXCH_C_N	78 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_R_C_P	83
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_R_C_N	83
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_C_P	83 86
NO_TEST=TRUE	DP_85D	DISPLAYPORT	DP_T29SRC_AUXCH_C_N	83 86
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_AUXCH_P	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_AUXCH_N	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_AUXCH_P	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_AUXCH_N	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_C_P<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_C_N<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_P<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_N<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_R_P<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVA_ML_R_N<3..0>	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_C_P<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_C_N<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_P<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_N<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_R_P<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	DP_SDRVB_ML_R_N<3..0>	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_A_RSVD_N	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_A_RSVD_P	84
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_B_RSVD_N	
NO_TEST=TRUE	T29_90D	DISPLAYPORT	T29_B_RSVD_P	

T29 NET PROPERTIES

ELECTRICAL_CONSTRAINT_SET	NET_TYPE PHYSICAL	NET_TYPE SPACING	NET	PAGES
	T29_SPI_55S	T29_SPI	JTAG_T29_TDI	15 21 86
	T29_SPI_55S	T29_SPI	JTAG_T29_TMS	15 18 86
	T29_SPI_55S	T29_SPI	JTAG_T29_TCK	15 21 25 86
	T29_SPI_55S	T29_SPI	JTAG_T29_TDO	15 21 86
I437	T29_SPI_55S	T29_SPI	T29_SPI_MOSI	86
I438	T29_SPI_55S	T29_SPI	T29_SPI_MISO	86
I439	T29_SPI_55S	T29_SPI	T29_SPI_CS_L	86
I440	T29_SPI_55S	T29_SPI	T29_SPI_CLK	86
	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	79 86 96
	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	86 96
	T29_SMB_55S	T29_SMB	I2C_T29_SDA	49 86
	T29_SMB_55S	T29_SMB	I2C_T29_SCL	49 86
I421	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_SB	79
I422	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_ENET	79
I423	CLK_25M_55S	CLK_25M	ENET_CLK25M_XTALI_OSC	36 79
I424	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_CLK	79
I433	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29	79 86 96
I425	CLK_25M_55S	CLK_25M	SYSCLK_CLK25M_T29_R	86 96
I426	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X2	79
I428	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X2_R	79
I427	T29_XTAL_100D	T29_XTAL	SYSCLK_CLK25M_X1	79
I431	T29_55S	T29_COMP	T29_RSENSE	86
I432	T29_55S	T29_COMP	T29_RBIAS	86
I434		T29_COMP	T29_A_BIAS	82 84 85
I435		T29_COMP	T29_B_BIAS	
I436		T29_COMP	DP_A_BIAS	84

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T29 CONSTRAINTS

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PM NET PROPERTIES
(PM, RESET, EN, PGOOD)

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PM	*	*	2:1_SPACING
PM_VTT	PM_VTT	*	2:1_SPACING
PM_VTT	*	*	3:1_SPACING
PM_VTT	GND	*	DEFAULT
PM	GND	*	DEFAULT

PHYSICAL	NET_TYPE SPACING	NET	PAGES
I357	PM	4V5_REG_EN	56
I358	PM	3V42G3H_SHDN_L	72
I359	PM	ALL_SYS_PWRGD_R	5 32 64
I361	PM	ALL_SYS_PWRGD_SMC	46 64
I360	PM	AP_PWR_EN	20 25 33
I362	PM	AP_MINI_RESET_L	33
I364	PM	AUD_I2C_INT_L	20 62
I363	PM	AUD_IP_PERIPHERAL_DET	20 61
I365	PM	AUD_IPHS_SWITCH_EN	21 62
I406	PM	AUD_SPDIF_IN	60 83 91
I407	PM	AUD_SPDIF_IN_CODEC	56 83
I404	PM	BDV_BKL_PWM	46 83 97
I401	PM	BL_PWM	6 83
I410	PM	BL_EN	6 81
I411	PM	BDV_BKL_PWM	46 83 97
I366	PM	CK505_27MHZ_EN	26
I367	PM	CPUVTT_REG_EN	
I369	PM_VTT	CPUVTT_REG_PGOOD_R	63
I368	PM	CPU_MEM_RESET_L	11 32
I372	PM	CPU_PECI_R	46
I370	PM_VTT	CPU_PWRGD	11 21 25
I371	PM	CPU_RESET_L	11 27
I374	PM	CPU_SKTOCC_L	11 63
I373	PM	CPU_CATERR_L	11
I375	PM	CPU_PECI	11 21 46
I376	PM	CPU_PROCHOT_L	11 47 65
I377	PM	CPU_THRMTRIP_L	11 47
I378	PM	CPU_PROC_SEL	11 19
I379	PM	DEBUG_RESET_L	27 48
I380	PM	DDRVTT_EN	
I382	PM	DP_INT_SPDIF_AUDIO	81 83
I381	PM	DP_INTPNL_HPD	81 83
I385	PM	3V3R2V9_DPAPWR_ADJ	82 95
I384	PM	DP_A_PWRDWN	84
I383	PM	DP_A_PWRDWN_FET_R	82
I386	PM	DP_A_PWRDWN_INV	82
I388	PM	DPAPWRSW_HVEN_L_R	82
I387	PM	DPAPWRSW_CT	82
I389	PM	DPAPWRSW_ILIM	82
I391	PM	DPAPWRSW_IFLT	82
I390	PM	T29_A_HV_EN	82 84
I392	PM	3V3R2V9_DPBPWR_ADJ	
I395	PM	DP_B_PWRDWN	
I393	PM	DP_B_PWRDWN_FET_R	
I394	PM	DP_B_PWRDWN_INV	
I398	PM	DPBPWRSW_HVEN_L_R	
I396	PM	DPBPWRSW_CT	
I397	PM	DPBPWRSW_ILIM	
I399	PM	DPBPWRSW_IFLT	
I400	PM	T29_B_HV_EN	
I420	PM	T29_PWR_EN	18 80 97
I421	PM	T29_RESET_RTR_L	80 86
I412	PM	LCD_BL_FILT	83
I402	PM	LCD_BLK_ON_DLY	
I403	PM	LCD_BL_PWM	83
I405	PM	MXM_PNL_BL_PWM	76 83

PHYSICAL	NET_TYPE SPACING	NET	PAGES
I250	PM	ENET_PWR_EN	20 25 36
I227	PM	ENET_LOW_PWR	15 21 37
I229	PM	FW_RESET_L	27 39
I231	PM	ENET_RESET_L	27 36
I236	PM	FW_PME_L	15 21 39
I237	PM	FW_PWR_EN	15 21
I238	PM	FW_CLKREQ_L	15 39
I248	PM	ISOLATE_CPU_MEM_L	21 25 32
I266	PM	LPC_PWRDWN_L	19 46 48
I265	PM	MEM_RESET_L	30 31 32 89
I269	PM	MINI_CLKREQ_L	15 33
I268	PM	MINI_RESET_L	27 33
I267	PM	MXM_CLKREQ_L	9 75
I270	PM	MXM_GOOD	5 21 25
I271	PM	ODD_PWR_EN_L	15 21 42
I68	PM	RTC_RESET_L	18 27 97
I48	PM	RSMRST_PWRGD	46 64
I47	PM	RTC_RESET_L	18 27 97
I319	PM	S4_ENABLES	63
I280	PM	SDCONN_STATE_RST_L	
I281	PM	SDCONN_DETECT_BUF_L	92
I282	PM	SDCONN_STATE_CHANGE	20 25 45
I283	PM	SDCARD_RESET	15 21 44 98
I315	PM	SDCARD_RESET_L	44
I284	PM	SDCARD_PLT_RST_L	27 44
I350	PM	SDCARD_PLT_RST_L_R	
I351	PM	SMC_PM_G2_EN	46 74
I352	PM	SMC_PM_G2_EN_R	74
I353	PM	SMC_PM_G2_EN_L	74
I354	PM	S5_DG_1	74
I355	PM	S5_MSFT_G1	74
I415	PM	USE_HDD_OOB_L	20 51
I414	PM	HDD_OOB_1V00_REF	51
I285	PM	SMC_ADAPTER_EN	19 46 47
I286	PM	SMC_RUNTIME_SCI_L	21 46 47
I287	PM	SMC_WAKE_SCI_L	15 18 21 46
I288	PM	SMC_DELAYED_PWRGD	47 64
I289	PM	SMC_LRESET_L	27 46
I290	PM	SMC_RESET_L	46 47 48
I416	PM	SMC_PROCHOT	46 47
I417	PM	SMC_PROCHOT_3_3_L	46 47
I419	PM	SMC_ONOFF_L	46 47
I418	PM	SMC_MANUAL_RST_L	47
I291	PM	SPI_DESCRIPTOR_OVERRIDE_L	18 46
I413	PM	T29_PWR_EN	18 80 97
I260	PM	T29_RESET_L	27 80
I408	PM	T29_DP_PORTA_PWR_EN	20 25 82 91
I409	PM	T29_DP_PORTA_PWR_EN_REG	82
I259	PM_VTT	XDP_CPUPWRGD	
I261	PM_VTT	XDP_DBRESET_L	11 25
I263	PM_VTT	XDP_PWRGD	
I262	PM	XDPPCH_PLTRST_L	25 27
I264	PM	USB_HUB_SOFT_RESET_L	20 25 34
I323	PM	VSYNC_DP_CONN	6 81
I324	PM	VSYNC_DP	81
I325	PM	VIDEO_ON	81
I344	PM	VTT_REG_PGOOD_L	63

NET_TYPE SPACING	NET	PAGES
PM	PLT_RESET_L	20 27
PM_VTT	PLT_RESET_LS1V05_L	11
PM	PM_BATLOW_L	15 19 46
PM	PM_CLK32K_SUSCLK	9 46 91
PM	PM_CLK32K_SUSCLK_R	9 19 91
PM	PM_CLKRUN_L	15 19 46 48
PM	PM_PWRBTN_L	19 25 46
PM	PM_RSMRST_L	27 46
PM	PM_RSMRST_PCH_L	19 27
PM	PCH_SRTCRST_L	18
PM	PCH_INTVRMEN_L	18
PM	PCH_DSWVRMEN	19
PM	PCH_DF_TVS	19
PM	PCH_PROCPWRGD	21
PM	PCIE_WAKE_L	19 33 36 78
PM	PM_DSW_PWRGD	19
PM	PM_ASW_PWRGD	19 64
PM	PM_MEM_PWRGD_R	11
PM	PM_EN_DDR1V5_S3_REG	63 71
PM	PM_EN_DDRVTT_S0_REG	32 63 71
PM	PM_EN_P12V_S0_FET	6 63
PM	PM_EN_P1V05_S0_REG	63 68
PM	PM_EN_P1V05_S3_REG	
PM	PM_EN_P1V5_S0_FET	63 73
PM	PM_EN_P1V8_S0_REG	63 71
PM	PM_EN_P3V3_S0_FET	63 73
PM	PM_EN_P3V3_S3_FET	63 73
PM	PM_EN_P3V3_S5_REG	78
PM	PM_EN_P5V_S0_FET	63 73
PM	PM_EN_P5V_S3_REG	63 78
PM	PM_EN_PVCCSA_S0_REG_L	64
PM	PM_EN_VCCSA_S0_CPU	
PM	PM_EN_PVCORE_CPU	63 65
PM_VTT	PM_MEM_PWRGD	11 19 97
PM	PM_MXM_EN	64 76
PM	PM_PCH_PWRGD_R	64
PM	PM_PECI_PWRGD	46 64
PM	PM_PECI_PWRGD_R	46
PM	PM_PGOOD_DDR1V5_S3_REG	5 63 71
PM	PM_PGOOD_P1V05_S0_REG	63 64 68
PM	PM_PGOOD_P1V5_S0_FET	11 64 73
PM	PM_PGOOD_P1V8_S0_REG	64 71
PM	PM_PGOOD_P3V3_S0_FET	63 64 73
PM	PM_PGOOD_P3V3_S3_FET	34 73
PM	PM_PGOOD_P3V3_S5_REG	27 64 78
PM	PM_PGOOD_P5V_S0_FET	63 64 73
PM	PM_PGOOD_MINI	33
PM	PM_PGOOD_PVCORE_CPU	5 25 64 65
PM	PM_PGOOD_PVCCSA_S0_REG	63 64
PM	PM_PGOOD_P5V_S3_REG	63 78 82
PM	PM_PGOOD_PVAXG	5 65
PM_VTT	PM_MEM_PWRGD	11 19 97
PM	PM_MEM_PWRGD_L	11
PM	PM_MXM_PGOOD	64 76
PM	PM_PCH_PWRGD	19 21 64
PM	PM_SLP_S3_5V	32
PM	PM_SLP_S3_5V_L	32
PM	PM_SLP_S3_5V_R2	32
PM	PM_SLP_S3_L	5 19 26 32 36 46 47 63 82
PM	PM_SLP_S4_L	5 19 32 46 47 63 97
PM	PM_SLP_S5_L	5 19 46 47 63
PM_VTT	PM_SYNC	11 19
PM	PM_SYSRST_L	19 25 27 46
PM	PM_SYS_PWRGD	19 32 64
PM_VTT	PM_THRMTRIP_L	21 47
PM	PM_SLP_S3_BUF_L	63
PM	PM_SLP_S4_1_L_R	63
PM	PM_SLP_S4_D_L	32
PM	PM_SLP_S4_L	5 19 32 46 47 63 97
PM	PGOOD_P1V5_S0_DLY	11
PM	PGOOD_1V8_S0_G1	64
PM	PGOOD_1V8_S0_G2	64
PM	PGOOD_P12V_S0	63 64
PM	PGOOD_P1V8_S0	64
PM	PGOOD_PCH_S0	5 64
PM	PGOOD_PCH_S0_R	64 91
PM	PGOOD_SYSPWROK	64
PM	PGOOD_SYSPWROK_R	64
PM	POWER_BUTTON_L	47
PM	PEG_RESET_L	9 27
PM	PGOOD_CPU_S0	64
PM	PGOOD_CPU_UNCORE	64 91
PM	PGOOD_5V_1V05_3V3	64 91
PM	PGOOD_3V3_1V05	64 91
PM	PGOOD_12V_S0_G1	64
PM	PGOOD_12V_S0_G2	64
PM	9V_COMP_REF	64
PM	12V_COMP_REF	64
PM	ALL_SYS_PWRGD	64 91

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PM RESETS ENABLES PGOOD CONST

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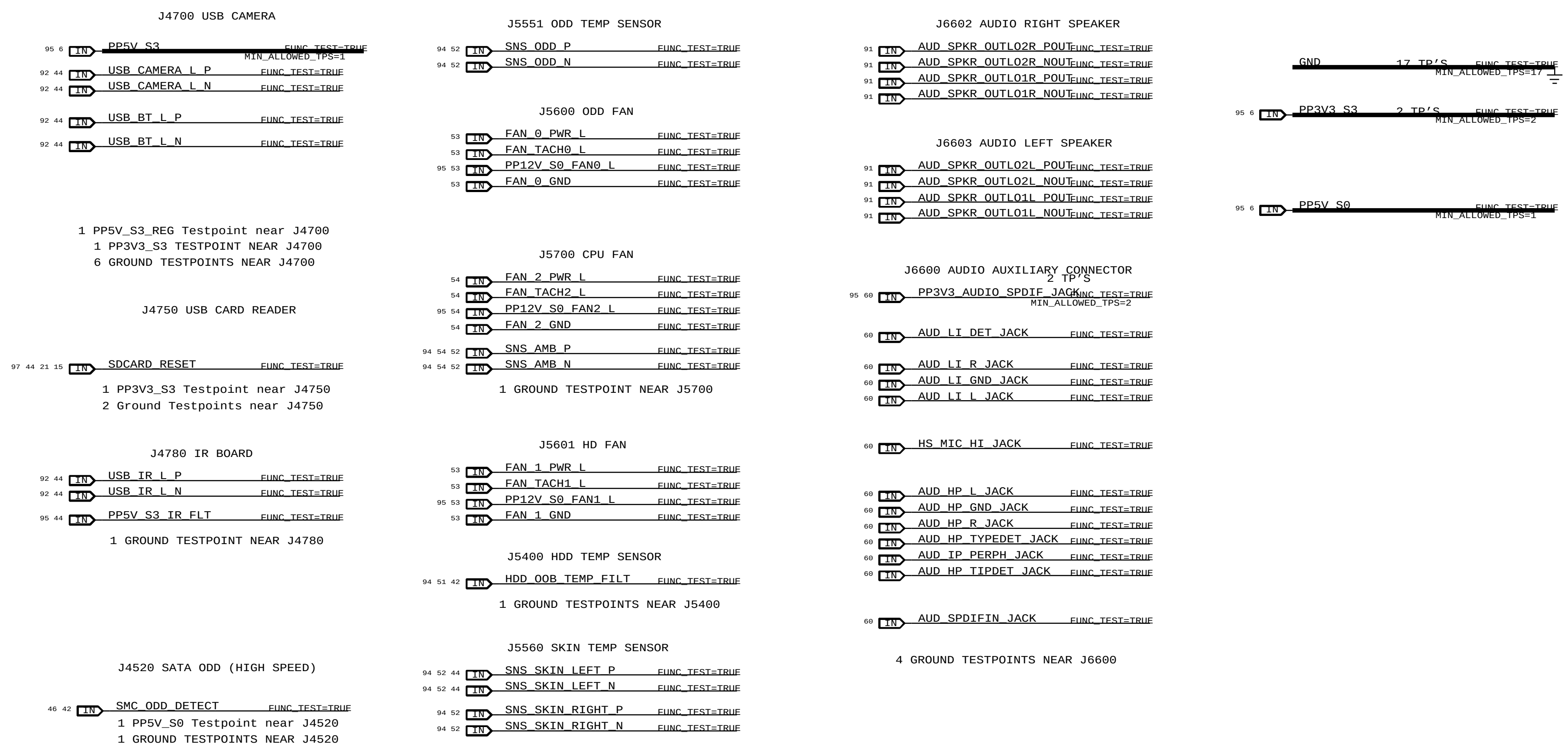
FUNCTIONAL TESTPOINTS FOR MAC-1 & ICT



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